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GMSL3/2 eDP Deserializers with Optional Decompression, HDCP, and Daisy-Chain

MAX96746/MAX96856/MAX96860/ MAX96862

General Description

The MAX96746/MAX96856/MAX96860/MAX96862 deserializers convert one or two GMSL™ serial input(s) to embedded DisplayPort (eDP) v1.4a. The devices support uncompressed 24bpp video up to 800MHz PCLK and 30bpp up to 640MHz.

The GMSL video input links operate at fixed rates of 12Gbps PAM4 in GMSL3™ mode and 6Gbps NRZ in GMSL2™ mode. The reverse link data rate is 187.5Mbps.

The devices also send and receive side-channel and peripheral control data to enable full-duplex single-wire transmission of video and bidirectional data.

Specific part numbers support DSC decompression, which supports maximum PCLKs of 1080MHz at 24bpp and 864MHz at 30bpp, over a single coax or twisted pair link.

Specific part numbers support HDCP 1.4 and HDCP 2.3 decryption of the local output video stream.

Specific part numbers support GMSL3/2 daisy-chaining of up to three additional video streams and HDCP pass-through to downstream deserializers. The video streams can have different resolutions and timings.

The concurrent control channel supports I^2C or UART. Two additional pass-through I^2C or UART channels and a pass-through SPI channel are provided for peripheral control. The audio channel supports forward I^2S stereo.

Applications

- Ultra-Wide Front Seat Infotainment Displays
- Rear-Seat Infotainment Displays

Ordering Information appears at end of data sheet.

Benefits and Features

- eDP Output Supports up to 1080MHz PCLK at 24bpp and 864MHz PCLK at 30bpp
- Multiple GMSL Link Rates for System Flexibility
  - 12Gbps or 6Gbps Single Lane Forward Link Rates
  - Up to 1200MHz PCLK over Daisy-Chain Output with DSC (Optional)
- Supports 24-Bit and 30-Bit Color Depth
- Decompression of 3:1 and 3.75:1 DSC (Optional)
- Bandwidth Support
  - Up to 215MHz/170MHz PCLK Uncompressed 24bpp/30bpp Video per GMSL2 Link
  - Up to 400MHz/320MHz PCLK Uncompressed 24bpp/30bpp Video per GMSL3 Link
  - Up to 800MHz/640MHz PCLK Uncompressed 24bpp/30bpp Video over Two GMSL3 links
  - Up to 600MHz PCLK Compressed 24bpp Video over Single GMSL2 Link
  - Up to 1080MHz PCLK Compressed 24bpp Video over Single GMSL3 Link
  - Up to 864MHz PCLK Compressed 30bpp Video over Single GMSL3 Link
  - 187Mbps Reverse Link Rate
- Synchronization of Two Devices for High Bandwidth Applications
- ASIL-B Compliant
- Full-Duplex Capability over a Single Wire
- HDCP 1.4 and 2.3 Decryption (Optional)
- Two Pass-Through I^2C/UART Channels, in Addition to Primary I^2C/UART Channel
- Local Forward I^2S Audio Output
- 8:10 Bit Color LUTs with FRC
- Concurrent Control Channel for Device Configuration and Communicating with Remote Peripherals
- Tunneled or Register-Programmable GPIOs
- Low-Current Sleep Mode
- Compact 8mm x 8mm TQFN Package with EP
- Operation Specified over Automotive Ambient Temperature Range of -40°C to +105°C
- AEC-Q100 Qualified
- Data can be Transmitted over Low-Cost 50Ω Coax or 100Ω STP Cables that meet the GMSL3/2 Channel Specification

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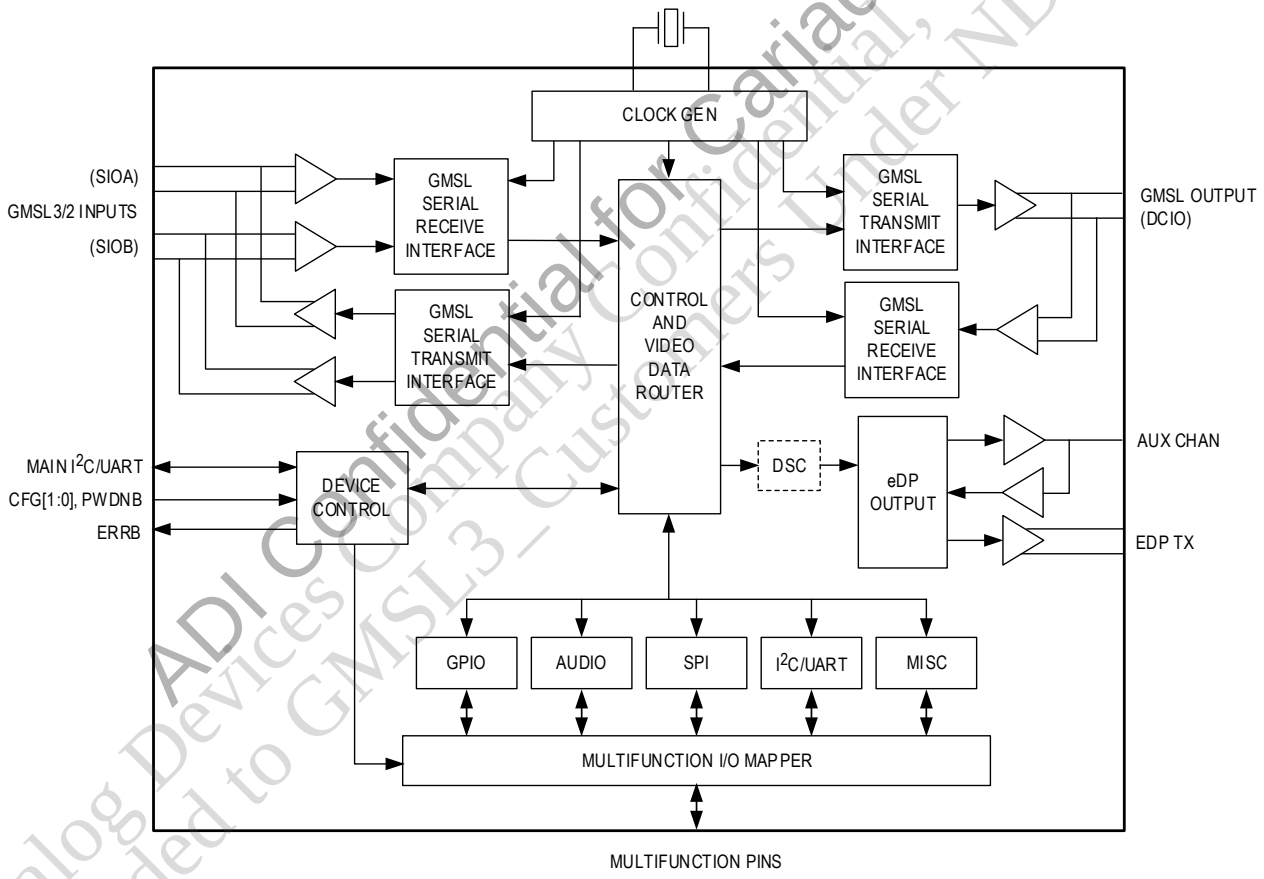
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Table 1. Typical Maximum Cable Length

	3.2mm Ø 50Ω Coax, Foam Dielectric	2.7mm Ø 50Ω Coax, Solid Dielectric	100Ω Shielded Twisted Pair, AWG26
Attenuation at 3GHz (Typ, Room Temp)	0.9dB/m	1.6dB/m	1.8dB/m
Attenuation at 3GHz (Max, Aged, 105°C)	1.1dB/m	2.0dB/m	2.2dB/m
GMSL Fwd/Rev Data Rate	**Typical Maximum Cable Length at 105°C**		
3Gbps/187.5Mbps	20m	10m	11m
6Gbps/187.5Mbps	15m	9m	8m
12Gbps/187.5Mbps	14m	8m	7m

Contact the factory to receive the GMSL2 and GMSL3 Channel Specification documents.

Simplified Block Diagram



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Absolute Maximum Ratings

(All voltages with respect to ground.) .....

V_{DDIO} .....-0.3V to +3.9V

V_{DD18} .....-0.3V to +2.0V

V_{DD} .....-0.3V to +1.1V

CAP_VDD .....-0.3V to +1.1V

SIO_ (Active State) (Note 1)..... (V_{DD18} - 1.1V) to V_{DD18}

SIO_ (Inactive State) (Note 1) .....-0.3V to +1.1V

DCIO_ .....-0.3V to (CAP_VDD + 0.3V)

eTX_ ..... -0.3V to +1.1V

AUXP/N .....−0.3V to (V_{DD18} + 0.3V)

XRES, X2, DIS_REM_CC, RSVD .....−0.3V to (V_{DD18} + 0.3V)

All Other Pins (Note 2).....−0.3V to (V_{DDIO} + 0.3V)

Continuous Power Dissipation (T_A = +70ºC) TQFN (Multilayer board, derate 48mW/ºC above +70ºC)..... 3800mW

Storage Temperature Range ..... -40°C to +150°C

Soldering Temperature (reflow) ..... +260°C

Note 1: Active state means the device is powered up and not in Sleep or Power-down modes. Inactive means the device is not powered up or powered up in Sleep or Power-down mode.

Note 2: Specified maximum voltage or 3.9V, whichever is lower.

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Package Information

56TQFN-EP

Package Code	T5688+5C
Outline Number	21-0135
Land Pattern Number	90-0046
Thermal Resistance, Four-Layer Board:	
Junction to Ambient (θ_{JA})	21ºC/W
Junction to Case (θ_{JA})	1ºC/W

SW-56TQFN-EP

Package Code	T5688Y+5C
Outline Number	21-100046
Land Pattern Number	90-100023
Thermal Resistance, Four-Layer Board:	
Junction to Ambient (θ_{JA})	21ºC/W
Junction to Case (θ_{JA})	1ºC/W

For the latest package outline information and land patterns (footprints), go to https://www.analog.com/en/resources/packaging-quality-symbols-footprints/package-index. Note that a "+", "#", or "-" in the package code indicates RoHS status only. Package drawings may show a different suffix character, but the drawing pertains to the package regardless of RoHS status.

Package thermal resistances were obtained using the method described in JEDEC specification JESD51-7, using a four-layer board in still air. For detailed information on package thermal considerations, refer to https://www.analog.com/en/resources/technical-articles/thermal-characterization-of-ic-packages.

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Electrical Characteristics

(V_{DD18} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, V_{DD} = 0.95V to 1.05V, T_A = -40°C to +105°C, EP connected to PCB ground. Typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DD} = 1.0V, T_A = +25°C, unless otherwise noted.) (*Note 3*)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
DC ELECTRICAL CHARACTERISTICS / REVERSE CHANNEL SERIAL OUTPUTS (SIO_)							
Output Voltage Swing (Single-Ended)	V_O	R_L = 50Ω ±1%	12Gbps Link Rate	110	150	190	mV
			6Gbps Link Rate	190	250	310	
Output Voltage Swing (Differential)	V_{ODT}	R_L = 100Ω ±1%, peak-to-peak differential voltage	12Gbps Link Rate	220	300	380	mV
			6Gbps Link Rate	380	500	620	
Change in V_{OD} between Complementary Output States	ΔV_{OD}	R_L = 100Ω ±1%, $\lvert V_{OD(H)}-V_{OD(L)}\rvert$				25	mV
Differential Output Offset Voltage	V_{OS}	R_L = 100Ω ±1%, offset voltage in each output state		V_{DD18} - 0.45		V_{DD18}-0.05	V
Change in V_{OS} between Complementary Output States	ΔV_{OS}	R_L = 100Ω ±1%, $\lvert V_{OS(H)}-V_{OS(L)}\rvert$				25	mV
Termination Resistance (Internal)	R_T	Any Pin to V_{DD18}		50	55	60	Ω
DC ELECTRICAL CHARACTERISTICS / GMSL DAISY-CHAIN FORWARD-CHANNEL SERIAL OUTPUTS (DCIO_)							
Output Voltage Swing (Single-Ended)	$\lvert V_O\rvert$	R_L = 50Ω ±1%, (V_{OH}-V_{OL}) for both outputs	PAM4	400	500	600	mV
			NRZ	300	400	500	
Output Voltage Swing (Differential)	V_{OD}	R_L = 100Ω ±1%, peak-to-peak differential voltage	PAM4	800	1000	1200	mV_{P-P}
			NRZ	600	800	1000	
Change in V_{OD} between Complementary Output States	$\Delta\lvert V_{OD}\rvert$	R_L = 100Ω ±1%, $\lvert V_{OD(H)}$-$V_{OD(L)}\rvert$				25	mV
Differential Output Offset Voltage	V_{OS}	R_L = 100Ω ±1%, offset voltage in each output state		0.4	0.5	0.6	V
Change in V_{OS} between Complementary Output States	ΔV_{OS}	R_L = 100Ω ±1%, $\lvert V_{OS(H)}$-$V_{OS(L)}\rvert$				25	mV
Termination Resistance (Internal)	R_T	Incremental resistance around V_{OS}		43	50	57	Ω
DC ELECTRICAL CHARACTERISTICS / eDP OUTPUTS (eTX__)							
Differential Peak-to-Peak Output Voltage	V_{OD}	R_L = 100Ω ±1% V_{DD} = 1.0V (*Note 9*)	Amplitude Setting 0	265	300	335	mV
			Amplitude Setting 1	305	350	395	
			Amplitude Setting 2	350	400	450	
			Amplitude Setting 3	395	450	505	
Termination Resistance (Internal)	R_T	Differential measurement		80	100	120	Ω
DC ELECTRICAL CHARACTERISTICS / eDP AUX CHANNEL (AUX__)							
Differential Peak-to-Peak Output Voltage	V_{OD}	R_L = 100Ω ±1%			1000		mV
Input Peak-to-Peak Voltage	V_{ID}			140		1360	mV

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(V_{DD18} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, V_{DD} = 0.95V to 1.05V, T_A = -40°C to +105°C, EP connected to PCB ground. Typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DD} = 1.0V, T_A = +25°C, unless otherwise noted.) (*Note 3*)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Input Common-Mode Voltage	V_{CMI}	Register setting at midscale		800		mV
Termination Resistance (Internal)	R_T	Differential measurement		100		Ω
DC ELECTRICAL CHARACTERISTICS / I/O PINS						
High-Level Input Voltage	V_{IH}		0.7 x V_{DDIO}			V
Low-Level Input Voltage	V_{IL}				0.3 x V_{DDIO}	V
High-Level Output Voltage	V_{OH}	I_{OH} = -4mA	V_{DDIO} - 0.4			V
Low-Level Output Voltage	V_{OL}	I_{OL} = 4mA			0.4	V
Input Current	I_{IN}	All pull-up/pull-down devices disabled, V_{IN} = 0V to V_{DDIO}			1	µA
Input Capacitance	C_{IN}			3		pF
Internal Pull-Up/Pull-Down Resistance	R_{IN}	40kΩ enabled		40		kΩ
		1MΩ enabled		1		MΩ
DC ELECTRICAL CHARACTERISTICS / OPEN-DRAIN PINS						
High-Level Input Voltage	V_{IH}		0.7 x V_{DDIO}			V
Low-Level Input Voltage	V_{IL}				0.3 x V_{DDIO}	V
Low-Level Open-Drain Output Voltage	V_{OL}	I_{OL} = 4mA			0.4	V
Input Current	I_{IN}	All pull-up/pull-down devices disabled, V_{IN} = 0V to V_{DDIO}			1	µA
Input Capacitance	C_{IN}			3		pF
Internal Pull-Up Resistance	R_{PU}	40kΩ enabled		40		kΩ
		1MΩ enabled		1		MΩ
DC ELECTRICAL CHARACTERISTICS / PWDNB INPUT						
High-Level Input Voltage	V_{IH}		0.7 x V_{DDIO}			V
Low-Level Input Voltage	V_{IL}				0.3 x V_{DDIO}	V
Input Current	I_{IN}	V_{IN} = 0V to V_{DDIO}			6	µA
Internal Pull-Down Resistance	R_{PD}			1		MΩ
Input Capacitance	C_{IN}			3		pF
PWDNB Hold Time	t_{HOLD_PWDNB}	The minimum duration PWDNB must be held low to reset the chip.		1		ms
DC ELECTRICAL CHARACTERISTICS / DIS_REM_CC INPUT						
High-Level Input Voltage	V_{IH}		1.6			V
Low-Level Input Voltage	V_{IL}				1.4	V
Input Current	I_{IN}	V_{IN} = 0V to V_{DD18}			30	µA

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(V_{DD18} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, V_{DD} = 0.95V to 1.05V, T_A = -40°C to +105°C, EP connected to PCB ground. Typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DD} = 1.0V, T_A = +25°C, unless otherwise noted.) (*Note 3*)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
Internal Pull-Down Resistance	R_{PD}				100		kΩ
Input Capacitance	C_{IN}				3		pF
DC ELECTRICAL CHARACTERISTICS / PUSH-PULL OUTPUTS							
High-Level Output Voltage	V_{OH}	I_{OH} = -4mA		V_{DDIO} - 0.4			V
Low-Level Output Voltage	V_{OL}	I_{OL} = 4mA				0.4	V
DC ELECTRICAL CHARACTERISTICS / LINE FAULT DETECTION INPUTS (LMN0, LMN1)							
Open Pin Voltage	V_{O0}	LMN0			1.25		V
	V_{O1}	LMN1			0.75		
DC ELECTRICAL CHARACTERISTICS / REFERENCE CLOCK INPUT REQUIREMENTS (CRYSTAL) (X1, X2)							
X1 Input Capacitance	C_{IN_X1}				3		pF
X2 Input Capacitance	C_{IN_X2}				1		pF
Internal X2 Limit Resistor	R_{LIM}				1.2		kΩ
Internal Feedback Resistor	R_{FB}				10		kΩ
Transconductance	g_m				28		mA/V
DC ELECTRICAL CHARACTERISTICS / REFERENCE CLOCK REQUIREMENTS (EXTERNAL INPUT ON X1, X2 UNCONNECTED)							
High-Level Input Voltage	V_{IH}			0.9			V
Low-Level Input Voltage	V_{IL}					0.4	V
Input Impedance	R_{IN}				10		kΩ
X1 Input Capacitance	C_{IN_X1}				3		pF
DC ELECTRICAL CHARACTERISTICS / POWER SUPPLY CURRENTS (MAX96860AS)							
Supply Current (*Note 5*)	I_{DD}	Single GMSL3 input with 11.8Gbps utilization, HDCP, eDP output 24bpp at 360MHz, 2xHBR2 lanes, 6Gbps GMSL2 daisy chain output with 1.2Gbps utilization.	V_{DD}		445	1110	mA
			V_{DD18}		165	190	
		Dual GMSL3 inputs each with 11.8Gbps utilization, HDCP, Even/Odd Pixel mode, eDP output 24bpp at 720MHz, 4xHBR2 lanes, 6Gbps GMSL2	V_{DD}		645	1370	
			V_{DD18}		250	290	

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(V_{DD18} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, V_{DD} = 0.95V to 1.05V, T_A = -40°C to +105°C, EP connected to PCB ground. Typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DD} = 1.0V, T_A = +25°C, unless otherwise noted.) (*Note 3*)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
		daisy chain output with 2.4Gbps utilization.					
DC ELECTRICAL CHARACTERISTICS / POWER SUPPLY CURRENTS (MAX96862RS)							
Supply Current (*Note 5*)	I_{DD}	Single GMSL3 input with 11.8Gbps utilization, HDCP, decompression, eDP output 24bpp at 1080MHz, 4xHBR3 lanes, 6Gbps GMSL2 daisy chain output with 1.2Gbps utilization.	V_{DD}		635	1355	mA
			V_{DD18}		165	190	
DC ELECTRICAL CHARACTERISTICS / POWER SUPPLY CURRENTS (MAX96856AC)							
Supply Current (*Note 5*)	I_{DD}	Single GMSL3 input with 11.7Gbps utilization, eDP output 30bpp at 320MHz, 4xHBR2 lanes.	V_{DD}		410	1110	mA
			V_{DD18}		142	165	
		Dual GMSL3 inputs each with 11.7Gbps utilization, even/odd pixel mode, eDP output 30bpp at 649.116MHz, 4xHBR3 lanes.	V_{DD}		580	1270	
			V_{DD18}		230	265	
DC ELECTRICAL CHARACTERISTICS / POWER SUPPLY CURRENTS (MAX96746RC)							
Supply Current (*Note 5*)	I_{DD}	Single GMSL2 input with 5.8Gbps utilization, eDP output 24bpp at 210MHz, 2xHBR2 lanes.	V_{DD}		340	990	mA
			V_{DD18}		148	170	
DC ELECTRICAL CHARACTERISTICS / POWER SUPPLY CURRENTS (V_{DDIO} SUPPLY CURRENTS)							
Supply Current	I_{DDIO}	Per toggling GPIO, C_L = 20pF	V_{DDIO} at 1.9V		44		µA/MHz
			V_{DDIO} at 3.6V		81		
DC ELECTRICAL CHARACTERISTICS / POWER-DOWN CURRENT							
Maximum Power-Down Current	I_{DD}	V_{DD} at 1.05V	T_A = +25°C		< 1		µA
			T_A = +105°C		< 1		
		V_{DD18} at 1.9V	T_A = +25°C		1		
			T_A = +105°C		17		

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(V_{DD18} = 1.7V to 1.9V, V_{DDIO} = 1.7V to 3.6V, V_{DD} = 0.95V to 1.05V, T_A = -40°C to +105°C, EP connected to PCB ground. Typical values are at V_{DD18} = V_{DDIO} = 1.8V, V_{DD} = 1.0V, T_A = +25°C, unless otherwise noted.) (*Note 3*)

PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
		V_{DDIO} at 3.6V	T_A = +25°C		6		
			T_A = +105°C		7		
DC ELECTRICAL CHARACTERISTICS / SLEEP CURRENT							
Maximum Sleep Current	I_{DD}	V_{DD} at 1.05V	T_A = +25°C		< 1		µA
			T_A = +105°C		< 1		
		V_{DD18} at 1.9V	T_A = +25°C		33		
			T_A = +105°C		51		
		V_{DDIO} at 3.6V	T_A = +25°C		10		
			T_A = +105°C		11		
AC ELECTRICAL CHARACTERISTICS / FORWARD CHANNEL SWITCHING CHARACTERISTICS (SIO_)							
Lock Time	t_{LOCK}	From power-up, release of RESET_LINK, or rising edge of PWDNB to rising edge of LOCK. (*Note 11*)			65	90	ms
Maximum Video Initialization Time	$t_{VIDEOSTART}$	Time from GMSL video packet input at SIO± input to when pixels appear at video outputs. Assumes link is already established and eDP link training is complete.			2		Frames
Maximum Video Latency	t_{VL}	Time from pixel within GMSL packet at SIO± to output at eDP interface.			25		µs
AC ELECTRICAL CHARACTERISTICS / REVERSE CHANNEL SERIAL OUTPUTS (SIO_)							
GMSL Reverse Channel Transmitter Rise/Fall Time	t_R, t_F	20% to 80%, V_O = 250mV, R_L = 100Ω			2300		ps
Total Serial Output P-P Jitter	t_{TSOJ}	PRBS7, single-ended or differential output			0.15		UI
Deterministic Serial Output P-P Jitter	t_{DSOJ}	PRBS7, single-ended or differential output			0.1		UI
AC ELECTRICAL CHARACTERISTICS / GMSL DAISY-CHAIN FORWARD-CHANNEL SERIAL OUTPUTS (DCIO_)							
Serial Output Rise Time	t_R	20% to 80%, V_O = 40mV, NRZ, R_L = 100Ω ±1%			50		ps
Serial Output Fall Time	t_F	80% to 20%, V_O = 40mV, NRZ, R_L = 100Ω ±1%			50		ps
Total Serial Output Jitter	t_{TSOJ}	PRBS7, single-ended or differential output			0.15		UI (p-p)
Deterministic Serial Output Jitter	t_{DSOJ}	PRBS7, single-ended or differential output			0.10		UI (p-p)
Lock Time	t_{LOCK}				65	90	ms
Maximum Video Latency	t_{VL}	Time from pixel within GMSL packet at SIO input to same pixel at DCIO output, FEC enabled.			620		ns
AC ELECTRICAL CHARACTERISTICS / eDP OUTPUTS (eDP__) (*Note 4*)							
Unit Interval	UI_Rate	1.62Gbps/lane (RBR)			617		ps
		2.7Gbps/lane (HBR)			370		

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PARAMETER	SYMBOL	CONDITIONS		MIN	TYP	MAX	UNITS
		5.4Gbps/lane (HBR2)			185		
		8.1Gbps/lane (HBR3)			123		
Preemphasis		All output voltage settings	Preemphasis Setting 0		0		dB
			Preemphasis Setting 1		2		
			Preemphasis Setting 2		4		
			Preemphasis Setting 3		6		
Total Jitter (*Note 7*)	t_{TJO}	Lane rate ≤ 5.4Gbps				0.27	UI
		Lane rate = 8.1Gbps				0.28	
Deterministic Jitter (*Note 7*)	t_{DJO}	Lane rate ≤ 5.4Gbps				0.17	UI
		Lane rate = 8.1Gbps				0.2	
Output Skew	t_{SKEW}	Intrapair				30	ps
		Lane-to-Lane				1250	
SSC Downspread						0.55	%
SSC Modulation Rate				30		33	kHz
AC ELECTRICAL CHARACTERISTICS / eDP AUX CHANNEL (AUX_) (*Note 4*)							
Unit Interval	UI			0.4		0.6	µs
Cycle-to-Cycle Jitter	t_{JCC}	UI variation within a single transaction				0.08	UI
		UI variation between adjacent bits				0.04	
AC ELECTRICAL CHARACTERISTICS / I²C / UART PORT TIMING (*Note 4*)							
Output Fall Time	t_f	70% to 30%, C_L = 20pF to 100pF, 1kΩ pull-up to V_{DDIO}		20 x V_{DDIO} /5.5V		150	ns
I²C/UART Wake Time	t_{GC_WAKE}	From power-up or rising edge of PWDNB to local register access. For remote register access, I²C/UART wake time is the same as lock time ($tPUL_{LOCK}$). (*Note 10*)				4	ms
AC ELECTRICAL CHARACTERISTICS / I²C TIMING							
SCL Clock Frequency	f_{SCL}	Mid f_{SCL} range		100		400	kHz
		High f_{SCL} range		400		1000	
Start Condition Hold Time	$t_{HD:STA}$	Mid f_{SCL} range		0.6			µs
		High f_{SCL} range		0.26			
Low Period of SCL Clock	t_{LOW}	Mid f_{SCL} range		1.3			µs
		High f_{SCL} range		0.5			
High Period of SCL Clock	t_{HIGH}	Mid f_{SCL} range		0.6			µs
		High f_{SCL} range		0.26			
Repeated Start Condition Setup Time	$t_{SU:STA}$	Mid f_{SCL} range		0.6			µs
		High f_{SCL} range		0.26			

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PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Data Hold Time	$t_{HD:DAT}$	Mid f_{SCL} range	0			ns
		High f_{SCL} range	0			
Data Setup Time	$t_{SU:DAT}$	Mid f_{SCL} range	100			ns
		High f_{SCL} range	50			
Setup Time for Stop Condition	$t_{SU:STO}$	Mid f_{SCL} range	0.6			µs
		High f_{SCL} range	0.26			
Bus Free Time	t_{BUF}	Mid f_{SCL} range	1.3			µs
		High f_{SCL} range	0.5			
Data Valid Time	$t_{VD:DAT}$	Mid f_{SCL} range			0.9	µs
		High f_{SCL} range			0.45	
Data Valid Acknowledge Time	$t_{VD:ACK}$	Mid f_{SCL} range			0.9	µs
		High f_{SCL} range			0.45	
Pulse Width of Spikes Suppressed	t_{SP}	Mid f_{SCL} range			50	ns
		High f_{SCL} range			50	
Capacitive Load on Each Bus Line	C_B			100		pF
AC ELECTRICAL CHARACTERISTICS / SPI MAIN						
Operating Frequency	f_{MCK}		0.588		50	MHz
SCLK Period	t_{MCK}			$1/f_{MCK}$		ns
SCLK Output Pulse-Width High/Low	t_{MCH}, t_{MCL}	(*Note 4*, *Note 8*)	$t_{MCK}/2$ - 3.4	$t_{MCK}/2$		ns
MOSI Data Output Delay	t_{MOD}	After SCLK falling edge (*Note 4*, *Note 8*)	-2.3		2.3	ns
MISO Input Setup Time	t_{MIS}	Before programmed sampling edge (*Note 4*, *Note 8*)	13.5			ns
MISO Input Hold Time	t_{MIH}	After programmed sampling edge (*Note 4*, *Note 8*)	-2			ns
AC ELECTRICAL CHARACTERISTICS / SPI SUBORDINATE						
Operating Frequency	f_{SCK}				50	MHz
SCLK Period	t_{SCK}			$1/f_{SCK}$		ns
SCLK Input Pulse-Width High/Low	t_{SCH}, t_{SCL}			$t_{SCK}/2$		ns
MISO Data Output Delay	t_{SOD}	After SCLK falling edge (*Note 4*, *Note 8*)	2.0		11.3	ns
MOSI Input Setup Time	t_{SIS}	Before SCLK rising edge (*Note 4*, *Note 8*)	5			ns
MOSI Input Hold Time	t_{SIH}	After SCLK rising edge (*Note 4*, *Note 8*)	3			ns
AC ELECTRICAL CHARACTERISTICS / I²S/TDM MAIN TIMING (*Note 4*, *Note 8*)						
WS Frequency	f_{WS}	C_L = 5pF	8		192	kHz
Sample Word Length	n_{WS}	C_L = 5pF	8		32	Bits
SCK Frequency	f_{SCK}	$f_{SCK} = f_{WS} \times n_{WS} \times (2 \text{ or } 8)$, C_L = 5pF	0.512		49.152	MHz
SCK Clock High Time	t_{HC}	$V_{SCK} \geq V_{IH}$, $t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.35 x t_{SCK}			ns

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PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
SCK Clock Low Time	t_{LC}	$V_{SCK} \le V_{IL}$, $t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.35 x t_{SCK}			ns
SD, WS Valid Time Before SCK	t_{MAVS}	$t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.2 x t_{SCK}	0.5 x t_{SCK}		ns
SD, WS Valid Time After SCK	t_{MAVH}	$t_{SCK} = 1/f_{SCK}$, C_L = 5pF	0.2 x t_{SCK}	0.5 x t_{SCK}		ns
AC ELECTRICAL CHARACTERISTICS / REFERENCE CLOCK INPUT REQUIREMENTS (CRYSTAL) (X1, X2) (*Note 4*)						
Frequency	f_{XTAL}			25		MHz
Frequency Stability + Frequency Tolerance	f_{TN}				±200	ppm
AC ELECTRICAL CHARACTERISTICS / REFERENCE CLOCK OUTPUT (RCLKOUT) (*Note 4*)						
Frequency	f_{REF}	Crystal or reference clock input 25MHz		25		MHz
Rise Time	t_R	20% to 80%, C_L=10pF, R_{term}=22Ω		1	3	ns
Fall Time	t_F	80% to 20%, C_L=10pF, R_{term}=22Ω		1	3	ns
Total Jitter	T_J	At BER=1e-12, C_L=10pF, R_{term}=22Ω			200	ps
Deterministic Jitter	D_J	C_L=10pF, R_{term}=22Ω			25	ps
AC ELECTRICAL CHARACTERISTICS / REFERENCE CLOCK REQUIREMENTS (EXTERNAL CLOCK INPUT ON X1/OSC, X2 UNCONNECTED) (*Note 4*)						
Frequency	f_{REF}			25		MHz
Frequency Stability + Frequency Tolerance	f_{TN}				±200	ppm
Input Duty Cycle	t_{DUTY}		40		60	%
Total Jitter	T_J	At BER=1e-12			300	ps
Deterministic Jitter	D_J	Measured with Sinusoidal jitter < 1MHz			150	ps

Note 3: Limits are 100% tested at T_A = +105°C. Limits over the operating temperature range and relevant supply voltage range are guaranteed by design and characterization.

Note 4: Not production tested. Guaranteed by design and characterization.

Note 5: Color Bar pattern, 60Hz frame rate.

Note 6: All parameters are measured with 100Ω ±1% load.

Note 7: 300mVp-p, 0dB preemphasis, CP2520 compliance pattern, measured at TP1.

Note 8: Measured at 50MHz.

Note 9: Amplitude settings are set at link training. Refer to AUX messages for selected amplitude settings.

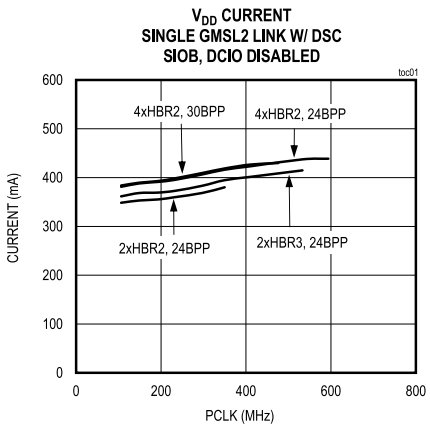
Note 10: Do not write to part until this is done.

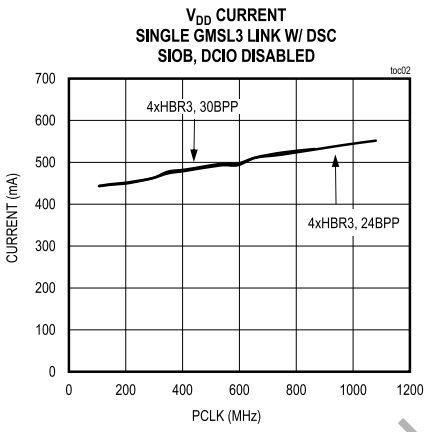
Note 11: From power-up, release of RESET_LINK, or rising edge of the PWDNB pin, to rising edge of the LOCK pin. t_{RD} must be < 70ms. For more information, see the *GMSL Link Lock* section.

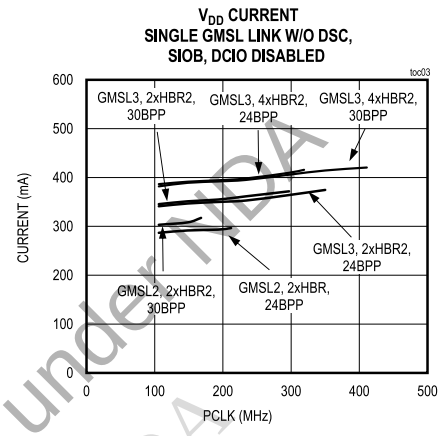
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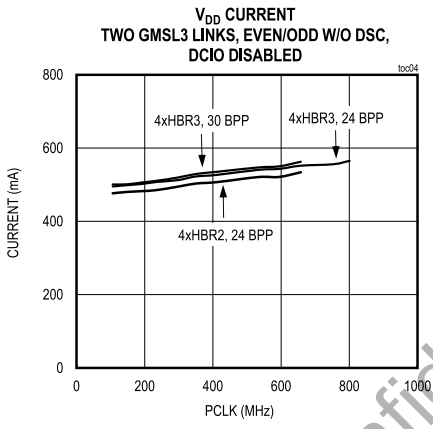
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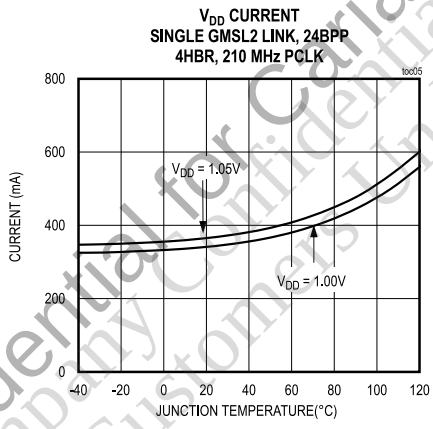
Typical Operating Characteristics

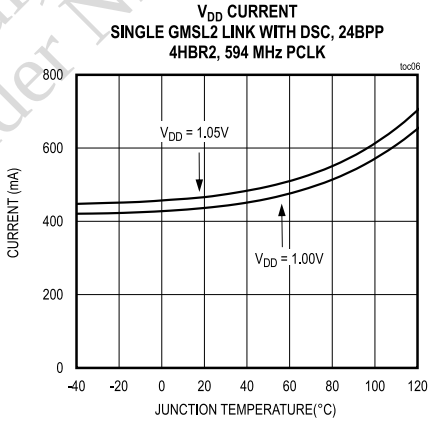


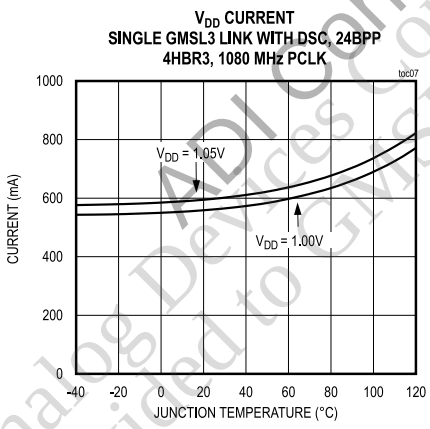


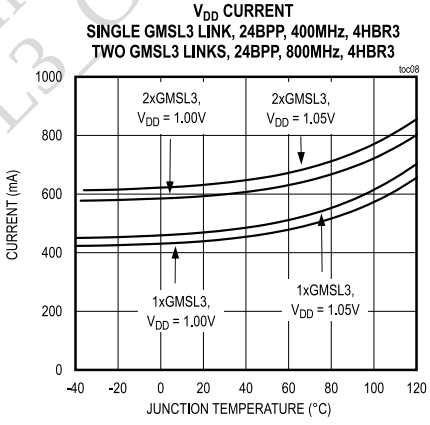


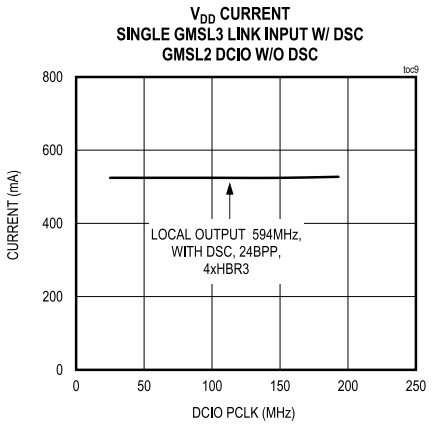






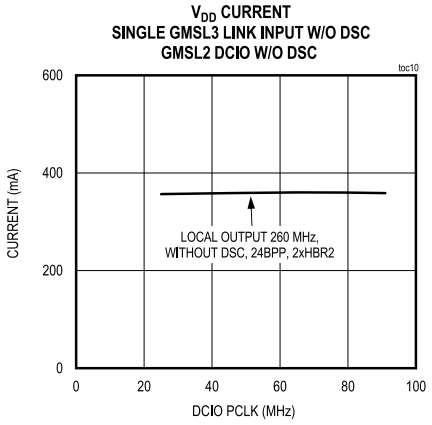


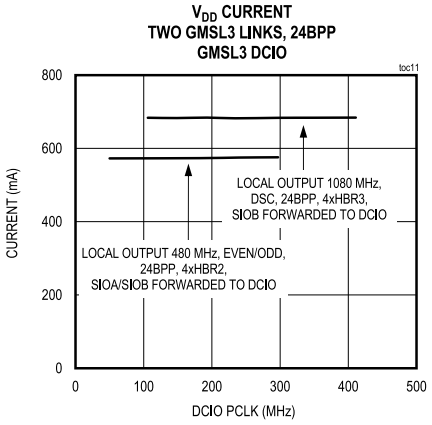


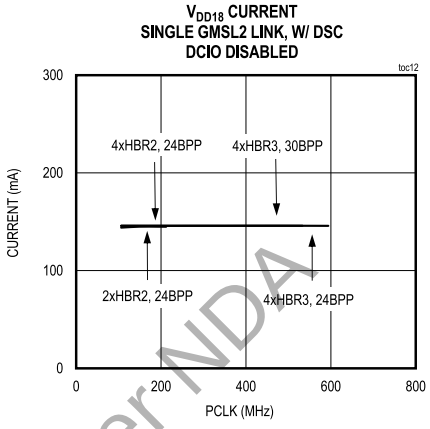


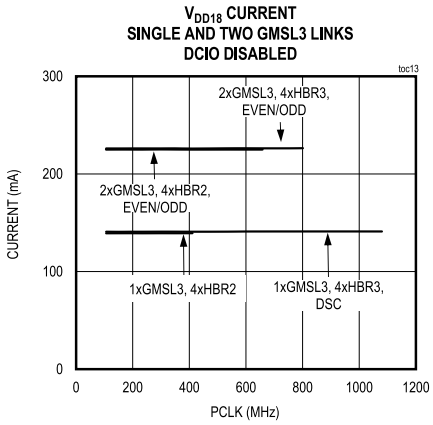
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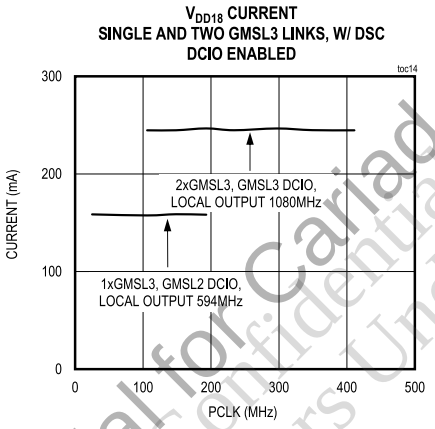
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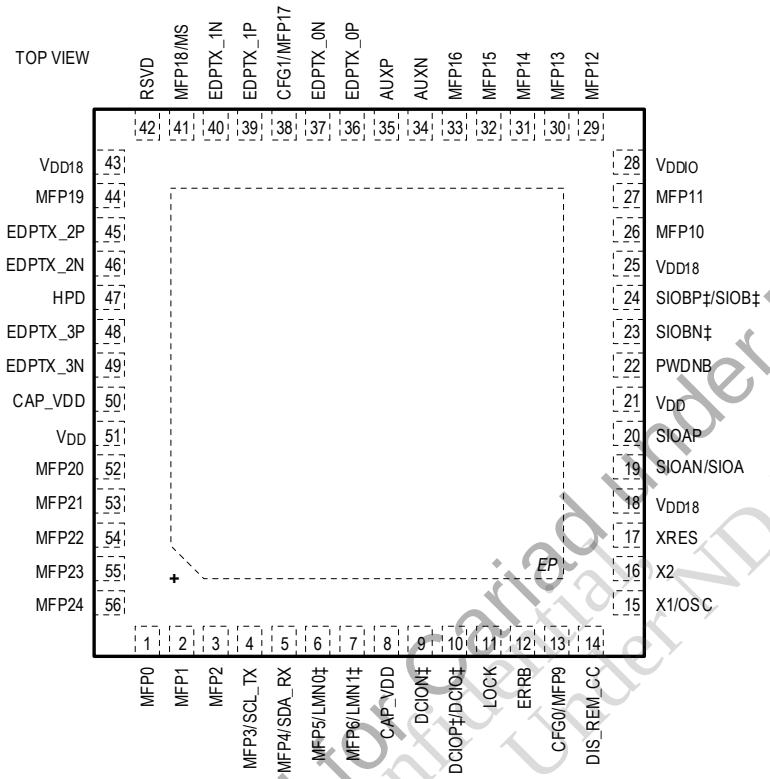




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Pin Configurations



Pin Descriptions

PIN	NAME	FUNCTION MODE SIGNAL NAME	FUNCTION
GMSL3/2 SERIAL LINK (LEAVE DCIO_OUTPUT PINS UNCONNECTED WHEN DAISY-CHAINING IS NOT USED)			
19	SIOAN/SIOA	SIOAN SIOA	SIOAN: GMSL3/2 Link A Inverted Twisted-Pair Serial-Data Input. State of CFG1 on power-up determines configuration. SIOA: GMSL3/2 Link A Coax Serial-Data Input. State of CFG1 on power-up determines configuration.
20	SIOAP	SIOAP	SIOAP: GMSL3/2 Link A Noninverted Twisted-Pair Serial-Data Input. State of CFG1 on power-up determines configuration.
24	SIOBP/SIOB	SIOBP SIOB	SIOBP: GMSL3/2 Link B Noninverted Twisted-Pair Serial-Data Input. State of CFG1 on power-up determines configuration. SIOB: GMSL3/2 Link B Coax Serial-Data Input. State of CFG1 on power-up determines configuration.
23	SIOBN	SIOBN	SIOBN: GMSL3/2 Link B Inverted Twisted-Pair Serial-Data Input. State of CFG1 on power-up determines configuration.

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9	DCION	DCION	DCION: GMSL3/2 Daisy-Chain Inverted Twisted-Pair Serial-Data Output. State of CFG[1:0] on power-up determines configuration.
10	DCIOP/DCIO	DCIOP DCIO	DCIOP: GMSL3/2 Daisy-Chain Noninverted Twisted-Pair Serial-Data Output. State of CFG[1:0] on power-up determines configuration. DCIO: GMSL3/2 Daisy-Chain Coax Serial-Data Output. State of CFG[1:0] on power-up determines configuration.
eDP INTERFACE			
47	HPD	HPD	eDP Hot-Plug Detect Input. Configured as open-drain I/O with 1MΩ pull-down to ground.
34	AUXN	AUXN	eDP Inverted AUX Channel.
35	AUXP	AUXP	eDP Nonnverted AUX Channel.
36	EDPTX_0P	EDPTX_0P	eDP Noninverted Data Output Channel 0.
37	EDPTX_0N	EDPTX_0N	eDP Inverted Data Output Channel 0.
39	EDPTX_1P	EDPTX_1P	eDP Noninverted Data Output Channel 1.
40	EDPTX_1N	EDPTX_1N	eDP Inverted Data Output Channel 1.
45	EDPTX_2P	EDPTX_2P	eDP Noninverted Data Output Channel 2.
46	EDPTX_2N	EDPTX_2N	eDP Inverted Data Output Channel 2.
48	EDPTX_3P	EDPTX_3P	eDP Noninverted Data Output Channel 3.
49	EDPTX_3N	EDPTX_3N	eDP Inverted Data Output Channel 3.
MULTIFUNCTION PINS (*DENOTES DEFAULT STATE AFTER POWER-UP)			
1	MFP0	SCLK GPIO0*	SCLK: SPI Clock. When configured as main, push-pull clock output. When configured as subordinate, clock input with internal 1MΩ pull-down to ground. GPIO0: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
2	MFP1	MOSI GPIO1*	MOSI: SPI Main-Out Subordinate-In. When configured as main, push-pull output that drives data to external subordinate. When configured as subordinate, input with internal 1MΩ pull-down to ground that receives data from external main. GPIO1: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
3	MFP2	MISO GPIO2*	MISO: SPI Main-In Subordinate-Out. When configured as main, input with internal 1MΩ pull-down to ground that receives data from external subordinate. When configured as subordinate, push-pull output that drives data to an external main. GPIO2: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.

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4	MFP3/SCL_TX	SCL* TX* GPIO3	SCL: I²C Clock Input/Open-Drain Output with Internal 40kΩ Pull-Up to V_{DDIO} (SCL or TX selected by CFG1 at power-up). TX: UART Open-Drain Output with Internal 40kΩ Pull-Up to V_{DDIO} (SCL or TX selected by CFG1 at power-up). GPIO3: Configurable General-Purpose Input/Output. Open-drain output with internal 40kΩ pull-up, CMOS input enabled, output disabled.
5	MFP4/SDA_RX	SDA* RX* GPIO4	SDA: I²C Data Input/Open-Drain Output with Internal 40kΩ Pull-Up to V_{DDIO} (SDA or RX selected by CFG1 at power-up). RX: UART Input (SDA or RX selected by CFG1 at power-up). GPIO4: Configurable General-Purpose Input/Output. Open-drain output with internal 40kΩ pull-up, CMOS input enabled, output disabled.
6	MFP5	LMN0 GPIO5*	LMN0: Line Fault Monitor Input. GPIO5: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
7	MFP6	LMN1 GPIO6*	LMN1: Line Fault Monitor Input. GPIO6: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
11	LOCK	LOCK*	LOCK: Open-Drain Lock Indication Output with Internal 40kΩ Pull-Up to V_{DDIO}. LOCK = HIGH indicates that GMSL link has locked and is ready to send/receive data. See LOCK_CFG register description. LOCK is HIGH when PWDNB is LOW.
12	ERRB	ERRB*	ERRB: Open-Drain Error Indication Output with Internal 40kΩ Pull-Up to V_{DDIO}. ERRB = LOW indicates a data error, line fault, or interrupt is detected. ERRB is HIGH when PWDNB is LOW.
13	CFG0/MFP9	CFG0* WMD GPO9	CFG0: Configuration Pin 1. Voltage at pin sets device mode, which is latched at power-up. Connect a resistor divider between V_{DDIO} and ground. WMD: Watermark Detect Push-Pull Output. High when watermark is detected. GPO9: Configurable General-Purpose Output.
26	MFP10	SDA1 RX1 GPIO10*	SDA1: Pass-Through Port 1 I²C Data Input/Open-Drain Output with Internal 40kΩ Pull-Up to V_{DDIO}. RX1: Pass-Through Port 1 UART Input. GPIO10: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
27	MFP11	SCL1 TX1 GPIO11*	SCL1: Pass-Through Port 1 I²C Clock Input/Open-Drain Output with Internal 40kΩ Pull-Up to V_{DDIO}.

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			TX1: Pass-Through Port 1 UART Output with Internal 40kΩ Pull-Up to V_{DDIO}. GPIO11: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
29	MFP12	ALIGN_LOCK_IN INITIAL_ALIGN_LOCK DE_OUT_X WS GPIO12*	ALIGN_LOCK_IN: The main deserializer receives information through this MFP that the subordinate deserializer has completed the external alignment once. INITIAL_ALIGN_LOCK: The subordinate deserializer sends information via this MFP indicating that it has completed the external alignment once. DE_OUT_X: Data Enable Output of Video Pipe X. WS: I^2S/TDM Word-Select Push-Pull Output. Supports forward-path audio. GPIO12: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
30	MFP13	HS_X SD GPIO13*	HS_X: Pipe X Horizontal Sync Input or Push-Pull Output. SD: I^2S/TDM FWD Audio Channel Serial-Data Push-Pull Output. Supports forward audio from serializer to deserializer. GPIO13: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
31	MFP14	VS_X SCK GPIO14*	VS_X: Pipe X Vertical Sync Input or Push-Pull Output. SCK: I^2S/TDM FWD Audio Channel Clock Push-Pull Output. Supports forward-path audio. GPIO14: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
32	MFP15	SDA2 RX2 GPIO15*	SDA2: Pass-Through Port 2 I^2C Data Input/Open-Drain Output with Internal 40kΩ Pull-Up to V_{DDIO}. RX2: Pass-Through Port 2 UART Input. GPIO15: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
33	MFP16	SCL2 TX2 GPIO16*	SCL2: Pass-Through Port 2 I^2C Clock Input/Open-Drain Output with Internal 40kΩ Pull-Up to V_{DDIO}. TX2: Pass-Through Port 2 UART Output with Internal 40kΩ Pull-Up to V_{DDIO}. GPIO16: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.

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38	CFG1/MFP17	CFG1* VA_DONE GPIO17	CFG1: Configuration Pin 2. Voltage at pin sets device modes, which are latched at power-up. Connect a resistor divider between V_{DDIO} and ground. VA_DONE: Video Authentication CRC/SHA Calculation Complete Push-Pull Interrupt Output. GPIO17: Configurable General-Purpose Input/Output.
41	MFP18/MS	VS_X_ALT MS GPIO18*	VS_X_ALT: Alternate Pipe X Vertical Sync Input or Push-Pull Output. MS: UART Mode Select with Internal 1MΩ Pull-Down to Ground. Set MS = low to select Base mode. Set MS = high to select Bypass mode. MS state may also be temporarily overwritten by a register write. GPIO18: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
44	MFP19	HS_X_ALT GPIO19	HS_X_ALT: Alternate Pipe X Horizontal Sync Input or Push-Pull Output. GPIO19: Configurable General-Purpose Input/Output. Power-up default.
52	MFP20	VS_OUT_Y GPIO20*	VS_OUT_Y: Vertical Sync Output of Video Pipe Y. GPIO20: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
53	MFP21	HS_OUT_Y GPIO21*	HS_OUT_Y: Horizontal Sync Output of Video Pipe Y. GPIO21: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
54	MFP22	ALIGN_LOCK_ALT_IN INITIAL_ALIGN_LOCK_ALT DE_OUT_Y RCLKOUT GPIO22*	ALIGN_LOCK_ALT_IN: The main deserializer receives information through this MFP that the subordinate deserializer has completed the external alignment once (Alternative MFP). INITIAL_ALIGN_LOCK_ALT: The subordinate deserializer sends information via this MFP indicating that it has completed the external alignment once (Alternative MFP). DE_OUT_Y: Data Enable Output of Video Pipe Y. RCLKOUT: 25MHz Reference Clock Push-Pull Output. GPIO22: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
55	MFP23	BNE SS1 GPIO23*	BNE: When Configured as Subordinate, SPI Buffer-Not-Empty Push-Pull Output. BNE = high indicates SPI data is available. SS1: SPI Subordinate Select. When configured as main, Subordinate 1 select push-pull output. GPIO23: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output

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			disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
56	MFP24	RO SS2 GPIO24*	RO: When Configured as Subordinate, SPI mode-Select Input with Internal 1MΩ Pull-Down to Ground. RO = high enables main read from MISO. RO = low enables main write to MOSI. SS2: SPI Subordinate Select. When configured as main, Subordinate 2 select push-pull output. GPIO24: Configurable General-Purpose Input/Output. Power-up default: GPI, CMOS input enabled, output disabled, GMSL GPIO tunnel disabled, local read enabled, 1MΩ pull-down. Note: Leave pin unconnected when not used.
MISCELLANEOUS			
14	DIS_REM_CC	DIS_REM_CC	Active-High Latch at Power-Up Input. Internal 100kΩ pull-down to ground. Disables the remote-control channel in multimaster configurations (only the local ECU (Electronic Control Unit) can control the deserializer). Connect to V_{DD18} to disable the remote -control channel (sets DIS_REM_CC register bits to logic-1). Connect to ground or leave floating to enable the remote-control channel function (sets DIS_REM_CC register bits to default logic-0 state). After power-up latch, writing to DIS_REM_CC can change the state of the remote-control channel enable. Note: Leave pin unconnected when not used.
15	X1/OSC	X1/OSC	X1/OSC: Crystal/Oscillator Input. Connect to either terminal of a ±200ppm 25MHz crystal or a 25MHz external clock source. If crystal is used, connect a load capacitor from X1/OSC to ground (load capacitor value depends on crystal used).
16	X2	X2	X2: Crystal Input. Connect to one terminal of a 25MHz ±200ppm crystal. If crystal is used, connect a load capacitor from X2 to ground (Load capacitor value depends on crystal used). If external oscillator is connected to X1/OSC, leave X2 floating.
17	XRES	XRES	Output Reference Resistor. Connect a 402Ω ±1% resistor between XRES and ground.
22	PWDNB	PWDNB	Active-Low, Power-Down Input with Internal 1MΩ Pull-Down to Ground. Apply logic low to place device in Power-down mode. Connect to V_{DDIO}/logic high for normal operation.
42	RSVD	RSVD	Reserved. Make no electrical connection to this pin.
POWER SUPPLIES (SEE *Table 4, Table 3*)			
8, 50	CAP_VDD	CAP_VDD	Decoupling Capacitor for 1V Core Supply. Place decoupling capacitor connected to PCB ground plane as close to pin as possible.
18, 25, 43	V_{DD18}	V_{DD18}	1.8V I/O Supply
21, 51	V_{DD}	V_{DD}	Core Supply. Connect 1.0V external power supply
28	V_{DDIO}	V_{DDIO}	1.8V-3.3V I/O Power Supply

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Timing Diagrams

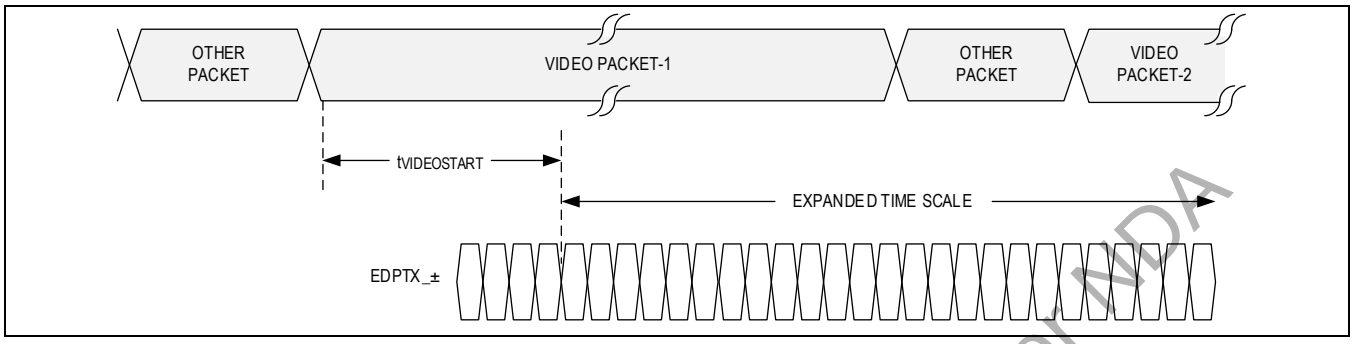


Figure 1. Deserializer Video Initialization Time

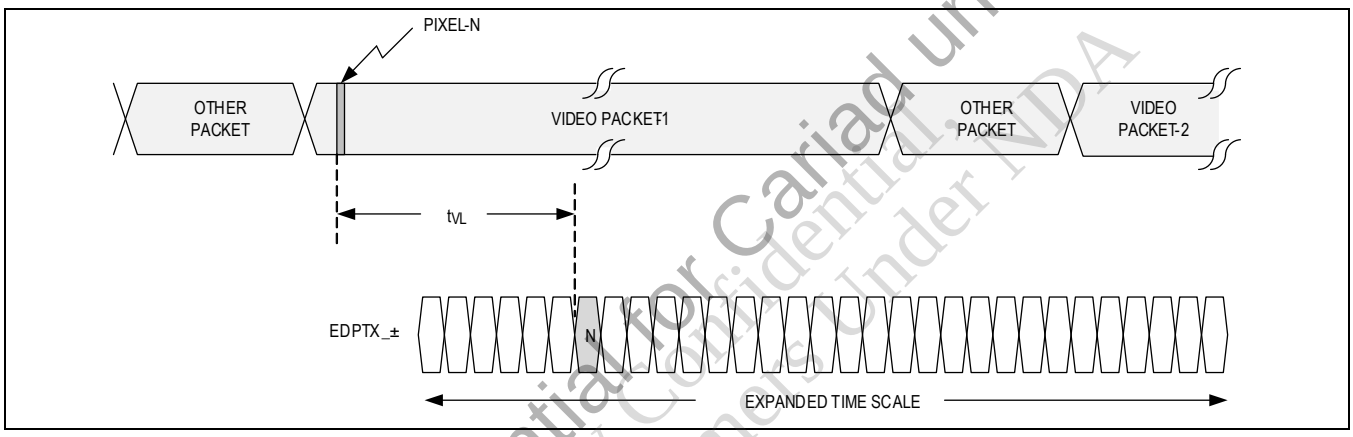


Figure 2. Deserializer Video Latency

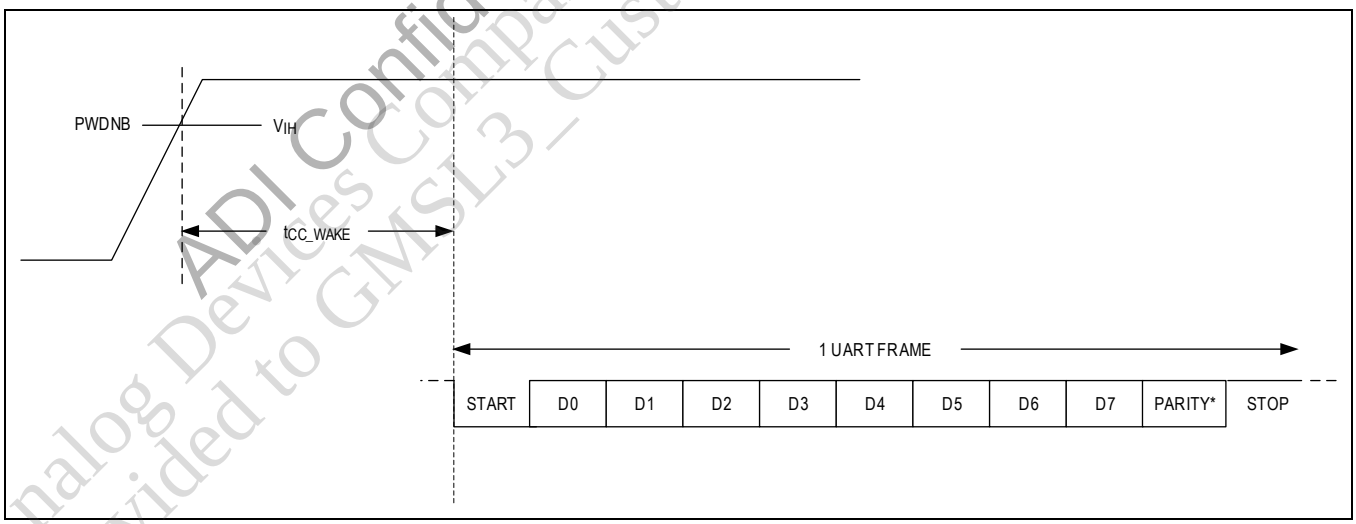


Figure 3. I²C/UART Wake Time

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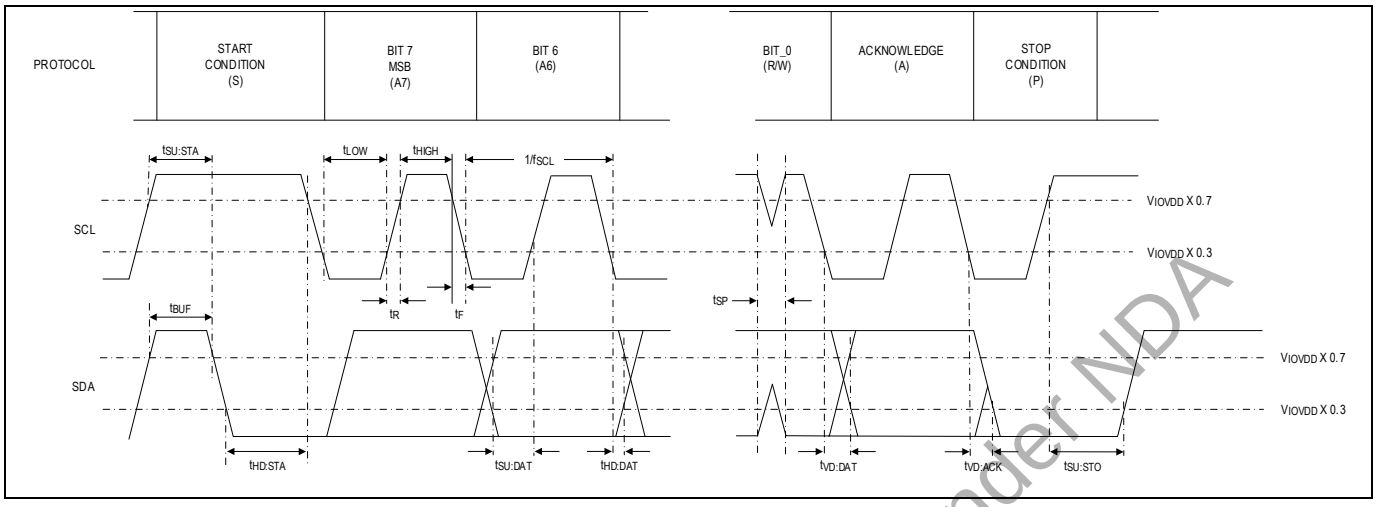


Figure 4. I²C Timing Parameters

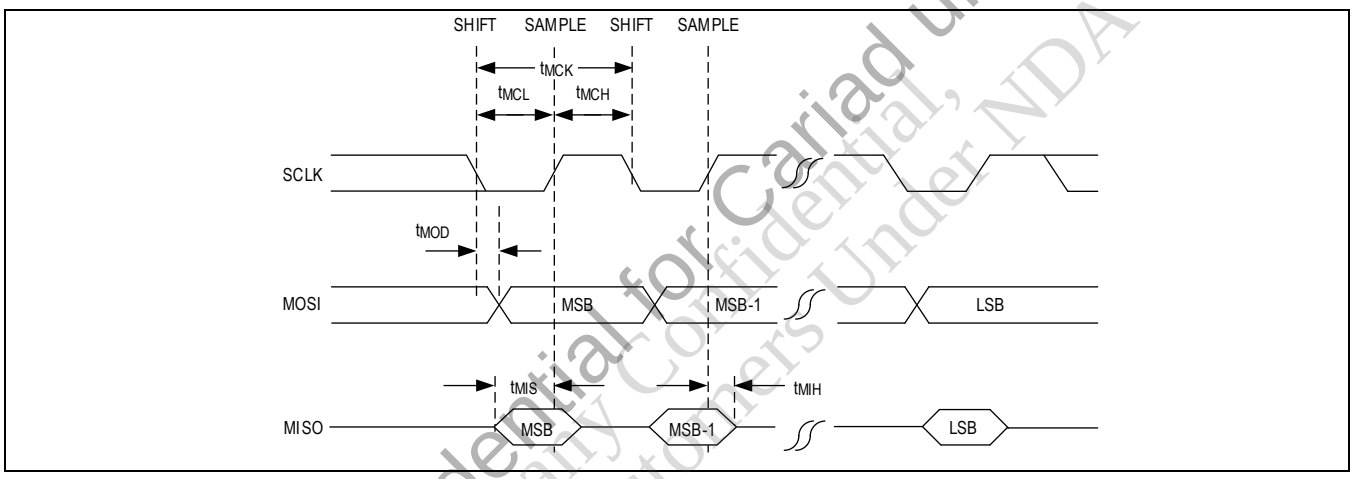


Figure 5. SPI Main-Mode Timing Parameters

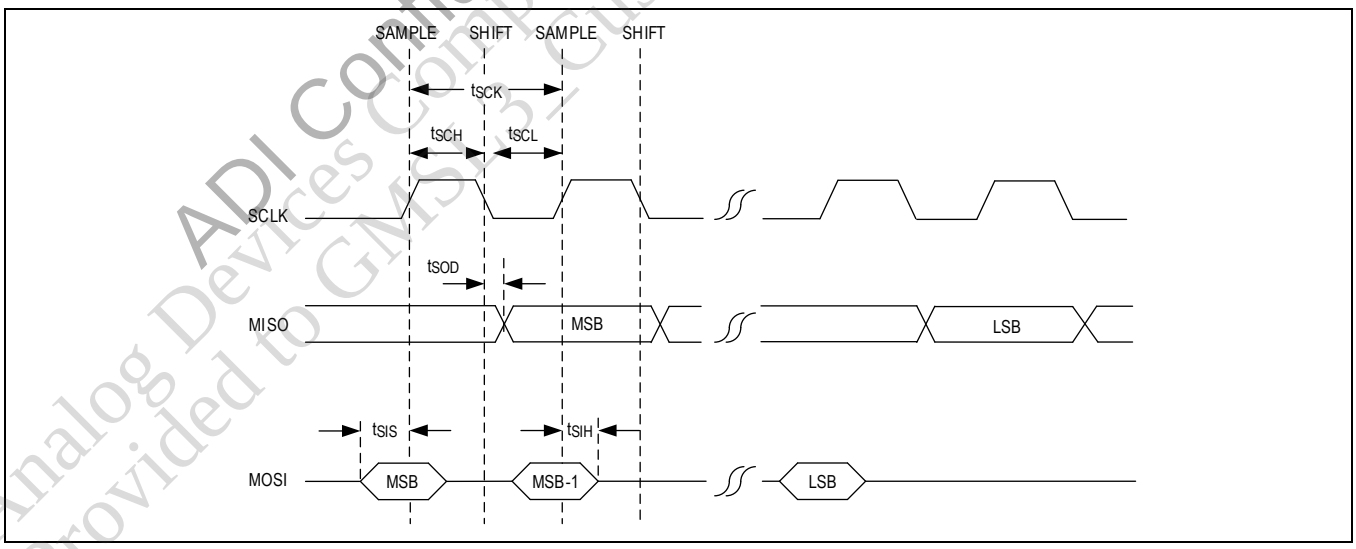


Figure 6. SPI Subordinate-Mode Timing Parameters

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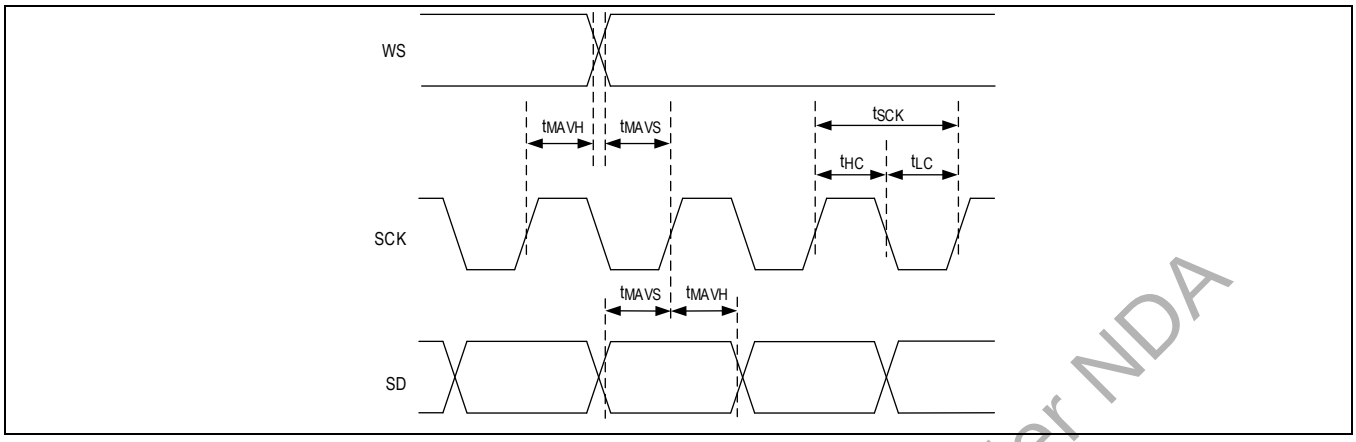


Figure 7. I^2S Main-Mode Timing Parameters

Product Specifications

Selection Guide

PART NUMBER	DSC	DAISY-CHAIN/END-POINT	GMSL INPUT	HDCP
MAX96746RC	No	End-Point	1x GMSL2	No
MAX96856AC	No	End-Point	2x GMSL3/2	No
MAX96860AS	No	Daisy-Chain	2x GMSL3/2	Yes
MAX96862RS	Yes	Daisy-Chain	1x GMSL3/2	Yes

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Detailed Description

Introduction

Analog Devices' GMSL3/2 serializers and deserializers provide sophisticated link management for high-speed, low-bit error rate, serial data transport. They support a comprehensive suite of display, camera, and communication interfaces over a single wire.

GMSL2 provides up to 6Gbps forward and 187.5Mbps reverse packetized non-return-to-zero (NRZ) data transmission over each fixed-speed link. Devices with two GMSL2 links provide a total capacity of up to 12Gbps and 375Mbps reverse in specific configurations.

GMSL3 provides 12Gbps forward packetized pulse amplitude modulation (PAM4) data transmission and 187.5Mbps reverse packetized NRZ data transmission over each fixed-speed link. Devices with two GMSL3 links provide a total capacity of up to 24Gbps and 375Mbps reverse.

The MAX96746/MAX96856/MAX96860/MAX96862 support both 24-bit and 30-bit color depth video streams.

The following sections provide a brief overview of the device's functions and features. Contact the factory for additional information and details on the configuration of each function and feature.

Additional Documentation

This data sheet contains electrical specifications, pin and functional descriptions, feature overviews and register definitions. Designers must also have the following information to correctly design using this device:

- The GMSL3/GMSL2 Channel Specifications contain physical layer requirements for the PCB traces, cables and connectors that constitute the GMSL link.
- The GMSL3/GMSL2 Hardware and Validation Guide contains recommendations for PCB design, applications circuits, selection of external components and guidelines for use of GMSL signal integrity tools.
- The GMSL3/GMSL2 User Guides contain detailed programming guidelines for GMSL3/2 device features.
- Errata sheets contain deviations from published device specifications and are specific to part number and revision ID.

Contact the factory to receive these documents and for additional guidance on MAX96746/MAX96856/MAX96860/MAX96862 features.

Other Functions

GMSL2 and GMSL3 serializers and deserializers have a primary I^2C/UART control channel interface that an ECU uses to access serializer and deserializer registers, as well as peripheral devices, from either end of the link. Each device also has 2 pass-through I^2C/UART channels available for local or remote peripheral control. The pass-through I^2C/UART channels do not have access to serializer and deserializer registers.

The MAX96746/MAX96856/MAX96860/MAX96862 devices support a single forward (serializer to deserializer) audio channel. It supports I^2S stereo and up to 8 channels in TDM mode. Sample rates of 32kHz to 192kHz are supported with sample depths from 8 bits to 32 bits. Product variants with daisy-chain capability support an additional, bidirectional audio channel with the same specifications.

The MAX96746/MAX96856/MAX96860/MAX96862 devices include an SPI interface for peripheral control. The SPI interface enables a host on one side of the GMSL link to control a peripheral on the opposite side. The host can be located at either end of the link or can swap ends by reprogramming the GMSL devices (a GMSL device can be configured as an SPI host or peripheral).

The MAX96746/MAX96856/MAX96860/MAX96862 devices provide up to 25 multifunction pins (MFPs). Multiple functions are assigned to each pin, including pass-through I^2C/UART, SPI, Audio, monitoring signals, and GPIOs. GPIOs are typically used to tunnel low-speed (< 100kbps) signals over the GMSL link. A GPIO tunnel can be set up in the forward or reverse direction. See the *Multifunction Pin (MFP) Assignments* section and *Pin Descriptions* section for more information.

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Use Case Overview

Deserializer Operating Modes

The serializers in the following use cases do not show the entire suite of capabilities. Contact the factory for information on all serializer capabilities.

Single Link Without Compression

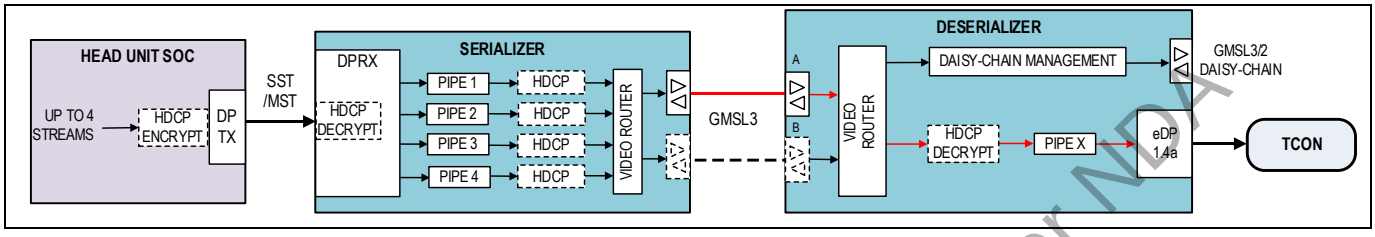


Figure 8. Uncompressed Single Link Configuration

A video stream of up to approximately 400MHz PCLK (24-bit color depth) or 320MHz (30-bit color depth) can be transported over a single GMSL3 link and locally output. Up to 215MHz PCLK (24-bit color depth only) can be transported over a single GMSL2 link. Deserializer GMSL Link A must be used. Video Pipe X is used in the deserializer. Excess bandwidth on this link, or on Link B (MAX96856/MAX96860 only), can be routed to the daisy-chain output.

Legacy GMSL2 Dual-Link Mode

The MAX96856/MAX96860 devices support even/odd splitter pixel mode, in which even and odd pixels are separated into two substreams by the serializer and routed over two GMSL links. They do not support legacy GMSL2 dual-link mode, in which video packets of a single video stream are routed over two links to divide the bandwidth loading.

Even/Odd Pixel Splitter Mode (MAX96856/MAX96860 only)

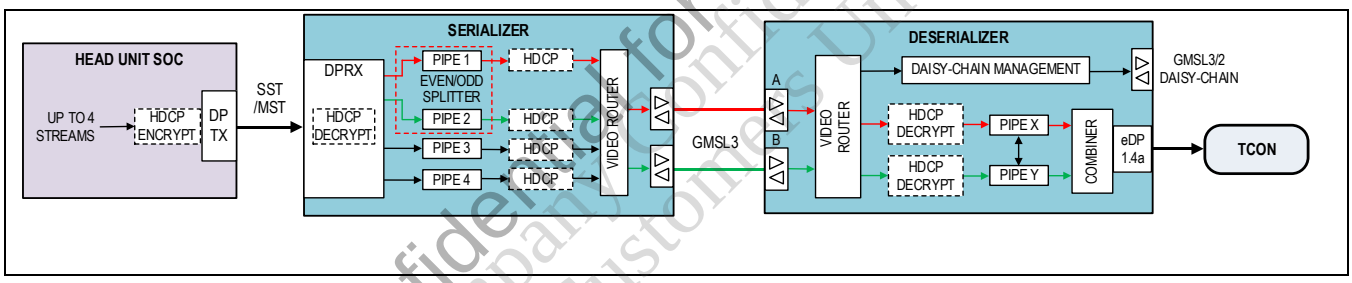


Figure 9. Even/Odd Pixel Splitter Configuration

A video stream of up to approximately 800MHz PCLK (24-bit color depth) or 640MHz (30-bit) can be split by specific serializers into even and odd pixel substreams, transported over two GMSL links, and recombined by the MAX96856/MAX96860. Even/Odd Pixel Splitter mode can only be used with an uncompressed video stream. The even pixel stream must be input on GMSL SIOA and the odd stream on SIOB. The definition of even and odd is such that the first pixel in a line is 0 and the second pixel is 1. Up to two other video streams can be daisy-chained.

Compressed Video (MAX96862 only)

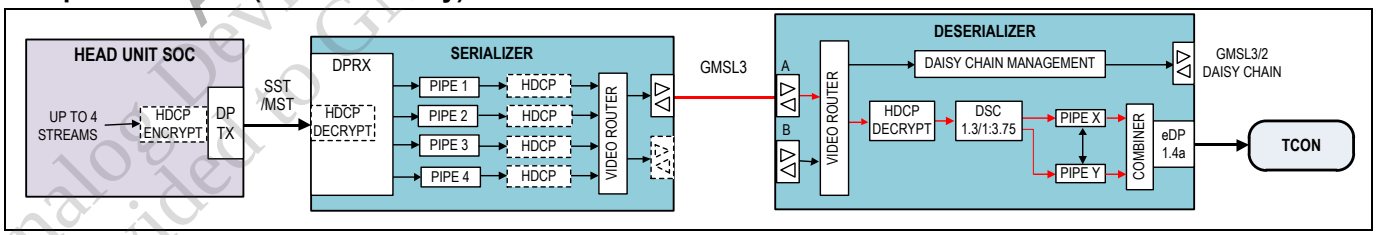


Figure 10. Compressed Single Link Configuration

A single video stream of up to 1080MHz PCLK (24bpp) or 864MHz (30bpp) can be output by the deserializer. The stream is compressed by the SOC using DSC 3:1 for 24-bit video and 3.75:1 for 30-bit video. The stream is decompressed by

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the deserializer. Deserializer GMSL Link A must be used. It can be configured as GMSL2 or GMSL3. Excess bandwidth on the link, or Link B (MAX96856/MAX96860 only), can be routed to the daisy-chain output. The other streams can be compressed or uncompressed.

Compressed Video with Parallel Deserializers (MAX96862 only)

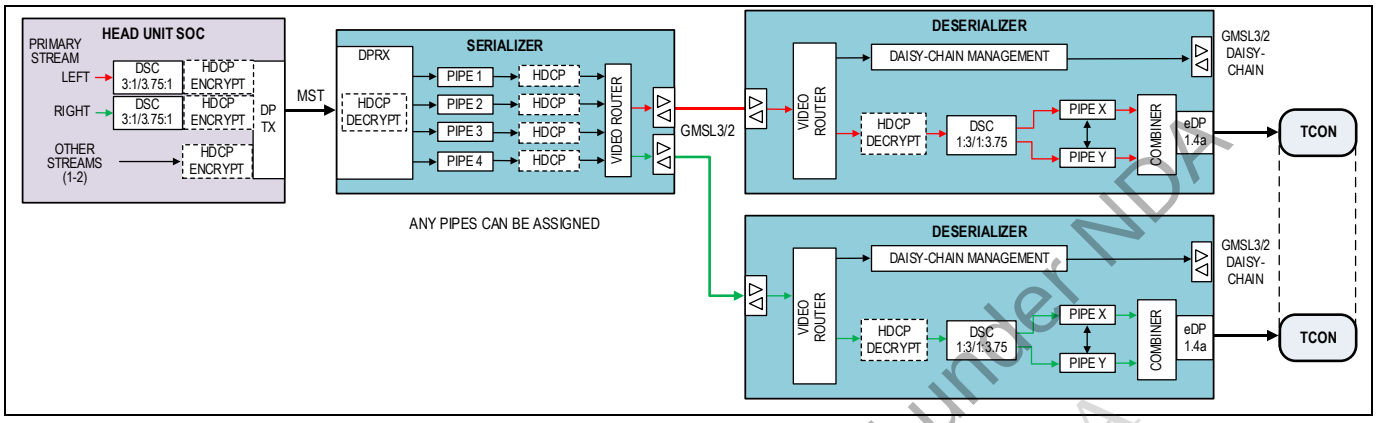


Figure 11. Parallel Deserializers with Compression

Using a single serializer, the MAX96862 supports up to approximately 2160MHz PCLK using compression, GMSL3, and paralleled deserializers. The limiting constraint is eDP output bandwidth. Separate substreams are compressed in the head unit system on chip (SoC) and are routed separately in the serializer and deserializers. Physically separate deserializers and timing controllers (TCONs) may be used to shorten the trace lengths from the TCONs to the panels. Alternatively, the deserializers can be colocated to drive a single two-port TCON. Up to two other video streams can be daisy-chained, either both from the same deserializer or one from each deserializer.

Daisy-Chaining

Compressed Video with Daisy-Chained Deserializer (MAX96862 only)

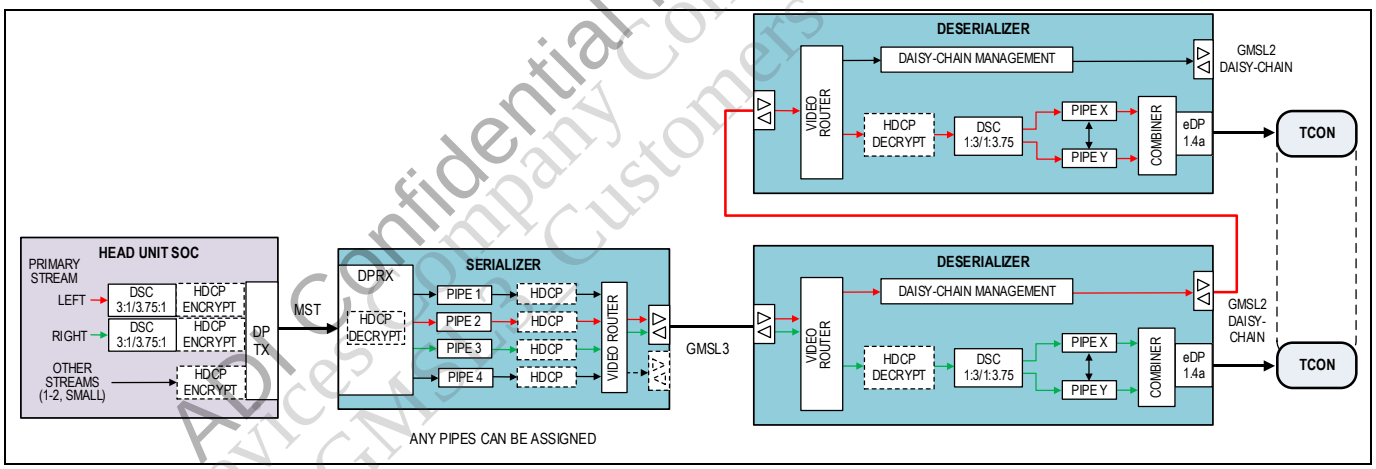


Figure 12. Daisy-Chain Configuration with Compression

To support ultra-high resolution displays, two MAX96862 deserializers can be daisy-chained. The advantage is that only one cable is needed between the head unit and display, with one less connector on the head unit.

The limiting constraint is the bandwidth of the GMSL3 link between the serializer and the first deserializer. This limits the maximum total PCLK to approximately 1200MHz. Up to two other video streams can be daisy-chained from the second deserializer, providing a total of four streams.

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Daisy-Chained Deserializer(s) Between Serializer and MAX96860/MAX96862

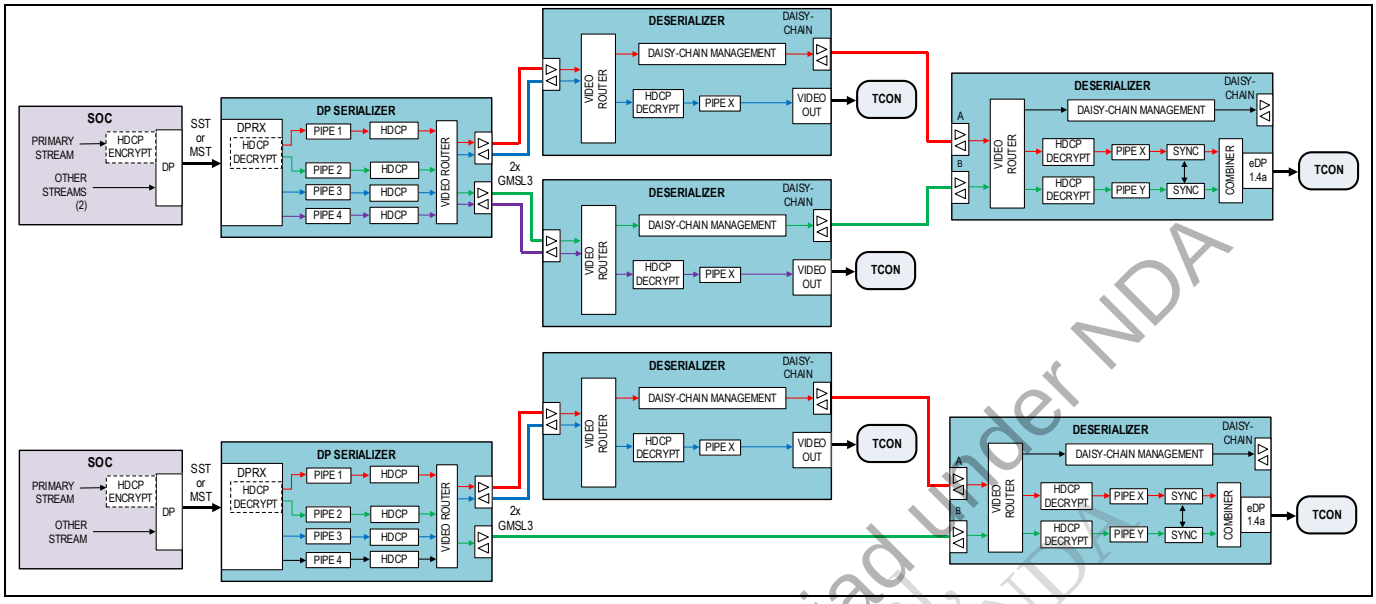


Figure 13. Daisy-Chain with Even/Odd Splitter Configuration

When in Even/Odd Pixel Splitter mode, MAX96856/MAX96860 will support one inline daisy-chain deserializer in either or both GMSL links. If only one link includes an inline deserializer, it must be in Link A (input to the MAX96856/MAX96860 on SIOA). If Even/Odd Pixel Splitter mode is not used, then a single inline deserializer may be inserted in either link. In a parallel configuration, the deserializer connected to SIOB of the serializer must have a local ECU for control channel access.

Video and Audio Path

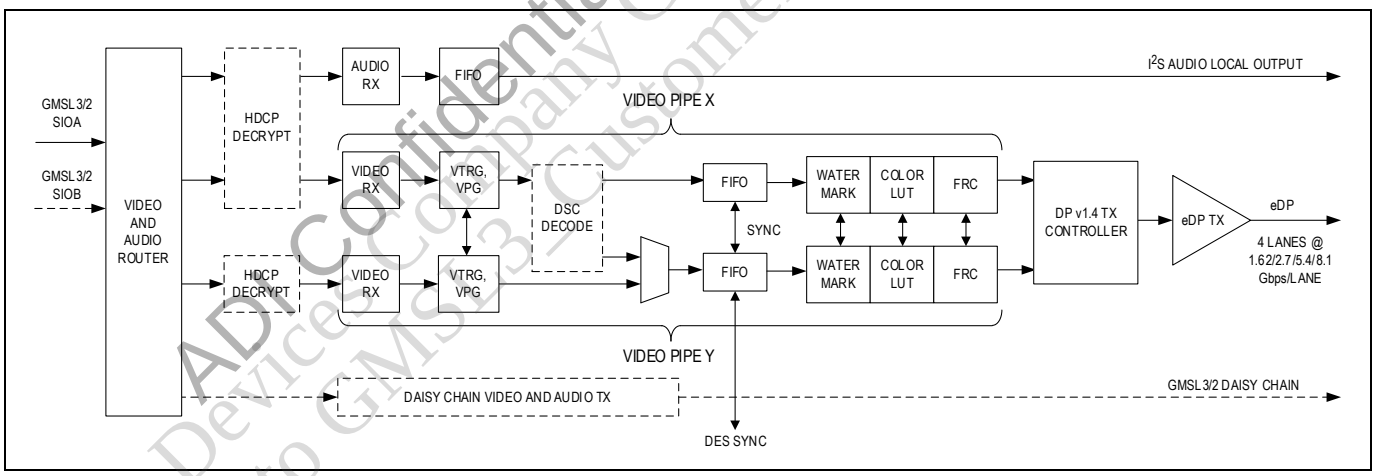


Figure 14. Video and Audio Signal Path

If two video streams with HDCP encryption are daisy-chained from MAX96856/MAX96860, they must route from the serializer to the MAX96856/MAX96860 on the same GMSL link. Both streams must be input on the same SIO port of MAX96856/MAX96860. The exception to this rule is if the second rank deserializer is a MAX96860/MAX96862, then the two HDCP-encrypted streams can arrive from the serializer on different GMSL links.

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CFG[1:0] Pins

The CFG[1:0] pins set initial register values and functional modes of the device at power-up. At power-up, the voltage levels at the CFG[1:0] input pins are latched. The decoded levels set initial register values and functional modes that might not be easily programmed through I2C or UART after the IC powers up. The configuration tables provide the details on how to choose resistor values for the CFG[1:0] pins.

The CFG[1:0] pins are shared with MFP pins. The chip does not drive the pin before the pin levels are latched. Connect a properly valued precision resistor network (according to the needed setting) to each CFG pin. Use the MFP pins (which double function as CFG pins) only as outputs after power-up.

See *Table 2*.

Table 2. CFG[1:0] Pin Configuration Table

CFG[1:0] INPUT VOLTAGE (PERCENTAGE OF V_{DDIO}) (NOTES A, B)			SUGGESTED RESISTOR VALUES (1% TOLERANCE) (NOTE C)		CFG0 MAPPED CONFIGURATION (NOTE D)			CFG1 MAPPED CONFIGURATION (NOTE E)		
MIN	TYP	MAX	R1 (Ω)	R2 (Ω)	DCIO OUTPUT RATE	DEVICE ADDRESS	GMSL STREAM ID	I2CSEL	CX_TP_SIO/ DCIO	SIO INPUT RATE
0.0%	0.0%	11.7%	OPEN	10,000	12G PAM4	0x90	0	I2C	Coax	12G PAM4
16.9%	20.2%	23.6%	80,600	20,500		0x94	1			6G NRZ
28.8%	32.1%	35.5%	68,100	32,400		0x98	2		STP	12G PAM4
40.7%	44.0%	47.4%	56,200	44,200		0xD0	3			6G NRZ
52.6%	56.0%	59.3%	44,200	56,200	6G NRZ	0xD4	0	UART	Coax	12G PAM4
64.5%	67.9%	71.2%	32,400	68,100		0xD8	1			6G NRZ
76.4%	79.8%	83.1%	20,500	80,600		0x50	2		STP	12G PAM4
88.3%	100%	100%	10,000	OPEN		0x54	3			6G NRZ

Notes:

A. Resistor divider tolerance, V_{DDIO} supply ripple, and external loading must not cause CFG[1:0] input voltage to exceed the maximum or minimum limits.

B. Other than the CFG[1:0] input resistor divider, any load on CFG[1:0] must be ≥ 25 x (R1 + R2).

C. Each resistor in the voltage divider must be ≤100kΩ.

D. DCIO OUTPUT RATE: GMSL daisy-chain output link rate.

DEVICE ADDRESS: Device I2C/UART address.

GMSL STREAM ID: Default GMSL stream ID associated with device address.

E. I2CSEL: I2C or UART interface for SDA_RX and SCL_TX.

CX_TP_SIO/DCIO: Shielded Twisted-Pair/Coax mode serial link.

TX_RATE_SIO/DCIO: Forward channel serial rate in bits-per-second.

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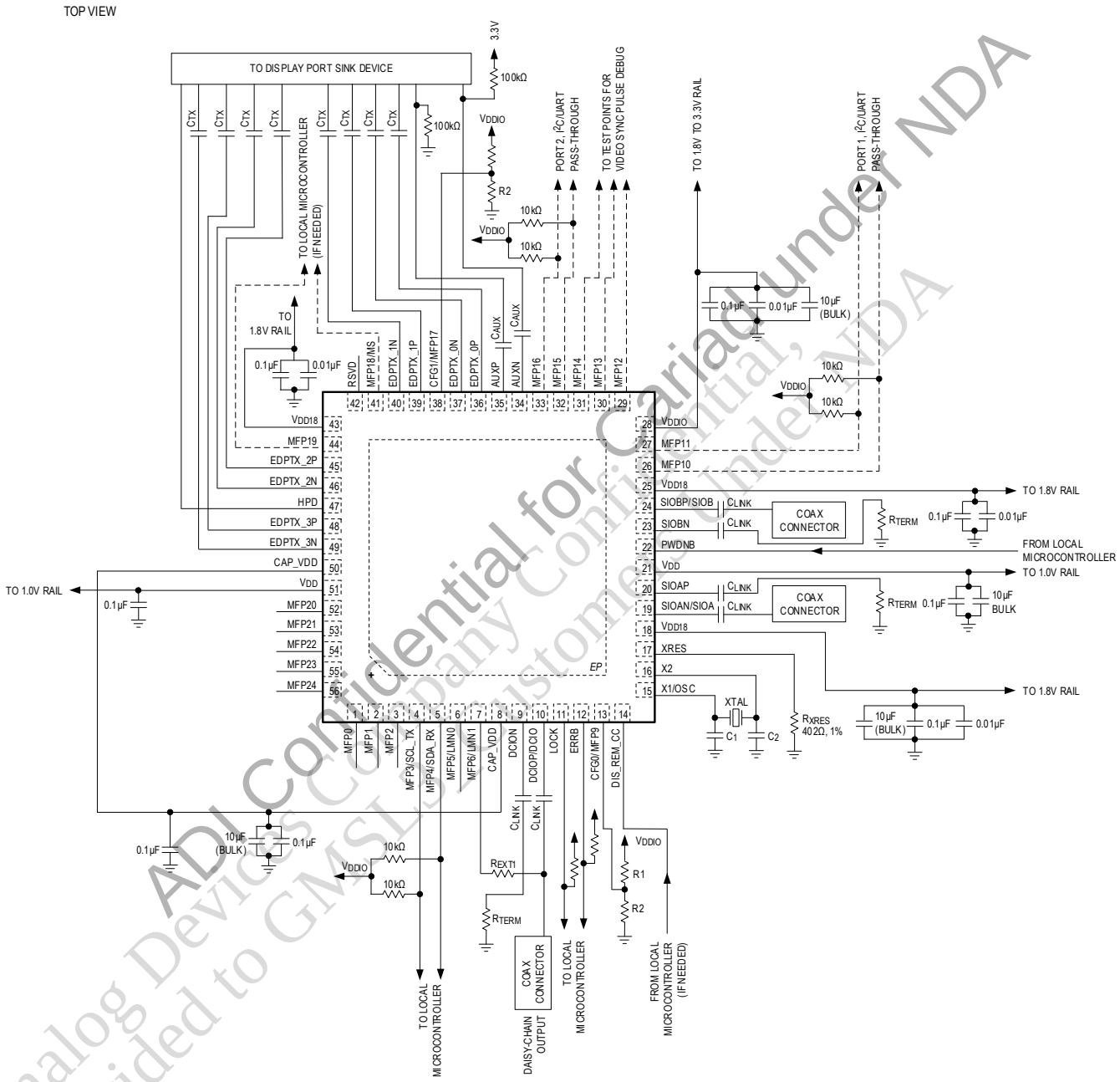
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Applications Information

Typical Application Circuits

Application Circuit (Coax)

See *Table 3* for additional information.

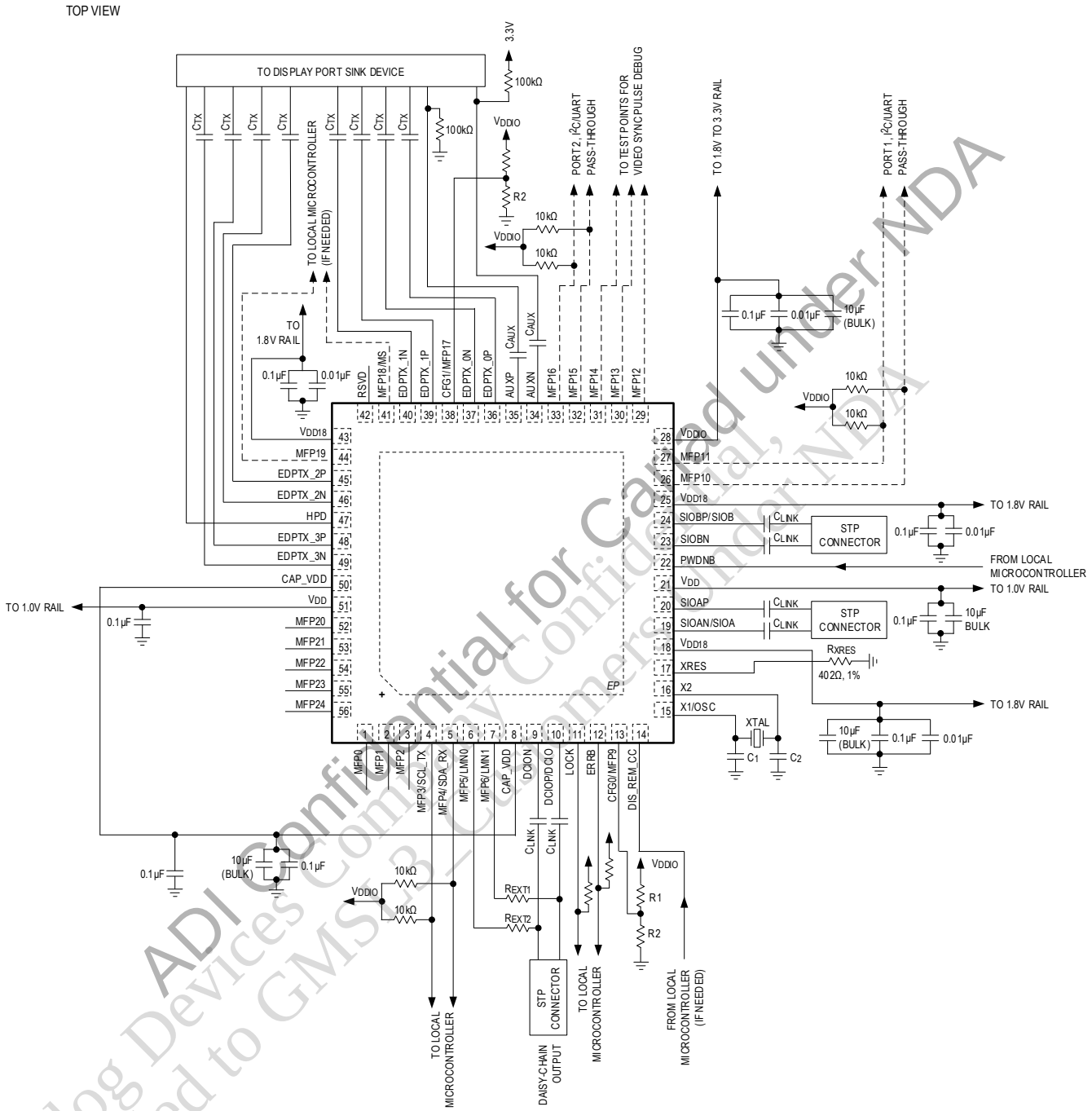


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Application Circuit (STP)

See *Table 3* for additional information.



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External Component Requirements

Table 3. External Component Requirements

COMPONENT	SYMBOL	CONDITION	VALUE	UNIT
XRES	R_{XRES}	Connect to pin 17.	402 ±1%. Use a single resistor	Ω
Crystal		Place as close as possible to pins 15 and 16.	25MHz ±200ppm	
Crystal Load Capacitors		Use crystal loading capacitor guidance from the crystal manufacturer. Select values which compensate for the X1 and X2 input and PCB node capacitances. Place the capacitors as close as possible to pins 15 (X1/OSC) and 16 (X2).	Application specific	
Link Isolation Capacitors	C_{LINK}	Place close to the SIO (19, 20, 23, 24) and DCIO pins (9, 10) used in the application.	0.1	µF
Coax Mode Termination Resistor	R_{TERM}	In Coax mode, connect R_{TERM} between unused SIO (20, 23) and DCIO (9) pin and ground on each coax link.	49.9 ±1%	Ω
SIO Line Fault Pull-Down Resistor	R_{PD}	Install at each used pin (19, 20, 23, 24) if attached serializer is using the Line Fault monitor.	49.9 ±1%	kΩ
DCIO Line Fault Resistor – Coax Mode	R_{EXT1}	LMN0 or LMN1	48.7 ±1%	kΩ
	R_{EXT2}		DNI	
DCIO Line Fault Resistor – STP Mode	R_{EXT1}	LMN0 or LMN1	48.7 ±1%	kΩ
	R_{EXT2}		42.2 ±1%	kΩ
V_{DD} Decoupling Capacitors*		Place 0.1µF capacitors as close as possible to V_{DD} pins 21 and 51. Include a minimum of 10µF bulk decoupling on the PCB.	(2x) 0.1 + 10	µF
CAP_VDD Decoupling Capacitors		Place 0.1µF capacitors as close as possible to CAP_VDD pins 8 and 50. Include a 10µF capacitor on the PCB.	(2x) 0.1 + 10	µF
V_{DD18} Decoupling Capacitors*		Place 0.01µF and 0.01µF capacitors as close as possible to V_{DD18} pins 18, 25, and 43. Include a minimum of 10µF bulk decoupling on the PCB.	(3x) 0.01 + (3x) 0.1 + 10	µF
V_{DDIO} Decoupling Capacitors*		Place 0.1µF and 0.01µF capacitors as close as possible to V_{DDIO} pin 28. Include a minimum of 10µF bulk decoupling on the PCB.	0.01 + 0.1 + 10	µF
eDP Source Isolation Capacitors	C_{TX}	Two required per lane.	75–265	nF
eDP Sink Isolation Capacitors	C_{RX}	Optional. Two required per lane if used.	75–265	nF
eDP Pull-Down Resistors	R_{EPD}	Required only if C_{RX} are used. Two required per lane if used.	100k–1M	Ω

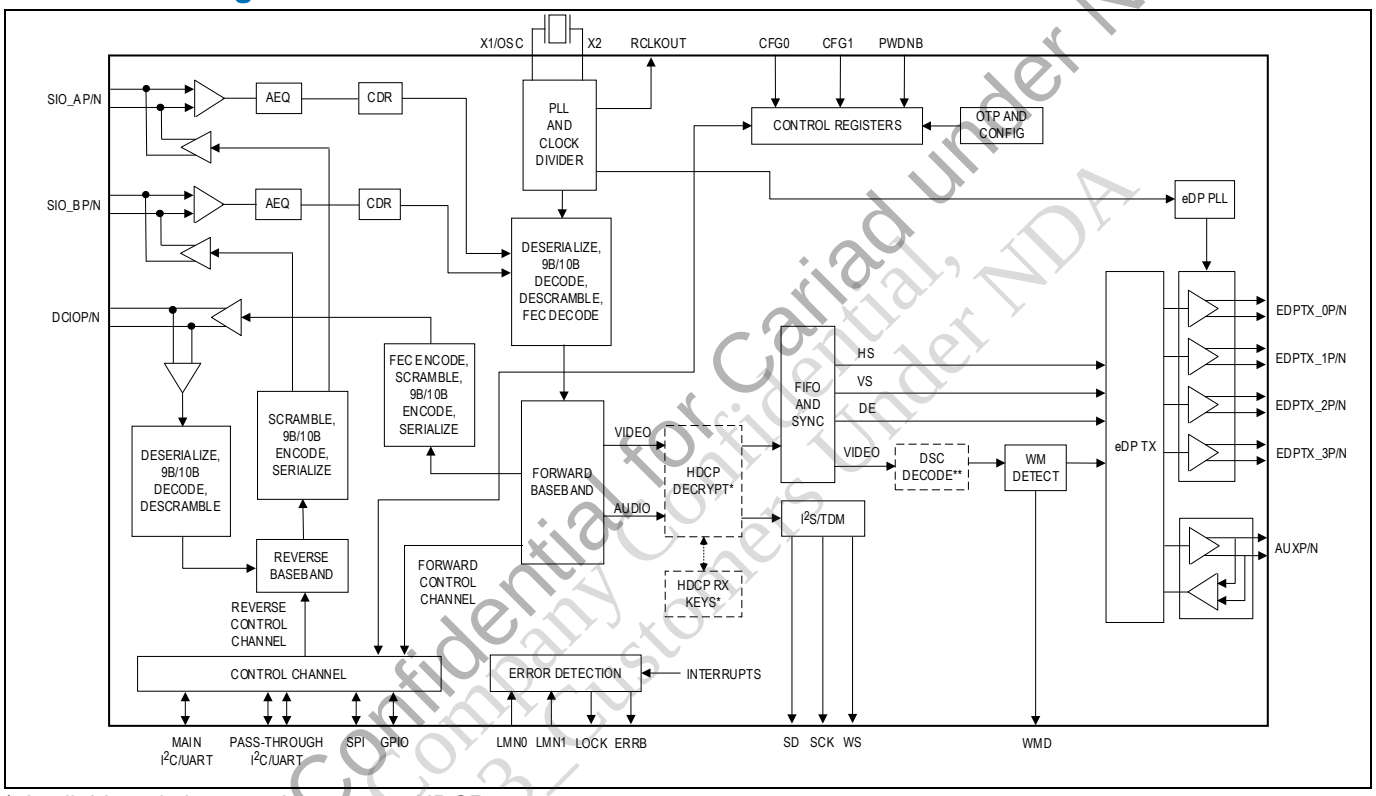
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COMPONENT	SYMBOL	CONDITION	VALUE	UNIT
eDP AUX Isolation Capacitors	C_{AUX}	Two required.	75–200	nF
Open-Drain Pull-Up Resistors		Application-specific. Quantity and values depend on multifunction GPIO pin configurations.	Application specific	

* With exception of CAP_VDD, power supply decoupling capacitor values are recommendations only. It is the responsibility of the board designer to determine what decoupling is necessary for the specific application.

Functional Diagram



* Available only in parts that support HDCP.

** Available only in parts that support DSC.

DCIO is optional. See the Selection Guide for part options.

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Hardware Design Recommendations

Layout Example

The PCB layout of the high-speed serial link traces from the connector to SIO pins is critical to a successful design with GMSL devices. See *Figure 15* and *Figure 16* for examples and design notes of single-ended and differential-pair layouts. In these figures, “P” and “N” refer to the positive and negative GMSL signals and apply to SIOAP/N, SIOBP/N, and DCIOP/N.

Refer to the GMSL Channel Specification for PCB requirements which must be met, and the GMSL Hardware Design and Validation Guide for detailed guidance on schematic and layout recommendations for PCB design to meet the channel specification. Contact the factory to obtain these documents.

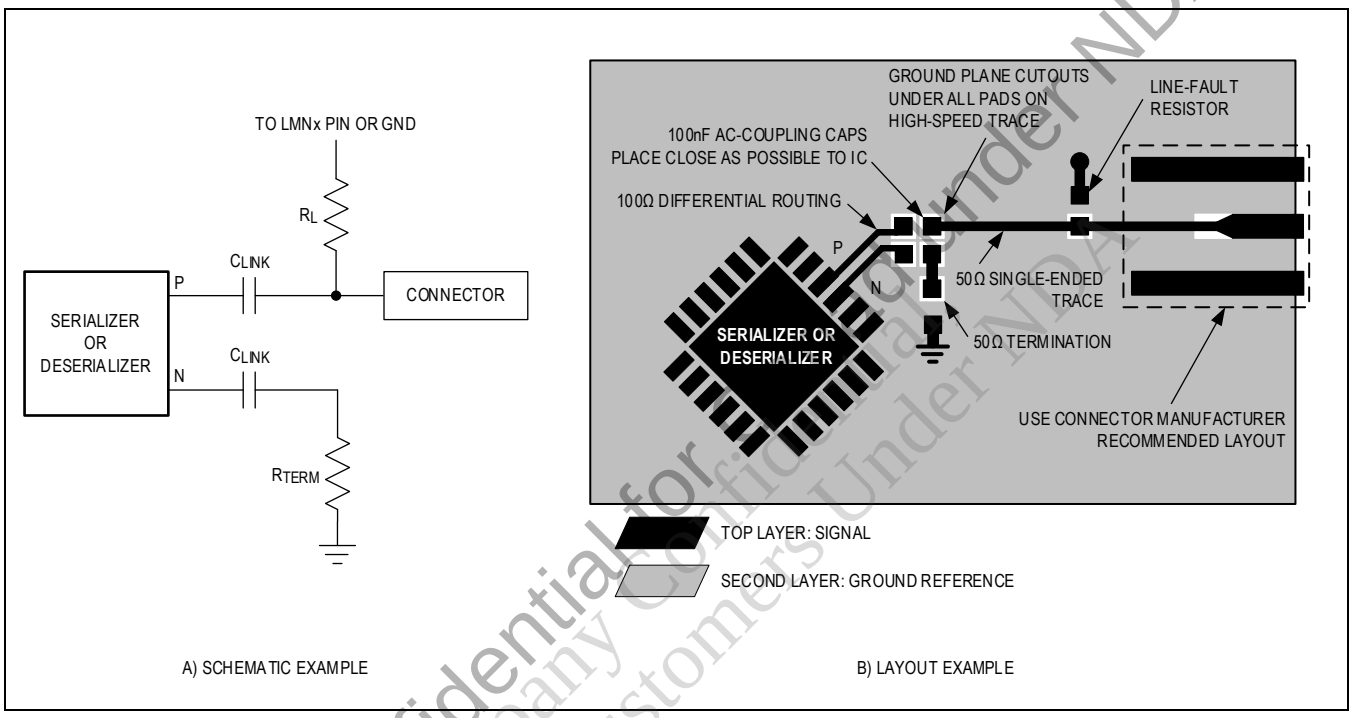


Figure 15. Schematic and Layout Example when Using a Single-Ended COAX Connector

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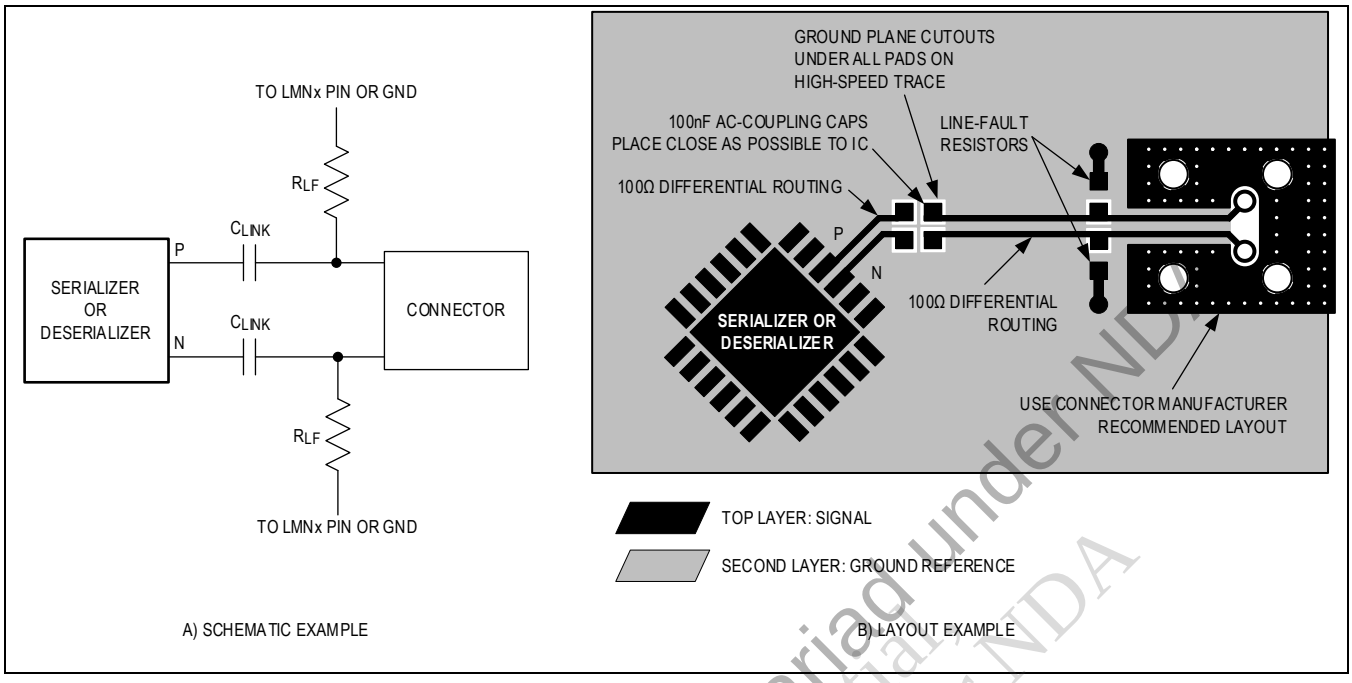


Figure 16. Schematic and Layout Example when Using a Differential STP Connector

Line Fault

The MAX96746/MAX96856/MAX96860/MAX96862 devices include a line-fault detection circuit that detects and reports open-circuit, short-to-battery, short-to-ground, and line-to-line short. The line-fault monitor requires external resistors R_{EXT1} and R_{PD} in Coax mode, and R_{EXT1}, R_{EXT2}, and R_{PD} in STP mode connected to the LMNx pins.

The line-fault monitor is disabled by default. Configuration options are available through the registers, and the status can be read by the register. If unmasked, a line-fault condition asserts ERRB. Line-fault detection cannot be used in conjunction with power-over-coax (PoC) or AC-coupled ground applications.

The line-fault monitor pins offer flexible connection and programming in either coax or STP applications.

Figure 17 illustrates the two configuration options to locate the line-fault detection.

- (A) The local-side serializer configuration is typically used in single SER/DES pair applications.
- (B) The local-side daisy-chain deserializer configuration is applicable to display use cases in a daisy-chain configuration.

All configurations apply to the Coax or STP modes.

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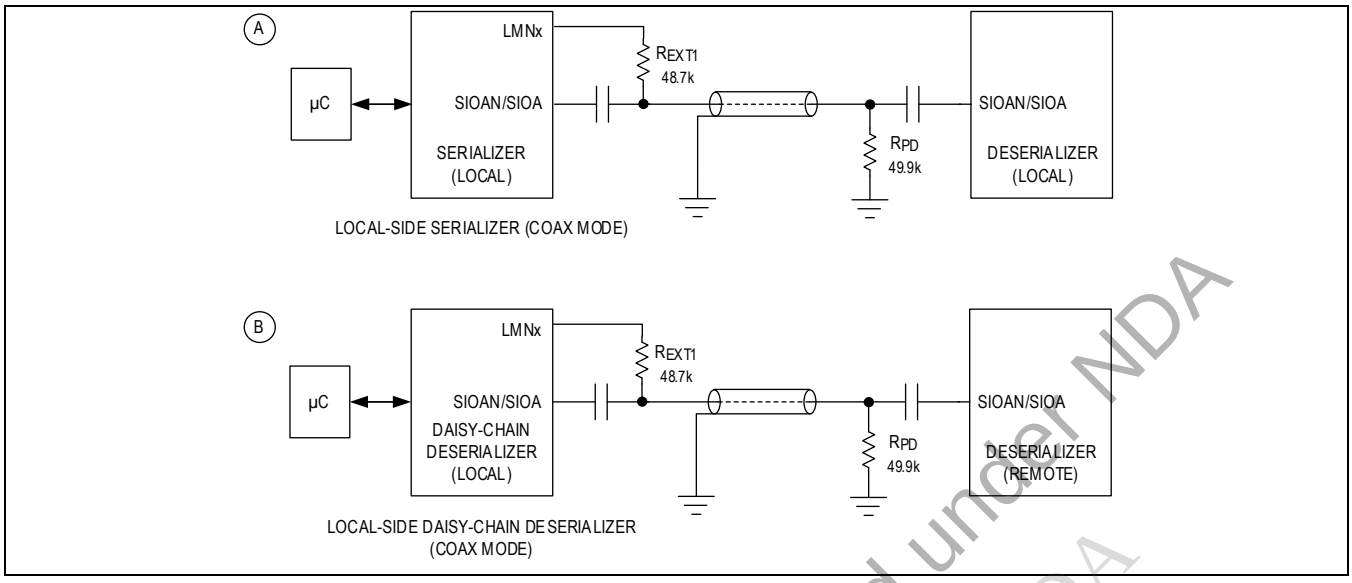


Figure 17. Line-Fault Detection Location Options

The LMNx pins are typically mapped to different multifunctional pins on each unique part and package option.

Coax Mode

In coax applications, as shown in *Figure 18*, the line-fault source (serializer or daisy-chain deserializer) connects a LMNx pin to the line using a 48.7kΩ resistor (R_{EXT1}). On the deserializer side, a 49.9kΩ resistor (R_{PD}) connected to GND is required.

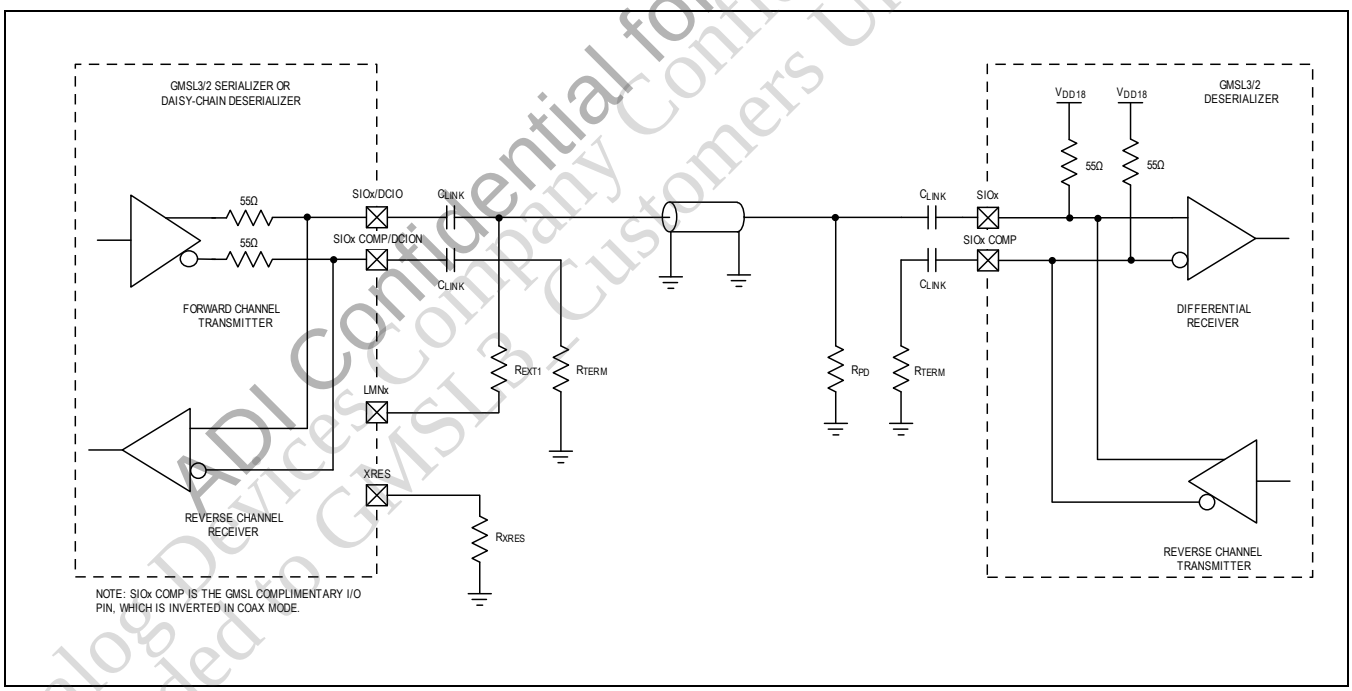


Figure 18. Typical GMSL Link Application Circuit for Coax Cable

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STP Mode

In twisted-pair applications, as shown in *Figure 19*, the line-fault source (serializer or daisy-chain deserializer) connects one LMNx pin to one of the twisted-pair lines using a 48.7kΩ resistor (R_{EXT1}) and to the other line using a 42.4kΩ resistor (R_{EXT2}). On the deserializer side, both lines require a 49.9kΩ resistor (R_{PD}) connected to GND.

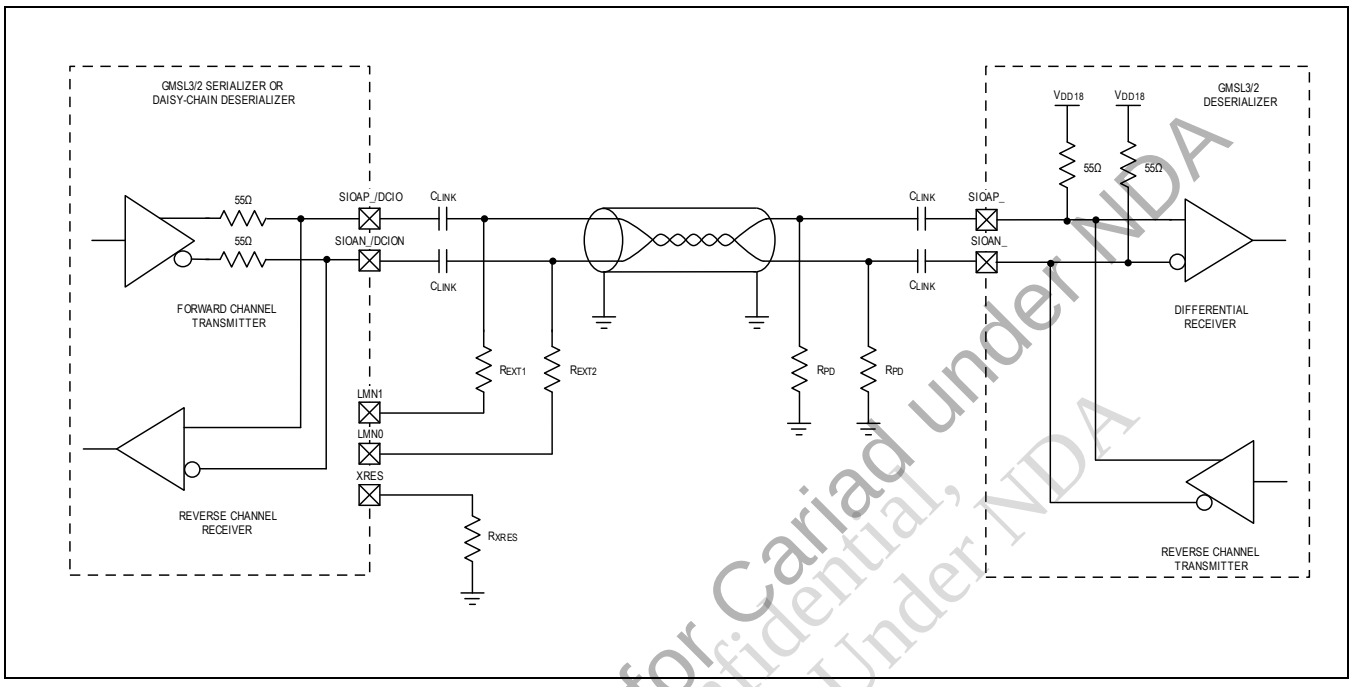


Figure 19. Typical GMSL Link Application Circuit for Twisted Pair

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Recommended Operating Conditions and ESD Protection

Table 4. Recommended Operating Conditions

PARAMETER	PIN NAME	NOMINAL VOLTAGE	MIN	TYP	MAX	UNIT
Supply Range	V_{DD18}		1.7	1.8	1.9	V
	V_{DD}		0.95	1.00	1.05	V
	V_{DDIO}	1.8V	1.7	1.8		V
		3.3V		3.3	3.6	V
Maximum Supply Noise Supply Noise < 1MHz	V_{DD18}			25		mVp-p
	V_{DD}			25		mVp-p
	V_{DDIO}	1.8V		50		mVp-p
		3.3V		100		mVp-p
Operating Junction Temperature, T_J			-40		+125	°C

Table 5. ESD Protection (Unpowered)

PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNIT
SIO, DCIO	V_{ESD}	Human Body Model (HBM), RD = 1.5kΩ, CS = 100pF		±8		kV
		ISO10605, RD = 330Ω, CS = 100pF, contact discharge, coax configuration		±6		
		ISO10605, RD = 330Ω, CS = 100pF, contact discharge, STP configuration		±6		
		ISO10605, RD = 330Ω, CS = 100pF, air discharge		±8		
All Other Pins	V_{ESD}	Human Body Model (HBM), RD = 1.5kΩ, CS = 100pF		±4		kV
SIO, DCIO Corner Pins All Other Pins	V_{ESD}	AEC-Q100-011, Classification C1, Charged Device Model (CDM)		750		V

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Power Supplies

V_{DD} is the digital supply and should be connected to a 1.0V power supply. V_{DD18} is the analog supply and should be connected to a 1.8V power supply. V_{DDIO} should be connected to a 1.8V-3.3V power supply. Supplies should comply with the supply minimums, maximums and noise specifications defined in *Table 3*.
Proper power supply bypassing of all supplies is essential for high-frequency circuit stability. See *Table 4*. Contact factory for guidance on sharing supplies and optimizing supply decoupling.
The power supply ramp-time recommendation is:

$$200\mu s < ramp\ time < 2ms$$

Power supply ramps should be monotonic. Once the supply voltage has reached the minimum supply voltage limit, it should not be allowed to drop below the specification.

Analog Devices provides power management ICs (PMICs) optimized for supporting serial link devices. Contact the factory for information.

Power sequencing requirements only apply to the MAX96856 and are:

- Power-up sequence:
  - V_{DD18} must be enabled first
  - V_{DDIO}, V_{DD}, and PWDNB must be enabled >1ms after V_{DD18} reaches its minimum supply voltage limit. The preferred sequence is:
    - V_{DDIO} must be enabled >1ms after V_{DD18} reaches its minimum supply voltage limit
    - V_{DD} enabled >1ms after V_{DDIO} reaches its minimum supply voltage limit
- Power-down sequence:
  - Disable power supplies in the reverse order from what's used in the power-up sequence
  - V_{DD18} supply must be disabled last.

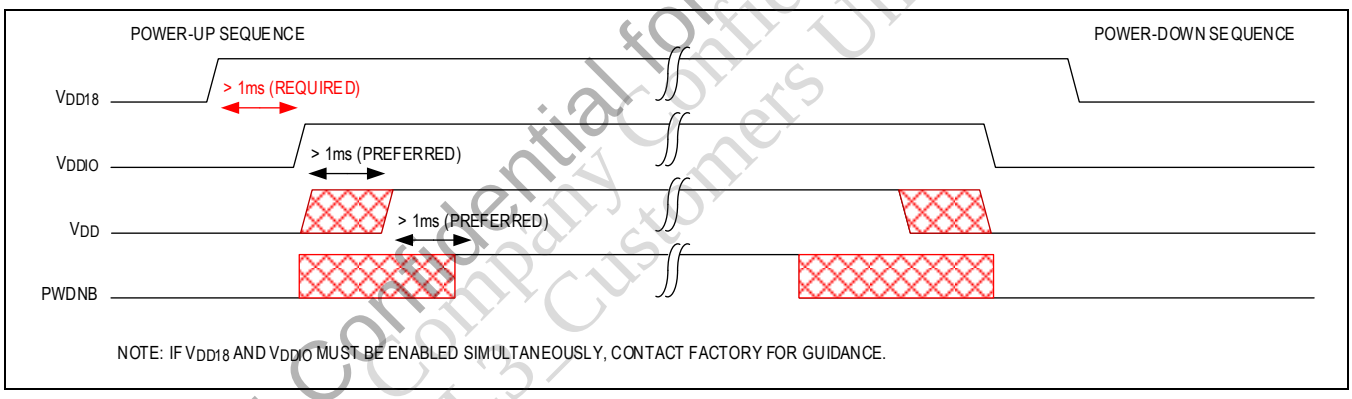


Figure 20. Power-Up/Power-Down Sequence Timing Diagram

Power Manager

The primary function of the power manager is to control the internal switched supply domains during the full sequence of power states. It serves as a guard before the chip powers up and allows it to do so smoothly. Once the internal supplies meet their valid levels, the power manager relinquishes control to the digital core for normal operations.

Another function of the power manager is to ensure the chip consumes minimum standby current during Sleep mode and can quickly restore configurations to full operation after waking up. Finally, the power manager monitors the power supplies for undervoltage (UV) and overvoltage (OV) conditions.

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Thermal Management

System thermal management must keep the operating junction temperature below 125ºC to avoid impacting device reliability. The target θ_{JA} value for the system designer is determined by the expected maximum power consumption and the cooling system efficiency, which varies based on use case. Heat dissipation of the MAX96746/MAX96856/MAX96860/MAX96862 is accomplished primarily through the exposed pad, so proper PCB design is essential for adequate thermal performance.

Contact the factory for more details.

Channel Specifications

Maximum cable length is limited by the frequency-dependent attenuation of the cable. Additionally, PCB and inline connectors degrade the return loss characteristic of the signal path. The GMSL3 or GMSL2 Channel Specification allows for up to four inline connectors (two on the PCBs and two between three cable segments) and provides detailed requirements for cable attenuation and return loss, as well as insertion loss and return loss requirements for PCB traces. In general, any physical channel implementation compliant with the GMSL3 or GMSL2 Channel Specification can be used with reliable results. Contact the factory for the GMSL3 or GMSL2 Channel Specification document.

Software Programming Model

Analog Devices' automotive serializers and deserializers follow a general software programming model. Except for features that require in-operation control channel accesses, such as ASIL safety measures and interrupt handling, the following programming model is used:

1. Set the impacted functional blocks to Disabled or Reset mode. A general method to place the part in IDLE state is to stop all side channel and video traffic, followed by a register write (RESET_LINK = 1) to stop the GMSL link.
2. Fully configure the settings for each feature before it is enabled.
3. Establish the link by setting RESET_LINK = 1 and wait for the link to lock.
4. Start video and side-channel traffic.

If changing the configuration of a feature during the operation of other features, disable the reconfigured feature, change its settings, and re-enable it.

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Features

GMSL Protocol Layer

GMSL3/2 is a fixed-rate transmission medium designed to carry multiple types of communication channels concurrently. The link bit rate is based on a constant-frequency link clock generated from the 25MHz crystal oscillator or an external reference frequency. The link clock does not have any relation to the video pixel clock.

GMSL3/2 uses a packet-based protocol to seamlessly share the link bandwidth between communication channels in a flexible way. Bandwidth allocation is dynamic so that if a certain channel is not active, it does not consume any link bandwidth, and all the remaining active channels can share the full link bandwidth. The maximum packet size is limited to prevent a single channel from monopolizing the link bandwidth for an extended time and to ensure that other channels are served. The user must ensure that the application's peak bandwidth requirement does not exceed the available link bandwidth. Idle packets are used to fill unused link bandwidth. The same data protocol is used on forward and reverse channels.

Link Modes

GMSL2 Mode

GMSL2 mode utilizes NRZ modulation. The MAX96746/MAX96856/MAX96860/MAX96862 support a forward link rate of 6Gbps. When set to GMSL2 mode by configuration pin CFG1 on power-up, forward error correction (FEC) is disabled by default. It can be enabled using register writes, if desired. FEC must be enabled in compressed video applications. Enabling FEC in uncompressed applications increases link robustness and is recommended if sufficient bandwidth is available (FEC consumes a fixed 6.7% bandwidth overhead when enabled).

The GMSL channel between the serializer output and deserializer input, including the PCB, connectors and cables, must meet the GMSL2 channel specification.

GMSL3 Mode

GMSL3 mode utilizes PAM4 modulation at a forward link rate of 12Gbps. When set to GMSL3 mode by configuration pin CFG1 on power-up, FEC is automatically enabled to provide robust communication. If GMSL3 is configured using register writes, FEC must be manually enabled. FEC is required in GMSL3 mode.

The GMSL channel between the serializer output and deserializer input, including PCB, connectors and cables, must meet the GMSL3 channel specification.

Even though GMSL2 and GMSL3 both use signal frequency of 3GHz, the GMSL3 channel specification differs from the GMSL2 channel specification. Typically, the same cables and connectors used in 6Gbps GMSL2 mode can be applied in 12Gbps GMSL3 applications, but it must be verified they meet the slightly stricter GMSL3 channel specification.

Link Integrity Features

GMSL provides link diagnostic tools for system bring-up and debug. These diagnostic tools leverage parts of the adaptive equalizer implementation and are neither laboratory instrumentation nor are they designed to demonstrate channel compliance.

The GMSL diagnostic toolset provides a high level of visibility of the link and system performance. The MAX96746/MAX96856/MAX96860/MAX96862 diagnostic tools include:

- Eye mapping
- Eye-opening monitor
- Link (voltage) margining
- FEC input BER rate

These tools are a mixture of hardware and supporting software for data collection and processing. Remote control channel access must be enabled to use these functions. The graphic user interface (GUI) provides the required software and processing.

Forward Error Correction (FEC)

FEC is used to reduce the GMSL link bit-error rate (BER) by detecting and correcting link errors. FEC is applied in the forward channel only and consumes approximately 6.7% additional link bandwidth. FEC is mandatory for operation in

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GMSL3 mode and when compressed video data is transmitted over a GMSL2 link. FEC is optional for uncompressed GMSL2 use cases (though it is recommended if the bandwidth is available).

Features and Notes:

- Error correction is applied to all data traffic in the GMSL forward channel.
- Burst-error correction capability equivalent to three FEC symbols (54 bits) per FEC block.
- Reports statistics which can be used to determine pre-correction and post-correction BERs.
- Optional ERRB interrupt output when uncorrectable errors exceed 32-bit programmable threshold (default threshold is one uncorrected error).
- Enabling FEC increases the required link bandwidth for video and all forward control and side-channel data by a factor of 128/120 (approximately 6.7%).
- FEC adds approximately 4300 UI latency to the link.

FEC Statistics Monitoring

The FEC block allows for the pre-correction and post-correction bit error rate (BER) to be calculated by the link. The FEC decoder reports the number of blocks processed, errors corrected and the number of uncorrectable errors. Contact the factory for details.

GMSL Reference Clock

GMSL devices require a reference clock source to generate the link-rate clock and associated internal clocks. The serializer can be clocked using an external 25MHz crystal or an external low jitter clock source.

Reference-Clock Generation

The 25MHz RCLKOUT can be used as a reference clock by a GMSL3/2 serializer in close proximity. This arrangement eliminates the need for an additional external crystal or oscillator for the second serializer.

Power Modes

The MAX96746/MAX96856/MAX96860/MAX96862 devices support the following power modes:

- ACTIVE: All required circuits are powered up.
- PWDNB: All blocks, including the control channel, are inactive.
- SLEEP: All blocks, including the control channel, except local and remote wake-up circuitry, are disabled.

There are two methods to go into low-power mode while all power supplies are active: assertion of the PWDNB pin (Power-Down mode) or invocation of the sleep state (Sleep mode). Both states provide very low power supply currents.

Power-Down (Standby) Mode

Asserting the PWDNB pin (active-low) places the device in Standby-Power mode and resets the digital registers and configurations to their default power-up condition. Any supply dropping below its internal threshold settings also places the device in Power-Down mode. If the power supply recovers, the device returns to the default power-up state with default register settings. Device start-up can then be reinitiated, and all of the registers reprogrammed.

Sleep Mode

The sleep state preserves a subset of register settings and configurations. The device can be put into the sleep state through an I^2C/UART command. When returning to active state, the device is restored to a condition where the GMSL configuration and control channel configuration is preserved. Video configuration is required.

Resume is invoked when the host processor initiates a dummy control channel transaction, which briefly wakes up the device. The initial temporary wakeup requires four falling edges on SDA. Depending on the device address used, there may need to be multiple consecutive dummy transactions issued by the main to achieve the required four SDA falling edges in I^2C mode. If the primary I^2C inputs are assigned to one of the pass-through I^2C input pins (SDA1/SCL1, SDA2/SCL2), the wake from sleep function can only be done through the main SDA_RX/SCL_TX pins, not the reassigned pass-through primary I^2C pins.

Daisy Chaining

The MAX96746/MAX96856/MAX96860/MAX96862 deserializers can be connected in daisy-chain configurations with up to four daisy-chained deserializers on each GMSL link when driven by a GMSL serializer that supports this feature. The MAX96746/MAX96856/MAX96860/MAX96862 devices select one of four possible video streams to output locally. By

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default, the stream that is output from the local device is removed from the data that is forwarded to the next deserializer. The deserializer can be programmed to not drop this video stream. The remaining video streams are repacketized and forwarded downstream to the additional deserializer in the chain.

All MAX96746/MAX96856/MAX96860/MAX96862 primary control and side channels (video, I^2C, UART, I^2S/TDM and DP audio, and GPIOs) can be directed to any of the deserializers in the daisy-chain. Deserializers can only return channel data to the serializer. The deserializers cannot communicate directly with each other. It is, however, possible for deserializers to communicate with one another through a turnaround of the data within the main ECU. For example, DESERIALIZER1 may send a message through I^2C to DESERIALIZER0 by first sending the request on the reverse link to the serializer. The serializer can then resend the message on the forward link to DESERIALIZER0.

The deserializer's lock, faults, and errors (ERRB) can be forwarded to the upstream serializer's lock and error functions or optionally to individual serializer GPOs.

The MAX96746/MAX96856/MAX96860/MAX96862 daisy-chain port supports 12Gbps GMSL3 and 6Gbps or 3Gbps GMSL2 operation.

Figure 21 illustrates a 2+2 daisy-chain configuration with four individual video streams. Two streams are routed to Link-A and two to Link-B. Each link can support a maximum of four daisy-chained deserializers.

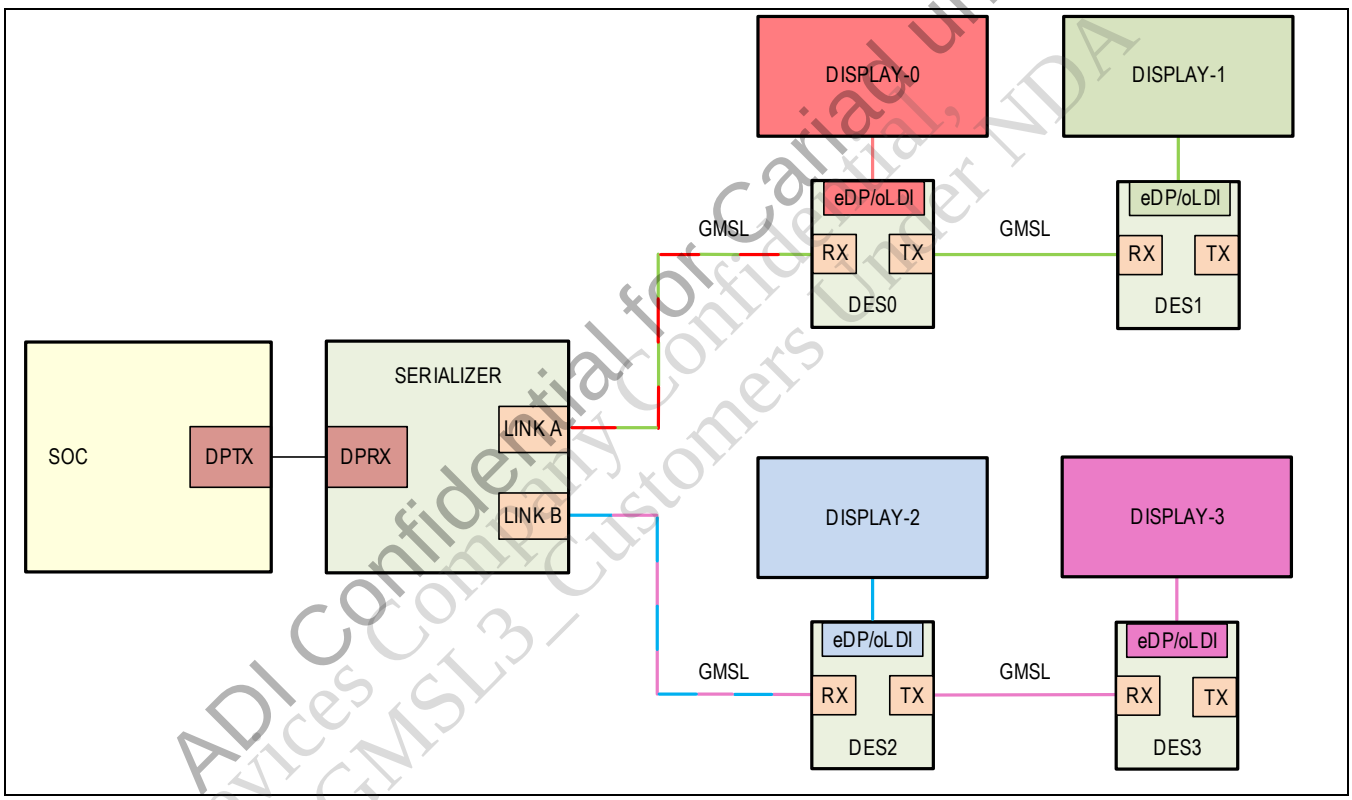


Figure 21. 2+2 Daisy-Chain

Even/Odd Pixel Splitter Mode

Select GMSL serializers include the ability to split the video coming from the source (typically an SOC) into two sub-streams by separating the even and odd pixels on each line of the video. The sub-streams are routed to two video pipes and are subsequently output to the GMSL links. The MAX96746/MAX96856/MAX96860/MAX96862 have two GMSL inputs and recombine the sub-streams into the original video stream. See *Figure 9* for a reference of the even/odd pixel splitter feature.

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Excess bandwidth on either or both two links can be forwarded over the DCIO. When using Even/Odd Pixel Splitter mode, the bandwidth limitations for the link configuration mode (GMSL3/2) must be considered. Contact the factory for support on using Even/Odd Splitter mode.

Adaptive Equalization (AEQ)

GMSL devices automatically adapt receiver characteristics to compensate for the insertion and return loss characteristics of the channel, which consists of the cables, connectors, and PCBs. For both the forward and reverse channel, the equalizer readapts on the live link (at a 1Hz rate) to continually track and adapt to changing channel characteristics, which can vary with temperature, cable bending, dynamic POC loads, cable aging, and/or other external factors.

For extremely short channels (i.e., insertion loss < 2dB), disable feed-forward equalization (FFE) by writing register RLMSCE (0x8CE) to 0x0.

GMSL Link Spread Spectrum

While typically not required to meet electromagnetic interference (EMI) requirements, optional spread-spectrum clocking (SSC) is available to mitigate EMI emitted from devices and interconnections, provides additional margin, and is recommended to be used when possible. SSC reduces peaks in the frequency spectrum by spreading the signal over a wider bandwidth. The spread has a 25kHz sawtooth modulation profile, programmable to deviate up to ±1250ppm from the center frequency. Contact the factory for more information.

GMSL Bandwidth Sharing

The GMSL forward bandwidth is shared between video, the primary I^2C/UART control channel, pass-through I^2C/UARTs, I^2S/TDM audio, SPI, and GPIOs. The reverse-channel bandwidth is also shared between all of the previously stated communication channels, with the exception of video packets. The total link bandwidth used by all communication channels cannot exceed the fixed available link bandwidth in either direction. If the max PCLK limit is being approached by an application, contact the factory.

GMSL3 devices use a bandwidth allocation scheduler that acts as an arbiter if multiple channels (i.e., video or control channels) attempt to simultaneously send packets over the link. The scheduler uses a channel-based priority setting and a calculated running average of the bandwidth usage of each type of communication channel to schedule transmissions over the link and throttle channels if needed.

The scheduler also uses a configurable priority level to modify the urgency at which packets are transmitted. The video and control channel packets can be assigned one of four priority levels: low, normal, high or urgent. Communication channels with maximum urgency or minimum latency requirements can be assigned an increased priority.

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GMSL Bandwidth Calculations

The GMSL3/2 forward link has a fixed link rate of 12Gbps or 6Gbps (or 3Gbps on the daisy-chain link). The reverse rate is fixed at 187.5Mbps. The GMSL3/2 protocol overhead is roughly 14%, which leaves approximately 10.4Gbps or 5.2Gbps of data throughput in the forward direction and 162Mbps in the reverse direction. FEC is optional in GMSL2 mode and mandatory in GMSL3 mode. If enabled, FEC will consume an additional 6.7% of overhead. FEC is also mandatory when video is compressed with VESA DSC.

Ensure that worst case utilization does not exceed the available throughput of the forward and reverse links. Analog Devices' evaluation kit (EV kit) GUI includes a bandwidth (BW) calculator that can be used for initial bandwidth requirement estimates. Analog Devices also has other tools useful for calculating link bandwidth utilization. Consult the factory when considering high bandwidth use cases to ensure error-free performance.

Table 6 provides estimates of the bandwidth utilization for each communication channel.

Table 6. Forward and Reverse Link Bandwidth Utilization

DATA	APPROXIMATE BANDWIDTH UTILIZATION
Display Video (Forward Path only)	Without VESA DSC compression PCLK x (bpp + 0.8 + packet_CRC) x 10/9 x 2048/2047 x 128/120* (*only applies when GMSL FEC is enabled. GMSL FEC must be used with GMSL3 PAM4 modulation) Without VESA DSC compression PCLK x (bpp + 0.8 + packet_CRC) x 10/9 x 2048/2047 x 128/120* x 1/3 (GMSL FEC must be used with VESA DSC compression)
I^2C	13 to 40 x I^2C clock rate
UART	6 x UART bit rate
SPI	1.7 to 3.1 x SPI rate, depending on SPI byte length
GPIO	60 x GPIO transition rate without delay compensation 80 x GPIO transition rate with delay compensation
I^2S/TDM	sample_rate x 20 x (roundup (channelcnt x sample_depth/18 + 0.5) + 2)

Note: Bandwidth utilization for all forward control channel and side channels communication increases by 6.7% when FEC is enabled.

Definitions:

PCLK = Total Horizontal pixels x total Vertical lines x frame rate. Horizontal includes blanking pixels, and vertical lines include blanking lines.

bpp = bits per pixel

packet_CRC = 0.5 if packet CRC is enabled, 0 if not enabled. Packet CRC is disabled by default.

sample_rate = audio sample rate (samples per second)

channelcnt = number of audio channels

sample_depth = bits per sample per audio channel (must be the same for all channels)

Video Lock

Video lock indicates that stable video is being received from the serializer, DP sink is attached to DP port (HPD signal is detected), DP link training started, and the video channel is configured and locked to incoming video. Optionally, the LOCK pin behavior in the deserializer can be changed by a register setting so that the LOCK pin is asserted when the deserializer is receiving video and has video lock (asserted with VIDEO_LOCK_OUTPUT_EN).

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GMSL Link Lock

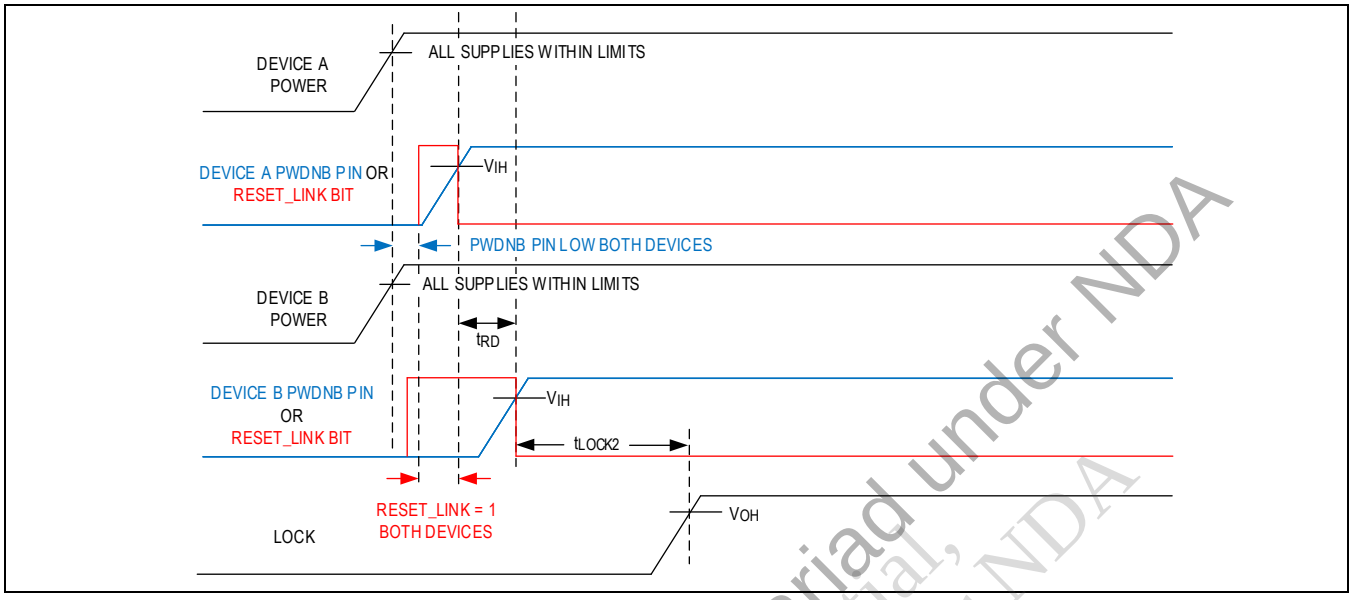


Figure 22. GMSL Lock Time

Note: In the descriptions below, “serializer” refers to either serializer or the daisy-chain output of a deserilizer.

Figure 22 illustrates the sequence that is used to characterize GMSL3/2 link-lock time. Device A is the first device (serializer or deserializer) to power-up or resume operation from a RESET_LINK state. Device B is the device (deserializer or serializer) at the other end of the GMSL link.

Link lock indicates that the data receive paths are locked (forward channel in the deserializer, reverse channel in the serializer). Video and control channel functions (I^2C/UART, SPI, GPIO, and audio) can be used immediately after link lock is asserted.

The device establishes single link GMSL connectivity and link lock automatically following power-up. This is an indication that the cable is plugged in, and the system is up and running. Lock is obtained with no interaction between the µC and GMSL devices. Both serializers and deserializers have an open-drain LOCK output pin and a related status register.

Configurations using SIOB require additional programming. For guidance on enabling these configurations and optimizing the lock time, contact the factory.

The GMSL link uses the crystal as the reference clock for GMSL, so a valid video input (PCLK) is not needed for the GMSL link to lock.

Notes:

- The lock sequence is initiated by the release of the PWDNB pin or the RESET_LINK bit in either the serializer or the deserializer.
- Lock time is measured from the later of PWDNB or RESET_LINK release in either the serializer or deserializer to LOCK being asserted.
- The PWDNB/REST_LINK states on the two sides of the link must have overlap when both devices are in PWDNB/RESET_LINK mode prior to the lock process starting.
- If RESET_LINK is used to initiate lock, PWDNB is assumed to be high after power-up (normal operation).
- If PWDNB is used to initiate lock, RESET_LINK is assumed to be low after power-up (normal operation).
- Lock time and maximum allowed t_{RD} vary between different families of GMSL devices. They depend on the characteristics of both the serializer and deserializer. The typical lock time of a specified link can be best estimated as the longest of the lock times specified in each device data sheet. Similarly, the maximum permissible t_{RD} for a specific

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link can be estimated as the smaller of the values specified in each device data sheet. For further guidance, contact the factory.

- To achieve the specified lock time, time delay t_{RD} (delay between release of the PWDNB/RESET_LINK on the two devices) must be less than the threshold specified in *Note 11* of the electrical characteristics. Contact the factory for guidance if this timing cannot be guaranteed.
- If there is an instantaneous interruption to link lock, at least 100ms following loss of lock should be provided to enable the link to automatically recover prior to any ECU initiated resets being issued. This will minimize any disruptions caused by a transient loss in connectivity.

Device Reset

There are general-reset options available through register writes:

- RESET_ALL resets all blocks, including all registers, digital blocks, and analog blocks. This process is similar to driving the PWDNB pin low and then high.
- Setting RESET_LINK_ALL resets all GMSL PHY-related digital logic and all data pipelines. After this bit is set, all control registers are still accessible through the local control channel. The link remains in RESET until RESET_LINK_ALL is cleared.

To configure the GMSL link, first set RESET_LINK_ALL = 1 and then program registers that affect GMSL link operation (i.e., RX_RATE_SIO/RX_RATE_DCIO, CX_TP_SIO/CX_TP/DCIO), followed by RESET_LINK_ALL = 0.

Link Start-Up

GMSL devices are in power-down mode when the PWDNB is low or when any of the power supplies are below their specified minimum voltage. Registers are set to default reset conditions. Refer to the Register table for the default reset conditions for this device.

The serializer and deserializer can power-up in any order. After PWDNB is released and all power supplies are up, each device starts its power-up sequence and performs the following actions in sequence:

1. Set internal registers according to the state of the multilevel configuration pins CFG[1:0].
2. The local control channel (I^2C or UART) is functional. The device registers are writable and readable. After PWDNB release, wait 4ms for the control channel to be ready.
3. Establish links based on device configuration.
4. Attempt to establish link on SIO pins. SIOA is on by default in the MAX96746/MAX96856/MAX96860/MAX96862, but SIOB requires additional configuration.
5. Attempt to establish link on DCIO pins when configured as a daisy-chain device. DCIO only starts to lock once SIOA is locked on the repeater device.
6. Each enabled PHY performs link calibration, equalizer adaptation, and data channel locking. The deserializer sets the LOCK pin high.
7. Control channel is available from the remote side.

The devices can be configured after they are linked. This can be done locally or over the control channel by an ECU on either the serializer or deserializer side.

Infoframe CC

Infoframe CC is an internal channel for the serializer and deserializer to communicate. It is used to communicate video timing and PPS parameters between the serializer and deserializer. The MAX96746/MAX96856/MAX96860/MAX96862 receive parameters that are used to configure the eDP output with the exact video timing parameters, PCLK, and frame rate. The PPS parameters used by the DSC compression engine are also received and used to configure the DSC decoder of the MAX96746/MAX96856/MAX96860/MAX96862 automatically. Only select GMSL serializers support Infoframe CC and the functionality that it provides.

Automatic Video Timing

Video timings can be automatically configured in the MAX96746/MAX96856/MAX96860/MAX96862 based on the DP MSA parameters that are received from the serializer over Infoframe CC. Once the eDP source begins outputting video to the serializer, the serializer transmits the received video timing parameters over Infoframe CC. The deserializer receives

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these video timing parameters and configures its eDP video output accordingly, there is no manual configuration that needs to be done. Only select GMSL serializers support this feature.

Automatic DSC Configuration

The DSC decoder of the MAX96746/MAX96856/MAX96860/MAX96862 can be automatically configured based on the PPS parameters that are received from the serializer over Infoframe CC. The DSC decoder is configured based on the properties of the stream including:

- Resolution
- Slice Width and Height
- 24-Bit vs. 30-Bit Color Depth
- Other DSC parameters

Deserializer to Deserializer Synchronization

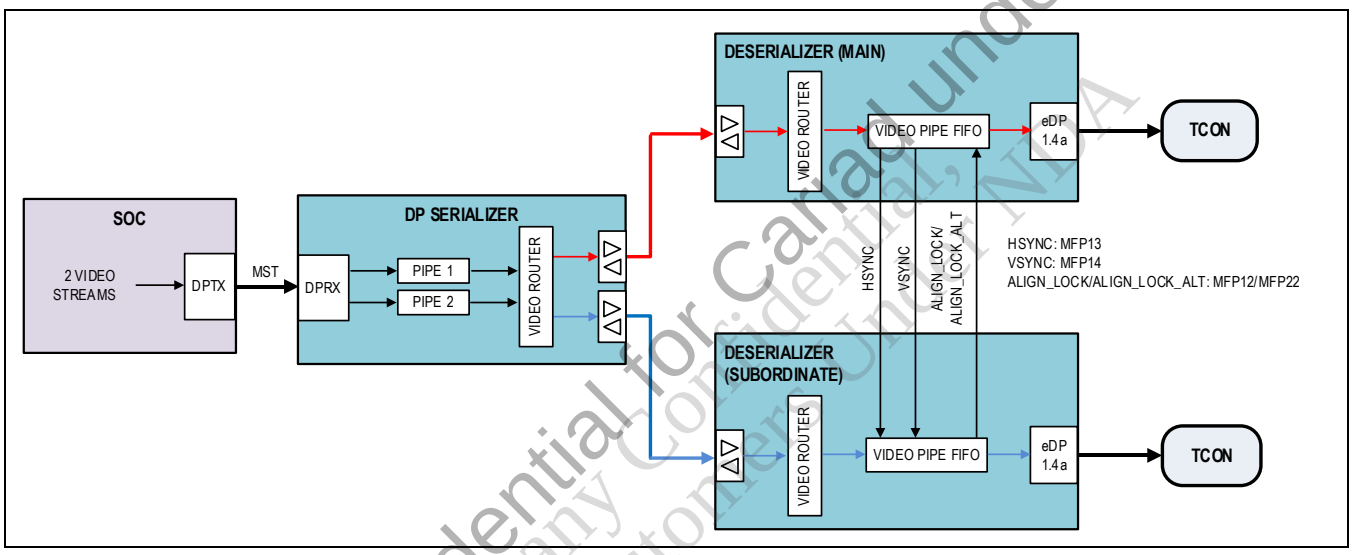


Figure 23. Synchronization of Parallel Deserializers

This mechanism enables synchronization of the video streams in two deserializers. It can be used for deserializers that are driven in parallel or daisy-chained.

The synchronization of two eDP deserializers can be achieved with two different methods:

- Manual register configuration/programming with fixed delay
  - Empirically determine VSYNC delay between two deserializers. In daisy-chain applications, the downstream MAX96746/MAX96856/MAX96860/MAX96862 will introduce approximately 550ns of delay.
  - Program appropriate device to set the delay.
- Automated Continuous Synchronization
  - Connect MFP pins for VSYNC, HSYNC, and ALIGN_LOCK/ALIGN_LOCK_ALT on both deserializers.
  - Program both devices to enable synchronization.
  - Deserializers will automatically perform continuous synchronization.

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Video Transmission

Video Pipeline

The MAX96746/MAX96856/MAX96860/MAX96862 video data path blocks contain functions including frame rate control (FRC) and spatial dithering, color lookup table (LUT), cyclic redundancy check (CRC) checker, video authenticator, watermark, and video pseudo random binary sequence (PRBS) checker.

Video Pipes

DSC Decoder

The MAX96862 implements a video decompression algorithm compliant with the VESA DSC v1.0/v1.2 specification. It provides 3:1 decompression for 24bpp video streams and 3.75:1 for 30bpp streams. This enables high-frame rate and high-resolution video to be passed through a single GMSL3/2 link together with other traffic.

The eDP output of the MAX96862 supports maximum PCLKs of 1080MHz at 24bpp and 864MHz at 30bpp with DSC decompression. The decoder supports a maximum of four slices with a maximum width of 5,120 pixels per slice and a maximum width of 11,520 pixels per line. The horizontal line length must be integer-divisible by the number of slices (HACTIVE / # of slices = integer number).

The decoder per-slice bandwidth is 400MP/s for both 24bpp and 30bpp. The required slice bandwidth is calculated as:

$$Slice\ Speed = HTOTAL\ x\ VTOTAL\ x\ Refresh\ Rate\ /\ number\ of\ slices$$

HTOTAL includes the horizontal blanking pixels and VTOTAL includes the vertical blanking lines. If the slice speed exceeds 400MP/s, one or more of the variables must be reduced, or more slices are needed. These calculations need to be done by the system designer. The deserializer's decoder operation will be programmed using the picture parameter set (PPS) values it receives from the host.

The decoder slice count must be identical to that used by the encoder. If the encoder's capabilities are lower than the deserializer decoder, then the encoder determines the maximum available performance.

The MAX96862 has these DSC decoder blanking limitations:

$$V_{fp} \geq 5, (V_{sw} + V_{bp}) > 5$$

$$HBLANK = (H_{fp} + H_{sw} + H_{bp}) \geq 30$$

The strictest limitations between the sterilizer and deserializer must be met for HBLANK (horizontal blanking). Refer to the data sheet of the paired serializer for video and blanking timing limitations.

High-Bandwidth Digital Content Protection (HDCP)

HDCP is an encryption scheme to protect digital audio and video content in various protocols. The MAX96860/MAX96862 devices support HDCP1.4 and 2.3 decryption of the GMSL input video stream(s).

HDCP is disabled by default. When HDCP is enabled, the function performs authentication, after which link integrity checks are periodically performed by the hardware through a dedicated control channel over the GMSL link. HDCP interrupts can be generated, and status registers provide the status of the authentication progress, indicating success and a ready state to the upstream video source.

Video-Timing Generator (VTG)

The VTG is an auxiliary function that can substitute HS, VS, and DE signals as defined by the configuration registers. With proper register programming, it is possible to generate any desired output frequency without an attached serializer. The VTG is used with the video-pattern generator to generate a test pattern at any resolution with internally generated clocks.

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Video-Pattern Generator (VPG)

The RGB888/RGB101010 VPG is also an auxiliary block that is capable of producing multiple pattern types with programable colors.

When enabled, the video pattern generated by the VPG can replace the video received from the GMSL input using the received timings, or it can generate a test pattern at a user-programmed resolution when used in conjunction with the VTG.

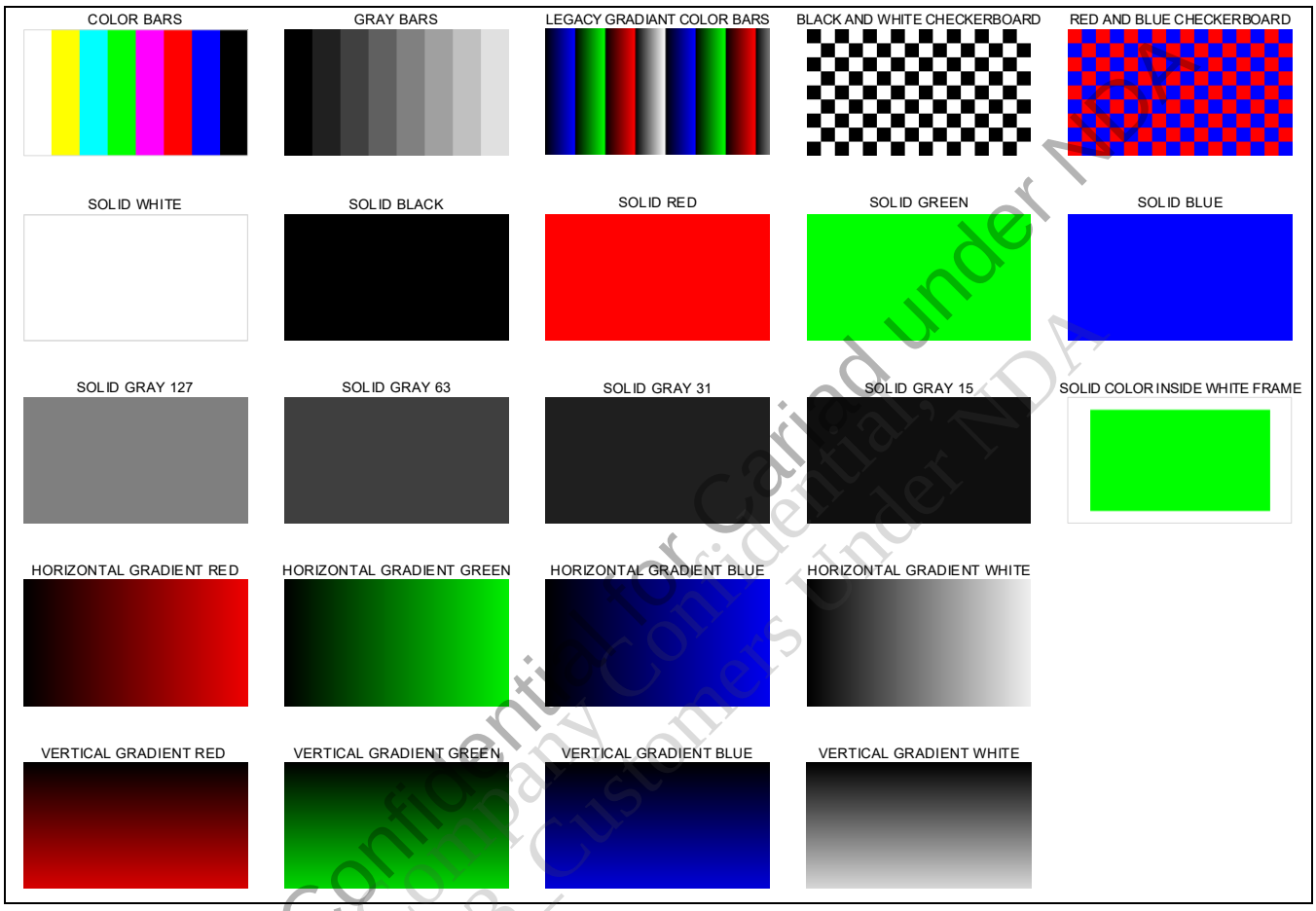


Figure 24. Video-Pattern Generator Test Patterns

Frame Rate Control and Spatial Dithering

The MAX96746/MAX96856/MAX96860/MAX96862 deserializers support deep-color and visually lossless video bandwidth reduction of video streams. Select GMSL display port (DP) serializers support video with up to 10-bits per color (bpc). Dropping the 2 least significant bits (LSBs) of the source video to match the display resolution leads to quantization error (Qe). Without additional processing, color information is lost.

Frame rate control (FRC) and spatial dithering (SD) are features that enable encoding this Qe into the upper bits in a visually lossless method. FRC and SD enable support for deep color (10bpc) and provide visually lossless video bandwidth reduction of 20% (10bpc to 8bpc). FRC and SD leverage the perceptive limitations of human vision to display higher resolution images on a lower resolution display by spreading the associated Qe temporally in subsequent frames (FRC) or spatially in adjacent pixels (SD). This processing is typically done in the display controller (TCON) but can also be done at either the serializer or deserializer. By implementing FRC or SD on the serializer, the bandwidth on the GMSL link is reduced by 20% for 10bpc to 8bpc and 25% for 8bpc to 6bpc. This enables the use of larger displays or reserving the bandwidth savings for other video streams on the same GMSL link.

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Color Lookup Table (LUT)

The MAX96746/MAX96856/MAX96860/MAX96862 deserializers contain color lookup tables (LUTs) to support automatic 1:1 translation of all RGB888 pixel color values to any desired RGB888/RGB101010 pixel color values. They are typically used for color filtering, gamma correction, etc.

The color LUT subsystem includes three separate LUTs, one for each color channel (R, G, B). Each LUT has an independent configuration register block to be filled, and independent enabling registers. Each LUT supports 256-entry depth, which enables 1:1 translation of 8-bit input values to any 8-bit or 10-bit output value. By default, the color LUTs are disabled and pixel values are not translated to another value. The lookup tables contain random values at power-up. Once the LUT registers are populated with the correct values, translation can be enabled using the COLOR_LUT_EN register bits. The 10bpc LUT outputs can be converted back to 8bpc using FRC.

Contact the factory for more information.

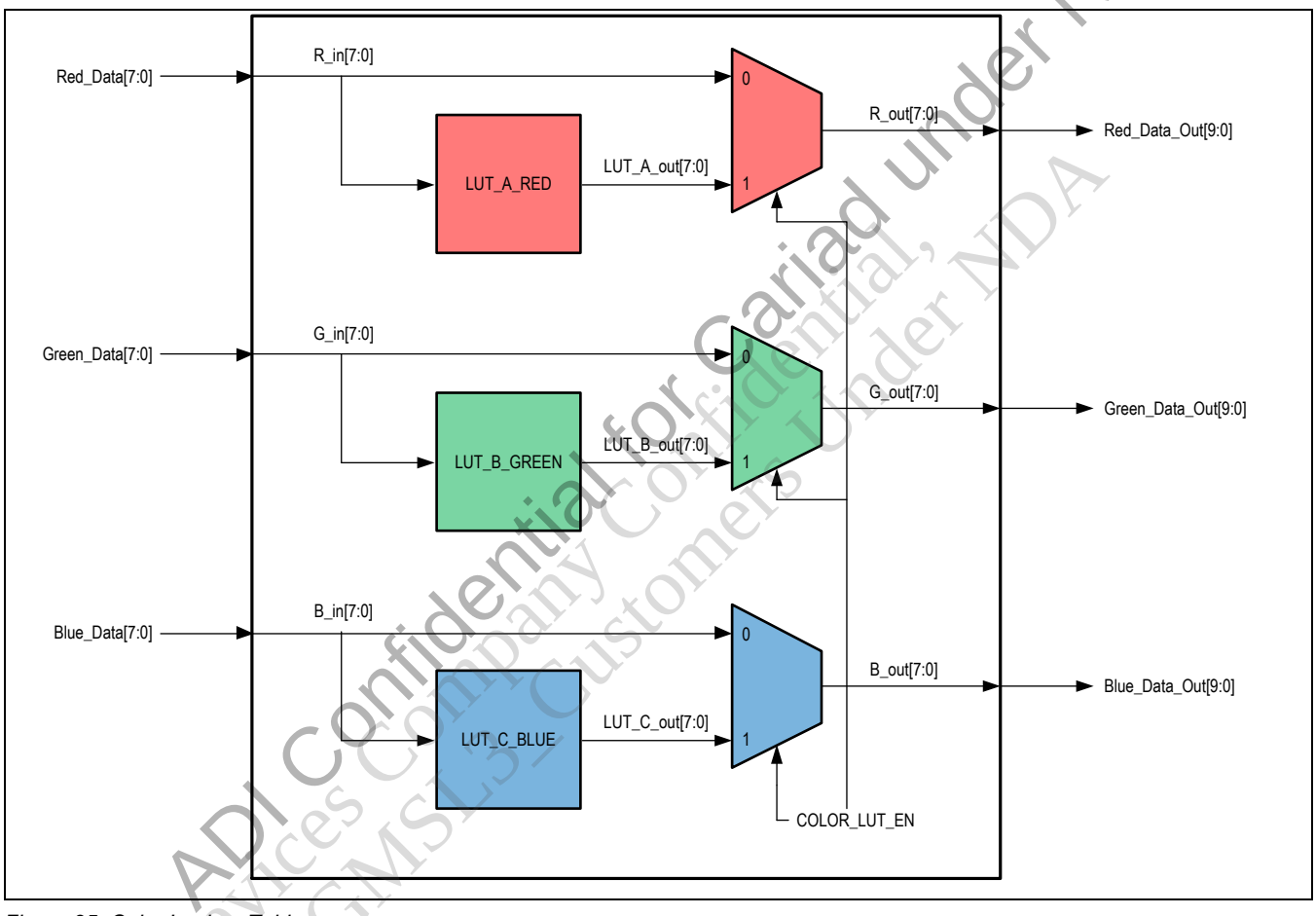


Figure 25. Color Lookup Tables

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DisplayPort Transmitter

Embedded DisplayPort (eDP) Output

The MAX96746/MAX96856/MAX96860/MAX96862 provide a subset of eDP features specifically to drive single displays in automotive applications. Only Single-stream transport (SST) mode is supported. Hot-plug detect (HPD) functionality is supported. For detailed information about eDP and DP, refer to the VESA® Embedded DisplayPort (eDP) Standard and the VESA DisplayPort (DP) Standard.

eDP Transmitter

The eDP transmitter sources isochronous data with 8-bit/10-bit encoding on one, two, or four lanes. It supports data rates of 1.62, 2.7, 4.5, and 8.1Gbps/lane.

Spread-spectrum clocking (SSC) can optionally be used to reduce radiated EMI. The drivers include internal 100Ω differential source termination. The transmitter includes a disable feature.

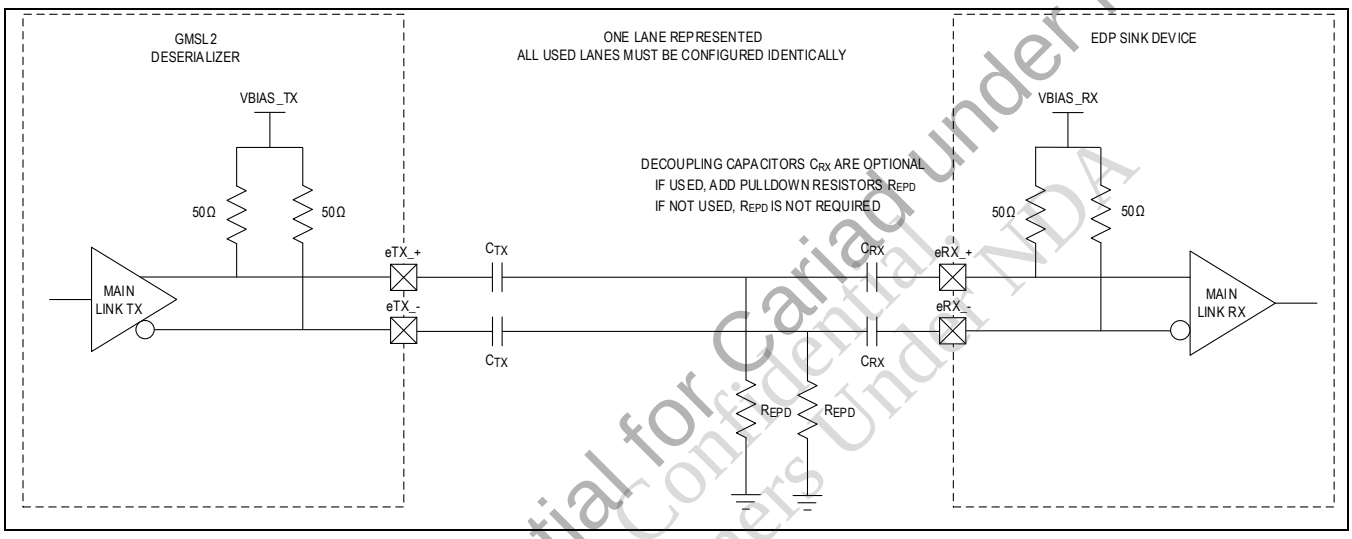


Figure 26. eDP Main Link Topology

Table 7. DP Tx Features Supported

DP FEATURE	SUPPORTED	DETAILS
DP/eDP 1.4 Compliance	No	DP output is compliant to DP1.4 PHY specifications but has not been tested to CTS compliance
DP Lanes and Lane Rates	Yes	One, two, or four lanes at RBR, HBR, HBR2, and HBR3 lane rates supported
eDP Transmit Voltage Levels	Yes	300mV, 350mV, 400mV, 450mV
EDID	No	Deserializer can optionally read DP sink's EDID and pass back to serializer (for select serializers). The intent of this feature is to support debugging and is not meant for use in production vehicles.
Embedded Audio	No	Embedded DP audio is not supported. Audio is supported on the I^2S pins
DSC Passthrough	No	DSC is not supported on the DP output
HDCP 1.4/2.3	No	HDCP is not supported on the DP output
HDR SDPs	No	HDR SDP and other metadata is not supported on the DP output
Panel Self Refresh (PSR) and Variable Frame Rate	No	Only fixed video timing is supported
DP Automotive Extensions (DPAE)	No	DP AE on DP output is not supported

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Video-Timing Regenerator (VTRG)

The VTRG is used to smooth data flow and reduce timing uncertainty that may be created by the fixed speed, packetized nature of data transmission across the GMSL link. The VTRG block is positioned in the video pipeline immediately after the video Rx block, where GMSL packets are converted to a parallel video stream consisting of pixels and sync signals.

Video Stream Selection

Any one of the video streams received by the GMSL link can be output to the local eDP output. The default stream is specified by the address set by CFG0 in *Table 2*. A different stream can be selected by STREAM_SEL_X/Y.

Any combination of the video streams (up to four) received by the GMSL link can also be output at the daisy-chain port DCIO. By default, the stream being locally output is dropped and all others are output at DCIO. Other streams can be dropped as programmed by DROP_VIDEO when AUTO_DROP_VIDEO_EN = 0.

The user must ensure there is sufficient DCIO bandwidth for the selected streams.

Stream ID to I²C/UART Address Association

I²C/UART addresses can be changed via register writes. Note that the device addresses of deserializers connected to a single serializer must be different.

Video Timing Status

The MAX96746/MAX96856/MAX96860/MAX96862 provides capability to help diagnose video problems, including if video is being received over the GMSL link (video lock), detected PCLK, HACTIVE, and VACTIVE.

Minimum Blanking and Other Video Limits

The MAX96746/MAX96856/MAX96860/MAX96862 devices do not have minimum blanking requirements for video. However, for video line-CRC, video frame-CRC, and SHA256-related functions (video authentication), there are some minimum blanking and line length constraints:

- Maximum line length for uncompressed stream is 16384 pixels.
- Line-CRC minimum line length is 256 pixels, and minimum horizontal blanking is 6 pixels.
- Frame-CRC minimum line length is 256 pixels, and minimum horizontal blanking is 6 pixels.
- Even/Odd Pixel Splitter mode has a maximum line length of 16394 pixels.
- SHA256 vertical blanking minimum is 1024 pixels.

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Bidirectional Channels

Overview

A µC or other controller can send and receive control and side-channel data over the GMSL serial link simultaneously with high-speed video data. GMSL2 and GMSL3 devices support the following interfaces:

- Primary I²C/UART (internal access)
- Pass-Through I²C/UART
- SPI
- GPIO
- I²S/TDM Audio

Data from all of these interfaces tunnel through the GMSL link. By default, the GMSL serializer and deserializer registers can only be accessed and configured through the primary I²C/UART interface.

The side channel, with its various interfaces, is accessed through multifunction GPIO pins. Multifunction pins have a default function and can be programmed to an alternate function after power-up. Due to practical limit in the number of pins available on a given device, not all interfaces can be simultaneously supported. See *Table 9* and the *Pin Descriptions* section for default and alternate multifunction pin functions, as well as available combinations of interfaces.

Primary I²C/UART

The primary I²C/UART is located on the SDA_RX and SCL_TX pins of each GMSL device. The I²C (SDA, SCL) or UART (Tx, Rx) interface is selected by the CFG configuration pin state at power-up (See the *GMSL Link Lock* sections). The selected interface provides main access to GMSL device registers, as well as peripheral device registers, from either end of the link. The I²C/UART communication is only over SIOA on the MAX96746/MAX96856/MAX96860/MAX96862 devices.

The main ECU can be located on either end of the link (usually the serializer side for display applications and the deserializer side for camera applications). Applications with an ECU in the display module can be supported by disabling the primary control channel connection across the GMSL link via a latch at power-up hardware pin (DIS_REM_CC) or via register write. The status of the DIS_REM_CC pin is only checked at power-up. After power-up, the state of DIS_REM_CC can be set via register write. Optionally, software arbitration (such as token passing) can be used to prevent packet collisions.

To configure peripheral devices over the link, the GMSL serializer and deserializer must use the same control-channel interface (both I²C or both UART). I²C/UART outputs are open-drain and require appropriately sized external pull-up resistors for proper operation.

I²C Multimain Arbitration

When a GMSL serializer is connected to more than one deserializer in daisy-chain and/or splitter modes, the serializer can arbitrate and process I²C packets from the I²C mains (ECUs) connected to each deserializer. The I²C packet from each deserializer-side ECU is processed at the serializer in the order it is received, and the non-selected I²C mains are put on hold by clock stretching by the respective deserializer.

Multimain I²C operation is available on the primary I²C channel and pass-through channels.

Primary I²C/UART Control-Channel CRC

Optionally, CRC is added to I²C and UART transactions to prevent hardware faults or random errors from misconfiguring the device. The MAX96746/MAX96856/MAX96860/MAX96862 I²C/UART subordinate controller expects and validates an 8-byte CRC value for all communication messages to maintain data integrity. This ensures the proper write is received and the device is configured properly.

For each serializer or deserializer device, there is a single counter incremented for both write and read transactions. For a system with both a serializer and deserializer, each of the two devices keep their own counter. The device's counter increments only when the ECU transaction is for that particular device and the I²C device address matches that device. Therefore, the ECU must maintain separate transaction counters for each serializer and deserializer and send the appropriate counter value when writing to each device.

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By default, the I²C/UART CRC and Message Counter features are disabled. When desired, these functions can be enabled by a register write. All subsequent packets then use the I²C/UART CRC and Message Counter features.

Pass-Through I²C/UART

The MAX96746/MAX96856/MAX96860/MAX96862 has two pass-through I²C/UART channels. By default, these channels do not have write access to registers in either the GMSL serializer or the deserializer; they simply tunnel the I²C or UART signal across the GMSL link. This can be useful for separating I²C channels so that multimain conflicts do not occur. I²C/UART outputs are open-drain and require appropriately sized external pull-up resistors for proper operation.

SPI

GMSL enables an SPI host (main) on one side of the GMSL link to control peripheral SPI subordinate on the opposite side. Communication can be in either direction across the GMSL link. Although multiple SPI peripherals may be connected, it is recommended that only one be communicated with at a time. Contact the factory for guidance on configuring SPI connection.

The SPI clock range is 600kHz to 50MHz. Ensure that setup and hold time requirements are met.

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Speed Programming for SPI

The SPI interface can be used over a wide range of frequencies. The MAX96746/MAX96856/MAX96860/MAX96862 provides flexible GPIO speed programming to maintain timing margins while minimizing radiated EMI.

Table 8 provides guidance on recommended speed settings for various SPI operating frequencies and VDDIO supply voltages. See *Table 11* and *Table 12* for GPIO pin speed programming information.

At lower frequencies, SPI data is typically latched on the opposite clock edge on which it is shifted (see *Figure 5* and *Figure 6*). However, at higher frequencies, the data must be latched on the same edge as the shift to meet the setup and hold time requirements. *Table 8* provides guidance to program the latching clock edge.

Table 8. Recommended SPI Pin Programming

FREQUENCY	VDDIO	LATCHING EDGE	RECOMMENDED SPEED SETTING
< 12.5MHz	1.7V to 2.24V	Opposite from Shift	01
	2.25V to 3.6V		10
12.5MHz to 25MHz	1.7V to 2.24V	Opposite from Shift	00
	2.25V to 3.6V		01
25MHz to 50MHz	1.7V to 2.24V	Same as Shift	00
	2.25V to 3.6V		01

I^2S/TDM Audio

GMSL devices for display applications support I^2S stereo and up to 8 channels of audio in TDM mode. The MAX96746/MAX96856/MAX96860/MAX96862 devices have 1 audio channel configurable in the forward (serializer to deserializer) direction. Sample rates of 8kHz to 192kHz and sample depths of 8 to 32 bits are supported. The maximum SCK frequency is 49.152MHz.

General-Purpose Inputs and Outputs (GPIO)

GPIOs are typically used to tunnel low speed (<100kbps) signals over the GMSL link. A GPIO tunnel can be set up in the forward (serializer to deserializer) or reverse (deserializer to serializer) direction. GPIO transmissions are transition-based; a GPIO packet is created and transmitted on the GMSL link when a rising or falling edge transition is detected at a GPI pin. The transition is regenerated at a GPO on the other end of the link.

The multifunction pins can be programmed as GPI (input), GPO (push-pull output or open-drain output), or GPIO (bidirectional input/output). Each GPIO can also be programmed with an internal 1MΩ or 4kΩ pull-up or pull-down resistor (or none). Although an internal pull-up is provided, high-speed open-drain outputs require an appropriate value external pull-up resistor to V_{DDIO}. Inputs cannot be left unconnected. Ensure that every pin configured as an input has a pull-up or pull-down programmed or is driven by another IC.

A GPI on one side of the serial link can be mapped to a single GPO or multiple GPOs on the other side of the link. Each GPI is assigned a pin ID with the destination GPO(s) on the other side of the link set to the same pin ID. By default, the ID mapping is GPIO0-GPIO0, GPIO1-GPIO1, GPIO2-GPIO2, etc. However, the GPIO mapping can be arbitrarily changed through register settings.

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Multifunction Pin (MFP) Assignments

Side-channel functions such as pass-through, I2C/UART, I2S/TDM audio, and SPI are enabled by programming multifunction pins (MFPs). Each pin has several possible functions, but only one can be used at a time.

Table 9 shows default and alternate functions for each MFP.

The transition time of each MFP can be changed from the default value by register programming. However, if the pass-through I2C/UART function is selected, the transition time of these specific pins is fixed independently of the setting. The primary I2C/UART channel has a fixed speed.

Table 9 indicates how latch-on power-up, I2C/UART, I2S/TDM audio, SPI, and other functions are mapped to the MFP pins.

MFP output rise and fall times depend on the transition time setting and VDDIO supply voltage. See *Table 12* for typical transition times.

Table 9. GMSL MFP Pin Function Map

PIN	LATCH ON POWER-UP	I2C/UART	AUDIO	SPI	OTHER FUNCTIONS	GPIO	POWER-UP DEFAULT	DEFAULT TRANSITION TIME SETTING (TTS)
MFP0				SCLK		GPIO0	GPI, CMOS in 1MΩ pull-down	10
MFP1				MOSI		GPIO1	GPI, CMOS in 1MΩ pull-down	10
MFP2				MISO		GPIO2	GPI, CMOS in 1MΩ pull-down	10
MFP3/SCL_TX		Main SCL, TX				GPIO3	SCL, TX CMOS in ODO 40kΩ pull-up	11
MFP4/SDA_RX		Main SDA, RX				GPIO4	SDA, RX CMOS in ODO 40kΩ pull-up	11
MFP5					LMN0	GPIO5	GPI, CMOS in 1MΩ pull-down	11
MFP6					LMN1	GPIO6	GPI, CMOS in 1MΩ pull-down	11
LOCK					LOCK		LOCK	11

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							ODO 40kΩ pull-up	
ERRB					ERRB		ERRB ODO 40kΩ pull-up	11
CFG0/MFP9	CFG0				WMD	GPO9	CFG0 Disabled, Hi-Z	11
MFP10		Pass-through SDA1, RX1				GPIO10	GPI, CMOS in 1MΩ pull-down	11
MFP11		Pass-through SCL1, TX1				GPIO11	GPI, CMOS in 1MΩ pull-down	11
MFP12			WS		ALIGN_LOCK_IN, INITIAL_ALIGN_LOCK, DE_OUT_X	GPIO12	GPI, CMOS in 1MΩ pull-down	10
MFP13			SD		HS_X	GPIO13	GPI, CMOS in 1MΩ pull-down	10
MFP14			SCK		VS_X	GPIO14	GPI, CMOS in 1MΩ pull-down	10
MFP15		Pass-through SDA2, RX2				GPIO15	GPI, CMOS in 1MΩ pull-down	11
MFP16		Pass-through SCL2, TX2				GPIO16	GPI, CMOS in 1MΩ pull-down	11
CFG1/MFP17	CFG1				VA_DONE	GPIO17	CFG1 Disabled, Hi-Z	11
MFP18/MS					VS_X_ALT, MS	GPIO18	GPI, CMOS in 1MΩ pull-down	11
MFP19					HS_X_ALT	GPIO19	GPI, CMOS in 1MΩ pull-down	11
MFP20					VS_OUT_Y	GPIO20	GPI, CMOS in 1MΩ pull-down	10
MFP21					HS_OUT_Y	GPIO21	GPI, CMOS in	10

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							1MΩ pull-down	
MFP22					ALIGN_LOCK_ALT_IN INITIAL_ALIGN_LOCK_ALT DE_OUT_Y, RCLKOUT	GPIO22	GPI, CMOS in 1MΩ pull-down	10
MFP23				BNE, SS1		GPIO23	GPI, CMOS in 1MΩ pull-down	10
MFP24				RO, SS2		GPIO24	GPI, CMOS in 1MΩ pull-down	10

Notes:

- All MFP pins are high-impedance in power-down state.
- All MFP I/Os are latched in Sleep mode.
- Disabled high-impedance – 1MΩ pull-up/pull-down is disabled and input/output is disabled.
- Latched – value of input before Sleep mode is entered remains latched in Sleep mode and after return to Power-up mode.

Multifunction GPIO Pin Assignments

Table 10 defines the pin I/O properties for each MFP function.

Table 10. MFP Pin I/O Properties

MFP FUNCTION	PIN I/O PROPERTIES
GPIOs	Push-pull, push, pull, or no_drive. Pull-up or pull-down with selectable 1MΩ or 40kΩ. Input, output, or input/output. I^2C-only I/O can only be open-drain output, all others can be open-drain or push-pull.
SCLK, RO, MOSI, MISO, MS, SCKIR, SDIR, WSIR	CMOS in with 1MΩ pull-down
SD, WS, SCK, SCLK, BNE, SS1, SS2, MISO, MOSI, VA_DONE, WMD, HS, VS, VBR_ERR	CMOS out
LOCK, ERRB	Open-drain output with 40kΩ pull-up
SDA_RX, SCL_TX, SDA1_RX1, SCL1_TX1, SDA2_RX2, SCL2_TX2	Open-drain I^2C-filtered output with 40kΩ pull-up, CMOS input
DIS_REM_CC	CMOS in with 100kΩ pull-down

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MFP Pin Equivalent Circuits

In general, MFP pins can all be classified as either programmable high-speed GPIO or programmable I²C/open-drain output (ODO) GPIOs. Equivalent circuit representations are illustrated in *Figure 27* and *Figure 28*.

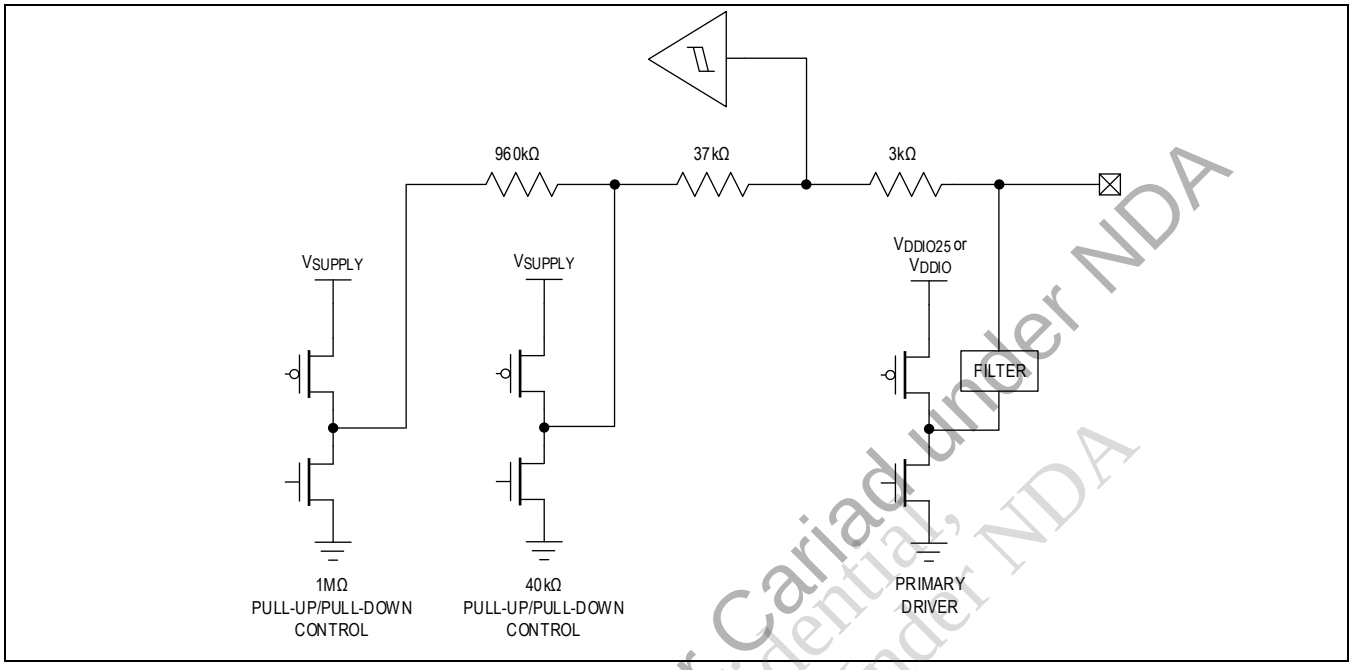


Figure 27. Standard High Speed GPIO

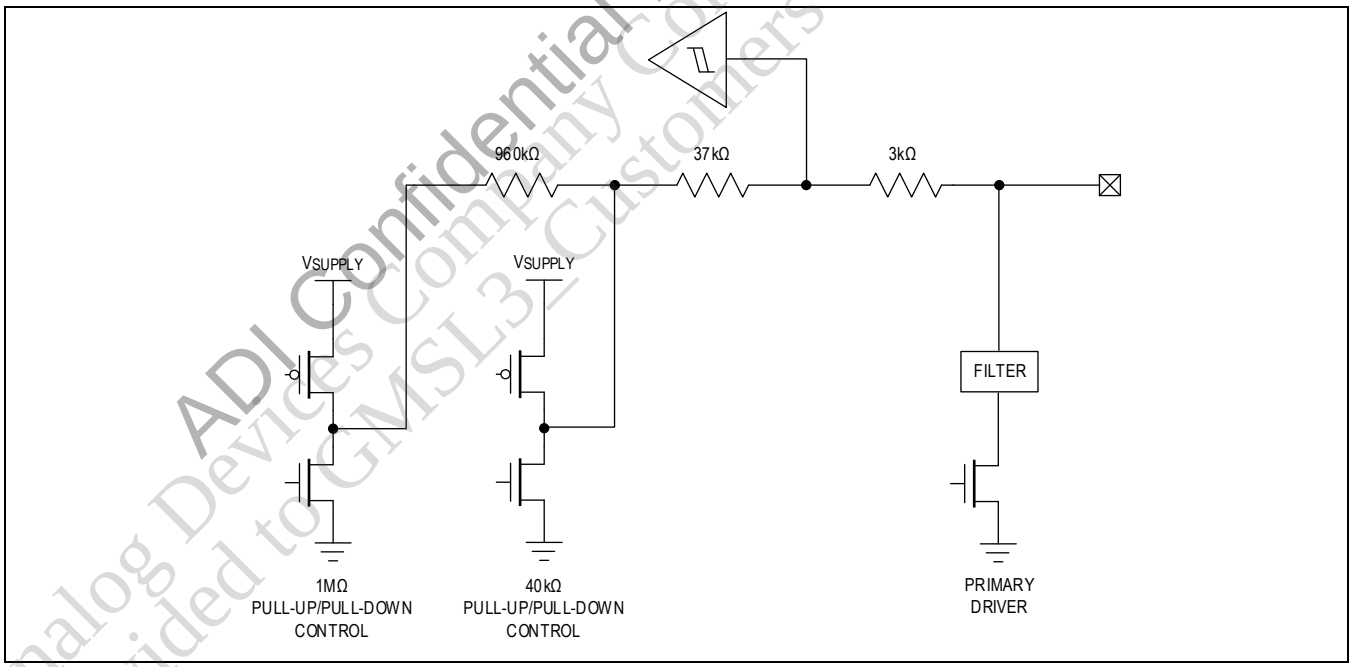


Figure 28. Standard I²C GPIO

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Table 11. Typical Control-Channel Latencies

FUNCTION	FORWARD	REVERSE	NOTES
I^2C	< 10µs	< 10µs	
UART	< 10µs	< 10µs	
Audio	< 100µs	< 100µs	For typical use cases
SPI	< 10µs	< 10µs	Round Trip

Table 12. Control-Channel and Side-Channel Typical Rise and Fall Times

PIN OR GPIO TRANSITION SETTING (TTS)	RISE TIME*		FALL TIME**	
	V_{DDIO} = 1.8V	V_{DDIO} = 3.3V	V_{DDIO} = 1.8V	V_{DDIO} = 3.3V
00	1.0ns	0.6ns	0.8ns	0.5ns
01	2.1ns	1.1ns	2.0ns	1.1ns
10	4.0ns	2.3ns	4.3ns	2.3ns
11	9ns	5ns	10ns	5ns
I^2C	***	***	40ns	30ns

*20% to 80%, 10pF load.

**70% to 30%, 10pF load.

*** Rise time for open-drain outputs depends on external pull-up resistor value.

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Safety

ERRB Pin Operation

All GMSL3/2 devices have a multipurpose open-drain error reporting and interrupt status output (ERRB). The configurable ERRB pin can output detected error status(es) and/or certain internal device events relevant to the host processor (i.e., interrupts). The MAX96746/MAX96856/MAX96860/MAX96862 ERRB function is assigned to MFP8. Additionally, the ERROR status is reported upstream to the serializer.

Most device errors/statuses can be output to the ERRB pin. Independent of ERRB reporting, the individual device error statuses are indicated in the local registers of the feature. Many of these errors have counters that track the cumulative number of errors that have occurred.

Errors are individually configured to be output to the ERRB pin. Each error eligible for ERRB pin routing has an output enable (OEN) bit that is used to send the error status to the ERRB OR gate. Not all errors are enabled by default. See the *Register Map* for OEN hardware defaults. If an error occurs and its ERR_OEN bit is enabled, the ERRB pin is driven low (i.e., active), and the ERROR bit reports a 1.

Figure 29 illustrates the OR gate controlling the ERRB status.

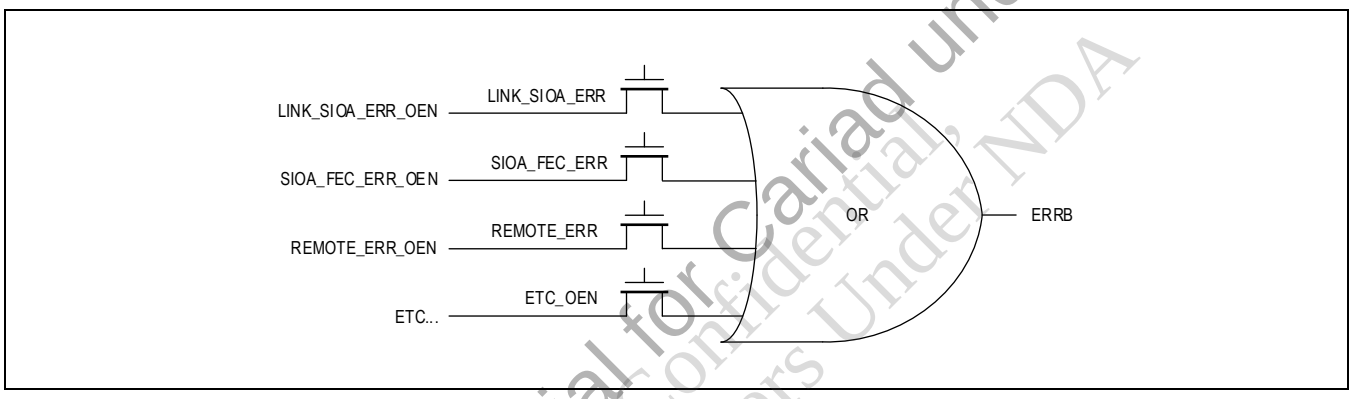


Figure 29. ERRB Pin Logic

Control Register CRCs

Control Register CRC is a functional safety feature to ensure the MAX96746/MAX96856/MAX96860/MAX96862 devices are not placed into an undesired state through a fault. Once the deserializer is programmed, the control and status register (CSR) bank is protected by a CRC calculation. After the deserializer is set up through the CSR, the values should not change while the device is operating. Any change is considered a fault, and ERRB is asserted if the OEN for register CRC flag is enabled. The deserializer can report if the CSR values change due to faults (i.e., soft errors). Alternately, the CRC value can be polled by the ECU to check for changes.

There are two modes: Basic CRC and User-Polling CRC. Contact the factory for details.

Functional Safety

The MAX96746/MAX96856/MAX96860/MAX96862 deserializers are composed of a number of functional blocks, with safety mechanisms defined to protect one or more of these functional blocks. Contact the factory for a list of these safety mechanisms and details on operating this device in safety-critical applications.

When using this device in a safety-critical application, the following is expected to be understood by the user:

- All safety measures offered by the devices in the system should be used as specified in a list of safety mechanisms. A system architect has the option to not use all of the safety measures offered but must account for the resultant loss in ASIL coverage by adding dedicated system level mechanisms. If specific safety measures are not used to protect each functional block, their malfunction will not be detected.
- The MAX96746/MAX96856/MAX96860/MAX96862 report errors via updating the ERROR bit internally and the ERRB pin externally. This allows the users to program an interrupt on the active low ERRB signal for quick fault

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response times. It is highly recommended to use ERRB interrupt in all applications. If ERRB is not connected through hardware, it is required that the user poll the internal ERROR bit. Polling must be faster than any system level fault detection interval.

- Some specific safety mechanisms do not report their failure condition to the ERRB pin. It is required that the user poll related register(s) to determine the status of those safety mechanisms.

Video Integrity Safety Functions

GMSL Frame CRC

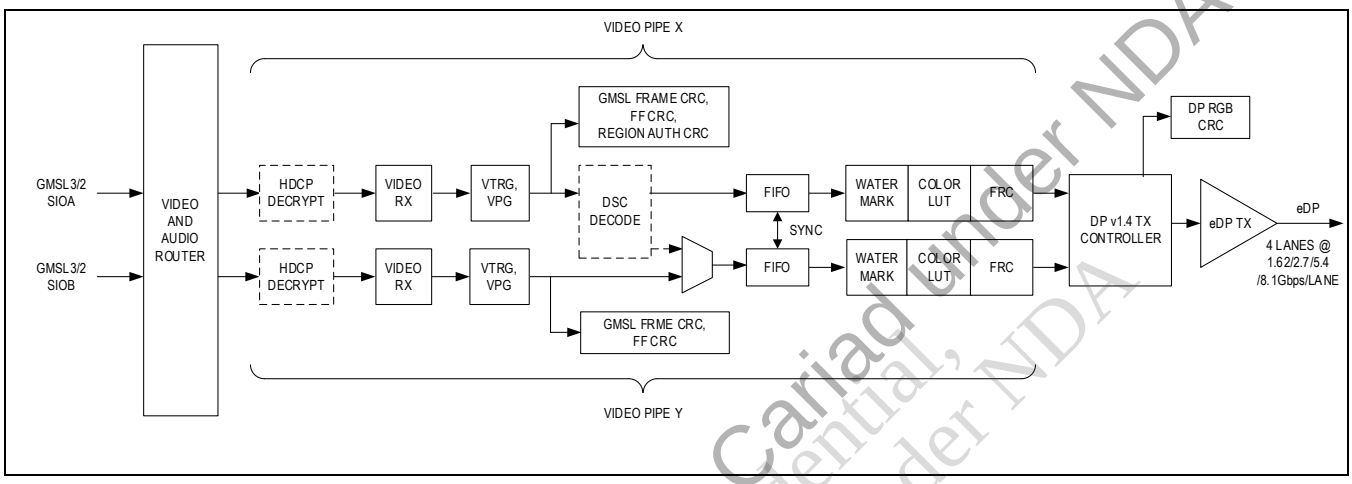


Figure 30. Video and Audio Signal Path with CRC

This mechanism supports full-frame CRC verification of video stream integrity between select serializers and the MAX96746/MAX96856/MAX96860/MAX96862 deserializers. The SOC calculates a CRC for each stream and provides it to the deserializer via the GMSL link. The deserializer checkpoint is prior to the FIFO blocks for uncompressed data and at the input to the DSC decoder for compressed data. Video Pipes X and Y have independent CRC checks. CRCs can be checked on compressed or uncompressed data in Pipe X and uncompressed data in Pipe Y when using Even/Odd Splitter mode. Both 24bpp and 30bpp streams are supported. The MAX96746/MAX96856/MAX96860/MAX96862 test the calculated CRC(s) against reference CRC(s) from the serializer. The polynomial is:

$$P = x^{32} + x^{31} + x^{28} + x^{25} + x^{24} + x^{23} + x^{21} + x^{18} + x^{11} + x^{8} + x^{7} + x^{6} + x^{5} + x^{3} + x + 1$$

Errors can be reported to ERRB.

Downstream DPTX RGB CRC

This mechanism verifies integrity of the eDP interface between the deserializer and the downstream TCON. Per the DisplayPort standard, the deserializer DPTX and the TCON DPRX perform an RGB frame CRC using the polynomial $f(x) = x^{16} + x^{15} + x^{2} + 1$. In the DP standard, this is defined to be a test mode function. The MAX96746/MAX96856/MAX96860/MAX96862 calculate the CRC on every frame. The TCON must also support calculation on every frame. At the end of each frame, the deserializer reads the TCON CRC results from its DPCD registers using the AUX channel and compares them to its own results. Additionally, a CRC error will be reported if the TCON CRC update counter is not being incremented.

Errors can be reported to ERRB.

Feedforward Frame CRC

This mechanism supports full-frame CRC verification of video stream integrity between the SOC and the MAX96746/MAX96856/MAX96860/MAX96862 deserializers. It covers the full signal chain from the SOC through the serializer, the GMSL link, and the deserializer up to the FIFOs or DSC decoder. Only select SOCs and GMSL serializers support this mechanism.

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Video Pipes X and Y have independent CRC checks. CRCs can be checked on compressed or uncompressed data in Pipe X, and uncompressed data in Pipes X and Y when using even/odd splitter mode. The MAX96746/MAX96856/MAX96860/MAX96862 test the calculated CRC(s) against reference CRC(s) transmitted by the SOC in an extra video line. The polynomial options are the following:

The following polynomial for 24bpp and 30bpp color:

$$P = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^6 + x^5 + x^4 + x^2 + x + 1$$

The GMSL frame CRC polynomial for 24bpp and 30bpp color:

$$P = x^{32} + x^{31} + x^{28} + x^{25} + x^{24} + x^{23} + x^{21} + x^{18} + x^{11} + x^8 + x^7 + x^6 + x^5 + x^3 + x + 1$$

Errors can be reported to ERRB.

I²C/UART Control-Channel Programming with CRC and Message Counter

By default, I²C/UART CRC and Message Counter features are disabled in the MAX96746/MAX96856/MAX96860/MAX96862. The conventional (legacy) I²C/UART packet format is the default. When desired, these functions are enabled by a register write by using the legacy packet format. All subsequent packets will use the new I²C/UART control-channel programming with CRC and Message Counter (ASIL-protected format).

Regional CRC Authentication

CRCs of defined regions of interest (ROI) in the video stream can be calculated by the MAX96746/MAX96856/MAX96860/MAX96862. This function supports authentication and can support telltale verification of uncompressed video using an external processor.

The region definitions must be programmed into the deserializer registers either by a local processor or one located at the serializer side of the GMSL link. Up to eight regions, comprised of two sets of four regions, can be defined and checked. The CRC engine can be changed on the fly to either of the region sets.

The MAX96746/MAX96856/MAX96860/MAX96862 does not test the CRC results. The CRC values must be read and checked by an external processor. Only video Pipe X supports this feature. The checkpoint is prior to the DSC decoder. Both compressed and uncompressed data can be checked. The feature works for both single and even/odd split video streams where only the even stream is checked.

The CRC polynomial is:

$$P = x^{32} + x^{31} + x^{28} + x^{25} + x^{24} + x^{23} + x^{21} + x^{18} + x^{11} + x^8 + x^7 + x^6 + x^5 + x^3 + x + 1$$

Errors can be reported to ERRB.

Color Recognition

Compression introduces challenges in verifying the integrity of the video data. Although the decompressed video is visually lossless, it is not identical to the data input to the compression engine.

Until decompressed CRC mechanisms are available for the entire signal chain, color recognition provides a method to determine with very high confidence that the correct telltale has been transmitted.

The SOC is responsible for ensuring that it has inserted the correct telltale into the video stream. Telltales consist of a solid color image with a contrasting monochromatic (typically black) background. To use color recognition, the SOC calculates hue, saturation, and luminosity (HSL) color metadata for the telltale ROI. This metadata must be programmed into the appropriate MAX96746/MAX96856/MAX96860/MAX96862 registers. The deserializer counts the number of pixels within the ROI that meet the HSL criteria and compares the number to the SOC value. The pass/fail criteria can be programmed to limits above and below the expected count.

Since the SOC ensures that the correct telltale has been sent, verifying that the color signature is a close match ensures with high confidence that the telltale was received.

Contact the factory for more information on color recognition.

Watermark Detector

The watermarking block detects a frozen-frame failure in a frame-based processing system between the generator and detector. Watermarking can only be applied to RGB888 data and may not be compatible with upstream image

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manipulation. This feature specifically detects frozen frames caused by ECUs in safety-critical applications. It does not detect frozen frames that occur before the watermark generator or after the watermark detector. GMSL serializers provide a watermark generator, while deserializers provide a watermark detector. This allows video generators like a surround view system to insert a watermark and a display to detect a watermark in a safety-relevant video stream after passing through an ECU that may have a lower ASIL rating level. The watermark generator inserts a time-varying watermark that is highly redundant and robust to image processing and display-stream compression. The watermark detector looks for this time-varying watermark, and if it fails to see all the generated watermarks, this indicates a frozen frame failure in a frame-based processing system between the generator and detector. Upon detection of this error condition, the watermark detector can generate an interrupt and/or blank the output video, returning the display to a safe state in less than 500ms.

Figure 31 shows the system-level block diagram with the watermark generation in a serializer upstream of the infotainment ECU and watermark detection in a deserializer downstream of the infotainment ECU.

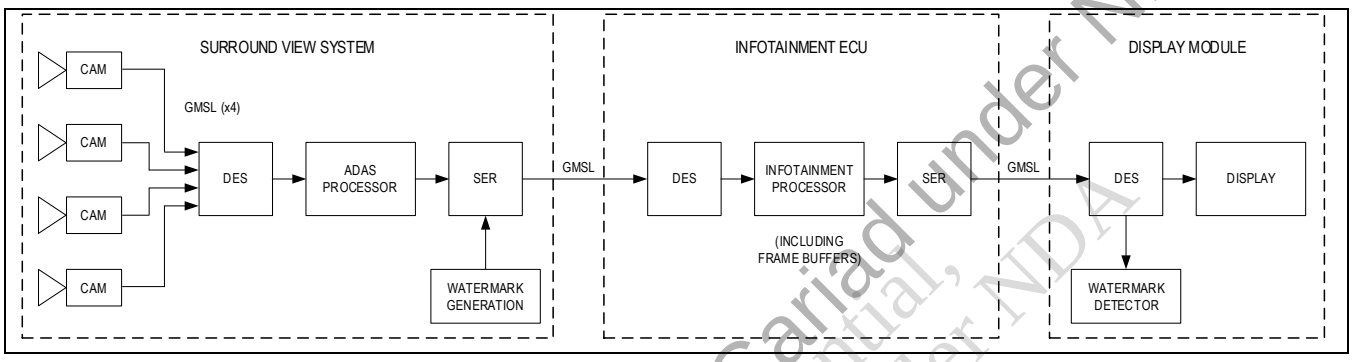


Figure 31. System-Level Block Diagram

Watermark detection can be configured on the whole video frame or a register-defined rectangular region. The region is programmed via register interface and can be any size smaller than the active video frame. The XMIN and XMAX values must be even numbers if DSC or Even/Odd Splitter mode are used. This allows the detection, removal, and optional blanking on error to occur just in the specified region, which should be used if the watermarked content is only displayed on a portion of the total video frame.

Video PRBS

The GMSL video channel has a PRBS generator in the serializer and a PRBS checker in the deserializer to test the video channel operation. Note that not all link bandwidth is used by the video channel alone.

Therefore, it is possible to have a bit error on the link that does not cause a video PRBS error. Video PRBS errors are detected in the deserializer.

Audio PRBS

The GMSL audio channel has a PRBS generator in the serializer and a PRBS checker in the deserializer to test the audio channel operation. The detected audio channel bit errors are reported in the MAX96746/MAX96856/MAX96860/MAX96862.

UV/OV Detection

The MAX96746/MAX96856/MAX96860/MAX96862 devices include UV and OV detection logic. Register programming is required to enable these features.

In the event of a UV condition, the ERRB pin is asserted when V_{DD18} or V_{DDIO} drops below an internal threshold level. The device continues to function unless V_{DD18} or V_{DDIO} drops further to an internal reset level, in which case the device is reset.

In a UV condition on V_{DD}, the ERRB pin is asserted when V_{DD} drops below an internal threshold level and the device is reset.

In the event of an OV condition on V_{DD18}, V_{DDIO}, or V_{DD} the ERRB pin is asserted, but no further action is taken. The device continues to remain powered with an OV condition and may suffer damage.

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Once the device is reset and the power supply voltage recovers, the device returns to its default power-up state with default register settings. The controller must reinitiate device startup and reprogram all registers.

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Ordering Information

PART NUMBER	TEMP RANGE	PIN-PACKAGE	HDCP
MAX96746RCGTN/V+*	-40°C to +105°C	56-lead TQFN-EP	No
MAX96746RCGTM/V+T*	-40°C to +105°C	56-lead TQFN-EP	No
MAX96746RCGTN/VY+*	-40°C to +105°C	56-lead SW TQFN-EP	No
MAX96746RCGTN/VY+T*	-40°C to +105°C	56-lead SW TQFN-EP	No
MAX96856ACGTN/V+	-40°C to +105°C	56-lead TQFN-EP	No
MAX96856ACGTN/V+T	-40°C to +105°C	56-lead TQFN-EP	No
MAX96856ACGTN/VY+*	-40°C to +105°C	56-lead SW TQFN-EP	No
MAX96856ACGTN/VY+T*	-40°C to +105°C	56-lead SW TQFN-EP	No
MAX96860ASGTN/V+	-40°C to +105°C	56-lead TQFN-EP	Yes**
MAX96860ASGTN/V+T	-40°C to +105°C	56-lead TQFN-EP	Yes**
MAX96860ASGTN/VY+	-40°C to +105°C	56-lead SW TQFN-EP	Yes**
MAX96860ASGTN/VY+T	-40°C to +105°C	56-lead SW TQFN-EP	Yes**
MAX96862RSGTN/V+	-40°C to +105°C	56-lead TQFN-EP	Yes**
MAX96862RSGTN/V+T	-40°C to +105°C	56-lead TQFN-EP	Yes**
MAX96862RSGTN/VY+	-40°C to +105°C	56-lead SW TQFN-EP	Yes**
MAX96862RSGTN/VY+T	-40°C to +105°C	56-lead SW TQFN-EP	Yes**

/V Denotes an automotive qualified part
Y Denotes Side-Wettable package
T Denotes tape and reel
+ Denotes a lead(Pb)-free/RoHS-compliant package
EP Denotes Exposed Pad
**Future Product - Contact factory for availability.*
***HDCP parts require registration with Digital Content Protection, LLC.*

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Register Map

Reserved and Unused Register Bits

Not all register bits in the register space are shown in the register table. Any bit not explicitly defined in the register table should be treated as reserved and should not be modified. When a write is required to a register with both defined and undefined register bits, first read the register's contents, then create a new register value by only changing the defined bits, and finally, write the new byte to the register (Read/Replace/Write).

In this document, default values are provided for read-only register bits. Read-only bit states are changed at power-up according to the actual state of the device. To avoid overwriting these bits, treat read-only bits as undefined.

Note §: Refer to Selection Guide table in data sheet as not all features are applicable to all part numbers.

HDCP (MAX96860 and MAX96862 only)

HDCP is enabled in the MAX96860 and MAX96862 and not in the MAX96746 and MAX96856. Register blocks (see the following table), registers, and bitfields that refer to HDCP are reserved in the MAX96856 and MAX96746 and should not be changed from the default state.

HDCP Register Blocks

BLOCK NAME	REGISTER ADDRESS RANGE
HDCP_STATUS	0x90
HDCP_CFG	0x1C7 - 0x1CA

ADDRESS	RESET	NAME	MSB							LSB
DEV_CFG										
0x00	0x90	DEV_CFG_0[7:0]	DEV_ADDR[6:0]							CFG_BLOCK
0x01	0x00	DEV_CFG_1[7:0]	RESET_ALL	RESET_LINK_ALL	RESET_LINK_SIOAB	RESET_LINK_SIOA	–	RESET_LINK_DCIO	–	–
0x02	0x00	DEV_CFG_2[7:0]	RESET_ONESHOT_ALL	RESET_ONESHOT_SIOAB	RESET_ONESHOT_SIOA	RESET_ONESHOT_SIOB	RESET_ONESHOT_DCIO	–	–	–
ERR_STATUS										

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ADDRESS	RESET	NAME	MSB							LSB
0x04	0x00	ERROR_STATUS_0[7:0]	ERR_OUTPUT_STATUS	LINK_SIOA_ERR	LINK_SIOB_ERR	LINK_DCIO_ERR	–	–	–	PSM_STATE_ERR
0x05	0x00	ERROR_STATUS_1[7:0]	SIOA_FEC_ERR	SIOB_FEC_ERR	RETENTION_MEMORY_CRC_ERR	CONTROL_CHANNEL_MSG_COUNT_ERR	–	LINEFAULT_0_ERR	LINEFAULT_1_ERR	CLOCK_MONITOR_ERR
0x06	0x00	ERROR_STATUS_2[7:0]	REMOTE_ERR	ARQ_ERR	INFOFRAME_CRC_ERR	VIDEO_PRBS_ERR	AUDIO_PRBS_ERR	OVERVOLTAGE_ERR	UNDERVOLTAGE_ERR	–
0x07	0x00	ERROR_STATUS_3[7:0]	VIDEO_SEQUENCE_ERR	LOSS_OF_VIDEO_LOCK_ERR	HDCP_ERR	REGISTER_CRC_ERR	DSC_ERR	WATERMARK_ERR	VIDEO_BLOCK_LENGTH_ERR	LOSS_OF_VIDEO_ALIGNMENT_ERR
0x08	0x00	ERROR_STATUS_4[7:0]	FRAME_CRC_ERR	FEEDFORWARD_CRC_ERR	FEEDFORWARD_COMBINED_FRAME_CRC_ERR	LINE_CRC_ERR	VIDEO_PACKET_CRC_ERR	–	DP_SINK_SYMBOL_ERR	–
0x09	0x00	ERROR_STATUS_5[7:0]	DP_INPUT_FIFO_CRC_ERR	WM_BYPASS_ERR	LUT_FRC_ERR	DPTX_FIFO_OVERFLOW_ERR	TELLTALE_ERR	LOSS_OF_LOCK_ERR	–	CONTROL_CHANNEL_CRC_ERR
0x0A	0x00	ERROR_STATUS_6[7:0]	OTP_CRC_ERR	–	VIDEO_CFG_ERR	DP_LINK_TRAINING_ERR	DP_SINK_CAPABILITY_ERR	AUX_ERR	HPD_IRQ_ERR	HPD_DISCONNECT_ERR
DEV_INFO										
0x0D	0xFF	DEV_INFO_0[7:0]	DEV_ID[7:0]							
0x0E	0x20	DEV_INFO_1[7:0]	DEV_REV[3:0]				–	–	–	–

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ADDRESS	RESET	NAME	MSB							LSB
0x0F	0x3F	DEV_INFO_2[7:0]	SPEED_CPBL[1:0]		SGMII_CPBL	DSC_CPBL	SIOB_CPBL	DAISYCHAIN_CPBL	GMSL_12G_CPBL	HDCP_CPBL
0x11	0x02	DEV_INFO_4[7:0]	OTP_REV[7:0]							
0x24	0xC3	DEV_ID16_LSB[7:0]	DEV_ID16_BYTE_0[7:0]							
0x25	0x00	DEV_ID16_MSB[7:0]	DEV_ID16_BYTE_1[7:0]							
LINK_STATUS										
0x30	0x00	LINK_STATUS_0[7:0]	LOCKED_SIOA	LOCKED_SIOB	LOCKED_DCIO	–	–	FEC_STATUS_SIOA	FEC_STATUS_SIOB	FEC_STATUS_DCIO
0x31	0x00	LINK_STATUS_1[7:0]	DEC_ERROR_COUNT_SIOA[7:0]							
0x32	0x00	LINK_STATUS_2[7:0]	DEC_ERROR_COUNT_SIOB[7:0]							
0x33	0x00	LINK_STATUS_3[7:0]	DEC_ERROR_COUNT_DCIO[7:0]							
FEC_STATUS										
0x37	0x	FEC_STATUS_0[7:0]	FEC_UNCORRECTABLE_ERRORS_SIOA[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
	00									
0x38	0x00	FEC_STATUS_1[7:0]	FEC_UNCORRECTABLE_ERRORS_SIOB[7:0]							
0x39	0x00	FEC_STATUS_2[7:0]	FEC_BLOCKS_PROCESSED_SIOA_BYTE_0[7:0]							
0x3A	0x00	FEC_STATUS_3[7:0]	FEC_BLOCKS_PROCESSED_SIOA_BYTE_1[7:0]							
0x3B	0x00	FEC_STATUS_4[7:0]	FEC_BLOCKS_PROCESSED_SIOA_BYTE_2[7:0]							
0x3C	0x00	FEC_STATUS_5[7:0]	FEC_BLOCKS_PROCESSED_SIOA_BYTE_3[7:0]							
0x3D	0x00	FEC_STATUS_6[7:0]	FEC_BLOCKS_PROCESSED_SIOB_BYTE_0[7:0]							
0x3E	0x00	FEC_STATUS_7[7:0]	FEC_BLOCKS_PROCESSED_SIOB_BYTE_1[7:0]							
0x3F	0x00	FEC_STATUS_8[7:0]	FEC_BLOCKS_PROCESSED_SIOB_BYTE_2[7:0]							
0x40	0x	FEC_STATUS_9[7:0]	FEC_BLOCKS_PROCESSED_SIOB_BYTE_3[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
	00									
0x41	0x00	FEC_STATUS_10[7:0]	FEC_CORRECTED_ERRORS_SIOA_BYTE_0[7:0]							
0x42	0x00	FEC_STATUS_11[7:0]	FEC_CORRECTED_ERRORS_SIOA_BYTE_1[7:0]							
0x43	0x00	FEC_STATUS_12[7:0]	FEC_CORRECTED_ERRORS_SIOA_BYTE_2[7:0]							
0x44	0x00	FEC_STATUS_13[7:0]	FEC_CORRECTED_ERRORS_SIOB_BYTE_0[7:0]							
0x45	0x00	FEC_STATUS_14[7:0]	FEC_CORRECTED_ERRORS_SIOB_BYTE_1[7:0]							
0x46	0x00	FEC_STATUS_15[7:0]	FEC_CORRECTED_ERRORS_SIOB_BYTE_2[7:0]							
COMM_STATUS										
0x48	0x00	CC_STATUS_0[7:0]	UART_TX_OVERFLOW	UART_RX_OVERFLOW	I2C_ACK_RECEIVED	I2C_TIMED_OUT	–	–	–	–
0x4A	0x00	PT_STATUS_0[7:0]	PT1_UART_TX_OVERFLOW	PT1_UART_RX_OVERFLOW	–	–	–	–	–	–

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ADDRESS	RESET	NAME	MSB							LSB
0x4B	0x00	PT_STATUS_1[7:0]	PT2_UART_TX_OVERFLOW	PT2_UART_RX_OVERFLOW	–	–	–	–	–	–
0x4D	0x00	SPI_STATUS_0[7:0]	SPI_TX_OVERFLOW	SPI_RX_OVERFLOW	–	SPI_BUFFER_LEVEL[4:0]				
0x54	0x00	AUDIO_STATUS_0[7:0]	–	–	–	–	AUDIO_LOCK	–	–	AUDIO_BLOCK_LENGTH_ERROR
0x55	0x00	AUDIO_STATUS_1[7:0]	AUDIO_RX_CLOCK_FREQUENCY[7:0]							
0x58	0x00	CFG_PIN_STATUS_0[7:0]	–	CFG_PIN_0_VALUE[2:0]			–	CFG_PIN_1_VALUE[2:0]		
0x59	0x00	GPIO_STATUS_0[7:0]	GPIO_VALUE_7	GPIO_VALUE_6	GPIO_VALUE_5	GPIO_VALUE_4	GPIO_VALUE_3	GPIO_VALUE_2	GPIO_VALUE_1	GPIO_VALUE_0
0x5A	0x00	GPIO_STATUS_1[7:0]	GPIO_VALUE_15	GPIO_VALUE_14	GPIO_VALUE_13	GPIO_VALUE_12	GPIO_VALUE_11	GPIO_VALUE_10	GPIO_VALUE_9	GPIO_VALUE_8
0x5B	0x00	GPIO_STATUS_2[7:0]	GPIO_VALUE_23	GPIO_VALUE_22	GPIO_VALUE_21	GPIO_VALUE_20	GPIO_VALUE_19	GPIO_VALUE_18	GPIO_VALUE_17	GPIO_VALUE_16
0x5C	0x00	GPIO_STATUS_3[7:0]	–	–	–	–	–	–	–	GPIO_VALUE_24
CRC_STATUS										

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ADDRESS	RESET	NAME	MSB							LSB
0x5E	0x00	REGIONAL_CRC_STATUS_0[7:0]	–	BANK_B_DONE	ACTIVE_A_OR_B	VA_DONE_STATUS	–	–	–	–
0x5F	0x00	REGIONAL_CRC_STATUS_1[7:0]	FRAME_COUNT_BYTE_0[7:0]							
0x60	0x00	REGIONAL_CRC_STATUS_2[7:0]	–	–	–	–	–	–	FRAME_COUNT_BYTE_1[1:0]	
0x61	0x00	REGIONAL_CRC_STATUS_3[7:0]	REGIONAL_CRC_A_BYTE_0[7:0]							
0x62	0x00	REGIONAL_CRC_STATUS_4[7:0]	REGIONAL_CRC_A_BYTE_1[7:0]							
0x63	0x00	REGIONAL_CRC_STATUS_5[7:0]	REGIONAL_CRC_A_BYTE_2[7:0]							
0x64	0x00	REGIONAL_CRC_STATUS_6[7:0]	REGIONAL_CRC_A_BYTE_3[7:0]							
0x65	0x00	REGIONAL_CRC_STATUS_7[7:0]	REGIONAL_CRC_B_BYTE_0[7:0]							
0x66	0x00	REGIONAL_CRC_STATUS_8[7:0]	REGIONAL_CRC_B_BYTE_1[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x67	0x00	REGIONAL_CRC_STATUS_9[7:0]	REGIONAL_CRC_B_BYTE_2[7:0]							
0x68	0x00	REGIONAL_CRC_STATUS_10[7:0]	REGIONAL_CRC_B_BYTE_3[7:0]							
TELLTALE_STATUS										
0x6D	0x00	TELLTALE_STATUS_0[7:0]	TELLTALE_BANK_B_ERR[3:0]				TELLTALE_BANK_A_ERR[3:0]			
CRC_MISC_STATUS										
0x6E	0x00	CRC_MISC_STATUS_0[7:0]	WM_BYPASS_ERROR_X	WM_BYPASS_ERROR_Y	LUT_FRC_ERROR_X	LUT_FRC_ERROR_Y	FRAME_CRC_ERROR_X	FRAME_CRC_ERROR_Y	–	–
0x6F	0x00	CRC_MISC_STATUS_1[7:0]	LINE_CRC_ERROR_X	LINE_CRC_ERROR_Y	VIDEO_PACKET_CRC_ERROR_X	VIDEO_PACKET_CRC_ERROR_Y	–	–	INFOFRAME_CRC_ERROR_SIOB	INFOFRAME_CRC_ERROR_SIOA
MISC_STATUS										
0x71	0x00	ARQ_STATUS_0[7:0]	MAX_RT_ERROR_INFOFRAME_CC	MAX_RT_ERR_CC	MAX_RT_ERR_PT_1	MAX_RT_ERR_PT_2	MAX_RT_ERR_SPI	MAX_RT_ERR_GPIO	–	MAX_RT_ERR_HDCP
0x73	0x48	LF_STATUS_0[7:0]	LINEFAULT_0_STATUS[2:0]			LINEFAULT_1_STATUS[2:0]			–	–
DP_STATUS										
0x75	0x	DP_STATUS_0[7:0]	DP_OVERFLOW	DP_UNDERFLOW	DP_INPUT_FIFO_OVERFLOW	DP_VBI_ERROR	DP_INPUT_FIFO_UNDERFLOW	–	–	–

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ADDRESS	RESET	NAME	MSB							LSB
	00									
0x76	0x000	DP_STATUS_1[7:0]	–	–	–	–	LINK_RETRAINED	HPD_EVENT_DETECTED	HPD_IRQ_DETECTED	HPD_CONNECT_STATE
0x77	0x0F	DP_STATUS_2[7:0]	DPTX_VIDEO_OUTPUT_ACTIVE	LINK_TRAINING_PASSED	LINK_TRAINING_FAILED	LINK_TRAINING_IN_PROGRESS	LINK_TRAINING_NOT_STARTED	WAITING_FOR_VIDEO_LOCK	WAITING_FOR_PSM_STATE	WAITING_FOR_HPD
0x78	0x000	DP_STATUS_3[7:0]	–	–	–	–	–	–	EDID_TRANSFER_TO_SER_DONE	SINK_EDID_READ_DONE
0x79	0x000	DP_STATUS_4[7:0]	MAJOR_REV[3:0]				MINOR_REV[3:0]			
0x7A	0x000	DP_STATUS_5[7:0]	EXTENDED_CAPS_FIELD	–	–	LOCAL_EDID_PRESENT	TPS4_SUPPORTED	TPS3_SUPPORTED	MAX_DOWNSPREAD	ENHANCED_FRAME_CAP
0x7B	0x000	DP_STATUS_6[7:0]	MAX_LINK_RATE[7:0]							
0x7C	0x000	DP_STATUS_7[7:0]	–	–	–	MAX_LANE_COUNT[4:0]				
0x7D	0x000	DP_STATUS_8[7:0]	–	TRAINING_AUX_RD_INTERVAL[6:0]						
0x7E	0x	DP_STATUS_9[7:0]	I2C_SPEED_CAP[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
	00									
0x80	0x000	DP_STATUS_12[7:0]	–	LANE1_SYMBOL_LOCKED	LANE1_CHANNEL_EQ_DONE	LANE1_CR_DONE	–	LANE0_SYMBOL_LOCKED	LANE0_CHANNEL_EQ_DONE	LANE0_CR_DONE
0x81	0x000	DP_STATUS_13[7:0]	–	LANE3_SYMBOL_LOCKED	LANE3_CHANNEL_EQ_DONE	LANE3_CR_DONE	–	LANE2_SYMBOL_LOCKED	LANE2_CHANNEL_EQ_DONE	LANE2_CR_DONE
0x82	0x000	DP_STATUS_14[7:0]	LINK_STATUS_UPDATED	–	–	–	–	–	–	INTERLANE_ALIGN_DONE
0x83	0x000	DP_STATUS_15[7:0]	AUX_HPD_DISCON_ERROR	–	DP_SINK_EDID_ERROR	I2C_OVER_AUX_ERROR	AUX_NACK_ERROR	AUX_TIMEOUT_ERROR	AUX_RETRY_ERROR	AUX_DEFER_ERROR
0x84	0x000	DP_STATUS_16[7:0]	OTP_CRC_ERROR_STATUS	–	DP_SINK_RGB_FRAME_CRC_STOPPED	DP_SINK_SYMBOL_ERROR_STATUS	VIDEO_CFG_ERROR_STATUS	DP_SINK_CHEQ_TPS_CAPABILITY_ERROR	DP_SINK_LANE_COUNT_CAPABILITY_ERROR	DP_SINK_LINK_RATE_CAPABILITY_ERROR
0x85	0x000	DP_STATUS_17[7:0]	–	–	–	–	DP_LINK_INTERLANE_ALIGN_ERROR	DP_LINK_SYMBOL_LOCK_ERROR	DP_LINK_CHANNEL_EQ_ERROR	DP_LINK_CR_ERROR
WM_STATUS										
0x88	0x000	WATERMARK_STATUS_0[7:0]	WM_ERROR_LIVE	WM_DETECT	–	–	–	–	–	–
DSC_STATUS										
0x8A	0x000	DSC_STATUS_0[7:0]	DSC_OVERFLOW	DSC_UNDERFLOW	DSC_PARITY_ERROR	DSC_TRANSPORT_CHUNK_SIZE_ERROR	–	–	–	–

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ADDRESS	RESET	NAME	MSB							LSB
PRBS_STATUS										
0x8C	0x00	PRBS_STATUS_0[7:0]	–	–	–	APRBS_VALID	–	–	–	VPRBS_FAIL
0x8D	0x00	PRBS_STATUS_1[7:0]	VPRBS_ERROR_COUNT[7:0]							
0x8E	0x00	PRBS_STATUS_2[7:0]	APRBS_ERROR_COUNT[7:0]							
HDCP_STATUS										
0x90	0x00	HDCP_STATUS_0[7:0]	DECRYPTION_ON_X	DECRYPTION_ON_Y	HDCP_LINK_INTEGRITY_X	HDCP_LINK_INTEGRITY_Y	HDCP2_AUTHENTICATED_X	HDCP2_AUTHENTICATED_Y	–	–
VIDEO_STATUS										
0x92	0x00	VIDEO_STATUS_0[7:0]	VIDEO_ALIGN_LINE_READ_ERR_DETECTED	VTRG_OVERFLOW_X	VTRG_UNDERFLOW_X	VIDEO_ALIGN_UNDERFLOW_X	VIDEO_ALIGN_OVERFLOW_X	VIDEO_SEQUENCE_ERROR_X	VIDEO_BLOCK_LENGTH_ERROR_X	LOSS_OF_VIDEO_PACKETS_LOCK_X
0x93	0x00	VIDEO_STATUS_1[7:0]	–	VTRG_OVERFLOW_Y	VTRG_UNDERFLOW_Y	VIDEO_ALIGN_UNDERFLOW_Y	VIDEO_ALIGN_OVERFLOW_Y	VIDEO_SEQUENCE_ERROR_Y	VIDEO_BLOCK_LENGTH_ERROR_Y	LOSS_OF_VIDEO_PACKETS_LOCK_Y
0x94	0x00	VIDEO_STATUS_2[7:0]	VIDEO_PACKETS_LOCK_Y	VIDEO_PACKETS_LOCK_X	VIDEO_LOCK_Y	VIDEO_LOCK_X	ODD_EVEN_SPLIT_STATUS	DETECTED_30_BPP	PPS_PARAMS_RECEIVED	AUTO_TIMING_RECEIVED
0x95	0x00	VIDEO_STATUS_3[7:0]	RECEIVED_PCLK_FREQUENCY_BYTE_0[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x96	0x00	VIDEO_STATUS_4[7:0]	RECEIVED_PCLK_FREQUENCY_BYTE_1[7:0]							
0x97	0x00	VIDEO_STATUS_5[7:0]	RECEIVED_PCLK_FREQUENCY_BYTE_2[7:0]							
0x98	0x00	VIDEO_STATUS_6[7:0]	RECEIVED_H_ACTIVE_BYTE_0[7:0]							
0x99	0x00	VIDEO_STATUS_7[7:0]	RECEIVED_H_ACTIVE_BYTE_1[7:0]							
0x9A	0x00	VIDEO_STATUS_8[7:0]	RECEIVED_H_FP_BYTE_0[7:0]							
0x9B	0x00	VIDEO_STATUS_9[7:0]	RECEIVED_H_FP_BYTE_1[7:0]							
0x9C	0x00	VIDEO_STATUS_10[7:0]	RECEIVED_H_SW_BYTE_0[7:0]							
0x9D	0x00	VIDEO_STATUS_11[7:0]	RECEIVED_H_SW_BYTE_1[7:0]							
0x9E	0x00	VIDEO_STATUS_12[7:0]	RECEIVED_H_BP_BYTE_0[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x9F	0x00	VIDEO_STATUS_13[7:0]	RECEIVED_H_BP_BYTE_1[7:0]							
0xA0	0x00	VIDEO_STATUS_14[7:0]	RECEIVED_V_ACTIVE_BYTE_0[7:0]							
0xA1	0x00	VIDEO_STATUS_15[7:0]	RECEIVED_V_ACTIVE_BYTE_1[7:0]							
0xA2	0x00	VIDEO_STATUS_16[7:0]	RECEIVED_V_FP[7:0]							
0xA3	0x00	VIDEO_STATUS_17[7:0]	RECEIVED_V_SW[7:0]							
0xA4	0x00	VIDEO_STATUS_18[7:0]	RECEIVED_V_BP[7:0]							
0xA5	0x00	VIDEO_STATUS_19[7:0]	DETECTED_PCLK_FREQUENCY_BYTE_0[7:0]							
0xA6	0x00	VIDEO_STATUS_20[7:0]	DETECTED_PCLK_FREQUENCY_BYTE_1[7:0]							
0xA7	0x00	VIDEO_STATUS_21[7:0]	–	–	–	–	DETECTED_PCLK_FREQUENCY_BYTE_2[3:0]			

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ADDRESS	RESET	NAME	MSB							LSB
0xA8	0x00	VIDEO_STATUS_22[7:0]	DETECTED_H_ACTIVE_BYTE_0[7:0]							
0xA9	0x00	VIDEO_STATUS_23[7:0]	DETECTED_H_ACTIVE_BYTE_1[7:0]							
0xAA	0x00	VIDEO_STATUS_24[7:0]	DETECTED_V_ACTIVE_BYTE_0[7:0]							
0xAB	0x00	VIDEO_STATUS_25[7:0]	DETECTED_V_ACTIVE_BYTE_1[7:0]							
0xAC	0x00	VIDEO_STATUS_26[7:0]	RECEIVED_HS_POLARITY	RECEIVED_VS_POLARITY	–	–	–	–	DETECTED_HS_POLARITY	DETECTED_VS_POLARITY
0xAD	0x00	VIDEO_STATUS_27[7:0]	RECEIVED_PPS_PIC_WIDTH_BYTE_0[7:0]							
0xAE	0x00	VIDEO_STATUS_28[7:0]	RECEIVED_PPS_PIC_WIDTH_BYTE_1[7:0]							
0xAF	0x00	VIDEO_STATUS_29[7:0]	RECEIVED_PPS_PIC_HEIGHT_BYTE_0[7:0]							
0xB0	0x00	VIDEO_STATUS_30[7:0]	RECEIVED_PPS_PIC_HEIGHT_BYTE_1[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xB1	0x00	VIDEO_STATUS_31[7:0]	RECEIVED_PPS_SLICE_WIDTH_BYTE_0[7:0]							
0xB2	0x00	VIDEO_STATUS_32[7:0]	RECEIVED_PPS_SLICE_WIDTH_BYTE_1[7:0]							
0xB3	0x00	VIDEO_STATUS_33[7:0]	RECEIVED_PPS_SLICE_HEIGHT_BYTE_0[7:0]							
0xB4	0x00	VIDEO_STATUS_34[7:0]	RECEIVED_PPS_SLICE_HEIGHT_BYTE_1[7:0]							
PSM_STATE_STATUS										
0xB6	0x00	PSM_STATE_STATUS_0[7:0]	–	PSM_ACCESS_ERROR	PSM_STATE_TIMER_ERROR	PSM_DATA_TIMER_ERROR	–	–	PSM_STARTUP_MBIST_FAILED	PSM_STARTUP_MBIST_DONE
POWER_STATUS										
0xBA	0x00	POWER_STATUS_0[7:0]	–	VDDIO_UNDERVOLTAGE_ERROR	VDD18_UNDERVOLTAGE_ERROR	VDD_BIAS_UNDERVOLTAGE_ERROR	VDD_UNDERVOLTAGE_ERROR	VDDIO_OVERVOLTAGE_ERROR	VDD18_OVERVOLTAGE_ERROR	VDD_OVERVOLTAGE_ERROR
ERR_LINK_CFG										
0x103	0x00	CFG_PIN_CFG_0[7:0]	CFG_PIN_0_OVR_EN	CFG_PIN_0_SET[2:0]			CFG_PIN_1_OVR_EN	CFG_PIN_1_SET[2:0]		
LINK_CFG										
0x105	0x	LINK_CFG_0[7:0]	LINK_EN_SIOA	LINK_EN_SIOB	LINK_EN_DCIO	–	–	RSVD	RSVD	RSVD

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ADDRESS	RESET	NAME	MSB							LSB
	A7									
0x106	0x22	LINK_CFG_1[7:0]	CX_TP_SIO	–	GMSL3_MODE_SIO	GMSL_FEC_EN_SIO	–	–	RX_RATE_SIO[1:0]	
0x107	0x38	LINK_CFG_2[7:0]	CX_TP_DCIO	–	GMSL3_MODE_DCIO	GMSL_FEC_EN_DCIO	TX_RATE_DCIO[1:0]		–	–
0x108	0x00	LINK_CFG_3[7:0]	–	–	–	–	–	–	FEC_STATS_RESET_SIOA	FEC_STATS_RESET_SIOB
ERRG_CFG										
0x10A	0x00	ERRG_CFG_0[7:0]	ERRG_EN_SIOA	ERRG_EN_SIOB	ERRG_EN_DCIO	–	–	–	–	–
0x10B	0x08	ERRG_CFG_1[7:0]	ERRG_MODE	ERRG_BURST[2:0]			ERRG_BER[1:0]		ERRG_COUNT[1:0]	
ERROR_CFG										
0x10D	0xF1	ERROR_CFG_0[7:0]	ERRB_PIN_OUTPUT_EN	LINK_SIOA_ERR_OEN	LINK_SIOB_ERR_OEN	LINK_DCIO_ERR_OEN	–	–	–	PSM_STATE_ERR_OEN
0x10E	0xE7	ERROR_CFG_1[7:0]	SIOA_FEC_ERR_OEN	SIOB_FEC_ERR_OEN	RETENTION_MEMORY_CRC_ERR_OEN	CONTROL_CHANNEL_MSG_COUNT_ERR_OEN	–	LINEFAULT_0_ERR_OEN	LINEFAULT_1_ERR_OEN	CLOCK_MONITOR_ERR_OEN
0x10F	0x7E	ERROR_CFG_2[7:0]	REMOTE_ERR_OEN	ARQ_ERR_OEN	INFOFRAME_CRC_ERR_OEN	VIDEO_PRBS_ERR_OEN	AUDIO_PRBS_ERR_OEN	OVERVOLTAGE_ERR_OEN	UNDERVOLTAGE_ERR_OEN	–

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ADDRESS	RESET	NAME	MSB							LSB
0x1110	0xCF	ERROR_CFG_3[7:0]	VIDEO_SEQUENCE_ERR_OEN	LOSS_OF_VIDEO_LOCK_ERR_OEN	HDCP_ERR_OEN	REG_CRC_ERR_OEN	DSC_ERR_OEN	WATERMARK_ERR_OEN	VIDEO_BLOCK_LENGTH_ERR_OEN	LOSS_OF_VIDEO_ALIGNMENT_ERR_OEN
0x1111	0x9A	ERROR_CFG_4[7:0]	FRAME_CRC_ERR_OEN	FEEDFORWARD_FRAME_CRC_ERR_OEN	COMBINED_FRAME_CRC_ERR_OEN	LINE_CRC_ERR_OEN	VIDEO_PACKET_CRC_ERR_OEN	–	DP_SINK_SYMBOL_ERR_OEN	–
0x1112	0xF0	ERROR_CFG_5[7:0]	DP_INPUT_FIFO_CRC_ERR_OEN	WM_BYPASS_ERR_OEN	LUT_FRC_ERR_OEN	DPTX_FIFO_OVERFLOW_ERR_OEN	TELLTALE_ERR_OEN	LOSS_OF_LOCK_OEN	–	CONTROL_CHANNEL_CRC_ERR_OEN
0x1113	0x3F	ERROR_CFG_6[7:0]	OTP_CRC_ERR_OEN	–	VIDEO_CFG_ERR_OEN	DP_LINK_TRAINING_ERR_OEN	DP_SINK_CAPABILITY_ERR_OEN	AUX_ERR_OEN	HPD_IRQ_ERR_OEN	HPD_DISCONNECT_ERR_OEN
CC_CFG										
0x1116	0x80	CC_CFG_0[7:0]	CC_LOCAL_OUT_EN	DIS_REM_CC	I2CSEL	UART_BYPASS_EN	–	–	CC_CRC_EN	CC_MESSAGE_COUNT_EN
0x1117	0x20	CC_CFG_1[7:0]	CC_LINK_SEL	CC_PRIORITY[1:0]		–	–	–	–	–
0x1118	0x2D	CC_CFG_2[7:0]	–	–	CC_I2C_SPEED_SLV[2:0]		CC_I2C_SPEED_MST[2:0]			
0x1119	0x00	CC_CFG_3[7:0]	CC_I2C_TRANSLATION_0_SOURCE[6:0]							–
0x111A	0x00	CC_CFG_4[7:0]	CC_I2C_TRANSLATION_0_DESTINATION[6:0]							–

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ADDRESS	RESET	NAME	MSB							LSB
0x11B	0x00	CC_CFG_5[7:0]	CC_I2C_TRANSLATION_1_SOURCE[6:0]							–
0x11C	0x00	CC_CFG_6[7:0]	CC_I2C_TRANSLATION_1_DESTINATION[6:0]							–
0x11D	0x00	CC_CFG_7[7:0]	–	–	–	RESET_MSGCNTR	–	–	RESET_MSGCNTR_ERR_CNT	RESET_CC_CRC_ERR_CNT
PT_CFG										
0x11E	0x00	PT_CFG_0[7:0]	PT1_EN	PT2_EN	PT1_I2CSEL	PT2_I2CSEL	PT1_LINK_SEL	PT2_LINK_SEL	PT1_MANUAL_MODE_EN	PT2_MANUAL_MODE_EN
0x120	0xDC	PT_CFG_2[7:0]	PT1_UART_BIT_LENGTH_BYTE_0[7:0]							
0x121	0x05	PT_CFG_3[7:0]	–	–	PT1_UART_BIT_LENGTH_BYTE_1[5:0]					
0x122	0x2D	PT_CFG_4[7:0]	–	–	PT1_I2C_SPEED_SLV[2:0]		PT1_I2C_SPEED_MST[2:0]			
0x123	0x00	PT_CFG_5[7:0]	PT1_I2C_TRANSLATION_0_SOURCE[6:0]							–
0x124	0x00	PT_CFG_6[7:0]	PT1_I2C_TRANSLATION_0_DESTINATION[6:0]							–

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ADDRESS	RESET	NAME	MSB							LSB
0x125	0x00	PT_CFG_7[7:0]	PT1_I2C_TRANSLATION_1_SOURCE[6:0]							–
0x126	0x00	PT_CFG_8[7:0]	PT1_I2C_TRANSLATION_1_DESTINATION[6:0]							–
0x128	0xDC	PT_CFG_10[7:0]	PT2_UART_BIT_LENGTH_BYTE_0[7:0]							
0x129	0x05	PT_CFG_11[7:0]	–	–	PT2_UART_BIT_LENGTH_BYTE_1[5:0]					
0x12A	0x2D	PT_CFG_12[7:0]	–	–	PT2_I2C_SPEED_SLV[2:0]		PT2_I2C_SPEED_MST[2:0]			
0x12B	0x00	PT_CFG_13[7:0]	PT2_I2C_TRANSLATION_0_SOURCE[6:0]							–
0x12C	0x00	PT_CFG_14[7:0]	PT2_I2C_TRANSLATION_0_DESTINATION[6:0]							–
0x12D	0x00	PT_CFG_15[7:0]	PT2_I2C_TRANSLATION_1_SOURCE[6:0]							–
0x12E	0x00	PT_CFG_16[7:0]	PT2_I2C_TRANSLATION_1_DESTINATION[6:0]							–
SPI_CFG										

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ADDRESS	RESET	NAME	MSB							LSB
0x1300	0x48	SPI_CFG_0[7:0]	SPI_EN	MST_SEL	SPI_MODE[1:0]		SPI_IGNORE_ID	–	SPI_LOCAL_ID[1:0]	
0x1301	0x00	SPI_CFG_1[7:0]	SPI_LINK_SEL	SPI_PRIORITY[1:0]		MISO_SAMPLE_EDGE	–	–	MST_SS_1_POL	MST_SS_2_POL
0x1302	0x0C	SPI_CFG_2[7:0]	–	–	–	–	MST_SS_1_EN	MST_SS_2_EN	RO_INPUT_EN	BNE_OUTPUT_EN
0x1303	0x01	SPI_CFG_3[7:0]	SPI_SLAVE_SEL_DELAY[7:0]							
0x1304	0x0F	SPI_CFG_4[7:0]	SPI_SCLK_LOW_CLKS[7:0]							
0x1305	0x0F	SPI_CFG_5[7:0]	SPI_SCLK_HIGH_CLKS[7:0]							
AUDIO_CFG										
0x1430	0x20	AUDIO_CFG_0[7:0]	AUDIO_LINK_SEL	AUDIO_PRIORITY[1:0]		I2S_MODE	AUDIO_RX_ID[1:0]		AUDIO_PRBS_CHECKER_EN	AUDIO_RX_EN
GPIO_CFG										
0x1450	0x00	GPIO_CFG_0[7:0]	GPIO_LINK_SEL	GPIO_PRIORITY[1:0]		–	–	–	–	–
GPIO_CFG 0										

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ADDRESS	RESET	NAME	MSB							LSB
0x146	0x80	GPIO_CFG_1[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x147	0xC4	GPIO_CFG_2[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
0x148	0x20	GPIO_CFG_3[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 1										
0x149	0x81	GPIO_CFG_4[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x14A	0xC4	GPIO_CFG_5[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
0x14B	0x21	GPIO_CFG_6[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 2										
0x14C	0x82	GPIO_CFG_7[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x14D	0xC4	GPIO_CFG_8[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE

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ADDRESS	RESET	NAME	MSB							LSB
0x14E	0x22	GPIO_CFG_9[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 5										
0x155	0x85	GPIO_CFG_10[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x156	0xC0	GPIO_CFG_11[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x157	0x25	GPIO_CFG_12[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 6										
0x158	0x86	GPIO_CFG_13[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x159	0xC0	GPIO_CFG_14[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x15A	0x26	GPIO_CFG_15[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 9										
0x161	0x09	GPIO_CFG_16[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				

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ADDRESS	RESET	NAME	MSB							LSB
0x162	0x80	GPIO_CFG_17[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x163	0x29	GPIO_CFG_18[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 10										
0x164	0x8A	GPIO_CFG_19[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x165	0xC0	GPIO_CFG_20[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x166	0x2A	GPIO_CFG_21[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 11										
0x167	0x8B	GPIO_CFG_22[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x168	0xC0	GPIO_CFG_23[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x169	0x2B	GPIO_CFG_24[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 12										

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ADDRESS	RESET	NAME	MSB							LSB
0x16A	0x8C	GPIO_CFG_25[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x16B	0xC0	GPIO_CFG_26[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x16C	0x2C	GPIO_CFG_27[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 13										
0x16D	0x8D	GPIO_CFG_28[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x16E	0xC4	GPIO_CFG_29[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
0x16F	0x2D	GPIO_CFG_30[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 14										
0x170	0x8E	GPIO_CFG_31[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x171	0xC4	GPIO_CFG_32[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE

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ADDRESS	RESET	NAME	MSB							LSB
0x172	0x2E	GPIO_CFG_33[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 15										
0x173	0x8F	GPIO_CFG_34[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x174	0xC0	GPIO_CFG_35[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x175	0x2F	GPIO_CFG_36[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 16										
0x176	0x90	GPIO_CFG_37[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x177	0xC6	GPIO_CFG_38[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
0x178	0x30	GPIO_CFG_39[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 17										
0x179	0x11	GPIO_CFG_40[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				

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ADDRESS	RESET	NAME	MSB							LSB
0x17A	0x80	GPIO_CFG_41[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x17B	0x31	GPIO_CFG_42[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 18										
0x17C	0x92	GPIO_CFG_43[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x17D	0xC0	GPIO_CFG_44[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x17E	0x32	GPIO_CFG_45[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 19										
0x17F	0x93	GPIO_CFG_46[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x180	0xC0	GPIO_CFG_47[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x181	0x33	GPIO_CFG_48[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 20										

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ADDRESS	RESET	NAME	MSB							LSB
0x182	0x94	GPIO_CFG_49[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x183	0xC0	GPIO_CFG_50[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x184	0x34	GPIO_CFG_51[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 21										
0x185	0x95	GPIO_CFG_52[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x186	0xC0	GPIO_CFG_53[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x187	0x35	GPIO_CFG_54[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 22										
0x188	0x96	GPIO_CFG_55[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x189	0xC0	GPIO_CFG_56[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE

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ADDRESS	RESET	NAME	MSB							LSB
0x18A	0x36	GPIO_CFG_57[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 23										
0x18B	0x97	GPIO_CFG_58[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x18C	0xC0	GPIO_CFG_59[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x18D	0x37	GPIO_CFG_60[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
GPIO_CFG 24										
0x18E	0x98	GPIO_CFG_61[7:0]	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
0x18F	0xC0	GPIO_CFG_62[7:0]	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
0x190	0x38	GPIO_CFG_63[7:0]	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
OUTPUT_CFG										
0x192	0x000	OUTPUT_CFG_0[7:0]	WMD_INT_EN	–	–	VIDEO_LOCK_OUTPUT_EN	LOCK_PIN_CONTROL[1:0]		–	RCLKOUT_EN

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ADDRESS	RESET	NAME	MSB							LSB
CRC_CFG										
0x195	0x05	CRC_CFG_0[7:0]	–	–	–	–	–	VIDEO_LINE_CRC_EN	VIDEO_PACKET_CRC_EN	FRAME_CRC_EN
0x196	0x00	CRC_CFG_1[7:0]	–	–	–	–	FCRC_INFOFRAME_CC_EN	–	–	DP_SINK_RGB_FRAME_CRC_EN
LF_TEMP_CFG										
0x199	0x00	LINEFAULT_CFG_0[7:0]	LINEFAULT_0_EN	LINEFAULT_1_EN	–	–	–	–	–	–
0x19B	0x00	ARQ_CFG_0[7:0]	–	–	–	–	–	–	–	MAX_RT_ERROR_RESET
VIDEO_PIPE_CFG										
0x1A0	0x80	VIDEO_PIPE_CFG_0[7:0]	VID_EN_X	STREAM_SEL_X[1:0]		–	SUPPORT_BLACK_VIDEO_ENABLE	BLANK_VIDEO	COLOR_LUT_EN	VPRBS_CHECKER_EN_X
0x1A1	0x30	VIDEO_PIPE_CFG_1[7:0]	VID_EN_Y	STREAM_SEL_Y[1:0]		LINK_SEL_Y	–	–	–	VPRBS_CHECKER_EN_Y
0x1A2	0x00	VIDEO_PIPE_CFG_2[7:0]	VS_HS_PIN_EN_X	VS_HS_INPUT_MODE_X	DE_OUTPUT_EN_X	–	VS_HS_OUTPUT_EN_Y	DE_OUTPUT_EN_Y	SYNC_LOCATION_SEL[1:0]	
0x1A3	0x00	VIDEO_PIPE_CFG_3[7:0]	VS_HS_ALT_PIN_EN_X	VS_HS_ALT_INPUT_MODE_X	–	ALIGN_LOCK_PIN_LOCATION	ALIGN_LOCK_INPUT_MODE	ALIGN_LOCK_PIN_EN	SYNC_LOCATION_ALT_SEL[1:0]	

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ADDRESS	RESET	NAME	MSB							LSB
DC_VIDEO_CFG										
0x1A5	0x09	DC_VIDEO_CFG_0[7:0]	DROP_VIDEO[3:0]				AUTO_DROP_VIDEO_EN	–	–	BLOCK_INFO_CC_FWD
DP_CFG										
0x1A8	0x84	DP_CFG_0[7:0]	AUTO_RETRAIN_ON_LOSS_OF_LINK_EN	–	DP_SINK_EDID_TIMING_EN	DP_SINK_READ_EDID_EN	DP_SPREADSPECTRUM_EN	AUTO_TIMING_EN	AUTO_DP_LINK_CONFIG	–
0x1A9	0x00	DP_CFG_1[7:0]	AUTO_RETRAIN_ON_DISCONNECT_EN	–	–	–	–	–	–	–
0x1AA	0x42	DP_CFG_2[7:0]	–	DP_LANE_COUNT[2:0]			–	DP_RATE[2:0]		
WM_CFG										
0x1B0	0x00	WATERMARK_LOCK_CFG_0[7:0]	WATERMARK_KEY_BYTE_0[7:0]							
0x1B1	0x00	WATERMARK_LOCK_CFG_1[7:0]	WATERMARK_KEY_BYTE_1[7:0]							
0x1B2	0x48	WATERMARK_CFG_0[7:0]	WATERMARK_DET_EN	WATERMARK_REMOVE	WATERMARK_REGIONAL_MODE	WATERMARK_BLANK	WATERMARK_REGIONAL_BLANK_MODE	WATERMARK_REGION_BLANK_EN	–	–
0x1B3	0x00	WATERMARK_CFG_1[7:0]	WM_REGION_XMIN_BYTE_0[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1B4	0x00	WATERMARK_CFG_2[7:0]	WM_REGION_XMIN_BYTE_1[7:0]							
0x1B5	0x00	WATERMARK_CFG_3[7:0]	WM_REGION_XMAX_BYTE_0[7:0]							
0x1B6	0x00	WATERMARK_CFG_4[7:0]	WM_REGION_XMAX_BYTE_1[7:0]							
0x1B7	0x00	WATERMARK_CFG_5[7:0]	WM_REGION_YMIN_BYTE_0[7:0]							
0x1B8	0x00	WATERMARK_CFG_6[7:0]	WM_REGION_YMIN_BYTE_1[7:0]							
0x1B9	0x00	WATERMARK_CFG_7[7:0]	WM_REGION_YMAX_BYTE_0[7:0]							
0x1BA	0x00	WATERMARK_CFG_8[7:0]	WM_REGION_YMAX_BYTE_1[7:0]							
VPG_CFG										
0x1BC	0x00	PATTERN_GENERATOR_CFG_0[7:0]	CUSTOM_COLOR_RED[7:0]							
0x1BD	0x00	PATTERN_GENERATOR_CFG_1[7:0]	CUSTOM_COLOR_GREEN[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1BE	0x00	PATTERN_GENERATOR_CFG_2[7:0]	CUSTOM_COLOR_BLUE[7:0]							
0x1BF	0x00	PATTERN_GENERATOR_CFG_3[7:0]	CUSTOM_COLOR_RED_30BIT_LSBS[1:0]		CUSTOM_COLOR_GREEN_30BIT_LSBS[1:0]		CUSTOM_COLOR_BLUE_30BIT_LSBS[1:0]		–	MODE_30BIT_EN
0x1C0	0x00	PATTERN_GENERATOR_CFG_4[7:0]	PATTERN_SIZE_X[7:0]							
0x1C1	0x00	PATTERN_GENERATOR_CFG_5[7:0]	PATTERN_SIZE_Y[7:0]							
0x1C2	0x00	PATTERN_GENERATOR_CFG_6[7:0]	GRADIENT_COLOR_STEP_SIZE[7:0]							
0x1C3	0x00	PATTERN_GENERATOR_CFG_7[7:0]	PATTERN_SELECT[7:0]							
DSC_CFG										
0x1C5	0x00	DSC_CFG_0[7:0]	DSC_DECODE_EN	MANUAL_DSC_EN	–	–	–	–	–	–
HDCP_CFG										
0x1C7	0xC1	HDCP_CFG_0[7:0]	HDCP_EN_X	HDCP_EN_Y	–	–	–	HDCP_GPIO_ID_Y[1:0]		HDCP_VER

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ADDRESS	RESET	NAME	MSB							LSB
0x1C8	0x94	HDCP_CFG_1[7:0]	HDCP_ADDRESS_Y[6:0]							–
0x1C9	0x88	HDCP_CFG_2[7:0]	HDCP_CONFIG_0[7]	HDCP_CONFIG_0[6]	HDCP_CONFIG_0[5]	HDCP_CONFIG_0[4]	HDCP_CONFIG_0[3]	HDCP_CONFIG_0[2]	HDCP_CONFIG_0[1]	HDCP_CONFIG_0[0]
0x1CA	0xA3	HDCP_CFG_3[7:0]	HDCP_CONFIG_1[5]	HDCP_CONFIG_1[4]	HDCP_CONFIG_1[3]	HDCP_CONFIG_1[2]	HDCP_CONFIG_1[1]	HDCP_CONFIG_1[0]	ARQ_AUTO_CFG_AHDCP_Y	ARQ_AUTO_CFG_AHDCP_X
FRC_CFG										
0x1D2	0x00	FRC_CFG_0[7:0]	FRC_EN	–	–	–	–	–	–	–
VIDEO_TIMING_CFG										
0x1D6	0x14	VIDEO_TIMING_CFG_0[7:0]	PCLK_FREQUENCY_BYTE_0[7:0]							
0x1D7	0x44	VIDEO_TIMING_CFG_1[7:0]	PCLK_FREQUENCY_BYTE_1[7:0]							
0x1D8	0x02	VIDEO_TIMING_CFG_2[7:0]	PCLK_FREQUENCY_BYTE_2[7:0]							
0x1D9	0x80	VIDEO_TIMING_CFG_3[7:0]	H_ACTIVE_BYTE_0[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1DA	0x07	VIDEO_TIMING_CFG_4[7:0]	H_ACTIVE_BYTE_1[7:0]							
0x1DB	0x58	VIDEO_TIMING_CFG_5[7:0]	H_FP_BYTE_0[7:0]							
0x1DC	0x00	VIDEO_TIMING_CFG_6[7:0]	H_FP_BYTE_1[7:0]							
0x1DD	0x2C	VIDEO_TIMING_CFG_7[7:0]	H_SW_BYTE_0[7:0]							
0x1DE	0x00	VIDEO_TIMING_CFG_8[7:0]	H_SW_BYTE_1[7:0]							
0x1DF	0x94	VIDEO_TIMING_CFG_9[7:0]	H_BP_BYTE_0[7:0]							
0x1E0	0x00	VIDEO_TIMING_CFG_10[7:0]	H_BP_BYTE_1[7:0]							
0x1E1	0x38	VIDEO_TIMING_CFG_11[7:0]	V_ACTIVE_BYTE_0[7:0]							
0x1E2	0x04	VIDEO_TIMING_CFG_12[7:0]	V_ACTIVE_BYTE_1[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E3	0x04	VIDEO_TIMING_CFG_13[7:0]	V_FP[7:0]							
0x1E4	0x05	VIDEO_TIMING_CFG_14[7:0]	V_SW[7:0]							
0x1E5	0x24	VIDEO_TIMING_CFG_15[7:0]	V_BP[7:0]							
0x1E6	0x03	VIDEO_TIMING_CFG_16[7:0]	–	FORCE_EVENODD_PIXEL_MODE_EN	COLOR_DEPTH_MANUAL_MODE	COLOR_DEPTH_30BIT_SELECT	–	HS_VS_POLARITY_MANUAL_MODE	VS_POLARITY	HS_POLARITY
PSM_STATE_CFG										
0x1E7	0x00	PSM_STATE_CFG_0[7:0]	PSM_STATE[7:0]							
0x1E8	0x00	PSM_STATUS_0[7:0]	PSM_STATUS[7:0]							
CC										
0x248	0x40	UART_0[7:0]	RSVD[1:0]		REM_MS_EN	LOC_MS_EN	RSVD	–	–	–
TCTRL										
0x2D3	0x00	INTR13[7:0]	–	–	–	–	–	EOM_ERR_FLAG_DCIO	EOM_ERR_FLAG_SIOB	EOM_ERR_FLAG_SIOA

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ADDRESS	RESET	NAME	MSB							LSB
CFGL AUDIO										
0x303	0x00	TR3[7:0]	–	–	–	–	–	TX_SRC_ID[2:0]		
VRX X										
0x502	0x08	LUT_CTRL[7:0]	–	–	LUT_MSB_DATA[1:0]		LUT_10_BIT_OUT	–	–	–
0x51F	0x00	VRX_ASIL_INJ[7:0]	–	–	FRC_DATAPATH_FAIL_INJ	LUT_CRC_BLUE_FAIL_INJ	LUT_CRC_GREEN_FAIL_INJ	LUT_CRC_RED_FAIL_INJ	LUT_DATAPATH_FAIL_INJ	WM_DATAPATH_FAIL_INJ
0x522	0x00	BLANK_REGION_CONTROL[7:0]	–	–	–	–	–	–	–	BLANK_REGION_EN
0x523	0x00	BLANK_REGION_XY_CFG_0[7:0]	BLANK_REGION_XMIN_BYTE_0[7:0]							
0x524	0x00	BLANK_REGION_XY_CFG_1[7:0]	BLANK_REGION_XMIN_BYTE_1[7:0]							
0x525	0x00	BLANK_REGION_XY_CFG_2[7:0]	BLANK_REGION_XMAX_BYTE_0[7:0]							
0x526	0x00	BLANK_REGION_XY_CFG_3[7:0]	BLANK_REGION_XMAX_BYTE_1[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x527	0x00	BLANK_REGION_XY_CFG_4[7:0]	BLANK_REGION_YMIN_BYTE_0[7:0]							
0x528	0x00	BLANK_REGION_XY_CFG_5[7:0]	BLANK_REGION_YMIN_BYTE_1[7:0]							
0x529	0x00	BLANK_REGION_XY_CFG_6[7:0]	BLANK_REGION_YMAX_BYTE_0[7:0]							
0x52A	0x00	BLANK_REGION_XY_CFG_7[7:0]	BLANK_REGION_YMAX_BYTE_1[7:0]							
VRX Y										
0x542	0x08	LUT_CTRL[7:0]	–	–	LUT_MSB_DATA[1:0]		LUT_10_BIT_OUT	–	–	–
0x55B	0x00	PRBS_ERR[7:0]	VPRBS_ERROR_COUNT[7:0]							
0x55F	0x00	VRX_ASIL_INJ[7:0]	–	–	FRC_DATAPATH_FAIL_INJ	LUT_CRC_BLUE_FAIL_INJ	LUT_CRC_GREEN_FAIL_INJ	LUT_CRC_RED_FAIL_INJ	LUT_DATAPATH_FAIL_INJ	WM_DATAPATH_FAIL_INJ
VIDEO_AUTH										
0x6000	0x05	AUTH0[7:0]	FRAME_CYCLE_MODULO_H[1:0]		LOCAL_CHALLENGE_MAX[1:0]		–	AUTH_GLOBAL_PERIODIC	AUTH_GLOBAL_TRIG	AUTH_EN

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ADDRESS	RESET	NAME	MSB							LSB
0x601	0x04	AUTH1[7:0]	FRAME_CYCLE_MODULO_L[7:0]							
0x602	0x00	XMINL0a[7:0]	XMIN0a_LSB[7:0]							
0x603	0x00	XMINH0a[7:0]	–	–	XMIN0a_MSB[5:0]					
0x604	0x01	XMAXL0a[7:0]	XMAX0a_LSB[7:0]							
0x605	0x00	XMAXH0a[7:0]	–	–	XMAX0a_MSB[5:0]					
0x606	0x00	XMINL1a[7:0]	XMIN1a_LSB[7:0]							
0x607	0x00	XMINH1a[7:0]	–	–	XMIN1a_MSB[5:0]					
0x608	0x00	XMAXL1a[7:0]	XMAX1a_LSB[7:0]							
0x609	0x00	XMAXH1a[7:0]	–	–	XMAX1a_MSB[5:0]					

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ADDRESS	RESET	NAME	MSB							LSB
0x60A	0x00	XMINL2a[7:0]	XMIN2a_LSB[7:0]							
0x60B	0x00	XMINH2a[7:0]	–	–	XMIN2a_MSB[5:0]					
0x60C	0x00	XMAXL2a[7:0]	XMAX2a_LSB[7:0]							
0x60D	0x00	XMAXH2a[7:0]	–	–	XMAX2a_MSB[5:0]					
0x60E	0x00	XMINL3a[7:0]	XMIN3a_LSB[7:0]							
0x60F	0x00	XMINH3a[7:0]	–	–	XMIN3a_MSB[5:0]					
0x610	0x00	XMAXL3a[7:0]	XMAX3a_LSB[7:0]							
0x611	0x00	XMAXH3a[7:0]	–	–	XMAX3a_MSB[5:0]					
0x612	0x00	XMINL0b[7:0]	XMIN0b_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x613	0x00	XMINH0b[7:0]	–	–	XMIN0b_MSB[5:0]					
0x614	0x01	XMAXL0b[7:0]	XMAX0b_LSB[7:0]							
0x615	0x00	XMAXH0b[7:0]	–	–	XMAX0b_MSB[5:0]					
0x616	0x00	XMINL1b[7:0]	XMIN1b_LSB[7:0]							
0x617	0x00	XMINH1b[7:0]	–	–	XMIN1b_MSB[5:0]					
0x618	0x00	XMAXL1b[7:0]	XMAX1b_LSB[7:0]							
0x619	0x00	XMAXH1b[7:0]	–	–	XMAX1b_MSB[5:0]					
0x61A	0x00	XMINL2b[7:0]	XMIN2b_LSB[7:0]							
0x61B	0x00	XMINH2b[7:0]	–	–	XMIN2b_MSB[5:0]					

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ADDRESS	RESET	NAME	MSB							LSB
0x61C	0x00	XMAXL2b[7:0]	XMAX2b_LSB[7:0]							
0x61D	0x00	XMAXH2b[7:0]	–	–	XMAX2b_MSB[5:0]					
0x61E	0x00	XMINL3b[7:0]	XMIN3b_LSB[7:0]							
0x61F	0x00	XMINH3b[7:0]	–	–	XMIN3b_MSB[5:0]					
0x620	0x00	XMAXL3b[7:0]	XMAX3b_LSB[7:0]							
0x621	0x00	XMAXH3b[7:0]	–	–	XMAX3b_MSB[5:0]					
0x622	0x00	YMINL0a[7:0]	YMIN0a_LSB[7:0]							
0x623	0x00	YMINH0a[7:0]	–	–	YMIN0a_MSB[5:0]					
0x624	0x01	YMAXL0a[7:0]	YMAX0a_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x625	0x00	YMAXH0a[7:0]	FRAME0a_MSB[1:0]		YMAX0a_MSB[5:0]					
0x626	0x01	frame0a[7:0]	FRAME0a_LSB[7:0]							
0x627	0x00	YMINL1a[7:0]	YMIN1a_LSB[7:0]							
0x628	0x00	YMINH1a[7:0]	–	–	YMIN1a_MSB[5:0]					
0x629	0x00	YMAXL1a[7:0]	YMAX1a_LSB[7:0]							
0x62A	0x00	YMAXH1a[7:0]	FRAME1a_MSB[1:0]		YMAX1a_MSB[5:0]					
0x62B	0x00	frame1a[7:0]	FRAME1a_LSB[7:0]							
0x62C	0x00	YMINL2a[7:0]	YMIN2a_LSB[7:0]							
0x62D	0x00	YMINH2a[7:0]	–	–	YMIN2a_MSB[5:0]					

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ADDRESS	RESET	NAME	MSB							LSB
0x62E	0x00	YMAXL2a[7:0]	YMAX2a_LSB[7:0]							
0x62F	0x00	YMAXH2a[7:0]	FRAME2a_MSB[1:0]		YMAX2a_MSB[5:0]					
0x630	0x00	frame2a[7:0]	FRAME2a_LSB[7:0]							
0x631	0x00	YMINL3a[7:0]	YMIN3a_LSB[7:0]							
0x632	0x00	YMINH3a[7:0]	–	–	YMIN3a_MSB[5:0]					
0x633	0x00	YMAXL3a[7:0]	YMAX3a_LSB[7:0]							
0x634	0x00	YMAXH3a[7:0]	FRAME3a_MSB[1:0]		YMAX3a_MSB[5:0]					
0x635	0x00	frame3a[7:0]	FRAME3a_LSB[7:0]							
0x636	0x00	YMINL0b[7:0]	YMIN0b_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x637	0x00	YMINH0b[7:0]	–	–	YMIN0b_MSB[5:0]					
0x638	0x01	YMAXL0b[7:0]	YMAX0b_LSB[7:0]							
0x639	0x00	YMAXH0b[7:0]	FRAME0b_MSB[1:0]		YMAX0b_MSB[5:0]					
0x63A	0x01	frame0b[7:0]	FRAME0b_LSB[7:0]							
0x63B	0x00	YMINL1b[7:0]	YMIN1b_LSB[7:0]							
0x63C	0x00	YMINH1b[7:0]	–	–	YMIN1b_MSB[5:0]					
0x63D	0x00	YMAXL1b[7:0]	YMAX1b_LSB[7:0]							
0x63E	0x00	YMAXH1b[7:0]	FRAME1b_MSB[1:0]		YMAX1b_MSB[5:0]					
0x63F	0x00	frame1b[7:0]	FRAME1b_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x6400	0x00	YMINL2b[7:0]	YMIN2b_LSB[7:0]							
0x6401	0x00	YMINH2b[7:0]	–	–	YMIN2b_MSB[5:0]					
0x6402	0x00	YMAXL2b[7:0]	YMAX2b_LSB[7:0]							
0x6403	0x00	YMAXH2b[7:0]	FRAME2b_MSB[1:0]		YMAX2b_MSB[5:0]					
0x6404	0x00	frame2b[7:0]	FRAME2b_LSB[7:0]							
0x6405	0x00	YMINL3b[7:0]	YMIN3b_LSB[7:0]							
0x6406	0x00	YMINH3b[7:0]	–	–	YMIN3b_MSB[5:0]					
0x6407	0x00	YMAXL3b[7:0]	YMAX3b_LSB[7:0]							
0x6408	0x00	YMAXH3b[7:0]	FRAME3b_MSB[1:0]		YMAX3b_MSB[5:0]					

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ADDRESS	RESET	NAME	MSB							LSB
0x649	0x00	frame3b[7:0]	FRAME3b_LSB[7:0]							
0x64A	0x00	out1[7:0]	FRAME_CNT_MSB	–	VA_DONE_FLAG	VA_DONE_INT_EN	VA_DONE_PIN	A_OR_B	AUTH_DONE_B	AUTH_DONE_A
0x64B	0x00	out1a[7:0]	FRAME_CNT_LSB[7:0]							
0x64C	0x00	out2[7:0]	CRC_OUT_A_0[7:0]							
0x64D	0x00	out3[7:0]	CRC_OUT_A_1[7:0]							
0x64E	0x00	out4[7:0]	CRC_OUT_A_2[7:0]							
0x64F	0x00	out5[7:0]	CRC_OUT_A_3[7:0]							
0x650	0x00	out6[7:0]	CRC_OUT_B_0[7:0]							
0x651	0x00	out7[7:0]	CRC_OUT_B_1[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x652	0x00	out8[7:0]	CRC_OUT_B_2[7:0]							
0x653	0x00	out9[7:0]	CRC_OUT_B_3[7:0]							
0x654	0x00	outa[7:0]	FULL_FRAME_CRC_0[7:0]							
0x655	0x00	outb[7:0]	FULL_FRAME_CRC_1[7:0]							
0x656	0x00	outc[7:0]	FULL_FRAME_CRC_2[7:0]							
0x657	0x00	outd[7:0]	FULL_FRAME_CRC_3[7:0]							
RLMS_SIO DC										
0x804	0x4B	RLMS4[7:0]	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
0x805	0x10	RLMS5[7:0]	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
0x806	0x80	RLMS6[7:0]	EOM_PV_MODE	EOM_RST_THR[6:0]						

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ADDRESS	RESET	NAME	MSB							LSB
0x807	0x00	RLMS7[7:0]	EOM_DONE	EOM[6:0]						
0x8CE	0x01	RLMSCE[7:0]	–	–	–	–	–	–	–	enffe
INFOFR_CC										
0xA00	0x00	CTRL0[7:0]	–	–	–	–	–	–	–	ENABLE_INFOFR_CC
FEEDFORWARD_CRC X										
0xC80	0x30	FFWD_CRC_CHK_REG0[7:0]	FRAME_CRC_CHECK_EN	FRAME_COUNT_CHKECK_EN	EN_VOTING_FOR_FCRC	EN_VOTING_FOR_FCNT	USE_FRAME_CRC_VALID_BIT	–	FRAME_CRC_CALC_EN	SUPPRESS_LAST_LINE
0xC81	0x00	FFWD_CRC_CHK_REG1[7:0]	–	TRIGGER_ONLY_ON_FCRC_RCVD	USE_DP_FCRC_INFO	USE_FCRC_FCNT_EXP_REG	USE_PREV_FRAME_TO_CHECK	FRAME_CRC_CHECK_SEL[2:0]		
0xC82	0x00	FFWD_CRC_CHK_REG2[7:0]	FRAME_CRC_EXP_BYTE_5[7:0]							
0xC83	0x00	FFWD_CRC_CHK_REG3[7:0]	FRAME_CRC_EXP_BYTE_4[7:0]							
0xC84	0x00	FFWD_CRC_CHK_REG4[7:0]	FRAME_CRC_EXP_BYTE_3[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xC85	0x00	FFWD_CRC_CHK_REG5[7:0]	FRAME_CRC_EXP_BYTE_2[7:0]							
0xC86	0x00	FFWD_CRC_CHK_REG6[7:0]	FRAME_CRC_EXP_BYTE_1[7:0]							
0xC87	0x00	FFWD_CRC_CHK_REG7[7:0]	FRAME_CRC_EXP_BYTE_0[7:0]							
0xC88	0x00	FFWD_CRC_CHK_REG8[7:0]	FRAME_COUNT_EXP[7:0]							
0xC89	0x00	FFWD_CRC_CHK_REG9[7:0]	–	–	–	–	–	RESET_FRAME_CNT	UPDATE_FCNT_EXP	UPDATE_FCRC_EXP
0xC8A	0x00	FFWD_CRC_CHK_REG10[7:0]	FRAME_CRC_CHECK_ERR	FRAME_COUNT_CHECK_ERR	VOTE_ERR_FCRC	VOTE_ERR_FCNT	–	–	–	–
0xC8B	0x00	FFWD_CRC_CHK_REG11[7:0]	FCRC_CHECK_ERR_LAST	FCNT_CHECK_ERR_LAST	FCNT_SYNCED	–	–	–	–	–
0xC8C	0x00	FFWD_CRC_CHK_REG12[7:0]	FRAME_CRC_RECEIVED_BYTE_5[7:0]							
0xC8D	0x00	FFWD_CRC_CHK_REG13[7:0]	FRAME_CRC_RECEIVED_BYTE_4[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xC8E	0x00	FFWD_CRC_CHK_REG14[7:0]	FRAME_CRC_RECEIVED_BYTE_3[7:0]							
0xC8F	0x00	FFWD_CRC_CHK_REG15[7:0]	FRAME_CRC_RECEIVED_BYTE_2[7:0]							
0xC90	0x00	FFWD_CRC_CHK_REG16[7:0]	FRAME_CRC_RECEIVED_BYTE_1[7:0]							
0xC91	0x00	FFWD_CRC_CHK_REG17[7:0]	FRAME_CRC_RECEIVED_BYTE_0[7:0]							
0xC92	0x00	FFWD_CRC_CHK_REG18[7:0]	FRAME_COUNT_RECEIVED_BYTE_3[7:0]							
0xC93	0x00	FFWD_CRC_CHK_REG19[7:0]	FRAME_COUNT_RECEIVED_BYTE_2[7:0]							
0xC94	0x00	FFWD_CRC_CHK_REG20[7:0]	FRAME_COUNT_RECEIVED_BYTE_1[7:0]							
0xC95	0x00	FFWD_CRC_CHK_REG21[7:0]	FRAME_COUNT_RECEIVED_BYTE_0[7:0]							
0xC96	0x00	FFWD_CRC_CHK_REG22[7:0]	SELECTED_CRC_VALUE_BYTE_3[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xC97	0x00	FFWD_CRC_CHK_REG23[7:0]	SELECTED_CRC_VALUE_BYTE_2[7:0]							
0xC98	0x00	FFWD_CRC_CHK_REG24[7:0]	SELECTED_CRC_VALUE_BYTE_1[7:0]							
0xC99	0x00	FFWD_CRC_CHK_REG25[7:0]	SELECTED_CRC_VALUE_BYTE_0[7:0]							
0xC9A	0x00	FFWD_CRC_CHK_REG26[7:0]	FRAME_COUNT[7:0]							
FEEDFORWARD_CRC Y										
0xCA0	0x30	FFWD_CRC_CHK_REG0[7:0]	FRAME_CRC_CHECK_EN	FRAME_COUNT_CHK_ECK_EN	EN_VOTING_FOR_FCRC	EN_VOTING_FOR_FCNT	USE_FRAME_CRC_VALID_BIT	–	FRAME_CRC_CALC_EN	SUPPRESS_LAST_LINE
0xCA1	0x00	FFWD_CRC_CHK_REG1[7:0]	–	TRIGGER_ONLY_ON_FCRC_RCVD	USE_DP_FCRC_INFO	USE_FCRC_FCNT_EXP_REG	USE_PREV_FRAME_TO_CHECK	FRAME_CRC_CHECK_SEL[2:0]		
0xCA2	0x00	FFWD_CRC_CHK_REG2[7:0]	FRAME_CRC_EXP_BYTE_5[7:0]							
0xCA3	0x00	FFWD_CRC_CHK_REG3[7:0]	FRAME_CRC_EXP_BYTE_4[7:0]							
0xCA4	0x00	FFWD_CRC_CHK_REG4[7:0]	FRAME_CRC_EXP_BYTE_3[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xCA5	0x00	FFWD_CRC_CHK_REG5[7:0]	FRAME_CRC_EXP_BYTE_2[7:0]							
0xCA6	0x00	FFWD_CRC_CHK_REG6[7:0]	FRAME_CRC_EXP_BYTE_1[7:0]							
0xCA7	0x00	FFWD_CRC_CHK_REG7[7:0]	FRAME_CRC_EXP_BYTE_0[7:0]							
0xCA8	0x00	FFWD_CRC_CHK_REG8[7:0]	FRAME_COUNT_EXP[7:0]							
0xCA9	0x00	FFWD_CRC_CHK_REG9[7:0]	–	–	–	–	–	RESET_FRAME_CNT	UPDATE_FCNT_EXP	UPDATE_FCRC_EXP
0xCAA	0x00	FFWD_CRC_CHK_REG10[7:0]	FRAME_CRC_CHECK_ERR	FRAME_COUNT_CHECK_ERR	VOTE_ERR_FCRC	VOTE_ERR_FCNT	–	–	–	–
0xCAB	0x00	FFWD_CRC_CHK_REG11[7:0]	FCRC_CHECK_ERR_LAST	FCNT_CHECK_ERR_LAST	FCNT_SYNCED	–	–	–	–	–
0xCAC	0x00	FFWD_CRC_CHK_REG12[7:0]	FRAME_CRC_RECEIVED_BYTE_5[7:0]							
0xCAD	0x00	FFWD_CRC_CHK_REG13[7:0]	FRAME_CRC_RECEIVED_BYTE_4[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xCAE	0x00	FFWD_CRC_CHK_REG14[7:0]	FRAME_CRC_RECEIVED_BYTE_3[7:0]							
0xCAF	0x00	FFWD_CRC_CHK_REG15[7:0]	FRAME_CRC_RECEIVED_BYTE_2[7:0]							
0xCB0	0x00	FFWD_CRC_CHK_REG16[7:0]	FRAME_CRC_RECEIVED_BYTE_1[7:0]							
0xCB1	0x00	FFWD_CRC_CHK_REG17[7:0]	FRAME_CRC_RECEIVED_BYTE_0[7:0]							
0xCB2	0x00	FFWD_CRC_CHK_REG18[7:0]	FRAME_COUNT_RECEIVED_BYTE_3[7:0]							
0xCB3	0x00	FFWD_CRC_CHK_REG19[7:0]	FRAME_COUNT_RECEIVED_BYTE_2[7:0]							
0xCB4	0x00	FFWD_CRC_CHK_REG20[7:0]	FRAME_COUNT_RECEIVED_BYTE_1[7:0]							
0xCB5	0x00	FFWD_CRC_CHK_REG21[7:0]	FRAME_COUNT_RECEIVED_BYTE_0[7:0]							
0xCB6	0x00	FFWD_CRC_CHK_REG22[7:0]	SELECTED_CRC_VALUE_BYTE_3[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xCB7	0x00	FFWD_CRC_CHK_REG23[7:0]	SELECTED_CRC_VALUE_BYTE_2[7:0]							
0xCB8	0x00	FFWD_CRC_CHK_REG24[7:0]	SELECTED_CRC_VALUE_BYTE_1[7:0]							
0xCB9	0x00	FFWD_CRC_CHK_REG25[7:0]	SELECTED_CRC_VALUE_BYTE_0[7:0]							
0xCBA	0x00	FFWD_CRC_CHK_REG26[7:0]	FRAME_COUNT[7:0]							
FEEDFORWARD_CRC_COMBINED										
0xCC0	0x30	FFWD_CRC_CHK_REG0[7:0]	COMBINED_FRAME_CRC_CHECK_EN	COMBINED_FRAME_COUNT_CHK_ECK_EN	EN_VOTING_FOR_FCRC	EN_VOTING_FOR_FCNT	USE_FRAME_CRC_VALID_BIT	–	COMBINED_FRAME_CRC_CALC_EN	SUPPRESS_LAST_LINE
0xCC1	0x00	FFWD_CRC_CHK_REG1[7:0]	–	TRIGGER_ONLY_ON_FCRC_RCVD	USE_DP_FCRC_INFO	USE_FCRC_FCNT_EXP_REG	USE_PREV_FRAME_TO_CHECK	FRAME_CRC_CHECK_SEL[2:0]		
0xCC2	0x00	FFWD_CRC_CHK_REG2[7:0]	COMBINED_FRAME_CRC_EXP_BYTE_5[7:0]							
0xCC3	0x00	FFWD_CRC_CHK_REG3[7:0]	COMBINED_FRAME_CRC_EXP_BYTE_4[7:0]							
0xCC4	0x00	FFWD_CRC_CHK_REG4[7:0]	COMBINED_FRAME_CRC_EXP_BYTE_3[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xCC5	0x00	FFWD_CRC_CHK_REG5[7:0]	COMBINED_FRAME_CRC_EXP_BYTE_2[7:0]							
0xCC6	0x00	FFWD_CRC_CHK_REG6[7:0]	COMBINED_FRAME_CRC_EXP_BYTE_1[7:0]							
0xCC7	0x00	FFWD_CRC_CHK_REG7[7:0]	COMBINED_FRAME_CRC_EXP_BYTE_0[7:0]							
0xCC8	0x00	FFWD_CRC_CHK_REG8[7:0]	COMBINED_FRAME_COUNT_EXP[7:0]							
0xCC9	0x00	FFWD_CRC_CHK_REG9[7:0]	–	–	–	–	–	RESET_FRAME_CNT	UPDATE_FCNT_EXP	UPDATE_FCRC_EXP
0xCCA	0x00	FFWD_CRC_CHK_REG10[7:0]	COMBINED_FRAME_CRC_CHECK_ERR	COMBINED_FRAME_COUNT_CHECK_ERR	VOTE_ERR_FCRC	VOTE_ERR_FCNT	–	–	–	–
0xCCB	0x00	FFWD_CRC_CHK_REG11[7:0]	FCRC_CHECK_ERR_LAST	FCNT_CHECK_ERR_LAST	FCNT_SYNCED	–	–	–	–	–
0xCCC	0x00	FFWD_CRC_CHK_REG12[7:0]	COMBINED_FRAME_CRC_RECEIVED_BYTE_5[7:0]							
0xCCD	0x00	FFWD_CRC_CHK_REG13[7:0]	COMBINED_FRAME_CRC_RECEIVED_BYTE_4[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xCCCE	0x00	FFWD_CRC_CHK_REG14[7:0]	COMBINED_FRAME_CRC_RECEIVED_BYTE_3[7:0]							
0xCCCF	0x00	FFWD_CRC_CHK_REG15[7:0]	COMBINED_FRAME_CRC_RECEIVED_BYTE_2[7:0]							
0xCCD0	0x00	FFWD_CRC_CHK_REG16[7:0]	COMBINED_FRAME_CRC_RECEIVED_BYTE_1[7:0]							
0xCCD1	0x00	FFWD_CRC_CHK_REG17[7:0]	COMBINED_FRAME_CRC_RECEIVED_BYTE_0[7:0]							
0xCCD2	0x00	FFWD_CRC_CHK_REG18[7:0]	COMBINED_FRAME_COUNT_RECEIVED_BYTE_3[7:0]							
0xCCD3	0x00	FFWD_CRC_CHK_REG19[7:0]	COMBINED_FRAME_COUNT_RECEIVED_BYTE_2[7:0]							
0xCCD4	0x00	FFWD_CRC_CHK_REG20[7:0]	COMBINED_FRAME_COUNT_RECEIVED_BYTE_1[7:0]							
0xCCD5	0x00	FFWD_CRC_CHK_REG21[7:0]	COMBINED_FRAME_COUNT_RECEIVED_BYTE_0[7:0]							
0xCCD6	0x00	FFWD_CRC_CHK_REG22[7:0]	SELECTED_CRC_VAL_5[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xCD7	0x00	FFWD_CRC_CHK_REG23[7:0]	SELECTED_CRC_VAL_4[7:0]							
0xCD8	0x00	FFWD_CRC_CHK_REG24[7:0]	SELECTED_CRC_VAL_3[7:0]							
0xCD9	0x00	FFWD_CRC_CHK_REG25[7:0]	SELECTED_CRC_VAL_2[7:0]							
0xCDA	0x00	FFWD_CRC_CHK_REG26[7:0]	SELECTED_CRC_VAL_1[7:0]							
0xCDB	0x00	FFWD_CRC_CHK_REG27[7:0]	SELECTED_CRC_VAL_0[7:0]							
0xCDC	0x00	FFWD_CRC_CHK_REG28[7:0]	FRAME_COUNT[7:0]							
ALIGN_FIFO										
0xD80	0x00	ALIGN_FIFO_0[7:0]	–	–	–	–	–	RST_ALIGN_FIFO	DELAY_PIPE_X_EN	ALIGN_FIFO_EN
0xD82	0x00	ALIGN_FIFO_2[7:0]	INITIAL_DELAY_VAL_L[7:0]							
0xD83	0x07	ALIGN_FIFO_3[7:0]	–	–	–	–	INITIAL_DELAY_VAL_H[3:0]			

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ADDRESS	RESET	NAME	MSB							LSB
0xD86	0x20	ALIGN_FIFO_6[7:0]	–	–	USE_EXT_HS_FOR_ALIGN	ENABLE_ALIGN_WITH_EXT_SIGNAL	–	–	–	–
0xD93	0x00	ALIGN_FIFO_19[7:0]	OFFSET_DETECTED_LIVE_SIGN	OFFSET_DETECTED_LIVE_MAG_H[6:0]						
0xD94	0x00	ALIGN_FIFO_20[7:0]	EXT_ALIGN_COMPLETED_IN_SUBORDINATE_DES	EXT_ALIGN_COMPLETED_OUT	OFFSET_DETECTED_TOGGLE	OFFSET_DETECTED_LIVE_MAG_L[4:0]				
0xD95	0x00	ALIGN_FIFO_21[7:0]	OFFSET_DETECTED_MAX_SIGN	OFFSET_DETECTED_MAX_MAG_H[6:0]						
0xD96	0x00	ALIGN_FIFO_22[7:0]	–	–	–	OFFSET_DETECTED_MAX_MAG_L[4:0]				
0xD98	0x00	ALIGN_FIFO_24[7:0]	–	–	EXT_ALIGNMENT_LOCKED_LIVE	–	–	–	–	–
0xD9A	0x00	ALIGN_FIFO_26[7:0]	–	–	–	–	INJECT_ERR_Y	INJECT_ERR_X	–	–
PSM_TASK_REGS										
0xE07	0x08	PSM_REG_7[7:0]	PSM_TASK_CTRL_7[7:0]							
0xE30	0xFF	IFCC_CTRL_0[7:0]	DEST_ADDRESS[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xE31	0x00	IFCC_CTRL_1[7:0]	REG_ADDR_L[7:0]							
0xE32	0x00	IFCC_CTRL_2[7:0]	REG_ADDR_H[7:0]							
0xE33	0x00	IFCC_CTRL_3[7:0]	IFCC_RW_BUFFER_0[7:0]							
0xE34	0x00	IFCC_CTRL_4[7:0]	IFCC_RW_BUFFER_1[7:0]							
0xE35	0x00	IFCC_CTRL_5[7:0]	IFCC_RW_BUFFER_2[7:0]							
0xE36	0x00	IFCC_CTRL_6[7:0]	IFCC_RW_BUFFER_3[7:0]							
0xE37	0x00	IFCC_CTRL_7[7:0]	IFCC_RW_BUFFER_4[7:0]							
0xE38	0x00	IFCC_CTRL_8[7:0]	IFCC_RW_BUFFER_5[7:0]							
0xE39	0x00	IFCC_CTRL_9[7:0]	IFCC_RW_BUFFER_6[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xE3A	0x00	IFCC_CTRL_10[7:0]	IFCC_RW_BUFFER_7[7:0]							
0xE3B	0x00	IFCC_CTRL_11[7:0]	IFCC_RW_BUFFER_8[7:0]							
0xE3C	0x00	IFCC_CTRL_12[7:0]	IFCC_RW_BUFFER_9[7:0]							
0xE3D	0x00	IFCC_CTRL_13[7:0]	IFCC_RW_BUFFER_10[7:0]							
0xE3E	0x00	IFCC_CTRL_14[7:0]	IFCC_RW_BUFFER_11[7:0]							
0xE3F	0x00	IFCC_CTRL_15[7:0]	IFCC_RW_BUFFER_12[7:0]							
0xE40	0x00	IFCC_CTRL_16[7:0]	IFCC_RW_BUFFER_13[7:0]							
0xE41	0x00	IFCC_CTRL_17[7:0]	IFCC_RW_BUFFER_14[7:0]							
0xE42	0x00	IFCC_CTRL_18[7:0]	IFCC_RW_BUFFER_15[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xE43	0x00	IFCC_CTRL_19[7:0]	NUM_OF_BYTES[7:0]							
0xE44	0x00	IFCC_CTRL_20[7:0]	EXECUTE	IFCC_COMMAND[6:0]						
0xE45	0x00	IFCC_USER_CMD_STATUS_0[7:0]	RD_DONE	WR_DONE	–	–	RD_DIS_ERR	WR_DIS_ERR	NUM_BYTE_ERR	COMMAND_ERR
0xE46	0x00	IFCC_USER_CMD_STATUS_1[7:0]	USER_READ_COUNT[7:0]							
0xE47	0x00	IFCC_USER_CMD_STATUS_2[7:0]	USER_WRITE_COUNT[7:0]							
0xE48	0x00	IFCC_RMT_CMD_STATUS_0[7:0]	RMT_RD_DONE	RMT_WR_DONE	–	–	RMT_RD_DIS_ERR	RMT_WR_DIS_ERR	RMT_NUM_BYTE_ERR	RMT_COMMAND_ERR
0xE49	0x00	IFCC_RMT_CMD_STATUS_1[7:0]	RMT_RD_COUNT[7:0]							
0xE4A	0x00	IFCC_RMT_CMD_STATUS_2[7:0]	RMT_WR_COUNT[7:0]							
0xE86	0x00	OTP_ERR_INJ_CTRL_2[7:0]	OTP_INJECT_ERROR_WORD_INDEX_MSB	–	OTP_ERROR_INJECT_EN	–	–	–	–	–
FRC X										

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ADDRESS	RESET	NAME	MSB							LSB
0x10C0	0x02	FRC_0[7:0]	–	–	–	–	–	FRCGAIN[1:0]		–
0x10C1	0x08	FRC_1[7:0]	–	–	–	–	FRCDATA WIDTH	–	–	–
RLMS_SIO A										
0x1404	0x4B	RLMS4[7:0]	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
0x1405	0x10	RLMS5[7:0]	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
0x1406	0x80	RLMS6[7:0]	EOM_PV_MODE	EOM_RST_THR[6:0]						
0x1407	0x00	RLMS7[7:0]	EOM_DONE	EOM[6:0]						
RLMS_SIO B										
0x1504	0x4B	RLMS4[7:0]	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
0x1505	0x10	RLMS5[7:0]	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						

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ADDRESS	RESET	NAME	MSB							LSB
0x1506	0x80	RLMS6[7:0]	EOM_PV_MODE	EOM_RST_THR[6:0]						
0x1507	0x00	RLMS7[7:0]	EOM_DONE	EOM[6:0]						
SHA256_A										
0x1E00	0x00	SHA_1[7:0]	SHA_A_1[7:0]							
0x1E01	0x00	SHA_2[7:0]	SHA_A_2[7:0]							
0x1E02	0x00	SHA_3[7:0]	SHA_A_3[7:0]							
0x1E03	0x00	SHA_4[7:0]	SHA_A_4[7:0]							
0x1E04	0x00	SHA_5[7:0]	SHA_A_5[7:0]							
0x1E05	0x00	SHA_6[7:0]	SHA_A_6[7:0]							
0x1E06	0x00	SHA_7[7:0]	SHA_A_7[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E07	0x00	SHA_8[7:0]	SHA_A_8[7:0]							
0x1E08	0x00	SHA_9[7:0]	SHA_A_9[7:0]							
0x1E09	0x00	SHA_10[7:0]	SHA_A_10[7:0]							
0x1E0A	0x00	SHA_11[7:0]	SHA_A_11[7:0]							
0x1E0B	0x00	SHA_12[7:0]	SHA_A_12[7:0]							
0x1E0C	0x00	SHA_13[7:0]	SHA_A_13[7:0]							
0x1E0D	0x00	SHA_14[7:0]	SHA_A_14[7:0]							
0x1E0E	0x00	SHA_15[7:0]	SHA_A_15[7:0]							
0x1E0F	0x00	SHA_16[7:0]	SHA_A_16[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E10	0x00	SHA_17[7:0]	SHA_A_17[7:0]							
0x1E11	0x00	SHA_18[7:0]	SHA_A_18[7:0]							
0x1E12	0x00	SHA_19[7:0]	SHA_A_19[7:0]							
0x1E13	0x00	SHA_20[7:0]	SHA_A_20[7:0]							
0x1E14	0x00	SHA_21[7:0]	SHA_A_21[7:0]							
0x1E15	0x00	SHA_22[7:0]	SHA_A_22[7:0]							
0x1E16	0x00	SHA_23[7:0]	SHA_A_23[7:0]							
0x1E17	0x00	SHA_24[7:0]	SHA_A_24[7:0]							
0x1E18	0x00	SHA_25[7:0]	SHA_A_25[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E19	0x00	SHA_26[7:0]	SHA_A_26[7:0]							
0x1E1A	0x00	SHA_27[7:0]	SHA_A_27[7:0]							
0x1E1B	0x00	SHA_28[7:0]	SHA_A_28[7:0]							
0x1E1C	0x00	SHA_29[7:0]	SHA_A_29[7:0]							
0x1E1D	0x00	SHA_30[7:0]	SHA_A_30[7:0]							
0x1E1E	0x00	SHA_31[7:0]	SHA_A_31[7:0]							
0x1E1F	0x00	SHA_32[7:0]	SHA_A_32[7:0]							
SHA256_B										
0x1E20	0x00	SHA_1[7:0]	SHA_B_1[7:0]							
0x1E21	0x00	SHA_2[7:0]	SHA_B_2[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E22	0x00	SHA_3[7:0]	SHA_B_3[7:0]							
0x1E23	0x00	SHA_4[7:0]	SHA_B_4[7:0]							
0x1E24	0x00	SHA_5[7:0]	SHA_B_5[7:0]							
0x1E25	0x00	SHA_6[7:0]	SHA_B_6[7:0]							
0x1E26	0x00	SHA_7[7:0]	SHA_B_7[7:0]							
0x1E27	0x00	SHA_8[7:0]	SHA_B_8[7:0]							
0x1E28	0x00	SHA_9[7:0]	SHA_B_9[7:0]							
0x1E29	0x00	SHA_10[7:0]	SHA_B_10[7:0]							
0x1E2A	0x00	SHA_11[7:0]	SHA_B_11[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E2B	0x00	SHA_12[7:0]	SHA_B_12[7:0]							
0x1E2C	0x00	SHA_13[7:0]	SHA_B_13[7:0]							
0x1E2D	0x00	SHA_14[7:0]	SHA_B_14[7:0]							
0x1E2E	0x00	SHA_15[7:0]	SHA_B_15[7:0]							
0x1E2F	0x00	SHA_16[7:0]	SHA_B_16[7:0]							
0x1E30	0x00	SHA_17[7:0]	SHA_B_17[7:0]							
0x1E31	0x00	SHA_18[7:0]	SHA_B_18[7:0]							
0x1E32	0x00	SHA_19[7:0]	SHA_B_19[7:0]							
0x1E33	0x00	SHA_20[7:0]	SHA_B_20[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E34	0x00	SHA_21[7:0]	SHA_B_21[7:0]							
0x1E35	0x00	SHA_22[7:0]	SHA_B_22[7:0]							
0x1E36	0x00	SHA_23[7:0]	SHA_B_23[7:0]							
0x1E37	0x00	SHA_24[7:0]	SHA_B_24[7:0]							
0x1E38	0x00	SHA_25[7:0]	SHA_B_25[7:0]							
0x1E39	0x00	SHA_26[7:0]	SHA_B_26[7:0]							
0x1E3A	0x00	SHA_27[7:0]	SHA_B_27[7:0]							
0x1E3B	0x00	SHA_28[7:0]	SHA_B_28[7:0]							
0x1E3C	0x00	SHA_29[7:0]	SHA_B_29[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1E3D	0x00	SHA_30[7:0]	SHA_B_30[7:0]							
0x1E3E	0x00	SHA_31[7:0]	SHA_B_31[7:0]							
0x1E3F	0x00	SHA_32[7:0]	SHA_B_32[7:0]							
FUNC_SAFE										
0x1F00	0x00	REGCRC0[7:0]	–	–	–	GEN_ROLLING_CRC	I2C_WR_COMPUTE	PERIODIC_COMPUTE	CHECK_CRC	RESET_CRC
0x1F01	0x00	REGCRC1[7:0]	CRC_PERIOD[7:0]							
0x1F02	0x00	REGCRC2[7:0]	REGCRC_LSB[7:0]							
0x1F03	0x00	REGCRC3[7:0]	REGCRC_MSB[7:0]							
0x1F20	0x00	CC_CRC0[7:0]	CC_CRC_VAL[7:0]							
0x1F21	0x00	CC_CRC1[7:0]	MSGCNTR_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x1F22	0x00	CC_CRC2[7:0]	MSGCNTR_MSB[7:0]							
0x1F23	0x00	CC_CRC3[7:0]	CC_CRC_ERR_CNT[7:0]							
0x1F24	0x00	CC_CRC4[7:0]	MSGCNTR_ERR_CNT[7:0]							
0x1F26	0x00	CC_RTTN_CRC_ERR[7:0]	–	–	–	–	–	–	–	INJECT_RTTN_CRC_ERR
COLOR_A_LUT X										
0x2000	0x00	LUT_A[7:0]	–	–	–	–	–	–	–	–
COLOR_B_LUT X										
0x2100	0x00	LUT_B[7:0]	–	–	–	–	–	–	–	–
COLOR_C_LUT X										
0x2200	0x00	LUT_C[7:0]	–	–	–	–	–	–	–	–
TELLTALE_DETECTOR_GEN_CSR										
0x3000	0x	TELLTALE_CTRL_REG_0[7:0]	–	–	–	TELLTALE_FORCE_BANK_B	TELLTALE_FORCE_BANK_A	TELLTALE_PERIODIC_MODE	TELLTALE_TRIGGER_MODE	TELLTALE_DET_EN

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ADDRESS	RESET	NAME	MSB							LSB
	04									
0x3001	0xFF	TELLTALE_CTRL_REG_1[7:0]	BANK_B_ROI_INTERRUPT_EN[3:0]				BANK_A_ROI_INTERRUPT_EN[3:0]			
0x3002	0x00	TELLTALE_STATUS_REG_0[7:0]	BANK_B_ROI_ERROR[3:0]				BANK_A_ROI_ERROR[3:0]			
0x3003	0x00	TELLTALE_STATUS_REG_1[7:0]	TELLTALE_ACTIVE	TELLTALE_ACTIVE_BANK_A_OR_B	–	–	–	–	–	–
REFERENCE_RED_COLOR 0										
0x3008	0x82	REF_COLOR_GENERAL[7:0]	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
0x3009	0x4A	REF_COLOR_HUE_MAX_L[7:0]	REF_COLOR_HUE_MAX_L[7:0]							
0x300A	0x0A	REF_COLOR_HUE_MIN[7:0]	REF_COLOR_HUE_MIN_L[7:0]							
0x300B	0x64	REF_COLOR_SAT_MAX[7:0]	–	REF_COLOR_SAT_MAX[6:0]						
0x300C	0x28	REF_COLOR_SAT_MIN[7:0]	–	REF_COLOR_SAT_MIN[6:0]						

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ADDRESS	RESET	NAME	MSB							LSB
0x300D	0x4B	REF_COLOR_LUM_MAX[7:0]	–	REF_COLOR_LUM_MAX[6:0]						
0x300E	0x19	REF_COLOR_LUM_MIN[7:0]	–	REF_COLOR_LUM_MIN[6:0]						
REFERENCE_ORANGE_COLOR 1										
0x3010	0x00	REF_COLOR_GENERAL[7:0]	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
0x3011	0x33	REF_COLOR_HUE_MAX_L[7:0]	REF_COLOR_HUE_MAX_L[7:0]							
0x3012	0x0E	REF_COLOR_HUE_MIN[7:0]	REF_COLOR_HUE_MIN_L[7:0]							
0x3013	0x64	REF_COLOR_SAT_MAX[7:0]	–	REF_COLOR_SAT_MAX[6:0]						
0x3014	0x28	REF_COLOR_SAT_MIN[7:0]	–	REF_COLOR_SAT_MIN[6:0]						
0x3015	0x4B	REF_COLOR_LUM_MAX[7:0]	–	REF_COLOR_LUM_MAX[6:0]						
0x3016	0x19	REF_COLOR_LUM_MIN[7:0]	–	REF_COLOR_LUM_MIN[6:0]						

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ADDRESS	RESET	NAME	MSB							LSB
REFERENCE_GREEN_COLOR 2										
0x3018	0x00	REF_COLOR_GENERAL[7:0]	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
0x3019	0xA7	REF_COLOR_HUE_MAX_L[7:0]	REF_COLOR_HUE_MAX_L[7:0]							
0x301A	0x57	REF_COLOR_HUE_MIN[7:0]	REF_COLOR_HUE_MIN_L[7:0]							
0x301B	0x64	REF_COLOR_SAT_MAX[7:0]	–	REF_COLOR_SAT_MAX[6:0]						
0x301C	0x28	REF_COLOR_SAT_MIN[7:0]	–	REF_COLOR_SAT_MIN[6:0]						
0x301D	0x4B	REF_COLOR_LUM_MAX[7:0]	–	REF_COLOR_LUM_MAX[6:0]						
0x301E	0x19	REF_COLOR_LUM_MIN[7:0]	–	REF_COLOR_LUM_MIN[6:0]						
REFERENCE_BLUE_COLOR 3										
0x3020	0x02	REF_COLOR_GENERAL[7:0]	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H

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ADDRESS	RESET	NAME	MSB							LSB
0x3021	0x0B	REF_COLOR_HUE_MAX_L[7:0]								
0x3022	0xBC	REF_COLOR_HUE_MIN[7:0]								
0x3023	0x64	REF_COLOR_SAT_MAX[7:0]	–							
0x3024	0x28	REF_COLOR_SAT_MIN[7:0]	–							
0x3025	0x4B	REF_COLOR_LUM_MAX[7:0]	–							
0x3026	0x19	REF_COLOR_LUM_MIN[7:0]	–							
REFERENCE_WHITE_COLOR 4										
0x3028	0x02	REF_COLOR_GENERAL[7:0]	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
0x3029	0x68	REF_COLOR_HUE_MAX_L[7:0]								
0x302A	0x00	REF_COLOR_HUE_MIN[7:0]								

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ADDRESS	RESET	NAME	MSB							LSB
0x302B	0x64	REF_COLOR_SAT_MAX[7:0]	–	REF_COLOR_SAT_MAX[6:0]						
0x302C	0x00	REF_COLOR_SAT_MIN[7:0]	–	REF_COLOR_SAT_MIN[6:0]						
0x302D	0x64	REF_COLOR_LUM_MAX[7:0]	–	REF_COLOR_LUM_MAX[6:0]						
0x302E	0x5F	REF_COLOR_LUM_MIN[7:0]	–	REF_COLOR_LUM_MIN[6:0]						
REFERENCE_GRAY_COLOR 5										
0x3030	0x02	REF_COLOR_GENERAL[7:0]	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
0x3031	0x68	REF_COLOR_HUE_MAX_L[7:0]								
0x3032	0x00	REF_COLOR_HUE_MIN_L[7:0]								
0x3033	0x19	REF_COLOR_SAT_MAX[7:0]	–	REF_COLOR_SAT_MAX[6:0]						
0x3034	0x00	REF_COLOR_SAT_MIN[7:0]	–	REF_COLOR_SAT_MIN[6:0]						

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ADDRESS	RESET	NAME	MSB							LSB
0x3035	0x4B	REF_COLOR_LUM_MAX[7:0]	–	REF_COLOR_LUM_MAX[6:0]						
0x3036	0x19	REF_COLOR_LUM_MIN[7:0]	–	REF_COLOR_LUM_MIN[6:0]						
TELLTALE_BANK_A										
0x3038	0x00	BANK_A_STATUS[7:0]	–	–	–	–	–	BANK_A_CALC_DONE	BANK_A_FRAME_COUNT[1:0]	
0x3039	0x00	BANK_A_ROI_1_0_ERROR[7:0]	–	BANK_A_ROI_1_COLOR_2_ERROR	BANK_A_ROI_1_COLOR_1_ERROR	BANK_A_ROI_1_COLOR_0_ERROR	–	BANK_A_ROI_0_COLOR_2_ERROR	BANK_A_ROI_0_COLOR_1_ERROR	BANK_A_ROI_0_COLOR_0_ERROR
0x303A	0x00	BANK_A_ROI_3_2_ERROR[7:0]	–	BANK_A_ROI_3_COLOR_2_ERROR	BANK_A_ROI_3_COLOR_1_ERROR	BANK_A_ROI_3_COLOR_0_ERROR	–	BANK_A_ROI_2_COLOR_2_ERROR	BANK_A_ROI_2_COLOR_1_ERROR	BANK_A_ROI_2_COLOR_0_ERROR
BANK_A_ROI_COORDINATES 0										
0x3040	0x00	BANK_A_ROI0_XMIN_LSB[7:0]	BANK_A_ROI0_XMIN_LSB[7:0]							
0x3041	0x00	BANK_A_ROI0_XMIN_MSB[7:0]	–	–	BANK_A_ROI0_XMIN_MSB[5:0]					
0x3042	0x00	BANK_A_ROI0_XMAX_LSB[7:0]	BANK_A_ROI0_XMAX_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x3043	0x00	BANK_A_ROI0_XMAX_MSB[7:0]	–	–	BANK_A_ROI0_XMAX_MSB[5:0]					
0x3044	0x00	BANK_A_ROI0_YMIN_LSB[7:0]	BANK_A_ROI0_YMIN_LSB[7:0]							
0x3045	0x00	BANK_A_ROI0_YMIN_MSB[7:0]	–	–	BANK_A_ROI0_YMIN_MSB[5:0]					
0x3046	0x00	BANK_A_ROI0_YMAX_LSB[7:0]	BANK_A_ROI0_YMAX_LSB[7:0]							
0x3047	0x00	BANK_A_ROI0_YMAX_MSB[7:0]	–	–	BANK_A_ROI0_YMAX_MSB[5:0]					
BANK_A_ROI_0_COLOR 0										
0x3048	0x00	BANK_A_ROI0_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI0_COLOR_SELECT[2:0]		
0x3049	0x00	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]							
0x304A	0x00	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]							
0x304B	0xFF	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x304C	0xFF	BANK_A_ROI0_COLOR_MAX_THR_MSB[7:0]								
0x304D	0x00	BANK_A_ROI0_COLOR_COUNT_LSB[7:0]								
0x304E	0x00	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]								
BANK_A_ROI_0_COLOR 1										
0x3050	0x00	BANK_A_ROI0_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI0_COLOR_SELECT[2:0]		
0x3051	0x00	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]								
0x3052	0x00	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]								
0x3053	0xFF	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]								
0x3054	0xFF	BANK_A_ROI0_COLOR_MAX_THR_MSB[7:0]								
0x3055	0x00	BANK_A_ROI0_COLOR_COUNT_LSB[7:0]								

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ADDRESS	RESET	NAME	MSB							LSB
0x3056	0x00	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_0_COLOR 2										
0x3058	0x00	BANK_A_ROI0_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI0_COLOR_SELECT[2:0]		
0x3059	0x00	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]							
0x305A	0x00	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]							
0x305B	0xFF	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]							
0x305C	0xFF	BANK_A_ROI0_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI0_COLOR_MAX_THR_MSB[7:0]							
0x305D	0x00	BANK_A_ROI0_COLOR_COUNT_LSB[7:0]	BANK_A_ROI0_COLOR_COUNT_LSB[7:0]							
0x305E	0x00	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_COORDINATES 1										

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ADDRESS	RESET	NAME	MSB							LSB
0x3060	0x00	BANK_A_ROI1_XMIN_LSB[7:0]	BANK_A_ROI1_XMIN_LSB[7:0]							
0x3061	0x00	BANK_A_ROI1_XMIN_MSB[7:0]	–	–	BANK_A_ROI1_XMIN_MSB[5:0]					
0x3062	0x00	BANK_A_ROI1_XMAX_LSB[7:0]	BANK_A_ROI1_XMAX_LSB[7:0]							
0x3063	0x00	BANK_A_ROI1_XMAX_MSB[7:0]	–	–	BANK_A_ROI1_XMAX_MSB[5:0]					
0x3064	0x00	BANK_A_ROI1_YMIN_LSB[7:0]	BANK_A_ROI1_YMIN_LSB[7:0]							
0x3065	0x00	BANK_A_ROI1_YMIN_MSB[7:0]	–	–	BANK_A_ROI1_YMIN_MSB[5:0]					
0x3066	0x00	BANK_A_ROI1_YMAX_LSB[7:0]	BANK_A_ROI1_YMAX_LSB[7:0]							
0x3067	0x00	BANK_A_ROI1_YMAX_MSB[7:0]	–	–	BANK_A_ROI1_YMAX_MSB[5:0]					
BANK_A_ROI_1_COLOR 0										
0x3068	0x00	BANK_A_ROI1_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI1_COLOR_SELECT[2:0]		

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ADDRESS	RESET	NAME	MSB							LSB
0x3069	0x00	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]							
0x306A	0x00	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]							
0x306B	0xFF	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]							
0x306C	0xFF	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]							
0x306D	0x00	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]							
0x306E	0x00	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_1_COLOR 1										
0x3070	0x00	BANK_A_ROI1_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI1_COLOR_SELECT[2:0]		
0x3071	0x00	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]							
0x3072	0x00	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x3073	0xFF	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]							
0x3074	0xFF	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]							
0x3075	0x00	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]							
0x3076	0x00	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_1_COLOR 2										
0x3078	0x00	BANK_A_ROI1_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI1_COLOR_SELECT[2:0]		
0x3079	0x00	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]							
0x307A	0x00	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]							
0x307B	0xFF	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]							
0x307C	0xFF	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x307D	0x00	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]							
0x307E	0x00	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_COORDINATES 2										
0x3080	0x00	BANK_A_ROI2_XMIN_LSB[7:0]	BANK_A_ROI2_XMIN_LSB[7:0]							
0x3081	0x00	BANK_A_ROI2_XMIN_MSB[7:0]	–	–	BANK_A_ROI2_XMIN_MSB[5:0]					
0x3082	0x00	BANK_A_ROI2_XMAX_LSB[7:0]	BANK_A_ROI2_XMAX_LSB[7:0]							
0x3083	0x00	BANK_A_ROI2_XMAX_MSB[7:0]	–	–	BANK_A_ROI2_XMAX_MSB[5:0]					
0x3084	0x00	BANK_A_ROI2_YMIN_LSB[7:0]	BANK_A_ROI2_YMIN_LSB[7:0]							
0x3085	0x00	BANK_A_ROI2_YMIN_MSB[7:0]	–	–	BANK_A_ROI2_YMIN_MSB[5:0]					
0x3086	0x00	BANK_A_ROI2_YMAX_LSB[7:0]	BANK_A_ROI2_YMAX_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
BANK_A_ROI_2_COLOR 0										
0x3088	0x00	BANK_A_ROI2_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI2_COLOR_SELECT[2:0]		
0x3089	0x00	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]							
0x308A	0x00	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]							
0x308B	0xFF	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]							
0x308C	0xFF	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]							
0x308D	0x00	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]							
0x308E	0x00	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_2_COLOR 1										
0x3090	0x00	BANK_A_ROI2_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI2_COLOR_SELECT[2:0]		

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ADDRESS	RESET	NAME	MSB							LSB
0x3091	0x00	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]							
0x3092	0x00	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]							
0x3093	0xFF	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]							
0x3094	0xFF	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]							
0x3095	0x00	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]							
0x3096	0x00	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_2_COLOR 2										
0x3098	0x00	BANK_A_ROI2_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI2_COLOR_SELECT[2:0]		
0x3099	0x00	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]							
0x309A	0x00	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x309B	0xFF	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]							
0x309C	0xFF	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]							
0x309D	0x00	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]							
0x309E	0x00	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_COORDINATES 3										
0x30A0	0x00	BANK_A_ROI3_XMIN_LSB[7:0]	BANK_A_ROI3_XMIN_LSB[7:0]							
0x30A1	0x00	BANK_A_ROI3_XMIN_MSB[7:0]	–	–	BANK_A_ROI3_XMIN_MSB[5:0]					
0x30A2	0x00	BANK_A_ROI3_XMAX_LSB[7:0]	BANK_A_ROI3_XMAX_LSB[7:0]							
0x30A3	0x00	BANK_A_ROI3_XMAX_MSB[7:0]	–	–	BANK_A_ROI3_XMAX_MSB[5:0]					
0x30A4	0x00	BANK_A_ROI3_YMIN_LSB[7:0]	BANK_A_ROI3_YMIN_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x30A5	0x00	BANK_A_ROI3_YMIN_MSB[7:0]	–	–	BANK_A_ROI3_YMIN_MSB[5:0]					
0x30A6	0x00	BANK_A_ROI3_YMAX_LSB[7:0]	BANK_A_ROI3_YMAX_LSB[7:0]							
0x30A7	0x00	BANK_A_ROI3_YMAX_MSB[7:0]	–	–	BANK_A_ROI3_YMAX_MSB[5:0]					
BANK_A_ROI_3_COLOR 0										
0x30A8	0x00	BANK_A_ROI3_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI3_COLOR_SELECT[2:0]		
0x30A9	0x00	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]							
0x30AA	0x00	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]							
0x30AB	0xFF	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]							
0x30AC	0xFF	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]							
0x30AD	0x00	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x30AE	0x00	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_3_COLOR 1										
0x30B0	0x00	BANK_A_ROI3_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI3_COLOR_SELECT[2:0]		
0x30B1	0x00	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]							
0x30B2	0x00	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]							
0x30B3	0xFF	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]							
0x30B4	0xFF	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]							
0x30B5	0x00	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]							
0x30B6	0x00	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]							
BANK_A_ROI_3_COLOR 2										

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ADDRESS	RESET	NAME	MSB							LSB
0x30B8	0x00	BANK_A_ROI3_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_A_ROI3_COLOR_SELECT[2:0]		
0x30B9	0x00	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]							
0x30BA	0x00	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]							
0x30BB	0xFF	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]							
0x30BC	0xFF	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]							
0x30BD	0x00	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]							
0x30BE	0x00	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]							
TELLTALE_BANK_B										
0x30C0	0x00	BANK_B_STATUS[7:0]	–	–	–	–	–	BANK_B_CALC_DONE	BANK_B_FRAME_COUNT[1:0]	
0x30C1	0x00	BANK_B_ROI_1_0_ERROR[7:0]	–	BANK_B_ROI_1_COLOR_2_ERROR	BANK_B_ROI_1_COLOR_1_ERROR	BANK_B_ROI_1_COLOR_0_ERROR	–	BANK_B_ROI_0_COLOR_2_ERROR	BANK_B_ROI_0_COLOR_1_ERROR	BANK_B_ROI_0_COLOR_0_ERROR

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ADDRESS	RESET	NAME	MSB							LSB
0x30C2	0x00	BANK_B_ROI_3_2_ERROR[7:0]	–	BANK_B_ROI_3_COLOR_2_ERROR	BANK_B_ROI_3_COLOR_1_ERROR	BANK_B_ROI_3_COLOR_0_ERROR	–	BANK_B_ROI_2_COLOR_2_ERROR	BANK_B_ROI_2_COLOR_1_ERROR	BANK_B_ROI_2_COLOR_0_ERROR
BANK_B_ROI_COORDINATES 0										
0x30C8	0x00	BANK_B_ROI0_XMIN_LSB[7:0]	BANK_B_ROI0_XMIN_LSB[7:0]							
0x30C9	0x00	BANK_B_ROI0_XMIN_MSB[7:0]	–	–	BANK_B_ROI0_XMIN_MSB[5:0]					
0x30CA	0x00	BANK_B_ROI0_XMAX_LSB[7:0]	BANK_B_ROI0_XMAX_LSB[7:0]							
0x30CB	0x00	BANK_B_ROI0_XMAX_MSB[7:0]	–	–	BANK_B_ROI0_XMAX_MSB[5:0]					
0x30CC	0x00	BANK_B_ROI0_YMIN_LSB[7:0]	BANK_B_ROI0_YMIN_LSB[7:0]							
0x30CD	0x00	BANK_B_ROI0_YMIN_MSB[7:0]	–	–	BANK_B_ROI0_YMIN_MSB[5:0]					
0x30CE	0x00	BANK_B_ROI0_YMAX_LSB[7:0]	BANK_B_ROI0_YMAX_LSB[7:0]							
0x30CF	0x00	BANK_B_ROI0_YMAX_MSB[7:0]	–	–	BANK_B_ROI0_YMAX_MSB[5:0]					

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ADDRESS	RESET	NAME	MSB							LSB
BANK_B_ROI_0_COLOR 0										
0x30D0	0x00	BANK_B_ROI0_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI0_COLOR_SELECT[2:0]		
0x30D1	0x00	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]							
0x30D2	0x00	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]							
0x30D3	0xFF	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]							
0x30D4	0xFF	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]							
0x30D5	0x00	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]							
0x30D6	0x00	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_0_COLOR 1										
0x30D8	0x00	BANK_B_ROI0_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI0_COLOR_SELECT[2:0]		

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ADDRESS	RESET	NAME	MSB							LSB
0x30D9	0x00	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]							
0x30DA	0x00	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]							
0x30DB	0xFF	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]							
0x30DC	0xFF	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]							
0x30DD	0x00	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]							
0x30DE	0x00	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_0_COLOR 2										
0x30E0	0x00	BANK_B_ROI0_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI0_COLOR_SELECT[2:0]		
0x30E1	0x00	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]							
0x30E2	0x00	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x30E3	0xFF	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]							
0x30E4	0xFF	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]							
0x30E5	0x00	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]							
0x30E6	0x00	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_COORDINATES 1										
0x30E8	0x00	BANK_B_ROI1_XMIN_LSB[7:0]	BANK_B_ROI1_XMIN_LSB[7:0]							
0x30E9	0x00	BANK_B_ROI1_XMIN_MSB[7:0]	–	–	BANK_B_ROI1_XMIN_MSB[5:0]					
0x30EA	0x00	BANK_B_ROI1_XMAX_LSB[7:0]	BANK_B_ROI1_XMAX_LSB[7:0]							
0x30EB	0x00	BANK_B_ROI1_XMAX_MSB[7:0]	–	–	BANK_B_ROI1_XMAX_MSB[5:0]					
0x30EC	0x00	BANK_B_ROI1_YMIN_LSB[7:0]	BANK_B_ROI1_YMIN_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x30ED	0x00	BANK_B_ROI1_YMIN_MSB[7:0]	–	–	BANK_B_ROI1_YMIN_MSB[5:0]					
0x30EE	0x00	BANK_B_ROI1_YMAX_LSB[7:0]	BANK_B_ROI1_YMAX_LSB[7:0]							
0x30EF	0x00	BANK_B_ROI1_YMAX_MSB[7:0]	–	–	BANK_B_ROI1_YMAX_MSB[5:0]					
BANK_B_ROI_1_COLOR 0										
0x30F0	0x00	BANK_B_ROI1_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI1_COLOR_SELECT[2:0]		
0x30F1	0x00	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]							
0x30F2	0x00	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]							
0x30F3	0xFF	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]							
0x30F4	0xFF	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]							
0x30F5	0x00	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x30F6	0x00	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_1_COLOR 1										
0x30F8	0x00	BANK_B_ROI1_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI1_COLOR_SELECT[2:0]		
0x30F9	0x00	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]							
0x30FA	0x00	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]							
0x30FB	0xFF	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]							
0x30FC	0xFF	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]							
0x30FD	0x00	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]							
0x30FE	0x00	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_1_COLOR 2										

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ADDRESS	RESET	NAME	MSB							LSB
0x3100	0x00	BANK_B_ROI1_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI1_COLOR_SELECT[2:0]		
0x3101	0x00	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]							
0x3102	0x00	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]							
0x3103	0xFF	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]							
0x3104	0xFF	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]							
0x3105	0x00	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]							
0x3106	0x00	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_COORDINATES 2										
0x3108	0x00	BANK_B_ROI2_XMIN_LSB[7:0]	BANK_B_ROI2_XMIN_LSB[7:0]							
0x3109	0x00	BANK_B_ROI2_XMIN_MSB[7:0]	–	–	BANK_B_ROI2_XMIN_MSB[5:0]					

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ADDRESS	RESET	NAME	MSB							LSB
0x310A	0x00	BANK_B_ROI2_XMAX_LSB[7:0]	BANK_B_ROI2_XMAX_LSB[7:0]							
0x310B	0x00	BANK_B_ROI2_XMAX_MSB[7:0]	–	–	BANK_B_ROI2_XMAX_MSB[5:0]					
0x310C	0x00	BANK_B_ROI2_YMIN_LSB[7:0]	BANK_B_ROI2_YMIN_LSB[7:0]							
0x310D	0x00	BANK_B_ROI2_YMIN_MSB[7:0]	–	–	BANK_B_ROI2_YMIN_MSB[5:0]					
0x310E	0x00	BANK_B_ROI2_YMAX_LSB[7:0]	BANK_B_ROI2_YMAX_LSB[7:0]							
0x310F	0x00	BANK_B_ROI2_YMAX_MSB[7:0]	–	–	BANK_B_ROI2_YMAX_MSB[5:0]					
BANK_B_ROI_2_COLOR 0										
0x3110	0x00	BANK_B_ROI2_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI2_COLOR_SELECT[2:0]		
0x3111	0x00	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]							
0x3112	0x00	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x3113	0xFF	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]							
0x3114	0xFF	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]							
0x3115	0x00	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]							
0x3116	0x00	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_2_COLOR 1										
0x3118	0x00	BANK_B_ROI2_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI2_COLOR_SELECT[2:0]		
0x3119	0x00	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]							
0x311A	0x00	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]							
0x311B	0xFF	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]							
0x311C	0xFF	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x311D	0x00	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]							
0x311E	0x00	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_2_COLOR 2										
0x3120	0x00	BANK_B_ROI2_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI2_COLOR_SELECT[2:0]		
0x3121	0x00	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]							
0x3122	0x00	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]							
0x3123	0xFF	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]							
0x3124	0xFF	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]							
0x3125	0x00	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]							
0x3126	0x00	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
BANK_B_ROI_COORDINATES 3										
0x3128	0x00	BANK_B_ROI3_XMIN_LSB[7:0]	BANK_B_ROI3_XMIN_LSB[7:0]							
0x3129	0x00	BANK_B_ROI3_XMIN_MSB[7:0]	–	–	BANK_B_ROI3_XMIN_MSB[5:0]					
0x312A	0x00	BANK_B_ROI3_XMAX_LSB[7:0]	BANK_B_ROI3_XMAX_LSB[7:0]							
0x312B	0x00	BANK_B_ROI3_XMAX_MSB[7:0]	–	–	BANK_B_ROI3_XMAX_MSB[5:0]					
0x312C	0x00	BANK_B_ROI3_YMIN_LSB[7:0]	BANK_B_ROI3_YMIN_LSB[7:0]							
0x312D	0x00	BANK_B_ROI3_YMIN_MSB[7:0]	–	–	BANK_B_ROI3_YMIN_MSB[5:0]					
0x312E	0x00	BANK_B_ROI3_YMAX_LSB[7:0]	BANK_B_ROI3_YMAX_LSB[7:0]							
0x312F	0x00	BANK_B_ROI3_YMAX_MSB[7:0]	–	–	BANK_B_ROI3_YMAX_MSB[5:0]					
BANK_B_ROI_3_COLOR 0										

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ADDRESS	RESET	NAME	MSB							LSB
0x3130	0x00	BANK_B_ROI3_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI3_COLOR_SELECT[2:0]		
0x3131	0x00	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]							
0x3132	0x00	BANK_B_ROI3_COLOR_MIN_THR_MSB[7:0]	BANK_B_ROI3_COLOR_MIN_THR_MSB[7:0]							
0x3133	0xFF	BANK_B_ROI3_COLOR_MAX_THR_LSB[7:0]	BANK_B_ROI3_COLOR_MAX_THR_LSB[7:0]							
0x3134	0xFF	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]							
0x3135	0x00	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]							
0x3136	0x00	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]							
BANK_B_ROI_3_COLOR 1										
0x3138	0x00	BANK_B_ROI3_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI3_COLOR_SELECT[2:0]		
0x3139	0x00	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0x313A	0x00	BANK_B_ROI3_COLOR_MIN_THR_MSB[7:0]								
0x313B	0xFF	BANK_B_ROI3_COLOR_MAX_THR_LSB[7:0]								
0x313C	0xFF	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]								
0x313D	0x00	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]								
0x313E	0x00	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]								
BANK_B_ROI_3_COLOR 2										
0x3140	0x00	BANK_B_ROI3_COLOR_SELECT[7:0]	–	–	–	–	–	BANK_B_ROI3_COLOR_SELECT[2:0]		
0x3141	0x00	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]								
0x3142	0x00	BANK_B_ROI3_COLOR_MIN_THR_MSB[7:0]								
0x3143	0xFF	BANK_B_ROI3_COLOR_MAX_THR_LSB[7:0]								

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ADDRESS	RESET	NAME	MSB							LSB
0x3144	0xFF	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]							
0x3145	0x00	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]							
0x3146	0x00	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]							
EDP_TX_LINK										
0x6944	0x00	MAIN_STREAM_MISC0_S0[7:0]	MS_BIT_DEPTH_S0[2:0]			–	–	–	–	–
0x695C	0x00	USER_DATA_COUNT_PER_LANE_S0_B0[7:0]	USER_DATA_COUNT_PER_LANE_S0_L[7:0]							
0x695D	0x00	USER_DATA_COUNT_PER_LANE_S0_B1[7:0]	USER_DATA_COUNT_PER_LANE_S0_M[7:0]							
VTRG_X										
0x7000	0x01	VTRG_CTRL_B0[7:0]	–		–	–	–	–	–	–	vtrg_en
VTRG_Y										
0x7080	0x01	VTRG_CTRL_B0[7:0]	–		–	–	–	–	–	–	vtrg_en

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ADDRESS	RESET	NAME	MSB							LSB
DSC_PPS PMX_RAM										
0xA260	0x00	DEC_PPS03_B0[7:0]	dsc_version_major[3:0]				dsc_version_minor[3:0]			
0xA261	0x00	DEC_PPS03_B1[7:0]	pps_identifier[7:0]							
0xA262	0x00	DEC_PPS03_B2[7:0]	RSVD[7:0]							
0xA263	0x00	DEC_PPS03_B3[7:0]	bits_per_component[3:0]				linebuf_depth[3:0]			
0xA264	0x00	DEC_PPS47_B0[7:0]	–	–	block_pred_enable	convert_rgb	simple_422	vbr_enable	bits_per_pixel_h[1:0]	
0xA265	0x00	DEC_PPS47_B1[7:0]	bits_per_pixel_l[7:0]							
0xA266	0x00	DEC_PPS47_B2[7:0]	pic_height_h[7:0]							
0xA267	0x00	DEC_PPS47_B3[7:0]	pic_height_l[7:0]							
0xA268	0x00	DEC_PPS811_B0[7:0]	pic_width_h[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xA269	0x00	DEC_PPS811_B1[7:0]	pic_width_l[7:0]							
0xA26A	0x00	DEC_PPS811_B2[7:0]	slice_height_h[7:0]							
0xA26B	0x00	DEC_PPS811_B3[7:0]	slice_height_l[7:0]							
0xA26C	0x00	DEC_PPS1215_B0[7:0]	slice_width_h[7:0]							
0xA26D	0x00	DEC_PPS1215_B1[7:0]	slice_width_l[7:0]							
0xA26E	0x00	DEC_PPS1215_B2[7:0]	chunk_size_h[7:0]							
0xA26F	0x00	DEC_PPS1215_B3[7:0]	chunk_size_l[7:0]							
0xA270	0x00	DEC_PPS1619_B0[7:0]	–	–	–	–	–	–	initial_xmit_delay_h[1:0]	
0xA271	0x00	DEC_PPS1619_B1[7:0]	initial_xmit_delay_l[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xA272	0x00	DEC_PPS16_19_B2[7:0]	initial_dec_delay_h[7:0]							
0xA273	0x00	DEC_PPS16_19_B3[7:0]	initial_dec_delay_l[7:0]							
0xA274	0x00	DEC_PPS20_23_B0[7:0]	RSVD[7:0]							
0xA275	0x00	DEC_PPS20_23_B1[7:0]	–	–	initial_scale_value[5:0]					
0xA276	0x00	DEC_PPS20_23_B2[7:0]	scale_increment_interval_h[7:0]							
0xA277	0x00	DEC_PPS20_23_B3[7:0]	scale_increment_interval_l[7:0]							
0xA278	0x00	DEC_PPS24_27_B0[7:0]	–	–	–	–	scale_decrement_interval_h[3:0]			
0xA279	0x00	DEC_PPS24_27_B1[7:0]	scale_decrement_interval_l[7:0]							
0xA27A	0x00	DEC_PPS24_27_B2[7:0]	RSVD[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xA27B	0x00	DEC_PPS2427_B3[7:0]	–	–	–	first_line_bpg_offset[4:0]				
0xA27C	0x00	DEC_PPS2831_B0[7:0]	nfl_bpg_offset_h[7:0]							
0xA27D	0x00	DEC_PPS2831_B1[7:0]	nfl_bpg_offset_l[7:0]							
0xA27E	0x00	DEC_PPS2831_B2[7:0]	slice_bpg_offset_h[7:0]							
0xA27F	0x00	DEC_PPS2831_B3[7:0]	slice_bpg_offset_l[7:0]							
0xA280	0x00	DEC_PPS3235_B0[7:0]	initial_offset_h[7:0]							
0xA281	0x00	DEC_PPS3235_B1[7:0]	initial_offset_l[7:0]							
0xA282	0x00	DEC_PPS3235_B2[7:0]	final_offset_h[7:0]							
0xA283	0x00	DEC_PPS3235_B3[7:0]	final_offset_l[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xA284	0x00	DEC_PPS36_39_B0[7:0]	–	–	–	flatness_min_qp[4:0]				
0xA285	0x00	DEC_PPS36_39_B1[7:0]	–	–	–	flatness_max_qp[4:0]				
0xA286	0x00	DEC_PPS36_39_B2[7:0]	rc_model_size_h[7:0]							
0xA287	0x00	DEC_PPS36_39_B3[7:0]	rc_model_size_l[7:0]							
0xA288	0x00	DEC_PPS40_43_B0[7:0]	–	–	–	–	rc_edge_factor[3:0]			
0xA289	0x00	DEC_PPS40_43_B1[7:0]	–	–	–	rc_quant_incr_limit0[4:0]				
0xA28A	0x00	DEC_PPS40_43_B2[7:0]	–	–	–	rc_quant_incr_limit1[4:0]				
0xA28B	0x00	DEC_PPS40_43_B3[7:0]	rc_tgt_offset_hi[3:0]				rc_tgt_offset_lo[3:0]			
0xA28C	0x00	DEC_PPS44_47_B0[7:0]	rc_buff_thresh0[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xA28D	0x00	DEC_PPS44_47_B1[7:0]	rc_buf_thresh1[7:0]							
0xA28E	0x00	DEC_PPS44_47_B2[7:0]	rc_buf_thresh2[7:0]							
0xA28F	0x00	DEC_PPS44_47_B3[7:0]	rc_buf_thresh3[7:0]							
0xA290	0x00	DEC_PPS48_51_B0[7:0]	rc_buf_thresh4[7:0]							
0xA291	0x00	DEC_PPS48_51_B1[7:0]	rc_buf_thresh5[7:0]							
0xA292	0x00	DEC_PPS48_51_B2[7:0]	rc_buf_thresh6[7:0]							
0xA293	0x00	DEC_PPS48_51_B3[7:0]	rc_buf_thresh7[7:0]							
0xA294	0x00	DEC_PPS52_55_B0[7:0]	rc_buf_thresh8[7:0]							
0xA295	0x00	DEC_PPS52_55_B1[7:0]	rc_buf_thresh9[7:0]							

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ADDRESS	RESET	NAME	MSB							LSB
0xA296	0x00	DEC_PPS52_55_B2[7:0]	rc_buf_thresh10[7:0]							
0xA297	0x00	DEC_PPS52_55_B3[7:0]	rc_buf_thresh11[7:0]							
0xA298	0x00	DEC_PPS56_59_B0[7:0]	rc_buf_thresh12[7:0]							
0xA299	0x00	DEC_PPS56_59_B1[7:0]	rc_buf_thresh13[7:0]							
0xA29A	0x00	DEC_PPS56_59_B2[7:0]	range_min_qp0[4:0]					range_max_qp0_h[2:0]		
0xA29B	0x00	DEC_PPS56_59_B3[7:0]	range_max_qp0_l[1:0]		range_bpg_offset0[5:0]					
0xA29C	0x00	DEC_PPS60_63_B0[7:0]	range_min_qp1[4:0]					range_max_qp1_h[2:0]		
0xA29D	0x00	DEC_PPS60_63_B1[7:0]	range_max_qp1_l[1:0]		range_bpg_offset1[5:0]					
0xA29E	0x00	DEC_PPS60_63_B2[7:0]	range_min_qp2[4:0]					range_max_qp2_h[2:0]		

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ADDRESS	RESET	NAME	MSB							LSB
0xA29F	0x00	DEC_PPS6063_B3[7:0]	range_max_qp2_l[1:0]		range_bpg_offset2[5:0]					
0xA2A0	0x00	DEC_PPS6467_B0[7:0]	range_min_qp3[4:0]					range_max_qp3_h[2:0]		
0xA2A1	0x00	DEC_PPS6467_B1[7:0]	range_max_qp3_l[1:0]		range_bpg_offset3[5:0]					
0xA2A2	0x00	DEC_PPS6467_B2[7:0]	range_min_qp4[4:0]					range_max_qp4_h[2:0]		
0xA2A3	0x00	DEC_PPS6467_B3[7:0]	range_max_qp4_l[1:0]		range_bpg_offset4[5:0]					
0xA2A4	0x00	DEC_PPS6871_B0[7:0]	range_min_qp5[4:0]					range_max_qp5_h[2:0]		
0xA2A5	0x00	DEC_PPS6871_B1[7:0]	range_max_qp5_l[1:0]		range_bpg_offset5[5:0]					
0xA2A6	0x00	DEC_PPS6871_B2[7:0]	range_min_qp6[4:0]					range_max_qp6_h[2:0]		
0xA2A7	0x00	DEC_PPS6871_B3[7:0]	range_max_qp6_l[1:0]		range_bpg_offset6[5:0]					

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ADDRESS	RESET	NAME	MSB							LSB
0xA2A8	0x00	DEC_PPS72_75_B0[7:0]	range_min_qp7[4:0]					range_max_qp7_h[2:0]		
0xA2A9	0x00	DEC_PPS72_75_B1[7:0]	range_max_qp7_l[1:0]		range_bpg_offset7[5:0]					
0xA2AA	0x00	DEC_PPS72_75_B2[7:0]	range_min_qp8[4:0]					range_max_qp8_h[2:0]		
0xA2AB	0x00	DEC_PPS72_75_B3[7:0]	range_max_qp8_l[1:0]		range_bpg_offset8[5:0]					
0xA2AC	0x00	DEC_PPS76_79_B0[7:0]	range_min_qp9[4:0]					range_max_qp9_h[2:0]		
0xA2AD	0x00	DEC_PPS76_79_B1[7:0]	range_max_qp9_l[1:0]		range_bpg_offset9[5:0]					
0xA2AE	0x00	DEC_PPS76_79_B2[7:0]	range_min_qp10[4:0]					range_max_qp10_h[2:0]		
0xA2AF	0x00	DEC_PPS76_79_B3[7:0]	range_max_qp10_l[1:0]		range_bpg_offset10[5:0]					
0xA2B0	0x00	DEC_PPS80_83_B0[7:0]	range_min_qp11[4:0]					range_max_qp11_h[2:0]		

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ADDRESS	RESET	NAME	MSB							LSB
0xA2B1	0x00	DEC_PPS80_83_B1[7:0]	range_max_qp11_l[1:0]		range_bpg_offset11[5:0]					
0xA2B2	0x00	DEC_PPS80_83_B2[7:0]	range_min_qp12[4:0]					range_max_qp12_h[2:0]		
0xA2B3	0x00	DEC_PPS80_83_B3[7:0]	range_max_qp12_l[1:0]		range_bpg_offset12[5:0]					
0xA2B4	0x00	DEC_PPS84_87_B0[7:0]	range_min_qp13[4:0]					range_max_qp13_h[2:0]		
0xA2B5	0x00	DEC_PPS84_87_B1[7:0]	range_max_qp13_l[1:0]		range_bpg_offset13[5:0]					
0xA2B6	0x00	DEC_PPS84_87_B2[7:0]	range_min_qp14[4:0]					range_max_qp14_h[2:0]		
0xA2B7	0x00	DEC_PPS84_87_B3[7:0]	range_max_qp14_l[1:0]		range_bpg_offset14[5:0]					

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Register Details

DEV_CFG_0 (0x0)

BIT	7	6	5	4	3	2	1	0
Field	DEV_ADDR[6:0]							CFG_BLOCK

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Reset	0b1001000	0b0
Access Type	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
DEV_ADDR	7:1	Device Address for I²C or UART addressing Default value is set by the CFG0 pin encoded voltage level as follows: CFG0 Voltage Level — Device Address 000 — 0b1001000 001 — 0b1001010 010 — 0b1001100 011 — 0b1101000 100 — 0b1101010 101 — 0b1101100 110 — 0b0101000 111 — 0b0101010	0b0000000: I²C write/read address is 0x00/0x01 0b0000001: I²C write/read address is 0x02/0x03 ... ... 0b1001000: I²C write/read address is 0x90/0x91 0b1001010: I²C write/read address is 0x94/0x95 0b1001100: I²C write/read address is 0x98/0x99 0b1101000: I²C write/read address is 0xD0/0xD1 0b1101010: I²C write/read address is 0xD4/0xD5 0b1101100: I²C write/read address is 0xD8/0xD9 0b0101000: I²C write/read address is 0x50/0x51 0b0101010: I²C write/read address is 0x54/0x55 ... ... 0b1111111: I²C write/read address is 0xFE/0xFF
CFG_BLOCK	0	Sets all registers as read-only to block further device configuration.	0: Configuration not blocked 1: Configuration blocked (all registers are read-only)

DEV_CFG_1 (0x1)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RESET_ALL	RESET_LINK_ALL	RESET_LINK_SIOAB	RESET_LINK_SIOA	–	RESET_LINK_DCIO	–	–
Reset	0b0	0b0	0b0	0b0	–	0b0	–	–
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	–	Write, Read	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
RESET_ALL	7	Resets all registers to default values and resets all blocks in device. Equivalent to toggling the PWDNB pin. This bit self-clears when written to a 1.	0b0: No action 0b1: Activate chip reset
RESET_LINK_ALL	6	Resets the GMSL links connected to SIOA, SIOB and DCIO. Register settings and other blocks are not reset. When set to 1, this bit holds the link in reset and does not self-clear. Write 1 to activate reset, write 0 to release reset.	0b0: Release link reset or no action 0b1: Hold GMSL links in reset
RESET_LINK_SIOAB	5	Resets the GMSL links connected to SIOA and SIOB. Register settings and other blocks are not reset. When set to 1, this bit holds the link in reset and does not self-clear.	0b0: Release link reset or no actiion 0b1: Hold GMSL links in reset

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BITFIELD	BITS	DESCRIPTION	DECODE
		Write 1 to activate reset, write 0 to release reset.	
RESET_LINK_SIOA	4	Resets the GMSL link connected to SIOA. Register settings and other blocks are not reset. When set to 1, this bit holds the link in reset and does not self-clear. Write 1 to activate reset, write 0 to release reset.	0b0: Release link reset or no action 0b1: Hold GMSL in reset
RESET_LINK_DCIO	2	Resets the GMSL link connected to DCIO. Register settings and other blocks are not reset. When set to 1, this bit holds the link in reset and does not self-clear. Write 1 to activate reset, write 0 to release reset.	0b0: Release link reset or no action 0b1: Hold GMSL link in reset

DEV_CFG_2 (0x2)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RESET_ONESHOT_ALL	RESET_ONESHOT_SIOAB	RESET_ONESHOT_SIOA	RESET_ONESHOT_SIOB	RESET_ONESHOT_DCIO	–	–	–
Reset	0b0	0b0	0b0	0b0	0b0	–	–	–
Access Type	Write Clears All, Read	Write Clears All, Read	Write Clears All, Read	Write Clears All, Read	Write Clears All, Read	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
RESET_ONESHOT_ALL	7	Resets the GMSL links connected to SIOA, SIOB and DCIO (same as RESET_LINK_ALL). Register settings and other blocks are not reset. When set to 1, this bit resets the link and then self-clears to automatically release reset.	0b0: No action 0b1: Reset GMSL links
RESET_ONESHOT_SIOAB	6	Resets the GMSL links connected to SIOA and SIOB (same as RESET_LINK_ALL). Register settings and other blocks are not reset. When set to 1, this bit resets the link and then self-clears to automatically release reset.	0b0: No action 0b1: Reset GMSL links
RESET_ONESHOT_SIOA	5	Resets the GMSL link connected to SIOA (same as RESET_LINK_ALL). Register settings and other blocks are not reset. When set to 1, this bit resets the link and then self-clears to automatically release reset.	0b0: No action 0b1: Reset GMSL link
RESET_ONESHOT_SIOB	4	Resets the GMSL link connected to SIOB (same as RESET_LINK_ALL). Register settings and other blocks are not reset. When set to 1, this bit resets the link and then self-clears to automatically release reset.	0b0: No action 0b1: Reset GMSL link
RESET_ONESHOT_DCIO	3	Resets the GMSL link connected to DCIO (same as RESET_LINK_ALL). Register settings and other blocks are not reset. When set to 1, this bit resets the link and then self-clears to automatically release reset.	0b0: No action 0b1: Reset GMSL link

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ERROR_STATUS_0 (0x4)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	ERR_OUTPUT_STATUS	LINK_SIOA_ERR	LINK_SIOB_ERR	LINK_DCIO_ERR	–	–	–	PSM_STATE_ERR
Reset	0b0	0b0	0b0	0b0	–	–	–	0b0
Access Type	Read Only	Read Only	Read Only	Read Only	–	–	–	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
ERR_OUTPUT_STATUS	7	Overall error status that is output to ERRB pin if enabled (if ERR_OUTPUT_OEN = 1). This status is calculated by ANDing each error status (XXXX_ERR) with its corresponding output enable (XXXX_ERR_OEN), and ORing together all the outcomes (i.e., if any error and its corresponding output enable is high, this value will be high.)	0b0: No overall errors, ERRB pin will be driven high (if ERR_OUTPUT_OEN = 1). 0b1: There is an error, ERRB pin will be driven low (if ERR_OUTPUT_OEN = 1).
LINK_SIOA_ERR	6	Decode error status on received data on SIOA link. Cleared when DEC_ERROR_COUNT_SIOA (0x31) is read or at rising edge of LOCK on SIOA.	0b0: No significant link errors on SIOA 0b1: Link errors exist on SIOA
LINK_SIOB_ERR	5	Decode error status on received data on SIOB link. Cleared when DEC_ERROR_COUNT_SIOB (0x32) is read or at rising edge of LOCK on SIOB.	0b0: No significant link errors on SIOB 0b1: Link errors exist on SIOB
LINK_DCIO_ERR	4	Decode error status on received data on DCIO link. Cleared when DEC_ERROR_COUNT_DCIO (0x33) is read or at rising edge of LOCK on DCIO link.	0b0: No significant link errors on DCIO 0b1: Link errors exist on DCIO
PSM_STATE_ERR	0	PSM (Programmable State Machine) Error Flag Asserted when programmable state machine signals it detected an error or when MBIST of ROM at startup failed. Cleared when PSM_STATE_STATUS_0 (0xB6) is read.	0x0: Error flag not asserted 0x1: Error flag asserted

ERROR_STATUS_1 (0x5)

See Note §.

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BIT	7	6	5	4	3	2	1	0
Field	SIOA_FEC_ERR	SIOB_FEC_ERR	RETENTION_MEMORY_CRC_ERR	CONTROL_CHANNEL_MSG_COUNT_ERR	–	LINEFAULT_0_ERR	LINEFAULT_1_ERR	CLOCK_MONITOR_ERR
Reset	0b0	0b0	0b0	0b0	–	0b0	0b0	0b0
Access Type	Read Clears All	Read Clears All	Read Clears All	Read Clears All	–	Read Only	Read Only	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
SIOA_FEC_ERR	7	FEC receiver error flag for SIOA link. Asserted when FEC receive errors exceed the threshold. Cleared on LOCK rising edge or by setting FEC_STATS_RESET_SIOA (0x108) high.	0b0: Flag not asserted 0b1: Flag asserted
SIOB_FEC_ERR	6	FEC receiver error flag for SIOB link. Asserted when FEC receive errors exceed the threshold. Cleared on LOCK rising edge or by setting FEC_STATS_RESET_SIOB (0x108) high.	0b0: Flag not asserted 0b1: Flag asserted
RETENTION_MEMORY_CRC_ERR	5	CRC check error detected on retention memory during restore operation	0b0: No error detected 0b1: Error detected
CONTROL_CHANNEL_MSG_COUNT_ERR	4	I²C/UART message count error flag. Asserted when MSGCNTR_ERR_CNT is non-zero. Cleared when RESET_MSGCNTR_ERR_CNT (0x11D) is written with 1.	0b0: No error detected 0b1: Error detected
LINEFAULT_0_ERR	2	Line-fault status of wire connected to LMN0 pin. See bitfield LINEFAULT_0_STATUS (0x73) for fault condition.	0b0: No line-fault error 0b1: Line-fault error active
LINEFAULT_1_ERR	1	Line-fault status of wire connected to LMN1 pin. See bitfield LINEFAULT_1_STATUS (0x73) for fault condition.	0b0: No line-fault error 0b1: Line-fault error active
CLOCK_MONITOR_ERR	0	Discrepancy error detected between 25MHz reference clock frequency and 150MHz CMU clock.	0b0: No error detected 0b1: Error detected

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ERROR_STATUS_2 (0x6)

BIT	7	6	5	4	3	2	1	0

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Field	REMOTE_ERR	ARQ_ERR	INFOFRAME_CRC_ERR	VIDEO_PRBS_ERR	AUDIO_PRBS_ERR	OVERVOLTAGE_ERR	UNDERVOLTAGE_ERR	–
Reset	0b0	0b0	0b0	0b0	0b0	0b0	0b0	–
Access Type	Read Only	Read Only	Read Only	Read Only	Read Only	Read Clears All	Read Only	–

BITFIELD	BITS	DESCRIPTION	DECODE
REMOTE_ERR	7	Received Remote GMSL Serializer Error Status (inverse of remote side ERRB pin level) Clear remote GMSL Serializer ERRB pin to clear this error status.	0b0: No remote side error 0b1: Remote side error
ARQ_ERR	6	ARQ retransmissions have exceeded the threshold for 1 of the ARQ-protected channels. See ARQ_STATUS_0 (0x71) register for details on which interface has caused this error. Cleared when MAX_RT_ERROR_RESET (0x19B) is written with 1.	0b0: No ARQ error 0b1: ARQ errors have exceeded the threshhold for at least 1 of the channels
INFOFRAME_CRC_ERR	5	INFOFR CRC Error Flag. Asserted when INFOFR CRC error count > 0. Cleared when CRC_MISC_STATUS_1 (0x6F) is read.	0b0: No CRC errors 0b1: CRC error detected
VIDEO_PRBS_ERR	4	Video PRBS Error Flag Asserted when VPRBS_ERR > 0. Cleared when VPRBS_ERROR_COUNT (0x8D) and VPRBS_ERROR_COUNT_Y (0x55B) are read.	0b0: Flag not asserted 0b1: Flag asserted
AUDIO_PRBS_ERR	3	Audio PRBS Error Flag Asserted when APRBS_ERR (0x143) > 0. Read APRBS_ERROR_COUNT (0x8E) to clear.	0b0: Flag not asserted 0b1: Flag asserted
OVERVOLTAGE_ERR	2	V_{DD} Overvoltage Indication This bit stays high when asserted and must be cleared by the user. It is set when V_{DD} exceeds the overvoltage threshold. It is cleared when POWER_STATUS_0 (0xBA) is read. Read this bitfield to clear. Reported when video is present.	0b0: Flag not asserted 0b1: Flag asserted
UNDERVOLTAGE_ERR	1	Undervoltage condition occurred on V_{DD} rail (V_{DD} < 0.806V). Cleared when POWER_STATUS_0 (0xBA) is read.	0b0: No undervoltage condition 0b1: Undervoltage condition occured on V_{DD} rail

ERROR_STATUS_3 (0x7)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	VIDEO_SEQUENCE_ERR	LOSS_OF_VIDEO_LOCK_ERR	HDCP_ERR	REGISTER_CRC_ERR	DSC_ERR	WATERMARK_ERR	VIDEO_BLOCK_LENGTH_ERR	LOSS_OF_VIDEO_ALIGNMENT_ERR

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Reset	0b0	0b0	0b0	0x0	0b0	0b0	0b0	0b0
Access Type	Read Only	Read Only	Read Only	Read Clears All	Read Only	Read Only	Read Clears All	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
VIDEO_SEQUENCE_ERR	7	Sequence error detected on received video packets. Cleared when VIDEO_STATUS_0 (0x92) and VIDEO_STATUS_1 (0x93) are read.	0b0: Error not detected 0b1: Error detected
LOSS_OF_VIDEO_LOCK_ERR	6	Loss of video error is detected. Cleared when VIDEO_STATUS_0 (0x92) is read.	0b0: No error detected 0b1: Error detected
HDCP_ERR	5	HDCP error detected. Triggered by video line CRC errors on three consecutive lines. Cleared when this ERROR_STATUS_3 is read.	0b0: No HDCP interrupt 0b1: HDCP interrupt
REGISTER_CRC_ERR	4	CRC check error detected on register block. Cleared when register RESET_CRC (0x1F00) is written with 1.	0b0: No error detected 0b1: Error detected
DSC_ERR	3	DSC Decoder Error Flag Asserted when any DSC decoder error occurs. This flag is set by assertion of any bit in DSC_STATUS. This flag is cleared when DSC_STATUS_0 (0x8A) is read.	0b0: Flag not asserted 0b1: Flag asserted
WATERMARK_ERR	2	Watermark Error Flag Asserted when a watermark error is detected. Read ERROR_STATUS_3 to clear.	0b0: Flag not asserted 0b1: Flag asserted
VIDEO_BLOCK_LENGTH_ERR	1	Combined Video Rx Block Length Error Detection If the received video packet length is shorter than expected, this interrupt is asserted. Read VIDEO_STATUS_0 (0x92) and VIDEO_STATUS_1 (0x93) to clear.	0b0: No error detected 0b1: Video Rx block length error detected
LOSS_OF_VIDEO_ALIGNMENT_ERR	0	Error detected due to loss of external video alignment between two different chips at the video align block. Cleared on read.	0x0: No error detected 0x1: Error detected

ERROR_STATUS_4 (0x8)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_ERR	FEEDFORWARD_CRC_ERR	FEEDFORWARD_COMBINED_FRAME_CRC_ERR	LINE_CRC_ERR	VIDEO_PACKET_CRC_ERR	_	DP_SINK_SYMBOL_ERR	_

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Reset	0b0	0b0	0x0	0b0	0b0	–	0x0	–
Access Type	Read Only	Read Only	Read Clears All	Read Only	Read Only	–	Read Clears All	–

BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CRC_ERR	7	Video Frame CRC Error Detected Asserted when a video frame CRC error is detected. This feature is enabled when the video region CRC block is programmed and enabled. Cleared when CRC_MISC_STATUS_0 (0x6E) is read. Also cleared by rising edge of video lock.	0b0: No frame CRC error detected 0b1: Frame CRC errors occurred
FEEDFORWARD_CRC_ERR	6	Feedforward Frame CRC Error Detected Cleared when CRC_MISC_STATUS_0 (0x6E) is read.	0b0: No error 0b1: Error
FEEDFORWARD_COMBINED_FRAME_CRC_ERR	5	CRC check error detected on feedforward CRC of combined X and Y frames. Cleared when CRC_MISC_STATUS_0 (0x6E) is read.	0b0: No CRC error detected 0b1: CRC error detected
LINE_CRC_ERR	4	Video Line CRC Error Flag. Read to clear. Asserted when a video line CRC error is detected. Cleared when CRC_MISC_STATUS_1 (0x6F) is read or at rising edge of video lock.	0b0: Flag not asserted 0b1: Flag asserted
VIDEO_PACKET_CRC_ERR	3	Video Pixel CRC Error Flag Asserted when video CRC error count > 0. Read CRC_MISC_STATUS_1 (0x6F) to clear. Also cleared by rising edge of video lock.	0b0: Flag not asserted 0b1: Flag asserted
DP_SINK_SYMBOL_ERR	1	Status indicating DP sink has detected symbol error. Cleared by writing 0.	0x0: No error detected 0x1: Error detected

ERROR_STATUS_5 (0x9)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	DP_INPUT_FIFO_CRC_ERR	WM_BYPASS_ERR	LUT_FRC_ERR	DPTX_FIFO_OVERFLOW_ERR	TELLTALE_ERR	LOSS_OF_LOCK_ERR	_	CONTROL_CHANNEL_CRC_ERR

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Reset	0b0	0b0	0b0	0b0	0b0	0b0	–	0b0
Access Type	Read Clears All	Read Clears All	Read Clears All	Read Clears All	Read Clears All	Read Clears All	–	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
DP_INPUT_FIFO_CRC_ERR	7	Video align block detected CRC error on its FIFO. Cleared when this ERROR_STATUS_5 is read.	0b0: No CRC error detected 0b1: CRC error detected
WM_BYPASS_ERR	6	Error detected on watermark redundant data path. Cleared when CRC_MISC_STATUS_0 (0x6E) is read.	0b0: Error not detected 0b1: Error detected
LUT_FRC_ERR	5	Error detected in LUT and FRC ASIL redundancy check. Cleared when CRC_MISC_STATUS_0 (0x6E) is read.	0b0: Error not detected 0b1: Error detected
DPTX_FIFO_OVERFLOW_ERR	4	DPTX controller detected overflow in its FIFO. Cleared when this ERROR_STATUS_5 is read.	0b0: FIFO overflow detected 0b1: No FIFO overflow detected
TELLTALE_ERR	3	Telltale color error detected. Cleared when TELLTALE_STATUS_0 (0x6D) is read.	0b0: No error detected 0b1: Error detected
LOSS_OF_LOCK_ERR	2	Loss of GMSL link lock detected. Cleared when this ERROR_STATUS_5 is read.	0b0: Loss of lock detected on GMSL link(s) 0b1: No loss of lock detected on GMSL link(s)
CONTROL_CHANNEL_CRC_ERR	0	Control Channel CRC error detected. Cleared when RESET_CC_CRC_ERR_CNT (0x11D) is written with 1.	0b0: No CRC check error detected 0b1: CRC check error detected

ERROR_STATUS_6 (0xA)

BIT	7	6	5	4	3	2	1	0
Field	OTP_CRC_ERR	–	VIDEO_CFG_ERR	DP_LINK_TRAINING_ERR	DP_SINK_CAPABILITY_ERR	AUX_ERR	HPD_IRQ_ERR	HPD_DISCONNECT_ERR
Reset	0x0	–	0x0	0b0	0x0	0x0	0x0	0x0
Access Type	Read Clears All	–	Read Clears All	Read Clears All	Read Clears All	Read Clears All	Read Clears All	Read Clears All

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BITFIELD	BITS	DESCRIPTION	DECODE
OTP_CRC_ERR	7	OTP CRC check error detected by PMX control state machine. Cleared when ERROR_STATUS_6 is read.	0b0: No error detected 0b1: Error detected
VIDEO_CFG_ERR	5	PSM Control State detected invalid video configuration. Cleared when ERROR_STATUS_6 is read.	0b0: No invalid video configuration detected 0b1: Invalid video configuration detected
DP_LINK_TRAINING_ERR	4	Error occurred during DP link training. See DP_STATUS_0 (0x75) and DP_STATUS_1 (0x76) registers for details. Cleared when ERROR_STATUS_6 is read.	0b0: Flag not asserted 0b1: Flag asserted
DP_SINK_CAPABILITY_ERR	3	Incompatible DP sink capability detected. Cleared when ERROR_STATUS_6 is read.	0b0: No error detected 0b1: Error detected
AUX_ERR	2	Error detected on Auxiliary channel communication with DP sink. Cleared when ERROR_STATUS_6 is read.	0b0: No error detected 0b1: Error detected
HPD_IRQ_ERR	1	Hot-plug interrupt from external DP sink detected. Cleared when ERROR_STATUS_6 is read.	0b0: No hot-plug interrupt detected 0b1: Hot-plug interrupt detected
HPD_DISCONNECT_ERR	0	Hot-plug disconnect error detected. Cleared when ERROR_STATUS_6 is read.	0b0: No error detected 0b1: Error detected

DEV_INFO_0 (0xD)

BIT	7	6	5	4	3	2	1	0
Field	DEV_ID[7:0]							
Reset	0xFF							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
DEV_ID	7:0	Use DEV_ID16 for device identifier.	0xFF: Fixed value

DEV_INFO_1 (0xE)

BIT	7	6	5	4	3	2	1	0
Field	DEV_REV[3:0]				–	–	–	–
Reset	0x2				–	–	–	–
Access Type	Read Only				–	–	–	–

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BITFIELD	BITS	DESCRIPTION	DECODE
DEV_REV	7:4	Device Revision	0xX: Revision number

DEV_INFO_2 (0xF)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	SPEED_CPBL[1:0]		SGMII_CPBL	DSC_CPBL	SIOB_CPBL	DAISYCHAIN_CPBL	GMSL_12G_CPBL	HDCP_CPBL
Reset	0b00		0b1	0b1	0b1	0b1	0b1	0b1
Access Type	Read Only		Read Only	Read Only	Read Only	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
SPEED_CPBL	7:6	Video Resolution PCLK Rate Capability	0b00: No PCLK frequency limit 0b01: Reserved 0b10: Reserved 0b11: Reserved
SGMII_CPBL	5	SGMII Capable	0b0: Not capable of SGMII operation 0b1: Capable of SGMII operation
DSC_CPBL	4	DSC Decoder Capable	0b0: Not capable of DSC decode 0b1: Capable of DSC decode
SIOB_CPBL	3	Dual GSML Link Capable	0b0: Not capable of Dual-link operation 0b1: Capable of Dual-link operation
DAISYCHAIN_CPBL	2	Daisy-Chain Capable	0b0: Not capable of daisy chain 0b1: Capable of daisy chain
GMSL_12G_CPBL	1	GMSL3 12G Rate Capable	0b0: Not capable of GMSL3 12G operation 0b1: Capable of GMSL3 12G operation
HDCP_CPBL	0	HDCP Capable	0b0: Not capable of HDCP 0b1: Capable of HDCP

DEV_INFO_4 (0x11)

BIT	7	6	5	4	3	2	1	0
Field	OTP_REV[7:0]							
Reset	0x02							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
OTP_REV	7:0	This register shows the revision of the data burned into the OTP. Every time a register default is changed through OTP, this revision is incremented as a notification. Default value seen in the document may be overwritten by OTP.

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DEV_ID16_LSB (0x24)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	DEV_ID16_BYTE_0[7:0]							
Reset	0xC3							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
DEV_ID16_BYTE_0	7:0	16-Bit Device Identifier See DEV_ID16_BYTE_1 for decode information.	0x00: Fixed value

DEV_ID16_MSB (0x25)

BIT	7	6	5	4	3	2	1	0
Field	DEV_ID16_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
DEV_ID16_BYTE_1	7:0	16-Bit Device Identifier	0xC0: MAX96860AS 0xC1: MAX96862RS 0xC4: MAX96746RC 0xC5: MAX96856AC

LINK_STATUS_0 (0x30)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	LOCKED_SIOA	LOCKED_SIOB	LOCKED_DCIO	–	–	FEC_STATUS_SIOA	FEC_STATUS_SIOB	FEC_STATUS_DCIO
Reset	0b0	0b0	0b0	–	–	0b0	0b0	0b0
Access Type	Read Only	Read Only	Read Only	–	–	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
LOCKED_SIOA	7	SIOA GMSL Link Locked This is a bidirectional lock and the LOCKED	0b0: GMSL link not locked 0b1: GMSL link locked

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BITFIELD	BITS	DESCRIPTION	DECODE
		status in the serializer connected to SIOA will reflect the same value.	
LOCKED_SIOB	6	SIOB GMSL Link Locked This is a bidirectional lock and the LOCKED status in the serializer connected to SIOA will reflect the same value.	0b0: GMSL link not locked 0b1: GMSL link locked
LOCKED_DCIO	5	DCIO GMSL Link Locked This is a bidirectional lock and the LOCKED status in the serializer connected to SIOA will reflect the same value.	0b0: GMSL link not locked 0b1: GMSL link locked
FEC_STATUS_SIOA	2	SIOA FEC Status	0b0: FEC not enabled 0b1: FEC enabled and active
FEC_STATUS_SIOB	1	SIOB FEC Status	0b0: FEC not enabled 0b1: FEC enabled and active
FEC_STATUS_DCIO	0	DCIO FEC Status	0b0: FEC not enabled 0b1: FEC enabled and active

LINK_STATUS_1 (0x31)

BIT	7	6	5	4	3	2	1	0
Field	DEC_ERROR_COUNT_SIOA[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
DEC_ERROR_COUNT_SIOA	7:0	Decode errors counted on SIOA forward channel. This field clears on read and when LOCKED_SIOA transitions from low to high (resets on rising edge of link lock).	0xXX: Number of Link SIO decoding errors detected

LINK_STATUS_2 (0x32)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	DEC_ERROR_COUNT_SIOB[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
DEC_ERROR_COUNT_SIOB	7:0	Decode errors counted on SIOB forward channel. This field clears on read and when	0xXX: Number of Link SIO decoding errors detected

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BITFIELD	BITS	DESCRIPTION	DECODE
		LOCKED_SIOB transitions from low to high (resets on rising edge of link lock).	

LINK_STATUS_3 (0x33)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	DEC_ERROR_COUNT_DCIO[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
DEC_ERROR_COUNT_DCIO	7:0	Decode errors counted on DCIO forward channel. This field clears on read and when LOCKED_DCIO transitions from low to high (resets on rising edge of link lock).	0xXX: Number of Link DCIO decoding errors detected

FEC_STATUS_0 (0x37)

BIT	7	6	5	4	3	2	1	0
Field	FEC_UNCORRECTABLE_ERRORS_SIOA[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_UNCORRECTABLE_ERRORS_SIOA	7:0	Number of errors the FEC block cannot correct on SIOA.	0xXX: Number of Link SIO decoding errors detected

FEC_STATUS_1 (0x38)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	FEC_UNCORRECTABLE_ERRORS_SIOB[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION	DECODE
FEC_UNCORRECTABLE_ERRORS_SIOB	7:0	Number of errors the FEC block cannot correct on SIOB.	0xXX: Number of Link SIO decoding errors detected

FEC_STATUS_2 (0x39)

BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOA_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOA_BYTE_0	7:0	Number of blocks processed by the SIOA FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOA, 0x108), divided by 32768. (Byte 0)	0xXX: Count of blocks processed [7:0]

FEC_STATUS_3 (0x3A)

BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOA_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOA_BYTE_1	7:0	Number of blocks processed by the SIOA FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOA, 0x108), divided by 32768. (Byte 1)	0xXX: Count of blocks processed [15:8]

FEC_STATUS_4 (0x3B)

BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOA_BYTE_2[7:0]							
Reset	0x00							

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Access Type	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOA_BYTE_2	7:0	Number of blocks processed by the SIOA FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOA, 0x108), divided by 32768. (Byte 2)	0xXX: Count of blocks processed [23:16]

FEC_STATUS_5 (0x3C)

BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOA_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOA_BYTE_3	7:0	Number of blocks processed by the SIOA FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOA, 0x108), divided by 32768. (Byte 3)	0xXX: Count of blocks processed [31:24]

FEC_STATUS_6 (0x3D)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOB_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOB_BYTE_0	7:0	Number of blocks processed by the SIOB FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOB, 0x108), divided by 32768. (Byte 0)	0xXX: Count of blocks processed [7:0]

FEC_STATUS_7 (0x3E)

See Note §.

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BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOB_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOB_BYTE_1	7:0	Number of blocks processed by the SIOB FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOB, 0x108), divided by 32768. (Byte 1)	0xXX: Count of blocks processed [15:8]

FEC_STATUS_8 (0x3F)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOB_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOB_BYTE_2	7:0	Number of blocks processed by the SIOB FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOB, 0x108), divided by 32768. (Byte 2)	0xXX: Count of blocks processed [23:16]

FEC_STATUS_9 (0x40)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	FEC_BLOCKS_PROCESSED_SIOB_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION	DECODE
FEC_BLOCKS_PROCESSED_SIOB_BYTE_3	7:0	Number of blocks processed by the SIOB FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOB, 0x108), divided by 32768. (Byte 3)	0xXX: Count of blocks processed [31:24]

FEC_STATUS_10 (0x41)

BIT	7	6	5	4	3	2	1	0
Field	FEC_CORRECTED_ERRORS_SIOA_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_CORRECTED_ERRORS_SIOA_BYTE_0	7:0	Number of bit errors corrected in the SIOA FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOA, 0x108). Note that this does not map directly to the number of bit errors on the serial link. (Byte 0)	0xXX: Count of bit errors corrected [7:0]

FEC_STATUS_11 (0x42)

BIT	7	6	5	4	3	2	1	0
Field	FEC_CORRECTED_ERRORS_SIOA_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_CORRECTED_ERRORS_SIOA_BYTE_1	7:0	Number of bit errors corrected in the SIOA FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOA, 0x108). Note that this does not map directly to the number of bit errors on the serial link. (Byte 1)	0xXX: Count of bit errors corrected [15:8]

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FEC_STATUS_12 (0x43)

BIT	7	6	5	4	3	2	1	0
Field	FEC_CORRECTED_ERRORS_SIOA_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_CORRECTED_ERRORS_SIOA_BYTE_2	7:0	Number of bit errors corrected in the SIOA FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOA, 0x108). Note that this does not map directly to the number of bit errors on the serial link. (Byte 2)	0xXX: Count of bit errors corrected [23:16]

FEC_STATUS_13 (0x44)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	FEC_CORRECTED_ERRORS_SIOB_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_CORRECTED_ERRORS_SIOB_BYTE_0	7:0	Number of bit errors corrected in the SIOB FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOB, 0x108). Note that this does not map directly to the number of bit errors on the serial link. (Byte 0)	0xXX: Count of bit errors corrected [7:0]

FEC_STATUS_14 (0x45)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	FEC_CORRECTED_ERRORS_SIOB_BYTE_1[7:0]							
Reset	0x00							

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Access Type	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_CORRECTED_ERRORS_SIOB_BYTE_1	7:0	Number of bit errors corrected in the SIOB FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOB, 0x108). Note that this does not map directly to the number of bit errors on the serial link. (Byte 1)	0xXX: Count of bit errors corrected [15-8]

FEC_STATUS_15 (0x46)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	FEC_CORRECTED_ERRORS_SIOB_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_CORRECTED_ERRORS_SIOB_BYTE_2	7:0	Number of bit errors corrected in the SIOB FEC decoder since the last FEC stats reset (FEC_STATS_RESET_SIOB, 0x108). Note that this does not map directly to the number of bit errors on the serial link. (Byte 2)	0xXX: Count of bit errors corrected [23:16]

CC_STATUS_0 (0x48)

BIT	7	6	5	4	3	2	1	0
Field	UART_TX_OVERFLOW	UART_RX_OVERFLOW	I2C_ACK_RECEIVED	I2C_TIMED_OUT	–	–	–	–
Reset	0b0	0b0	0b0	0b0	–	–	–	–
Access Type	Read Clears All	Read Clears All	Read Only	Read Only	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
UART_TX_OVERFLOW	7	UART Transmit Buffer Overflow	0b0: Buffer not overflowed 0b1: Buffer overflowed
UART_RX_OVERFLOW	6	UART Receive Buffer Overflow	0b0: Buffer not overflowed 0b1: Buffer overflowed

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BITFIELD	BITS	DESCRIPTION	DECODE
I2C_ACK_RECEIVED	5	I²C acknowledge bit for any I²C byte has been received from the remote side for the previous I²C packet.	0b0: I²C acknowledge bit not received 0b1: I²C acknowledge bit received
I2C_TIMED_OUT	4	Internal I²C-to-I²C subordinate or main has timed out while receiving packet from remote device.	0b0: I²C not timed out 0b1: I²C timeout has occurred

PT_STATUS_0 (0x4A)

BIT	7	6	5	4	3	2	1	0
Field	PT1_UART_TX_OVERFLOW	PT1_UART_RX_OVERFLOW	–	–	–	–	–	–
Reset	0b0	0b0	–	–	–	–	–	–
Access Type	Read Clears All	Read Clears All	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
PT1_UART_TX_OVERFLOW	7	Pass-Through UART Channel 1 Transmit FIFO Overflow Flag	0b0: No overflow occurred 0b1: Overflow occurred
PT1_UART_RX_OVERFLOW	6	Pass-Through UART Channel 1 Receive FIFO Overflow Flag	0b0: No overflow occurred 0b1: Overflow occurred

PT_STATUS_1 (0x4B)

BIT	7	6	5	4	3	2	1	0
Field	PT2_UART_TX_OVERFLOW	PT2_UART_RX_OVERFLOW	–	–	–	–	–	–
Reset	0b0	0b0	–	–	–	–	–	–
Access Type	Read Clears All	Read Clears All	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
PT2_UART_TX_OVERFLOW	7	Pass-Through UART Channel 2 Transmit FIFO Overflow Flag	0b0: No overflow occurred 0b1: Overflow occurred
PT2_UART_RX_OVERFLOW	6	Pass-Through UART Channel 2 Receive FIFO Overflow Flag	0b0: No overflow occurred 0b1: Overflow occurred

SPI_STATUS_0 (0x4D)

BIT	7	6	5	4	3	2	1	0

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Field	SPI_TX_OVERFLOW	SPI_RX_OVERFLOW	–	SPI_BUFFER_LEVEL[4:0]
Reset	0b0	0b0	–	0b00000
Access Type	Read Clears All	Read Clears All	–	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_TX_OVERFLOW	7	SPI Transmit Buffer Overflow	0b0: Buffer not overflowed 0b1: Buffer overflowed
SPI_RX_OVERFLOW	6	SPI Receive Buffer Overflow	0b0: Buffer not overflowed 0b1: Buffer overflowed
SPI_BUFFER_LEVEL	4:0	Number of SPI data bytes available for reading from Rx buffer.	0bXXXXX: Number of bytes available

AUDIO_STATUS_0 (0x54)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	AUDIO_LOCK	–	–	AUDIO_BLOCK_LENGTH_ERROR
Reset	–	–	–	–	0b0	–	–	0b0
Access Type	–	–	–	–	Read Only	–	–	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
AUDIO_LOCK	3	Indicates whether or not audio pipeline frequency is locked to audio clock.	0b0: Audio pipeline frequency is not locked 0b1: Audio pipeline frequency is locked
AUDIO_BLOCK_LENGTH_ERROR	0	Detected packet length error in received audio packet.	0b0: No error detected 0b1: Error detected

AUDIO_STATUS_1 (0x55)

BIT	7	6	5	4	3	2	1	0
Field	AUDIO_RX_CLOCK_FREQUENCY[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION	DECODE
AUDIO_RX_CLOCK_FREQUENCY	7:0	Audio clock frequency detected in audio receive block from connected serializer, measured to nearest MHz.	0x0: 0MHz audio clock detected (no clock detected) 0x1: 1MHz audio clock detected 0x2: 2MHz audio clock detected 0x3: 3MHz audio clock detected ... ... 0xFD: 253MHz audio clock detected 0xFE: 254MHz audio clock detected 0xFF: 255MHz audio clock detected

CFG_PIN_STATUS_0 (0x58)

BIT	7	6	5	4	3	2	1	0
Field	–	CFG_PIN_0_VALUE[2:0]			–	CFG_PIN_1_VALUE[2:0]		
Reset	–	0b0			–	0b0		
Access Type	–	Read Clears All			–	Read Clears All		

BITFIELD	BITS	DESCRIPTION	DECODE
CFG_PIN_0_VALUE	6:4	Status of the CFG0 pin on device power-up.	0b000: CFG State 0 0b001: CFG State 1 0b010: CFG State 2 0b011: CFG State 3 0b100: CFG State 4 0b101: CFG State 5 0b110: CFG State 6 0b111: CFG State 7
CFG_PIN_1_VALUE	2:0	Status of the CFG1 pin on device power-up.	0b000: CFG State 0 0b001: CFG State 1 0b010: CFG State 2 0b011: CFG State 3 0b100: CFG State 4 0b101: CFG State 5 0b110: CFG State 6 0b111: CFG State 7

GPIO_STATUS_0 (0x59)

BIT	7	6	5	4	3	2	1	0
Field	GPIO_VALUE_7	GPIO_VALUE_6	GPIO_VALUE_5	GPIO_VALUE_4	GPIO_VALUE_3	GPIO_VALUE_2	GPIO_VALUE_1	GPIO_VALUE_0
Reset	0b0	0b0	0b0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_VALUE_7	7	Value of GPIO 7. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_6	6	Value of GPIO 6. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_5	5	Value of GPIO 5. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_4	4	Value of GPIO 4. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_3	3	Value of GPIO 3. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_2	2	Value of GPIO 2. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_1	1	Value of GPIO 1. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_0	0	Value of GPIO 0. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured.	0b0: This GPIO pin is low 0b1: This GPIO pin is high

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BITFIELD	BITS	DESCRIPTION	DECODE
		If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	

GPIO_STATUS_1 (0x5A)

BIT	7	6	5	4	3	2	1	0
Field	GPIO_VALUE_15	GPIO_VALUE_14	GPIO_VALUE_13	GPIO_VALUE_12	GPIO_VALUE_11	GPIO_VALUE_10	GPIO_VALUE_9	GPIO_VALUE_8
Reset	0b0	0b0	0b0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_VALUE_15	7	Value of GPIO 15. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_14	6	Value of GPIO 14. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_13	5	Value of GPIO 13. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_12	4	Value of GPIO 12. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_11	3	Value of GPIO 11. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1

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BITFIELD	BITS	DESCRIPTION	DECODE
		GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	
GPIO_VALUE_10	2	Value of GPIO 10. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_9	1	Value of GPIO 9. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_8	0	Value of GPIO 8. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1

GPIO_STATUS_2 (0x5B)

BIT	7	6	5	4	3	2	1	0
Field	GPIO_VALUE_23	GPIO_VALUE_22	GPIO_VALUE_21	GPIO_VALUE_20	GPIO_VALUE_19	GPIO_VALUE_18	GPIO_VALUE_17	GPIO_VALUE_16
Reset	0b0	0b0	0b0	0b0	0b0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_VALUE_23	7	Value of GPIO 23. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_22	6	Value of GPIO 22. If GPIO is configured as a GPIO input, this is the state of the local pin and is the	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1

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BITFIELD	BITS	DESCRIPTION	DECODE
		value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	
GPIO_VALUE_21	5	Value of GPIO 21. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_20	4	Value of GPIO 20. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_19	3	Value of GPIO 19. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_18	2	Value of GPIO 18. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_17	1	Value of GPIO 17. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_VALUE_16	0	Value of GPIO 16. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1

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GPIO_STATUS_3 (0x5C)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	GPIO_VALUE_24
Reset	–	–	–	–	–	–	–	0b0
Access Type	–	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_VALUE_24	0	Value of GPIO 24. If GPIO is configured as a GPIO input, this is the state of the local pin and is the value transmitted across the link at the GPIO_TX_ID_x[4:0] that is configured. If GPIO is configured as a GPIO output, this is the value that is received over the link and will be driven to the pin if GPIO_OUTPUT_CONFIG_x[1:0] = 0b01 or 0b10.	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1

REGIONAL_CRC_STATUS_0 (0x5E)

BIT	7	6	5	4	3	2	1	0
Field	–	BANK_B_DONE	ACTIVE_A_OR_B	VA_DONE_STATUS	–	–	–	–
Reset	–	0b0	0b0	0b0	–	–	–	–
Access Type	–	Read Only	Read Only	Read Only	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_DONE	6	Regional CRC Bank B calculation done.	0b0: Not finished 0b1: Finished
ACTIVE_A_OR_B	5	Which Bank is actively calculating regional CRC.	0b0: Bank A 0b1: Bank B
VA_DONE_STATUS	4	Video authentication done status flag. This flag is automatically cleared when the next set of authentication challenge starts.	0b0: Running 0b1: Challenge complete

REGIONAL_CRC_STATUS_1 (0x5F)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_COUNT_BYTE_0	7:0	Frame Count for Regional CRC	0xXX: LSB bits for frame cycle counter value

REGIONAL_CRC_STATUS_2 (0x60)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	FRAME_COUNT_BYTE_1[1:0]	
Reset	–	–	–	–	–	–	0x00	
Access Type	–	–	–	–	–	–	Read Only	

BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_COUNT_BYTE_1	1:0	Frame Count for Regional CRC for Bits 9:8	0bXX: Frame count for regional CRC for bits 9:8

REGIONAL_CRC_STATUS_3 (0x61)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_A_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_A_BYTE_0	7:0	Value of Regional CRC Bank A. (Byte 0)

REGIONAL_CRC_STATUS_4 (0x62)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_A_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_A_BYTE_1	7:0	Value of Regional CRC Bank A. (Byte 1)

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REGIONAL_CRC_STATUS_5 (0x63)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_A_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_A_BYTE_2	7:0	Value of Regional CRC Bank A. (Byte 2)

REGIONAL_CRC_STATUS_6 (0x64)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_A_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_A_BYTE_3	7:0	Value of Regional CRC Bank A. (Byte 3)

REGIONAL_CRC_STATUS_7 (0x65)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_B_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_B_BYTE_0	7:0	Value of Regional CRC Bank B. (Byte 0)

REGIONAL_CRC_STATUS_8 (0x66)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_B_BYTE_1[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_B_BYTE_1	7:0	Value of Regional CRC Bank B. (Byte 1)

REGIONAL_CRC_STATUS_9 (0x67)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_B_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_B_BYTE_2	7:0	Value of Regional CRC Bank B. (Byte 2)

REGIONAL_CRC_STATUS_10 (0x68)

BIT	7	6	5	4	3	2	1	0
Field	REGIONAL_CRC_B_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGIONAL_CRC_B_BYTE_3	7:0	Value of Regional CRC Bank B. (Byte 3)

TELLTALE_STATUS_0 (0x6D)

BIT	7	6	5	4	3	2	1	0
Field	TELLTALE_BANK_B_ERR[3:0]				TELLTALE_BANK_A_ERR[3:0]			
Reset	0x0				0x0			
Access Type	Read Only				Read Only			

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BITFIELD	BITS	DESCRIPTION
TELLTALE_BANK_B_ERR	7:4	Telltale Bank B color error detected. Cleared when this read.
TELLTALE_BANK_A_ERR	3:0	Telltale Bank A color error detected. Cleared when this read.

CRC_MISC_STATUS_0 (0x6E)

BIT	7	6	5	4	3	2	1	0
Field	WM_BYPASS_ERROR_X	WM_BYPASS_ERROR_Y	LUT_FRC_ERROR_X	LUT_FRC_ERROR_Y	FRAME_CRC_ERROR_X	FRAME_CRC_ERROR_Y	–	–
Reset	0b0	0b0	0x0	0x0	0x0	0x0	–	–
Access Type	Read Only	Read Only	Read Clears All	Read Clears All	Read Clears All	Read Clears All	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
WM_BYPASS_ERROR_X	7	Pipe X watermark bypass redundancy error detected. Cleared when this CRC_MISC_STATUS_0 is read.	0b0: No error detected 0b1: Error detected
WM_BYPASS_ERROR_Y	6	Pipe Y watermark bypass redundancy error detected. Cleared when this CRC_MISC_STATUS_0 is read.	0b0: No error detected 0b1: Error detected
LUT_FRC_ERROR_X	5	Pipe X ASIL error detected through LUT and FRC. Cleared when this CRC_MISC_STATUS_0 is read.	0b0: No error detected 0b1: Error detected
LUT_FRC_ERROR_Y	4	Pipe Y ASIL error detected through LUT and FRC. Cleared when this CRC_MISC_STATUS_0 is read.	0b0: No error detected 0b1: Error detected
FRAME_CRC_ERROR_X	3	Pipe X video frame CRC check error detected. Cleared when this CRC_MISC_STATUS_0 is read.	0b0: No error detected 0b1: Error detected
FRAME_CRC_ERROR_Y	2	Pipe Y video frame CRC check error detected. Cleared when this CRC_MISC_STATUS_0 is read.	0b0: No error detected 0b1: Error detected

CRC_MISC_STATUS_1 (0x6F)

See Note §.

BIT	7	6	5	4	3	2	1	0

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Field	LINE_CRC_ERROR_X	LINE_CRC_ERROR_Y	VIDEO_PACKET_CRC_ERROR_X	VIDEO_PACKET_CRC_ERROR_Y	–	–	INFOFRAME_CRC_ERROR_SIOB	INFOFRAME_CRC_ERROR_SIOA
Reset	0x0	0x0	0x0	0x0	–	–	0x0	0x0
Access Type	Read Clears All	Read Clears All	Read Clears All	Read Clears All	–	–	Read Clears All	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
LINE_CRC_ERROR_X	7	Pipe X video line CRC error detected. Cleared when this CRC_MISC_STATUS_1 is read. Also cleared at rising edge of video lock.	0b0: No error detected 0b1: Error detected
LINE_CRC_ERROR_Y	6	Pipe Y video line CRC error detected. Cleared when this CRC_MISC_STATUS_1 is read. Also cleared at rising edge of video lock.	0b0: No error detected 0b1: Error detected
VIDEO_PACKET_CRC_ERROR_X	5	Pipe X video packet CRC error detected. Cleared when this CRC_MISC_STATUS_1 is read. Also cleared at rising edge of video lock.	0b0: No error detected 0b1: Error detected
VIDEO_PACKET_CRC_ERROR_Y	4	Pipe Y video packet CRC error detected. Cleared when this CRC_MISC_STATUS_1 is read. Also cleared at rising edge of video lock.	0b0: No error detected 0b1: Error detected
INFOFRAME_CRC_ERROR_SIOB	1	Link SIOB Infoframe CRC Error Cleared when this CRC_MISC_STATUS_1 is read.	0b0: No error detected 0b1: Error detected
INFOFRAME_CRC_ERROR_SIOA	0	Link SIOA Infoframe CRC Error Cleared when this CRC_MISC_STATUS_1 is read.	0b0: No error detected 0b1: Error detected

ARQ_STATUS_0 (0x71)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	MAX_RT_ERROR_INFOFRAME_CC	MAX_RT_ERR_CC	MAX_RT_ERR_PT_1	MAX_RT_ERR_PT_2	MAX_RT_ERR_SPI	MAX_RT_ERR_GPIO	_	MAX_RT_ERR_HDCP
Reset	0x0	0b0	0b0	0b0	0b0	0b0	–	0b0

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Access Type	Write, Read	Read Only	Read Only	Read Only	Read Only	Read Only	–	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
MAX_RT_ERROR_INFOFRAME_CC	7	Maximum retry attempts reached on Infoframe CC transmission. Write access is for internal use only.	0b0: Maximum retry attempts not reached 0b1: Maximum retry attempts reached
MAX_RT_ERR_CC	6	Control channel (CC) packets retransmission error. CRC has detected errors, and ARQ block has retransmitted more times than the threshold (X).	0b0: Retransmission threshold not exceeded 0b1: Retransmission threshold exceeded
MAX_RT_ERR_PT_1	5	Pass-through Channel 1 (I^2C or UART) packets retransmission error. CRC has detected errors, and ARQ block has retransmitted more times than the threshold (X).	0b0: Retransmission threshold not exceeded 0b1: Retransmission threshold exceeded
MAX_RT_ERR_PT_2	4	Pass-through Channel 2 (I^2C or UART) packets retransmission error. CRC has detected errors, and ARQ block has retransmitted more times than the threshold (X).	0b0: Retransmission threshold not exceeded 0b1: Retransmission threshold exceeded
MAX_RT_ERR_SPI	3	SPI packets retransmission error. CRC has detected errors, and ARQ block has retransmitted more times than the threshold (X).	0b0: Retransmission threshold not exceeded 0b1: Retransmission threshold exceeded
MAX_RT_ERR_GPIO	2	GPIO packets retransmission error. CRC has detected errors and ARQ block has retransmitted more times than the threshold (X).	0b0: Retransmission threshold not exceeded 0b1: Retransmission threshold exceeded
MAX_RT_ERR_HDCP	0	HDCP packets retransmission error. CRC has detected errors, and ARQ block has retransmitted more times than the threshold (X).	0b0: Retransmission threshold not exceeded 0b1: Retransmission threshold exceeded

LF_STATUS_0 (0x73)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	LINEFAULT_0_STATUS[2:0]			LINEFAULT_1_STATUS[2:0]			–	–
Reset	0b010			0b010			–	–
Access Type	Read Only			Read Only			–	–

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BITFIELD	BITS	DESCRIPTION	DECODE
LINEFAULT_0_STATUS	7:5	Line-fault status of GMSL link connected to LMN0 pin.	0b000: Short-to-battery 0b001: Short-to-GND 0b010: Normal operation 0b011: Line open 0b1XX: Line-to-line short
LINEFAULT_1_STATUS	4:2	Line-fault status of GMSL link connected to LMN1 pin.	0b000: Short-to-battery 0b001: Short-to-GND 0b010: Normal operation 0b011: Line open 0b1XX: Line-to-line short

DP_STATUS_0 (0x75)

BIT	7	6	5	4	3	2	1	0
Field	DP_OVERFLOW	DP_UNDERFLOW	DP_INPUT_FIFO_OVERFLOW	DP_VBI_ERROR	DP_INPUT_FIFO_UNDERFLOW	–	–	–
Reset	0b0	0b0	0b0	0b0	0b0	–	–	–
Access Type	Read Only	Read Only	Read Only	Read Only	Read Only	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
DP_OVERFLOW	7	DP Tx Controller Buffer Overflow Detected	0b0: Buffer not overflowed 0b1: Buffer overflowed
DP_UNDERFLOW	6	DP Tx Controller Buffer Underflow	0b0: Buffer not underflowed 0b1: Buffer underflowed
DP_INPUT_FIFO_OVERFLOW	5	DP Input FIFO Buffer Overflow	0b0: Buffer not overflowed 0b1: Buffer overflowed
DP_VBI_ERROR	4	DP Vertical Blanking Interval Timing Error Detected	0b0: No error detected 0b1: Error detected
DP_INPUT_FIFO_UNDERFLOW	3	DP Input FIFO Buffer Underflow	0b0: Buffer not underflowed 0b1: Buffer underflowed

DP_STATUS_1 (0x76)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	LINK_RETRAINED	HPD_EVENT_DETECTED	HPD_IRQ_DETECTED	HPD_CONNECT_STATE
Reset	–	–	–	–	0x0	0x0	0x0	0x0
Access Type	–	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read

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BITFIELD	BITS	DESCRIPTION	DECODE
LINK_RETRAINED	3	DP Link Retraining has started	0b0: DP link retraining has not started 0b1: DP link retraining has started
HPD_EVENT_DETECTED	2	HPD Event Detected	0b0: No event detected 0b1: Event detected
HPD_IRQ_DETECTED	1	DP HPD Interrupt Event Detected	0b0: No interrupt detected 0b1: Interrupt detected
HPD_CONNECT_STATE	0	DP HPD Connection State	0b0: Not connected 0b1: Connected

DP_STATUS_2 (0x77)

BIT	7	6	5	4	3	2	1	0
Field	DPTX_VIDEO_OUTPUT_ACTIVE	LINK_TRAINING_PASSED	LINK_TRAINING_FAILED	LINK_TRAINING_IN_PROGRESS	LINK_TRAINING_NOT_STARTED	WAITING_FOR_VIDEO_LOCK	WAITING_FOR_PSM_STATE	WAITING_FOR_HPD
Reset	0x0	0x0	0x0	0x0	0b1	0b1	0b1	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
DPTX_VIDEO_OUTPUT_ACTIVE	7	DPTX Controller Transmitting Video to DP Sink	0b0: Not transmitting video 0b1: Transmitting video
LINK_TRAINING_PASSED	6	DP Link Training Passed	0b0: Link training did not pass 0b1: Link training passed
LINK_TRAINING_FAILED	5	DP Link Training has Failed	0b0: Link training did not fail 0b1: Link training failed
LINK_TRAINING_IN_PROGRESS	4	DP Link Training is in Progress	0b0: Link training not in progress 0b1: Link training in progress
LINK_TRAINING_NOT_STARTED	3	State of DP Link Training	0b0: Link training started 0b1: Link training not started
WAITING_FOR_VIDEO_LOCK	2	Waiting for video configuration and video lock before starting DP link training.	0b0: Not waiting 0b1: Waiting
WAITING_FOR_PSM_STATE	1	Waiting for training PSM state before proceeding with video/DP link training configuration.	0b0: Not waiting 0b1: Waiting
WAITING_FOR_HPD	0	DPTX Controller waiting for HPD connection from DP sink.	0b0: Not waiting for HPD 0b1: Waiting for HPD

DP_STATUS_3 (0x78)

BIT	7	6	5	4	3	2	1	0

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Field	–	–	–	–	–	–	EDID_TRANSFER_TO_SER_DONE	SINK_EDID_READ_DONE
Reset	–	–	–	–	–	–	0x0	0x0
Access Type	–	–	–	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
EDID_TRANSFER_TO_SER_DONE	1	Transfer of DisplayID EDID data read from DP sink to GMSL Serializer is done.	0b0: Transfer not done 0b1: Transfer done
SINK_EDID_READ_DONE	0	Read of DisplayID EDID data from DP sink completed.	0b0: EDID read not completed 0b1: EDID read completed

DP_STATUS_4 (0x79)

BIT	7	6	5	4	3	2	1	0
Field	MAJOR_REV[3:0]				MINOR_REV[3:0]			
Reset	0x0				0x0			
Access Type	Write, Read				Write, Read			

BITFIELD	BITS	DESCRIPTION	DECODE
MAJOR_REV	7:4	DPCD major revision number readback from attached DP sink.	0x1: DPCD r1.*
MINOR_REV	3:0	DPCD minor revision number readback from attached DP sink.	0x0: DPCD r*.0 0x1: DPCD r*.1 0x2: DPCD r*.2 0x3: DPCD r*.3 0x4: DPCD r*.4

DP_STATUS_5 (0x7A)

BIT	7	6	5	4	3	2	1	0
Field	EXTENDED_CAPS_FIELD	–	–	LOCAL_EDID_PRESENT	TPS4_SUPPORTED	TPS3_SUPPORTED	MAX_DOWNSPREAD	ENHANCED_FRAME_CAP
Reset	0x0	–	–	0x0	0x0	0x0	0x0	0x0
Access Type	Write, Read	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

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BITFIELD	BITS	DESCRIPTION	DECODE
EXTENDED_CAPS_FIELD	7	DPCD Extended Receiver Capability Field Present	0b0: Not present 0b1: Present
LOCAL_EDID_PRESENT	4	DPCD Local EDID Present	0b0: This receiver port does not have a local DisplayID or legacy EDID 0b1: This receiver port has a local DisplayID or legacy EDID
TPS4_SUPPORTED	3	DPCD bit which indicates link Training Pattern Sequence 4 (TPS4) support.	0b0: TPS4 is not supported 0b1: TPS4 is supported
TPS3_SUPPORTED	2	DPCD bit which indicates Link Training Pattern Sequence 3 (TPS3) support.	0b0: TPS3 is not supported 0b1: TPS3 is supported
MAX_DOWNSPREAD	1	DPCD Maximum Downspread	0b0: No downspread supported 0b1: Up to 0.5% downspread supported
ENHANCED_FRAME_CAP	0	DPCD Enhanced Frame CAP	0b0: Enhanced framing symbol sequence for BS and SR is not supported 0b1: Enhanced framing symbol sequence for BS and SR is supported

DP_STATUS_6 (0x7B)

BIT	7	6	5	4	3	2	1	0
Field	MAX_LINK_RATE[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
MAX_LINK_RATE	7:0	DPCD MAX_LINK_RATE Four values are supported.	0x06h: 1.62Gbps/lane (RBR) 0x0Ah: 2.7Gbps/lane (HBR) 0x14h: 5.4Gbps/lane (HBR2) 0x1Eh: 8.1Gbps/lane (HBR3) All others: Reserved

DP_STATUS_7 (0x7C)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	MAX_LANE_COUNT[4:0]				
Reset	–	–	–	0x0				
Access Type	–	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
MAX_LANE_COUNT	4:0	DPCD MAX_LANE_COUNT	0x01h: One lane (Lane 0 only) 0x02h: Two lanes (Lane 0 and 1 only)

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BITFIELD	BITS	DESCRIPTION	DECODE
			0x04h: Four lanes (Lane 0, 1, 2, and 3) All others: Reserved

DP_STATUS_8 (0x7D)

BIT	7	6	5	4	3	2	1	0
Field	–	TRAINING_AUX_RD_INTERVAL[6:0]						
Reset	–	0x0						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
TRAINING_AUX_RD_INTERVAL	6:0	DPCD TRAINING_AUX_RD_INTERVAL Link status/adjust request read interval during main-link training.	0x00h: 100µs for the Main-Link LANEx_CR_DONE sequence; 400µs for the Main-Link LANEx_CHANNEL_EQ_DONE sequence. 0x01h: 100µs for the Main-Link LANEx_CR_DONE sequence; 4ms for the Main-Link LANEx_CHANNEL_EQ_DONE sequence. 0x02h: 100µs for the Main-Link LANEx_CR_DONE sequence; 8ms for the Main-Link LANEx_CHANNEL_EQ_DONE sequence. 0x03h: 100µs for the Main-Link LANEx_CR_DONE sequence; 12ms for the Main-Link LANEx_CHANNEL_EQ_DONE sequence. 0x04h: 100µs for the Main-Link LANEx_CR_DONE sequence; 16ms for the Main-Link LANEx_CHANNEL_EQ_DONE sequence. All others: Reserved

DP_STATUS_9 (0x7E)

BIT	7	6	5	4	3	2	1	0
Field	I2C_SPEED_CAP[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
I2C_SPEED_CAP	7:0	DPCD I^2C Speed Control Capabilities Bit Map Bits to indicated I^2C speed control capabilities.	0x01h: 1Kbps 0x02h: 5Kbps 0x04h: 10Kbps 0x08h: 100Kbps 0x10h: 400Kbps 0x20h: 1Mbps 0x40h: Reserved 0x80h: Reserved

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DP_STATUS_12 (0x80)

BIT	7	6	5	4	3	2	1	0
Field	–	LANE1_SYMBOL_LOCKED	LANE1_CHANNEL_EQ_DONE	LANE1_CR_DONE	–	LANE0_SYMBOL_LOCKED	LANE0_CHANNEL_EQ_DONE	LANE0_CR_DONE
Reset	–	0x0	0x0	0x0	–	0x0	0x0	0x0
Access Type	–	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
LANE1_SYMBOL_LOCKED	6	DPCD LANE1_SYMBOL_LOCKED	0x0: Not locked 0x1: Locked
LANE1_CHANNEL_EQ_DONE	5	DPCD LANE1_CHANNEL_EQ_DONE	0x0: Not done 0x1: Done
LANE1_CR_DONE	4	DPCP LANE1_CR_DONE DP Lane 1 clock recovery done status.	0x0: Not done 0x1: Done
LANE0_SYMBOL_LOCKED	2	DPCD LANE0_SYMBOL_LOCKED	0x0: Not locked 0x1: Locked
LANE0_CHANNEL_EQ_DONE	1	DPCD LANE0_CHANNEL_EQ_DONE	0x0: Not done 0x1: Done
LANE0_CR_DONE	0	DPCP LANE0_CR_DONE DP Lane 0 clock recovery done status.	0x0: Not done 0x1: Done

DP_STATUS_13 (0x81)

BIT	7	6	5	4	3	2	1	0
Field	–	LANE3_SYMBOL_LOCKED	LANE3_CHANNEL_EQ_DONE	LANE3_CR_DONE	–	LANE2_SYMBOL_LOCKED	LANE2_CHANNEL_EQ_DONE	LANE2_CR_DONE
Reset	–	0x0	0x0	0x0	–	0x0	0x0	0x0
Access Type	–	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
LANE3_SYMBOL_LOCKED	6	DPCD LANE3_SYMBOL_LOCKED	0x0: Not locked 0x1: Locked
LANE3_CHANNEL_EQ_DONE	5	DPCD LANE3_CHANNEL_EQ_DONE	0x0: Not done 0x1: Done
LANE3_CR_DONE	4	DPCP LANE3_CR_DONE DP Lane 3 clock recovery done status	0x0: Not done 0x1: Done

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BITFIELD	BITS	DESCRIPTION	DECODE
LANE2_SYMBOL_LOCKED	2	DPCD LANE2_SYMBOL_LOCKED	0x0: Not locked 0x1: Locked
LANE2_CHANNEL_EQ_DONE	1	DPCD LANE2_CHANNEL_EQ_DONE	0x0: Not done 0x1: Done
LANE2_CR_DONE	0	DPCP LANE2_CR_DONE. DP Lane 2 clock recovery done status	0x0: Not done 0x1: Done

DP_STATUS_14 (0x82)

BIT	7	6	5	4	3	2	1	0
Field	LINK_STATUS_UPDATED	–	–	–	–	–	–	INTERLANE_ALIGN_DONE
Reset	0x0	–	–	–	–	–	–	0x0
Access Type	Write, Read	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
LINK_STATUS_UPDATED	7	DPCD LINK_STATUS_UPDATED Link status and adjust request updated since the last read. Set when updated.	0x0: Not updated 0x1: Updated
INTERLANE_ALIGN_DONE	0	DPCD INTERLANE_ALIGN_DONE	0x0: Not done 0x1: Done

DP_STATUS_15 (0x83)

BIT	7	6	5	4	3	2	1	0
Field	AUX_HPD_DISCON_ERROR	–	DP_SINK_EDID_ERROR	I2C_OVER_AUX_ERROR	AUX_NACK_ERROR	AUX_TIMEOUT_ERROR	AUX_RETRY_ERROR	AUX_DEFER_ERROR
Reset	0b0	–	0b0	0x0	0x0	0x0	0x0	0x0
Access Type	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
AUX_HPD_DISCON_ERROR	7	AUX transaction status indicating error detected due to HPD disconnect for longer than 2ms. Write 0 to clear.	0x0: No error detected 0x1: Error detected
DP_SINK_EDID_ERROR	5	Error detected during read of DP sink EDID. Write 0 to clear.	0b0: No error detected 0b1: Error detected

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BITFIELD	BITS	DESCRIPTION	DECODE
I2C_OVER_AUX_ERROR	4	Error detected in I²C-over-Aux transaction. Write 0 to clear.	0x0: No error detected 0x1: Error detected
AUX_NACK_ERROR	3	Error detected in AUX_NACK. Write 0 to clear.	0x0: No error detected 0x1: Error detected
AUX_TIMEOUT_ERROR	2	Aux response timeout detected. Write 0 to clear.	0x0: No timeout detected 0x1: Timeout detected
AUX_RETRY_ERROR	1	Error detected on AUX_RETRY. Write 0 to clear.	0x0: No error detected 0x1: Error detected
AUX_DEFER_ERROR	0	Error detected on AUX_DEFER. Write 0 to clear.	0x0: No error detected 0x1: Error detected

DP_STATUS_16 (0x84)

BIT	7	6	5	4	3	2	1	0
Field	OTP_CRC_ERROR_STATUS	–	DP_SINK_RGB_FRAME_CRC_STOPPED	DP_SINK_SYMBOL_ERROR_STATUS	VIDEO_CFG_ERROR_STATUS	DP_SINK_CHEQ_TPS_CAPABILITY_ERROR	DP_SINK_LANE_COUNT_CAPABILITY_ERROR	DP_SINK_LINK_RATE_CAPABILITY_ERROR
Reset	0x0	–	0b0	0b0	0x0	0x0	0x0	0x0
Access Type	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
OTP_CRC_ERROR_STATUS	7	Error detected on CRC of PMX control OTP.	0x0: No error detected 0x1: Error detected
DP_SINK_RGB_FRAME_CRC_STOPPED	5	This bit indicates that CRC error DP_SINK_RGB_FRAME_CRC_ERROR_STATUS is due to the DP sink no longer calculating CRC.	0x0: DP sink still calculating CRC 0x1: DP sink stopped calculating CRC
DP_SINK_SYMBOL_ERROR_STATUS	4	Status indicating DP sink has detected symbol error.	0x0: No error detected 0x1: Error detected
VIDEO_CFG_ERROR_STATUS	3	PSM detected an error in video configuration.	0x0: No error detected 0x1: Error detected
DP_SINK_CHEQ_TPS_CAPABILITY_ERROR	2	Error detected with DP sink CHEQ, TPS capabilities.	0x0: No error detected 0x1: Error detected
DP_SINK_LANE_COUNT_CAPABILITY_ERROR	1	Error detected with DP sink lane count capability.	0x0: No error detected 0x1: Error detected
DP_SINK_LINK_RATE_CAPABILITY_ERROR	0	Error detected with DP sink link rate capability.	0x0: No error detected 0x1: Error detected

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DP_STATUS_17 (0x85)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	DP_LINK_INTERLANE_ALIGN_ERROR	DP_LINK_SYMBOL_LOCK_ERROR	DP_LINK_CHANNEL_EQ_ERROR	DP_LINK_CR_ERROR
Reset	–	–	–	–	0x0	0x0	0x0	0x0
Access Type	–	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
DP_LINK_INTERLANE_ALIGN_ERROR	3	Error detected in interlane alignment during DP link training.	0x0: No error detected 0x1: Error detected
DP_LINK_SYMBOL_LOCK_ERROR	2	Error detected in symbol lock during DP link training.	0x0: No error detected 0x1: Error detected
DP_LINK_CHANNEL_EQ_ERROR	1	Error detected in channel equalization during DP link training.	0x0: No error detected 0x1: Error detected
DP_LINK_CR_ERROR	0	Error detected on clock recovery during DP link training.	0x0: No error detected 0x1: Error detected

WATERMARK_STATUS_0 (0x88)

BIT	7	6	5	4	3	2	1	0
Field	WM_ERROR_LIVE	WM_DETECT	–	–	–	–	–	–
Reset	0b0	0b0	–	–	–	–	–	–
Access Type	Read Only	Read Only	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
WM_ERROR_LIVE	7	Live Watermark Error	0b0: No watermark error 0b1: Watermark error active, bit automatically clears when error clears.
WM_DETECT	6	Watermark Detected Status If WMD_INT_EN is 1, this value is output to GPIO9.	0b0: Watermark detected 0b1: Watermark not detected

DSC_STATUS_0 (0x8A)

See Note §.

BIT	7	6	5	4	3	2	1	0

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Field	DSC_OVERFLOW	DSC_UNDERFLOW	DSC_PARITY_ERROR	DSC_TRANSPORT_CHUNK_SIZE_ERROR	–	–	–	–
Reset	0b0	0b0	0b0	0b0	–	–	–	–
Access Type	Read Only	Read Only	Read Only	Read Only	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
DSC_OVERFLOW	7	DSC Buffer Overflow	0b0: Buffer not overflowed 0b1: Buffer overflowed
DSC_UNDERFLOW	6	DSC Buffer Underflow	0b0: Buffer not underflowed 0b1: Buffer underflowed
DSC_PARITY_ERROR	5	DSC decode memories detected parity error.	0b0: Parity error not detected 0b1: Parity error detected
DSC_TRANSPORT_CHUNK_SIZE_ERROR	4	DSC decode engine detected transport chunk size error.	0b0: No error detected 0b1: Error detected

PRBS_STATUS_0 (0x8C)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	APRBS_VALID	–	–	–	VPRBS_FAIL
Reset	–	–	–	0b0	–	–	–	0b0
Access Type	–	–	–	Read Only	–	–	–	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
APRBS_VALID	4	Indicates whether or not audio PRBS is running.	0b0: Audio PRBS is not running 0b1: Audio PRBS is running
VPRBS_FAIL	0	Video PRBS Synchronization Check Pass/Fail. Video pipeline should be locked for proper operation of this register.	0b0: Video PRBS check passed 0b1: Video check failed

PRBS_STATUS_1 (0x8D)

BIT	7	6	5	4	3	2	1	0
Field	VPRBS_ERROR_COUNT[7:0]							
Reset	0x00							
Access Type	Read Clears All							

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BITFIELD	BITS	DESCRIPTION	DECODE
VPRBS_ERROR_COUNT	7:0	Video PRBS Error Counter Clears on read.	0xXX: Number of video PRBS errors since last read

PRBS_STATUS_2 (0x8E)

BIT	7	6	5	4	3	2	1	0
Field	APRBS_ERROR_COUNT[7:0]							
Reset	0x00							
Access Type	Read Clears All							

BITFIELD	BITS	DESCRIPTION	DECODE
APRBS_ERROR_COUNT	7:0	Audio PRBS Error Counter Clears on read.	0xXX: Number of audio PRBS errors since last read

HDCP_STATUS_0 (0x90)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	DECRYPTION_ON_X	DECRYPTION_ON_Y	HDCP_LINK_INTEGRITY_X	HDCP_LINK_INTEGRITY_Y	HDCP2_AUTHENTICATED_X	HDCP2_AUTHENTICATED_Y	–	–
Reset	0x0	0x0	0x0	0x0	0x0	0x0	–	–
Access Type	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
DECRYPTION_ON_X	7	Decryption active for audio and video for Pipe X.	0b0: Decryption is not active 0b1: Decryption is active
DECRYPTION_ON_Y	6	Decryption active for audio and video for Pipe Y.	0b0: Decryption is not active 0b1: Decryption is active
HDCP_LINK_INTEGRITY_X	5	HDCP Link integrity status for Pipe X. Indicates decryption is active and no video line CRC errors have occurred.	0b0: Link integrity is bad (video line CRC errors have occurred) or decryption is not active 0b1: Link integrity is good
HDCP_LINK_INTEGRITY_Y	4	HDCP Link integrity status for Pipe Y. Indicates decryption is active and no video line CRC errors have occurred.	0b0: Link integrity is bad (video line CRC errors have occurred) or decryption is not active 0b1: Link integrity is good

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BITFIELD	BITS	DESCRIPTION	DECODE
HDCP2_AUTHENTICATED_X	3	Authentication status (only valid for HDCP2) for Pipe X.	0b0: HDCP2 is not authenticated 0b1: HDCP2 is authenticated
HDCP2_AUTHENTICATED_Y	2	Authentication status (only valid for HDCP2) for Pipe Y.	0b0: HDCP2 is not authenticated 0b1: HDCP2 is authenticated

VIDEO_STATUS_0 (0x92)

BIT	7	6	5	4	3	2	1	0
Field	VIDEO_ALIGN_LINE_READ_ERR_DETECTED	VTRG_OVERFLOW_X	VTRG_UNDERFLOW_X	VIDEO_ALIGN_UNDERFLOW_X	VIDEO_ALIGN_OVERFLOW_X	VIDEO_SEQUENCE_ERROR_X	VIDEO_BLOCK_LENGTH_ERROR_X	LOSS_OF_VIDEO_PACKETS_LOCK_X
Reset	0b0	0x0	0x0	0x0	0x0	0b0	0b0	0x0
Access Type	Read Only	Read Clears All	Read Clears All	Read Only	Read Only	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
VIDEO_ALIGN_LINE_READ_ERR_DETECTED	7	Indicates that a video line was not read in time in video align block	0x0: No error detected 0x1: Error detected
VTRG_OVERFLOW_X	6	Pipe X video timing regenerator (VTRG) buffer overflow.	0b0: Buffer not overflowed 0b1: Buffer overflowed
VTRG_UNDERFLOW_X	5	Pipe X video timing regenerator (VTRG) buffer underflow.	0b0: Buffer not underflowed 0b1: Buffer underflowed
VIDEO_ALIGN_UNDERFLOW_X	4	Sticky flag of video align block underflow detected.	0b0: No underflow detected 0b1: Underflow detected
VIDEO_ALIGN_OVERFLOW_X	3	Sticky flag of video align block overflow detected.	0b0: No overflow detected 0b1: Overflow detected
VIDEO_SEQUENCE_ERROR_X	2	Pipe X video packet sequence error detected. Asserted when any received video packets arrive out of order.	0b0: No video packet sequence error 0b1: Video packet sequence error
VIDEO_BLOCK_LENGTH_ERROR_X	1	Pipe X video block length error detected. Asserted when a received video block has too many or too few pixels.	0b0: No video block length error 0b1: Video block length error
LOSS_OF_VIDEO_PACKETS_LOCK_X	0	Indicates that the deserializer has failed to detect incoming video packets or overflow/underflow has occurred in the video path. Sticky bit is set on event.	0b0: Video not lost 0b1: Video lost

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VIDEO_STATUS_1 (0x93)

BIT	7	6	5	4	3	2	1	0
Field	–	VTRG_OVERFLOW_Y	VTRG_UNDERFLOW_Y	VIDEO_ALIGN_UNDERFLOW_Y	VIDEO_ALIGN_OVERFLOW_Y	VIDEO_SEQUENCE_ERROR_Y	VIDEO_BLOCK_LENGTH_ERROR_Y	LOSS_OF_VIDEO_PACKETS_LOCK_Y
Reset	–	0x0	0x0			0b0	0b0	0x0
Access Type	–	Read Clears All	Read Clears All	Read Only	Read Only	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
VTRG_OVERFLOW_Y	6	Pipe Y video timing regenerator (VTRG) buffer overflow.	0b0: Buffer not overflowed 0b1: Buffer overflowed
VTRG_UNDERFLOW_Y	5	Pipe Y video timing regenerator (VTRG) buffer underflow.	0b0: Buffer not underflowed 0b1: Buffer underflowed
VIDEO_ALIGN_UNDERFLOW_Y	4	Sticky flag of video align block overflow detected.	0b0: No underflow detected 0b1: Underflow detected
VIDEO_ALIGN_OVERFLOW_Y	3	Sticky flag of video align block overflow detected.	0b0: No overflow detected 0b1: Overflow detected
VIDEO_SEQUENCE_ERROR_Y	2	Pipe Y video packet sequence error detected. Asserted when any received video packets arrive out of order.	0b0: No video packet sequence error 0b1: Video packet sequence error
VIDEO_BLOCK_LENGTH_ERROR_Y	1	Pipe X video block length error detected. Asserted when a received video block has too many or too few pixels.	0b0: No video block length error 0b1: Video block length error
LOSS_OF_VIDEO_PACKETS_LOCK_Y	0	Indicates that the deserializer has failed to detect incoming video packets or overflow/underflow has occurred in the video path. Sticky bit is set on event.	0x0: Video not lost 0x1: Video lost

VIDEO_STATUS_2 (0x94)

BIT	7	6	5	4	3	2	1	0
Field	VIDEO_PACKETS_LOCK_Y	VIDEO_PACKETS_LOCK_X	VIDEO_LOCK_Y	VIDEO_LOCK_X	ODD_EVEN_SPLIT_STATUS	DETECTED_30_BPP	PPS_PARAMS_RECEIVED	AUTO_TIMING_RECEIVED
Reset	0b0	0b0	0b0	0b0	0x0	0x0	0b0	0b0
Access	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	Write, Read	Write, Read

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Type								

BITFIELD	BITS	DESCRIPTION	DECODE
VIDEO_PACKETS_LOCK_Y	7	Video pipe receiving valid video packets with the selected STREAM_SEL_Y ID from the serializer.	0b0: Video not locked 0b1: Video locked
VIDEO_PACKETS_LOCK_X	6	Video pipe receiving valid video packets with the selected STREAM_SEL_X ID from the serializer.	0b0: Video not locked 0b1: Video locked
VIDEO_LOCK_Y	5	Stable video is being received from the serializer, DP sink is attached to DP port (HPD signal is detected), DP link training started by writing PSM_STATE register, and Video channel is configured and locked to incoming video.	0b0: Video not stable 0b1: Video stable
VIDEO_LOCK_X	4	Stable video is being received from the serializer, DP sink is attached to DP port (HPD signal is detected), DP link training started by writing PSM_STATE register, and Video channel is configured and locked to incoming video.	0b0: Video not stable 0b1: Video stable
ODD_EVEN_SPLIT_STATUS	3	Status of even/odd split combiner usage.	0b0: Even/odd split combiner not in use 0b1: Even/odd split combiner in use
DETECTED_30_BPP	2	Detected 30bpp video received from GMSL serializer.	0b0: 24bpp 0b1: 30bpp
PPS_PARAMS_RECEIVED	1	DSC PPS parameters received from the serializer. Write access for internal use only.	0b0: No PPS parameters received 0b1: PPS parameters received
AUTO_TIMING_RECEIVED	0	Auto timing received from the serializer, if supported and enabled. Values are populated in registers VIDEO_STATUS_2 to VIDEO_STATUS_18. Write access for internal use only.	0b0: Video timing not received from serializer 0b1: Video timing received from serializer

VIDEO_STATUS_3 (0x95)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PCLK_FREQUENCY_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RECEIVED_PCLK_FREQUENCY_BYTE_0	7:0	Received pixel clock (PCLK) frequency from serializer (to the nearest kHz). (Byte 0). Write access for internal use only.

VIDEO_STATUS_4 (0x96)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PCLK_FREQUENCY_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PCLK_FREQUENCY_BYTE_1	7:0	Received pixel clock (PCLK) frequency from serializer (to the nearest kHz). (Byte 1) Write access for internal use only.

VIDEO_STATUS_5 (0x97)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PCLK_FREQUENCY_BYTE_2[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PCLK_FREQUENCY_BYTE_2	7:0	Received pixel clock (PCLK) frequency from serializer (to the nearest kHz). (Byte 2) Write access for internal use only.

VIDEO_STATUS_6 (0x98)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_ACTIVE_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RECEIVED_H_ACTIVE_BYTE_0	7:0	Received horizontal active (HACTIVE) resolution from serializer. Measured in pixels. (Byte 0) Write access for internal use only.

VIDEO_STATUS_7 (0x99)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_ACTIVE_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_H_ACTIVE_BYTE_1	7:0	Received horizontal active (HACTIVE) resolution from serializer. Measured in pixels. (Byte 1) Write access for internal use only.

VIDEO_STATUS_8 (0x9A)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_FP_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_H_FP_BYTE_0	7:0	Received horizontal front porch (HFP) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_9 (0x9B)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_FP_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RECEIVED_H_FP_BYTE_1	7:0	Received horizontal front porch (HFP) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_10 (0x9C)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_SW_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_H_SW_BYTE_0	7:0	Received horizontal sync width (HSW) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_11 (0x9D)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_SW_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_H_SW_BYTE_1	7:0	Received horizontal sync width (HSW) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_12 (0x9E)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_BP_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RECEIVED_H_BP_BYTE_0	7:0	Received horizontal back porch (HFP) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_13 (0x9F)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_H_BP_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_H_BP_BYTE_1	7:0	Received horizontal back porch (HFP) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_14 (0xA0)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_V_ACTIVE_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_V_ACTIVE_BYTE_0	7:0	Received vertical active (VACTIVE) resolution from serializer. Measured in lines. (Byte 0) Write access for internal use only.

VIDEO_STATUS_15 (0xA1)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_V_ACTIVE_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RECEIVED_V_ACTIVE_BYTE_1	7:0	Received vertical active (VACTIVE) resolution from serializer. Measured in lines. (Byte 1) Write access for internal use only.

VIDEO_STATUS_16 (0xA2)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_V_FP[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_V_FP	7:0	Received vertical front porch (VFP) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_17 (0xA3)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_V_SW[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_V_SW	7:0	Received vertical sync width (VSW) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_18 (0xA4)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_V_BP[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RECEIVED_V_BP	7:0	Received vertical back porch (VBP) from serializer. Measured in pixels. Write access for internal use only.

VIDEO_STATUS_19 (0xA5)

BIT	7	6	5	4	3	2	1	0
Field	DETECTED_PCLK_FREQUENCY_BYTE_0[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
DETECTED_PCLK_FREQUENCY_BYTE_0	7:0	Detected pixel clock (PCLK) frequency (in kHz) in deserializer video pipe. (Byte 0)

VIDEO_STATUS_20 (0xA6)

BIT	7	6	5	4	3	2	1	0
Field	DETECTED_PCLK_FREQUENCY_BYTE_1[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
DETECTED_PCLK_FREQUENCY_BYTE_1	7:0	Detected pixel clock (PCLK) frequency (in kHz) in deserializer video pipe. (Byte 1)

VIDEO_STATUS_21 (0xA7)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	DETECTED_PCLK_FREQUENCY_BYTE_2[3:0]			
Reset	–	–	–	–	0x0			
Access Type	–	–	–	–	Read Only			

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BITFIELD	BITS	DESCRIPTION
DETECTED_PCLK_FREQUENCY_BYTE_2	3:0	Detected pixel clock (PCLK) frequency (in kHz) in deserializer video pipe. (Byte 2)

VIDEO_STATUS_22 (0xA8)

BIT	7	6	5	4	3	2	1	0
Field	DETECTED_H_ACTIVE_BYTE_0[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
DETECTED_H_ACTIVE_BYTE_0	7:0	Detected horizontal active (HACTIVE) resolution from serializer. Measured in pixels. (Byte 0)

VIDEO_STATUS_23 (0xA9)

BIT	7	6	5	4	3	2	1	0
Field	DETECTED_H_ACTIVE_BYTE_1[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
DETECTED_H_ACTIVE_BYTE_1	7:0	Detected horizontal active (HACTIVE) resolution from serializer. Measured in pixels. (Byte 1)

VIDEO_STATUS_24 (0xAA)

BIT	7	6	5	4	3	2	1	0
Field	DETECTED_V_ACTIVE_BYTE_0[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
DETECTED_V_ACTIVE_BYTE_0	7:0	Detected vertical active (VACTIVE) resolution from serializer. Measured in lines. (Byte 0)

VIDEO_STATUS_25 (0xAB)

BIT	7	6	5	4	3	2	1	0
Field	DETECTED_V_ACTIVE_BYTE_1[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
DETECTED_V_ACTIVE_BYTE_1	7:0	Detected vertical active (VACTIVE) resolution from serializer. Measured in lines. (Byte 1)

VIDEO_STATUS_26 (0xAC)

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_HS_POLARITY	RECEIVED_VS_POLARITY	–	–	–	–	DETECTED_HS_POLARITY	DETECTED_VS_POLARITY
Reset	0x0	0x0	–	–	–	–	0x0	0x0
Access Type	Write, Read	Write, Read	–	–	–	–	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
RECEIVED_HS_POLARITY	7	HSYNC polarity of received video stream from GMSL serializer. Write access for internal use only.	0b0: Negative HSYNC polarity 0b1: Positive HSYNC polarity
RECEIVED_VS_POLARITY	6	VSYNC polarity of received video stream from GMSL serializer. Write access for internal use only.	0b0: Negative VSYNC polarity 0b1: Positive VSYNC polarity
DETECTED_HS_POLARITY	1	Detected HS signal polarity.	0b0: Negative HS polarity 0b1: Positive HS polarity
DETECTED_VS_POLARITY	0	Detected VS pulse polarity.	0b0: Negative VS polarity 0b1: Positive VS polarity

VIDEO_STATUS_27 (0xAD)

See Note §.

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BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PPS_PIC_WIDTH_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_PIC_WIDTH_BYTE_0	7:0	Received PPS value for picture width from serializer. This is the horizontal active resolution measured in pixels used by the compression engine and should match the expected horizontal resolution (HACTIVE). (Byte 0) Write access for internal use only.

VIDEO_STATUS_28 (0xAE)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PPS_PIC_WIDTH_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_PIC_WIDTH_BYTE_1	7:0	Received PPS value for picture width from serializer. This is the horizontal active resolution measured in pixels used by the compression engine and should match the expected horizontal resolution (HACTIVE). (Byte 1) Write access for internal use only.

VIDEO_STATUS_29 (0xAF)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PPS_PIC_HEIGHT_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_PIC_HEIGHT_BYTE_0	7:0	Received PPS value for picture height from serializer. This is the horizontal active resolution measured in lines used by the compression engine and should match the expected vertical resolution (VACTIVE). (Byte 0) Write access for internal use only.

VIDEO_STATUS_30 (0xB0)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PPS_PIC_HEIGHT_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_PIC_HEIGHT_BYTE_1	7:0	Received PPS value for picture height from serializer. This is the horizontal active resolution measured in lines used by the compression engine and should match the expected vertical resolution (VACTIVE). (Byte 1) Write access for internal use only.

VIDEO_STATUS_31 (0xB1)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PPS_SLICE_WIDTH_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_SLICE_WIDTH_BYTE_0	7:0	Received PPS value for slice width from serializer. This is the horizontal slice width measured in pixels used by the compression engine. (Byte 0) Write access for internal use only.

VIDEO_STATUS_32 (0xB2)

See Note §.

BIT	7	6	5	4	3	2	1	0

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Field	RECEIVED_PPS_SLICE_WIDTH_BYTE_1[7:0]
Reset	0x0
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_SLICE_WIDTH_BYTE_1	7:0	Received PPS value for picture width from serializer. This is the horizontal slice width measured in pixels used by the compression engine. (Byte 1) Write access for internal use only.

VIDEO_STATUS_33 (0xB3)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PPS_SLICE_HEIGHT_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_SLICE_HEIGHT_BYTE_0	7:0	Received PPS value for slice height from serializer. This is the horizontal slice height measured in lines used by the compression engine. (Byte 0) Write access for internal use only.

VIDEO_STATUS_34 (0xB4)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	RECEIVED_PPS_SLICE_HEIGHT_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RECEIVED_PPS_SLICE_HEIGHT_BYTE_1	7:0	Received PPS value for slice height from serializer. This is the horizontal slice height measured in lines used by the compression engine. (Byte 0) Write access for internal use only.

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PSM_STATE_STATUS_0 (0xB6)

BIT	7	6	5	4	3	2	1	0
Field	–	PSM_ACCESS_ERROR	PSM_STATE_TIMER_ERROR	PSM_DATA_TIMER_ERROR	–	–	PSM_STARTUP_MBIST_FAILED	PSM_STARTUP_MBIST_DONE
Reset	–	0x0	0x0	0x0	–	–	0x0	0x0
Access Type	–	Read Only	Read Only	Read Only	–	–	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
PSM_ACCESS_ERROR	6	PSM Access Error	0b0: No error occurred 0b1: Error occurred
PSM_STATE_TIMER_ERROR	5	PSM State Timer Error	0b0: No timeout occurred 0b1: Timeout occurred
PSM_DATA_TIMER_ERROR	4	PSM Data Access Timer Error	0b0: No timeout occurred 0b1: Timeout occurred
PSM_STARTUP_MBIST_FAILED	1	MBIST of PSM Memory at Start-Up Error Status	0b0: No error detected 0b1: Error detected
PSM_STARTUP_MBIST_DONE	0	MBIST of PSM ROM at Start-Up Completion Status	0b0: Not done 0b1: Done

POWER_STATUS_0 (0xBA)

BIT	7	6	5	4	3	2	1	0
Field	–	VDDIO_UNDERVOLTAGE_ERROR	VDD18_UNDERVOLTAGE_ERROR	VDD_BIAS_UNDERVOLTAGE_ERROR	VDD_UNDERVOLTAGE_ERROR	VDDIO_OVERVOLTAGE_ERROR	VDD18_OVERVOLTAGE_ERROR	VDD_OVERVOLTAGE_ERROR
Reset	–	0x0	0x0	0x0	0x0	0x0	0x0	0x0
Access Type	–	Read Only	Read Only	Read Clears All	Read Only	Read Clears All	Read Clears All	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
VDDIO_UNDERVOLTAGE_ERROR	6	Latched when switched V_{DDIO} supply < 1.625V. Cleared on read of this POWER_STATUS_0.	0b0: No error detected 0b1: Error detected
VDD18_UNDERVOLTAGE_ERROR	5	Latched when V_{DD18} < 1.625V. Cleared on read of this POWER_STATUS_0.	0b0: No error detected 0b1: Error detected

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BITFIELD	BITS	DESCRIPTION	DECODE
VDD_BIAS_UNDERVOLTAGE_ERROR	4	Power Manager switched 1V supply comparator value fell below 0.806V. Cleared on read of this POWER_STATUS_0.	0b0: V_{DD} undervoltage detected 0b1: V_{DD} undervoltage detected
VDD_UNDERVOLTAGE_ERROR	3	Latched when V_{DD_sw} < 0.806V. Cleared on read of this POWER_STATUS_0.	0b0: No error detected 0b1: Error detected
VDDIO_OVERVOLTAGE_ERROR	2	V_{DDIO} overvoltage detected. Cleared on read of this POWER_STATUS_0.	0b0: No overvoltage detected 0b1: Overvoltage detected
VDD18_OVERVOLTAGE_ERROR	1	V_{DD18} overvoltage detected. Cleared on read of this POWER_STATUS_0.	0b0: No overvoltage detected 0b1: Overvoltage detected
VDD_OVERVOLTAGE_ERROR	0	V_{DD} overvoltage detected. Cleared on read of this POWER_STATUS_0.	0b0: No overvoltage detected 0b1: Overvoltage detected

CFG_PIN_CFG_0 (0x103)

BIT	7	6	5	4	3	2	1	0
Field	CFG_PIN_0_OVR_EN	CFG_PIN_0_SET[2:0]			CFG_PIN_1_OVR_EN	CFG_PIN_1_SET[2:0]		
Reset	0x0	0x000			0x0	0x000		
Access Type	Write, Read	Write, Read			Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
CFG_PIN_0_OVR_EN	7	Override enable of CFG0 value	0b0: Override not enabled 0b1: Override enabled
CFG_PIN_0_SET	6:4	Value to override CFG0, if enabled	
CFG_PIN_1_OVR_EN	3	Override enable of CFG1 value	0b0: Override not enabled 0b1: Override enabled
CFG_PIN_1_SET	2:0	Value to override CFG1, if enabled	

LINK_CFG_0 (0x105)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	LINK_EN_SIOA	LINK_EN_SIOB	LINK_EN_DCIO	–	–	RSVD	RSVD	RSVD
Reset	0b1	0b0	0x1	–	–	0b1	0b1	0b1
Access Type	Write, Read	Write, Read	Write, Read	–	–	Write, Read	Write, Read	Write, Read

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BITFIELD	BITS	DESCRIPTION	DECODE
LINK_EN_SIOA	7	Enables GMSL Link SIOA	0b0: Link A disabled 0b1: Link A enabled
LINK_EN_SIOB	6	Enables GMSL Link SIOB	0b0: Link B disabled 0b1: Link B enabled
LINK_EN_DCIO	5	Enable DCIO Link	0b0: Daisy-chain link disabled 0b1: Daisy-chain link enabled
RSVD	2	Selects which SIO pin is used when COAX mode is selected (CX_TP_SIO = 1)	0: SIOxN pin used for COAX 1: SIOxP pin used for COAX
RSVD	1	Selects which SIO pin is used when COAX mode is selected (CX_TP_SIO = 1)	0: SIOxN pin used for COAX 1: SIOxP pin used for COAX
RSVD	0	Selects which DCIO pin is used when COAX mode is selected (CX_TP_DCIO = 1)	0: DCION pin used for COAX 1: DCIOP pin used for COAX

LINK_CFG_1 (0x106)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	CX_TP_SIO	–	GMSL3_MODE_SIO	GMSL_FEC_EN_SIO	–	–	RX_RATE_SIO[1:0]	
Reset	0b0	–	0x1	0b0	–	–	0b10	
Access Type	Write, Read	–	Write, Read	Write, Read	–	–	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
CX_TP_SIO	7	Coax vs. Twisted Pair (single ended vs. differential) selection for the SIO PHYs (SIOA and SIOB). Bit is set according to the latched CFG pin value at power-up.	0b0: Shielded twisted-pair drive 0b1: Coax drive
GMSL3_MODE_SIO	5	Selects which mode SIOA and SIOB PHYs are operating in. Different modes cannot be selected for SIOA vs. SIOB.	0b0: GMSL2 mode 0b1: GMSL3 mode
GMSL_FEC_EN_SIO	4	Enables FEC on the SIOA and SIOB PHYs. FEC is mandatory for GMSL3 operation and is optional (but recommended) for GMSL2 operation if the corresponding serializer supports FEC. This bit is set high when the device is put into GMSL3 mode via CFG pins. Change activated upon link reset.	0x0: Disable FEC receive operation 0x1: Enable FEC receive operation
RX_RATE_SIO	1:0	Selects the Rx rate for the SIOA and SIOB GMSL links (forward channel). SIOA and SIOB are set to the same Rx rate. When changed, becomes active after next link reset).	00: Reserved 01: Reserved 10: 6Gbps 11: 12Gbps

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LINK_CFG_2 (0x107)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	CX_TP_DCIO	–	GMSL3_MODE_DCIO	GMSL_FEC_EN_DCIO	TX_RATE_DCIO[1:0]		–	–
Reset	0b0	–	0x1	0b1	0b10		–	–
Access Type	Write, Read	–	Write, Read	Write, Read	Write, Read		–	–

BITFIELD	BITS	DESCRIPTION	DECODE
CX_TP_DCIO	7	Coax/twisted-pair cable select for Link DCIO. Bit is set according to the latched CFG pin value at power-up.	0b0: Shielded twisted-pair drive 0b1: Coax drive
GMSL3_MODE_DCIO	5	Selects which mode DCIO PHY is operating in.	0x0: GMSL2 mode 0x1: GMSL3 mode
GMSL_FEC_EN_DCIO	4	Enables FEC on the DCIO PHY. FEC is mandatory for GMSL3 operation and is optional (but recommended) for GMSL2 operation if the corresponding serializer supports FEC. This bit is set high when the device is put into GMSL3 mode via CFG pins. Change activated upon link reset.	0b0: FEC is off 0b1: FEC is on
TX_RATE_DCIO	3:2	Transmit Rate on DCIO Link When changing this value, restart the link with a write to the RESET_ONESHOT or RESET_LINK register for the new value to take effect. Default value is set by CFG pin at power-up.	00: Reserved 01: 3Gbps 10: 6Gbps 11: 12Gbps

LINK_CFG_3 (0x108)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	FEC_STATS_RESET_SIOA	FEC_STATS_RESET_SIOB
Reset	–	–	–	–	–	–	0b0	0b0
Access Type	–	–	–	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
FEC_STATS_RESET_SIOA	1	Reset the FEC statistics on blocks processed, corrected bit errors and uncorrectable bit errors on SIOA link.	0b0: No action 0b1: Reset the FEC statistics for SIOA

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BITFIELD	BITS	DESCRIPTION	DECODE
		This bit is self-clearing.	
FEC_STATS_RESET_SIOB	0	Reset the FEC statistics on blocks processed, corrected bit errors and uncorrectable bit errors on SIOB link. This bit is self-clearing.	0b0: No action 0b1: Reset the FEC statistics for SIOB

ERRG_CFG_0 (0x10A)

BIT	7	6	5	4	3	2	1	0
Field	ERRG_EN_SIOA	ERRG_EN_SIOB	ERRG_EN_DCIO	–	–	–	–	–
Reset	0b0	0b0	0b0	–	–	–	–	–
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
ERRG_EN_SIOA	7	Enable error generator on SIOA, using settings defined in register ERRG_CFG_1	0b0: Link SIOA error generator disabled 0b1: Link SIOA error generator enabled
ERRG_EN_SIOB	6	Enable error generator on SIOB, using settings defined in register ERRG_CFG_1	0b0: Link SIOB error generator disabled 0b1: Link SIOB error generator enabled
ERRG_EN_DCIO	5	Enable error generator on DCIO, using settings defined in register ERRG_CFG_1	0b0: Link DCIO error generator disabled 0b1: Link DCIO error generator enabled

ERRG_CFG_1 (0x10B)

BIT	7	6	5	4	3	2	1	0
Field	ERRG_MODE	ERRG_BURST[2:0]			ERRG_BER[1:0]		ERRG_COUNT[1:0]	
Reset	0b0	0b000			0b10		0b00	
Access Type	Write, Read	Write, Read			Write, Read		Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
ERRG_MODE	7	Selects the error generation distribution between a pseudo random distribution vs. a periodic distribution	0b0: Pseudo random 0b1: Periodic
ERRG_BURST	6:4	For each error trigger (at rate selected by ERRG_BER[1:0]), selects the number of burst errors to generate. For a burst error window, the	0b000: 1 bit 0b001: 2 bits 0b010: 3 bits 0b011: 4 bits

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BITFIELD	BITS	DESCRIPTION	DECODE
		first and last bit are flipped and the middle bits in the window have a 50% chance of being flipped. For example, if the bitfield value is 0b100, the burst-error length is 8 bits. Thus, each time an 8-bit error burst is generated, there is an error in the first and last bit of the 8-bit window, and the probability of an error in each of the remaining 6 bits is 50%.	0b100: 8 bits 0b101: 12 bits 0b110: 16 bits 0b111: 20 bits
ERRG_BER	3:2	Bit Error Rate for the Error Generator	0x0: 1 error triggered in 5120 bits (BER approx. 2.0×10^{-4}) 0x1: 1 error triggered in 81920 bits (BER approx. 1.2×10^{-5}) 0x2: 1 error triggered in 1310720 bits (BER approx. 7.6×10^{-7}) 0x3: 1 error triggered in 20971520 bits (BER approx. 4.8×10^{-8})
ERRG_COUNT	1:0	Number of errors to be generated when error generator is enabled (ERRG_EN_XXX transitions from 0 to 1).	0b00: Continuous error generation. Use this mode to get continuous errors at the BER defined in ERRG_BER[1:0]. 0b01: 16 errors 0b10: 128 errors 0b11: 1024 errors

ERROR_CFG_0 (0x10D)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	ERRB_PIN_OUTPUT_EN	LINK_SIOA_ERR_OEN	LINK_SIOB_ERR_OEN	LINK_DCIO_ERR_OEN	–	–	–	PSM_STATE_ERR_OEN
Reset	0b1	0x1	0x1	0x1	–	–	–	0x1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
ERRB_PIN_OUTPUT_EN	7	Enables overall error status (reported in ERR_OUTPUT) to the ERRB pin.	0b0: ERRB output disabled 0b1: ERRB output enabled
LINK_SIOA_ERR_OEN	6	Output enable to report link SIOA errors to overall error status. NOTE: The reset value of this register changes based on whether FEC on the link is enabled or not. If FEC is enabled, the reset value of this register is 0. User must explicitly write 1 to this register to enable SIOA link error to be reported to ERRB pin. If FEC is not enabled, the reset value of this register is 1.	0b0: Error not output to overall error status 0b1: Error output to overall error status
LINK_SIOB_ERR_OEN	5	Output enable to report link SIOB errors to overall error status.	0b0: Error not output to overall error status 0b1: Error output to overall error status

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BITFIELD	BITS	DESCRIPTION	DECODE
		NOTE: The reset value of this register changes based on whether FEC on the link is enabled or not. If FEC is enabled, the reset value of this register is 0. User must explicitly write 1 to this register to enable SIOB link error to be reported to ERRB pin. If FEC is not enabled, the reset value of this register is 1.	
LINK_DCIO_ERR_OEN	4	Output enable to report link DCIO errors to overall error status.	0b0: Error not output to overall error status 0b1: Error output to overall error status
PSM_STATE_ERR_OEN	0	Output enable to report PSM state errors to overall error status.	0b0: Error not output to overall error status 0b1: Error output to overall error status

ERROR_CFG_1 (0x10E)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	SIOA_FEC_ERR_OEN	SIOB_FEC_ERR_OEN	RETENTION_MEMORY_CRC_ERR_OEN	CONTROL_CHANNEL_MSG_COUNT_ERR_OEN	–	LINEFAULT_0_ERR_OEN	LINEFAULT_1_ERR_OEN	CLOCK_MONITOR_ERR_OEN
Reset	0b1	0b1	0x1	0b0	–	0b1	0b1	0x1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
SIOA_FEC_ERR_OEN	7	Output enable to report SIOA FEC errors to overall error status. When FEC is enabled, this bit is automatically set.	0b0: Error not output to overall error status 0b1: Error output to overall error status
SIOB_FEC_ERR_OEN	6	Output enable to report SIOB FEC errors to overall error status. When FEC is enabled, this bit is automatically set.	0b0: Error not output to overall error status 0b1: Error output to overall error status
RETENTION_MEMORY_CRC_ERR_OEN	5	Enables reporting on retention memory CRC check error to ERRB pin	0b0: Error not output to overall error status 0b1: Error output to overall error status
CONTROL_CHANNEL_MSG_COUNT_ERR_OEN	4	Enables reporting of I^2C/UART message count errors	0b0: Error not output to overall error status 0b1: Error output to overall error status

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BITFIELD	BITS	DESCRIPTION	DECODE
		(I2C_UART_MSGCNTR_ERR_FLAG) at ERRB pin.	
LINEFAULT_0_ERR_OEN	2	Output enable to report Line Fault 0 errors to overall error status	0b0: Error not output to overall error status 0b1: Error output to overall error status
LINEFAULT_1_ERR_OEN	1	Output enable to report Line Fault 1 errors to overall error status	0b0: Error not output to overall error status 0b1: Error output to overall error status
CLOCK_MONITOR_ERR_OEN	0	Enable reporting of clock monitor error on ERRB pin	0b0: Error not output to overall error status 0b1: Error output to overall error status

ERROR_CFG_2 (0x10F)

BIT	7	6	5	4	3	2	1	0
Field	REMOTE_ERR_OEN	ARQ_ERR_OEN	INFOFRAME_CRC_ERR_OEN	VIDEO_PRBS_ERR_OEN	AUDIO_PRBS_ERR_OEN	OVERVOLTAGE_ERR_OEN	UNDERVOLTAGE_ERR_OEN	–
Reset	0b0	0b1	0x1	0b1	0b1	0b1	0b1	–
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–

BITFIELD	BITS	DESCRIPTION	DECODE
REMOTE_ERR_OEN	7	Output enable to report remote device ERRB to overall error status.	0b0: Error not output to overall error status 0b1: Error output to overall error status
ARQ_ERR_OEN	6	Output enable to report ARQ errors to overall error status.	0b0: Error not output to overall error status 0b1: Error output to overall error status
INFOFRAME_CRC_ERR_OEN	5	Enable reporting of info frame CRC error to ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
VIDEO_PRBS_ERR_OEN	4	Enables reporting of video PRBS error at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
AUDIO_PRBS_ERR_OEN	3	Enables reporting of audio PRBS errors at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
OVERVOLTAGE_ERR_OEN	2	Enables V_{DD} overvoltage status on ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
UNDERVOLTAGE_ERR_OEN	1	Enables reporting of V_{DD} undervoltage interrupt at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status

ERROR_CFG_3 (0x110)

See Note §.

BIT	7	6	5	4	3	2	1	0

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Field	VIDEO_SEQUENCE_ERR_OEN	LOSS_OF_VIDEO_LOCK_ERR_OEN	HDCP_ERR_OEN	REG_CRC_ERR_OEN	DSC_ERR_OEN	WATERMARK_ERR_OEN	VIDEO_BLOCK_LENGTH_ERR_OEN	LOSS_OF_VIDEO_ALIGNMENT_ERR_OEN
Reset	0b1	0b1	0b0	0b0	0x1	0b1	0b1	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
VIDEO_SEQUENCE_ERR_OEN	7	Enables reporting of combined video sequence error flag at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
LOSS_OF_VIDEO_LOCK_ERR_OEN	6	Enables reporting loss of video lock on ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
HDCP_ERR_OEN	5	Enables reporting of GMSL HDCP error (HDCP_ERR) to ERRB pin.	0b0: HDCP error not propagated to overall error status 0b1: HDCP error propagate to overall error status
REG_CRC_ERR_OEN	4	Enable reporting register CRC at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
DSC_ERR_OEN	3	Enables reporting of DSC decoder errors (DSC_ERR) at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
WATERMARK_ERR_OEN	2	Enables reporting of watermark errors (WM_ERR) at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
VIDEO_BLOCK_LENGTH_ERR_OEN	1	Enables reporting of video block length error flag at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
LOSS_OF_VIDEO_ALIGNMENT_ERR_OEN	0	Enables reporting of error due to loss of video alignment to ERRB pin.	0x0: Disable reporting 0x1: Enable reporting

ERROR_CFG_4 (0x111)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_ERR_OEN	FEEDFORWARD_FRAME_CRC_ERR_OEN	COMBINED_FRAME_CRC_ERR_OEN	LINE_CRC_ERR_OEN	VIDEO_PACKET_CRC_ERR_OEN	–	DP_SINK_SYMBOL_ERR_OEN	–
Reset	0b1	0b0	0x0	0b1	0b1	–	0x1	–
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–	Write, Read	–

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BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CRC_ERR_OEN	7	Enables reporting of video frame CRC errors (FRAME_CRC_ERR) at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
FEEDFORWARD_FRAME_CRC_ERR_OEN	6	Enables reporting of video frame CRC errors (FEEDFORWARD_FRAME_CRC_ERR) at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
COMBINED_FRAME_CRC_ERR_OEN	5	Enables reporting of feedforward frame CRC error detected on combined even and odd pixel frames to ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
LINE_CRC_ERR_OEN	4	Enables reporting of video line CRC errors (LINE_CRC_ERR) at ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
VIDEO_PACKET_CRC_ERR_OEN	3	Enables reporting of video pixel CRC errors (VIDEO_PACKET_CRC_ERR) at ERRB pin.	0x0: Error not output to overall error status 0x1: Error output to overall error status
DP_SINK_SYMBOL_ERR_OEN	1	Enables reporting of DP_SINK_SYMBOL_ERR to ERRB pin.	0x0: Do not enable to ERRB 0x1: Enable to ERRB

ERROR_CFG_5 (0x112)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	DP_INPUT_FIFO_CRC_ERR_OEN	WM_BYPASS_ERR_OEN	LUT_FRC_ERR_OEN	DPTX_FIFO_OVERFLOW_ERR_OEN	TELLTALE_ERR_OEN	LOSS_OF_LOCK_OEN	_	CONTROL_CHANNEL_CRC_ERR_OEN
Reset	0x1	0x1	0x1	0b1	0b0	0b0	–	0x0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
DP_INPUT_FIFO_CRC_ERR_OEN	7	Enables reporting on ASIL CRC error detected on the align FIFOs to ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
WM_BYPASS_ERR_OEN	6	Enables reporting of watermark ASIL bypass error on ERRB.	0x0: Error not output to overall error status 0x1: Error output to overall error status
LUT_FRC_ERR_OEN	5	Enables reporting of LUT and FRC ASIL data path errors to ERBB pin.	0x0: Error not output to overall error status 0x1: Error output to overall error status
DPTX_FIFO_OVERFLOW_ERR_OEN	4	Enables reporting of DPTX controller FIFO overflow error on ERRB pin.	0x0: Error not output to overall error status 0x1: Error output to overall error status
TELLTALE_ERR_OEN	3	Enables reporting of telltale error on ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status

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BITFIELD	BITS	DESCRIPTION	DECODE
LOSS_OF_LOCK_OEN	2	Enables reporting loss of lock on ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
CONTROL_CHANNEL_CRC_ERR_OEN	0	Enables reporting of detected CRC check error on control channel packets to ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status

ERROR_CFG_6 (0x113)

BIT	7	6	5	4	3	2	1	0
Field	OTP_CRC_ERR_OEN	_	VIDEO_CFG_ERR_OEN	DP_LINK_TRAINING_ERR_OEN	DP_SINK_CAPABILITY_ERR_OEN	AUX_ERR_OEN	HPD_IRQ_ERR_OEN	HPD_DISCONNECT_ERR_OEN
Reset	0x0	–	0x1	0b1	0x1	0x1	0x1	0x1
Access Type	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
OTP_CRC_ERR_OEN	7	Enable reporting of OTP CRC check to ERRB pin.	0b0: Error not output to overall error status 0b1: Error output to overall error status
VIDEO_CFG_ERR_OEN	5	Enables reporting of video configuration error detected by control state.	0b0: Error not output to overall error status 0b1: Error output to overall error status
DP_LINK_TRAINING_ERR_OEN	4	When this bit is enabled, ERRB will be asserted when DP link training error occurs.	0b0: Error not output to overall error status 0b1: Error output to overall error status
DP_SINK_CAPABILITY_ERR_OEN	3	Enable reporting of DP_SINK_CAPABILITY_ERR on ERRB.	0b0: Error not output to overall error status 0b1: Error output to overall error status
AUX_ERR_OEN	2	Enable reporting of DP AUX_ERR on ERRB.	0b0: Error not output to overall error status 0b1: Error output to overall error status
HPD_IRQ_ERR_OEN	1	Enable reporting of HPD_IRQ_ERR on ERRB.	0b0: Error not output to overall error status 0b1: Error output to overall error status
HPD_DISCONNECT_ERR_OEN	0	Enable reporting of HPD_DISCONNECT_ERR on ERRB.	0b0: Error not output to overall error status 0b1: Error output to overall error status

CC_CFG_0 (0x116)

BIT	7	6	5	4	3	2	1	0
Field	CC_LOCAL_OUT_EN	DIS_REM_CC	I2CSEL	UART_BYPASS_EN	–	–	CC_CRC_EN	CC_MESSAGE_COUNT_EN

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Reset	0x1	0b0	0b0	0b0	–	–	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
CC_LOCAL_OUT_EN	7	Enable local control-channel output to SDA_RX and SCL_TX pins. Note that when this bit is enabled, it prevents any further local communication with the device until this bit is set back high, which can only be done by the remote device.	0x0: Control channel disabled to local pins 0x1: Control channel enabled to local pins
DIS_REM_CC	6	Disable remote control-channel connection to serializer across the SIO link.	0b0: Control channel enabled to remote serializer 0b1: Control channel disabled to remote serializer
I2CSEL	5	I²C/UART selection Bit is set according to the latched CFG pin value at power-up.	0b0: UART 0b1: I²C
UART_BYPASS_EN	4	Enables UART soft-bypass mode. Bypass mode remains active as long as there is UART activity. When there is no UART activity for the selected duration (configured by BYPASS_TO), device exits Bypass mode, and the bit is automatically cleared.	0b0: UART soft-bypass mode disabled 0b1: UART soft-bypass mode enabled
CC_CRC_EN	1	Enable I²C/UART CRC	0x0: I²C/UART CRC disabled 0x1: I²C/UART CRC enabled
CC_MESSAGE_COUNT_EN	0	Enable I²C/UART message counter	0x0: I²C/UART Message Counter disabled 0x1: I²C/UART Message Counter enabled

CC_CFG_1 (0x117)

BIT	7	6	5	4	3	2	1	0
Field	CC_LINK_SEL	CC_PRIORITY[1:0]		–	–	–	–	–
Reset	0x0	0b01		–	–	–	–	–
Access Type	Write, Read	Write, Read		–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
CC_LINK_SEL	7	Selects the link that the control channel is connected to	0x0: Connect to SIOA 0x1: Connect to SIOB
CC_PRIORITY	6:5	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority

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CC_CFG_2 (0x118)

BIT	7	6	5	4	3	2	1	0
Field	–	–	CC_I2C_SPEED_SLV[2:0]			CC_I2C_SPEED_MST[2:0]		
Reset	–	–	0b101			0b101		
Access Type	–	–	Write, Read			Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
CC_I2C_SPEED_SLV	5:3	I²C-to-I²C Subordinate Speed Select	0x0: 9.92Kbps (I²C Standard) 0x1: 33.2Kbps (I²C Standard) 0x2: 99.2Kbps (I²C Standard or Fast mode) 0x3: 123Kbps (I²C Fast mode) 0x4: 203Kbps (I²C Fast mode) 0x5: 397Kbps (I²C Fast mode or Fast-mode Plus) 0x6: 625Kbps (I²C Fast-mode Plus) 0x7: 980Kbps (I²C Fast-mode Plus)
CC_I2C_SPEED_MST	2:0	I²C-to-I²C Main Bit-Rate Setting Configures the I²C bit rate used by the internal I²C main (in the device on the remote side from the external I²C main). Set this according to the I²C speed mode.	0b000: 9.92Kbps - Set for I²C Standard mode speed 0b001: 33.2Kbps - Set for I²C Standard mode speed 0b010: 99.2Kbps - Set for I²C Standard or Fast-mode speed 0b011: 123Kbps - Set for I²C Fast-mode speed 0b100: 203Kbps - Set for I²C Fast-mode speed 0b101: 397Kbps - Set for I²C Fast or Fast-mode Plus speed 0b110: 625Kbps - Set for I²C Fast-mode Plus speed 0b111: 980Kbps - Set for I²C Fast-mode Plus speed

CC_CFG_3 (0x119)

BIT	7	6	5	4	3	2	1	0
Field	CC_I2C_TRANSLATION_0_SOURCE[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
CC_I2C_TRANSLATION_0_SOURCE	7:1	I²C Address Translator Source 1 When I²C device address matches I2C_TRANSLATE_0_SOURCE, internal I²C main (on remote side) replaces the device address by I2C_TRANSLATION_0_DESTINATION.	0bXXXXXXX: Value of I2C_TRANSLATION_0_SOURCE

CC_CFG_4 (0x11A)

BIT	7	6	5	4	3	2	1	0

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Field	CC_I2C_TRANSLATION_0_DESTINATION[6:0]	–
Reset	0b0000000	–
Access Type	Write, Read	–

BITFIELD	BITS	DESCRIPTION	DECODE
CC_I2C_TRANSLATION_0_DESTINATION	7:1	I²C Address Translator Destination 0 See the description of I2C_TRANSLATION_0_SOURCE.	0bXXXXXXX: Value of I2C_TRANSLATION_0_DESTINATION

CC_CFG_5 (0x11B)

BIT	7	6	5	4	3	2	1	0
Field	CC_I2C_TRANSLATION_1_SOURCE[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
CC_I2C_TRANSLATION_1_SOURCE	7:1	I²C Address Translator Source 1 When I²C device address matches I2C_TRANSLATE_1_SOURCE, internal I²C main (on remote side) replaces the device address by I2C_TRANSLATION_1_DESTINATION.	0bXXXXXXX: Value of I2C_TRANSLATION_1_DESTINATION

CC_CFG_6 (0x11C)

BIT	7	6	5	4	3	2	1	0
Field	CC_I2C_TRANSLATION_1_DESTINATION[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
CC_I2C_TRANSLATION_1_DESTINATION	7:1	I²C Address Translator Destination 1 See the description of I2C_TRANSLATION_1_SOURCE	0bXXXXXXX: Value of I²C TRANSLATION_1_DESTINATION

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CC_CFG_7 (0x11D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	RESET_MSGCNTR	–	–	RESET_MSGCNTR_ERR_CNT	RESET_CC_CRC_ERR_CNT
Reset	–	–	–	0b0	–	–	0b0	0b0
Access Type	–	–	–	Write Clears All, Read	–	–	Write Clears All, Read	Write Clears All, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RESET_MSGCNTR	4	Reset MSGCNTR_MSB (0x1F22) and MSGCNTR_LSB (0x1F21) to 0x00 (this bit automatically clears itself).	0b0: No action 0b1: Issue reset
RESET_MSGCNTR_ERR_CNT	1	Reset MSGCNTR_ERR_CNT (0x1D24) to 0x00 (this bit automatically clears).	0b0: No action 0b1: Issue reset
RESET_CC_CRC_ERR_CNT	0	Reset CC_CRC_ERR_CNT (0x1D23) to 0x00 (this bit automatically clears).	0b0: No action 0b1: Issue reset

PT_CFG_0 (0x11E)

BIT	7	6	5	4	3	2	1	0
Field	PT1_EN	PT2_EN	PT1_I2CSEL	PT2_I2CSEL	PT1_LINK_SEL	PT2_LINK_SEL	PT1_MANUAL_MODE_EN	PT2_MANUAL_MODE_EN
Reset	0b0	0b0	0b0	0b0	0x0	0x0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
PT1_EN	7	Enables pass-through Channel 1 (SDA1/RX1, SCL1/TX1).	0b0: Disable I^2C pass-through Channel 1 0b1: Enable I^2C pass-through Channel 1
PT2_EN	6	Enables pass-through Channel 2 (SDA2/RX2, SCL2/TX2).	0b0: Disable I^2C pass-through Channel 2 0b1: Enable I^2C pass-through Channel 2
PT1_I2CSEL	5	Configures I^2C or UART pass-through Channel 1 (SDA1/RX1, SCL1/TX1).	0b0: Pass-through Channel 1 in UART mode 0b1: Pass-through Channel 1 in I^2C mode
PT2_I2CSEL	4	Configures I^2C or UART pass-through Channel 2 (SDA2/RX2, SCL2/TX2).	0b0: Pass-through Channel 2 in UART mode 0b1: Pass-through Channel 2 in I^2C mode
PT1_LINK_SEL	3	Selects the link that the PT1 is connected to.	0x0: Connect to SIOA 0x1: Connect to SIOB
PT2_LINK_SEL	2	Selects the link that the PT2 is connected to.	0x0: Connect to SIOA 0x1: Connect to SIOB

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BITFIELD	BITS	DESCRIPTION	DECODE
PT1_MANUAL_MODE_EN	1	Uses the custom UART bit rate (selected by the PT1_UART_BIT_LENGTH_BYTE_0/1 registers) in pass-through UART Channel 1.	0b0: Use standard bit rate 0b1: Use custom bit rate
PT2_MANUAL_MODE_EN	0	Uses the custom UART bit rate (selected by the PT2_UART_BIT_LENGTH_BYTE_0/1 registers) in pass-through UART Channel 2.	0b0: Use standard bit rate 0b1: Use custom bit rate

PT_CFG_2 (0x120)

BIT	7	6	5	4	3	2	1	0
Field	PT1_UART_BIT_LENGTH_BYTE_0[7:0]							
Reset	0xDC							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PT1_UART_BIT_LENGTH_BYTE_0	7:0	Low byte of custom UART bit length for pass-through UART Channel 1. Set this register to the UART bit length divided by 6.666ns (LSb 8 bits). Set PT1_MANUAL_MODE to 1 to use this value.	0xXX: Low byte of custom UART bit length for pass-through UART Channel 1

PT_CFG_3 (0x121)

BIT	7	6	5	4	3	2	1	0
Field	–	–	PT1_UART_BIT_LENGTH_BYTE_1[5:0]					
Reset	–	–	0b000101					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
PT1_UART_BIT_LENGTH_BYTE_1	5:0	High byte of custom UART bit length for pass-through UART Channel 1. Set this register to the UART bit length divided by 6.666ns (LSb 8 bits). Set PT1_MANUAL_MODE to 1 to use this value.	0xXX: High byte of custom UART bit length for pass-through UART Channel 1

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PT_CFG_4 (0x122)

BIT	7	6	5	4	3	2	1	0
Field	–	–	PT1_I2C_SPEED_SLV[2:0]			PT1_I2C_SPEED_MST[2:0]		
Reset	–	–	0b101			0b101		
Access Type	–	–	Write, Read			Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
PT1_I2C_SPEED_SLV	5:3	Pass-through channel I²C-to-I²C subordinate speed select.	0x0: 9.92Kbps (I²C Standard) 0x1: 33.2Kbps (I²C Standard) 0x2: 99.2Kbps (I²C Standard or Fast mode) 0x3: 123Kbps (I²C Fast mode) 0x4: 203Kbps (I²C Fast mode) 0x5: 397Kbps (I²C Fast mode or Fast-mode Plus) 0x6: 625Kbps (I²C Fast-mode Plus) 0x7: 980Kbps (I²C Fast-mode Plus)
PT1_I2C_SPEED_MST	2:0	Pass-through I²C-to-I²C main bit rate setting. Configures the I²C bit rate used by the internal I²C main in the device on remote side from the external I²C main. Set this according to the I²C Speed mode: Fast-mode Plus = 101 to 111 Fast-mode = 010 to 101 Standard-mode = 000 to 010	0b000: 9.92Kbps - Set for I²C Standard mode speed 0b001: 33.2Kbps - Set for I²C Standard mode speed 0b010: 99.2Kbps - Set for I²C Standard or Fast-mode speed 0b011: 123Kbps - Set for I²C Fast-mode speed 0b100: 203Kbps - Set for I²C Fast-mode speed 0b101: 397Kbps - Set for I²C Fast or Fast-mode Plus speed 0b110: 625Kbps - Set for I²C Fast-mode Plus speed 0b111: 980Kbps - Set for I²C Fast-mode Plus speed

PT_CFG_5 (0x123)

BIT	7	6	5	4	3	2	1	0
Field	PT1_I2C_TRANSLATION_0_SOURCE[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION
PT1_I2C_TRANSLATION_0_SOURCE	7:1	I²C address translation 0 source address. Program this bitfield with the address to be translated from. Keep at default when not using address translation.

PT_CFG_6 (0x124)

BIT	7	6	5	4	3	2	1	0
Field	PT1_I2C_TRANSLATION_0_DESTINATION[6:0]							–

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Reset	0b0000000	–
Access Type	Write, Read	–

BITFIELD	BITS	DESCRIPTION
PT1_I2C_TRANSLATION_0_DESTINATION	7:1	I²C address translation 0 destination address. Program this bitfield with the address to be translated to. Keep at default when not using address translation.

PT_CFG_7 (0x125)

BIT	7	6	5	4	3	2	1	0
Field	PT1_I2C_TRANSLATION_1_SOURCE[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION
PT1_I2C_TRANSLATION_1_SOURCE	7:1	I²C address translation 1 source address. Program this bitfield with the address to be translated from. Keep at default when not using address translation.

PT_CFG_8 (0x126)

BIT	7	6	5	4	3	2	1	0
Field	PT1_I2C_TRANSLATION_1_DESTINATION[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION
PT1_I2C_TRANSLATION_1_DESTINATION	7:1	I²C address translation 1 destination address. Program this bitfield with the address to be translated to. Keep at default when not using address translation.

PT_CFG_10 (0x128)

BIT	7	6	5	4	3	2	1	0
Field	PT2_UART_BIT_LENGTH_BYTE_0[7:0]							
Reset	0xDC							

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Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
PT2_UART_BIT_LENGTH_BYTE_0	7:0	Low byte of custom UART bit length for pass-through UART Channel 2. Set this register to the UART bit length divided by 6.666ns (LSb 8 bits). Set PT2_MANUAL_MODE to 1 to use this value.	0xXX: Low byte of custom UART bit length for pass-through UART Channel 2

PT_CFG_11 (0x129)

BIT	7	6	5	4	3	2	1	0
Field	–	–	PT2_UART_BIT_LENGTH_BYTE_1[5:0]					
Reset	–	–	0b000101					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION	DECODE
PT2_UART_BIT_LENGTH_BYTE_1	5:0	High byte of custom UART bit length for pass-through UART Channel 2. Set this register to the UART bit length divided by 6.666ns (LSb 8 bits). Set PT2_MANUAL_MODE to 2 to use this value.	0xXX: High byte of custom UART bit length for pass-through UART Channel 2

PT_CFG_12 (0x12A)

BIT	7	6	5	4	3	2	1	0
Field	–	–	PT2_I2C_SPEED_SLV[2:0]			PT2_I2C_SPEED_MST[2:0]		
Reset	–	–	0b101			0b101		
Access Type	–	–	Write, Read			Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
PT2_I2C_SPEED_SLV	5:3	Pass-through channel I²C-to-I²C subordinate speed select.	0x0: 9.92Kbps (I²C Standard) 0x1: 33.2Kbps (I²C Standard) 0x2: 99.2Kbps (I²C Standard) or Fast mode) 0x3: 123Kbps (I²C Fast mode) 0x4: 203Kbps (I²C Fast mode) 0x5: 397Kbps (I²C Fast mode or Fast-mode Plus) 0x6: 625Kbps (I²C Fast-mode Plus) 0x7: 980Kbps (I²C Fast-mode Plus)

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BITFIELD	BITS	DESCRIPTION	DECODE
PT2_I2C_SPEED_MST	2:0	Pass-through I²C-to-I²C main bit rate setting. Configures the I²C bit rate used by the internal I²C main in the device on the remote side from the external I²C main. Set this according to the I²C Speed mode: Fast-mode Plus = 101 to 111 Fast-mode = 010 to 101 Standard-mode = 000 to 010	0b000: 9.92Kbps - Set for I²C Standard mode speed 0b001: 33.2Kbps - Set for I²C Standard mode speed 0b010: 99.2Kbps - Set for I²C Standard or Fast-mode speed 0b011: 123Kbps - Set for I²C Fast-mode speed 0b100: 203Kbps - Set for I²C Fast-mode speed 0b101: 397Kbps - Set for I²C Fast or Fast-mode Plus speed 0b110: 625Kbps - Set for I²C Fast-mode Plus speed 0b111: 980Kbps - Set for I²C Fast-mode Plus speed

PT_CFG_13 (0x12B)

BIT	7	6	5	4	3	2	1	0
Field	PT2_I2C_TRANSLATION_0_SOURCE[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION
PT2_I2C_TRANSLATION_0_SOURCE	7:1	I²C address translation 0 source address. Program this bitfield with the address to be translated from. Keep at default when not using address translation.

PT_CFG_14 (0x12C)

BIT	7	6	5	4	3	2	1	0
Field	PT2_I2C_TRANSLATION_0_DESTINATION[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION
PT2_I2C_TRANSLATION_0_DESTINATION	7:1	I²C address translation 0 destination address. Program this bitfield with the address to be translated to. Keep at default when not using address translation.

PT_CFG_15 (0x12D)

BIT	7	6	5	4	3	2	1	0
Field	PT2_I2C_TRANSLATION_1_SOURCE[6:0]							–

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Reset	0b0000000	–
Access Type	Write, Read	–

BITFIELD	BITS	DESCRIPTION
PT2_I2C_TRANSLATION_1_SOURCE	7:1	I²C address translation 1 source address. Program this bitfield with the address to be translated from. Keep at default when not using address translation.

PT_CFG_16 (0x12E)

BIT	7	6	5	4	3	2	1	0
Field	PT2_I2C_TRANSLATION_1_DESTINATION[6:0]							–
Reset	0b0000000							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION
PT2_I2C_TRANSLATION_1_DESTINATION	7:1	I²C address translation 1 destination address. Program this bitfield with the address to be translated to. Keep at default when not using address translation.

SPI_CFG_0 (0x130)

BIT	7	6	5	4	3	2	1	0
Field	SPI_EN	MST_SEL	SPI_MODE[1:0]		SPI_IGNORE_ID	–	SPI_LOCAL_ID[1:0]	
Reset	0b0	0b1	0b00		0b1	–	0b00	
Access Type	Write, Read	Write, Read	Write, Read		Write, Read	–	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_EN	7	Enables SPI Channel	0b0: SPI channel disabled 0b1: SPI channel enabled
MST_SEL	6	Selects if SPI is Main or Subordinate	0b0: SPI subordinate 0b1: SPI main
SPI_MODE	5:4	Bit 0 selects mode 0 or 3. Bit 1 enables the suppression of an extra SCK prior to SS deassertion when SPI mode 3 is selected.	0b01: SPI mode 3 with extra SCK present prior to SS deassertion 0b10: Reserved 0b11: SPI mode 3 with extra SCK suppressed prior to SS deassertion
SPI_IGNORE_ID	3	Selects if SPI should use or ignore header ID to determine packet acceptance.	0b0: Accept only packets with proper ID 0b1: Ignore ID and accept all packets

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BITFIELD	BITS	DESCRIPTION	DECODE
SPI_LOCAL_ID	1:0	Program to local ID if filtering packets based on header ID.	0b00: ID = 0 0b01: ID = 1 0b10: ID = 2 0b11: ID = 3

SPI_CFG_1 (0x131)

BIT	7	6	5	4	3	2	1	0
Field	SPI_LINK_SEL	SPI_PRIORITY[1:0]		MISO_SAMPLE_EDGE	–	–	MST_SS_1_POL	MST_SS_2_POL
Reset	0x0	0b00		0b0	–	–	0x0	0x0
Access Type	Write, Read	Write, Read		Write, Read	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_LINK_SEL	7	Selects the link that the adapter is connected to.	0x0: Connect to SIOA 0x1: Connect to SIOB
SPI_PRIORITY	6:5	Sets the priority for this channel's packet requests.	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
MISO_SAMPLE_EDGE	4	Sample MISO after half or full SCK period.	0b0: MISO sampled after half SCK period 0b1: MISO sampled after full SCK period
MST_SS_1_POL	1	Select the polarity for SS1 when SPI is a main.	0x0: SS1 is active low 0x1: SS1 is active high
MST_SS_2_POL	0	Select the polarity for SS2 when SPI is a main.	0x0: SS2 is active low 0x1: SS2 is active high

SPI_CFG_2 (0x132)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	MST_SS_1_EN	MST_SS_2_EN	RO_INPUT_EN	BNE_OUTPUT_EN
Reset	–	–	–	–	0b1	0b1	0b0	0b0
Access Type	–	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
MST_SS_1_EN	3	Enables GPIO for use as Subordinate Select 1 output.	0b0: GPIO not used for SPI SS1 function 0b1: GPIO used for SPI SS1 function
MST_SS_2_EN	2	Enables GPIO for use as Subordinate Select 2 output.	0b0: GPIO not used for SPI SS2 function 0b1: GPIO used for SPI SS2 function

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BITFIELD	BITS	DESCRIPTION	DECODE
RO_INPUT_EN	1	Enables GPIO for use as RO input for control of SPI data movement.	0b0: GPIO not used for SPI RO function 0b1: GPIO used for SPI RO function
BNE_OUTPUT_EN	0	Enables GPIO for use as BNE output for SPI data available status.	0b0: GPIO not used for SPI BNE function 0b1: GPIO used for SPI BNE function

SPI_CFG_3 (0x133)

BIT	7	6	5	4	3	2	1	0
Field	SPI_SLAVE_SEL_DELAY[7:0]							
Reset	0x01							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_SLAVE_SEL_DELAY	7:0	Number of 300MHz clocks to delay between: 1. Assertion of SS and start of SCK pulses. 2. End of SCK pulses and deassertion of SS. 3. Deassertion of SS and reassertion of SS, if necessary.	0xXX: Number of clock delays

SPI_CFG_4 (0x134)

BIT	7	6	5	4	3	2	1	0
Field	SPI_SCLK_LOW_CLKS[7:0]							
Reset	0x0F							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
SPI_SCLK_LOW_CLKS	7:0	Number of 300MHz clocks for SCK low time	0xXX: Number of clocks for SCK low time

SPI_CFG_5 (0x135)

BIT	7	6	5	4	3	2	1	0
Field	SPI_SCLK_HIGH_CLKS[7:0]							
Reset	0x0F							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION	DECODE
SPI_SCLK_HIGH_CLKS	7:0	Number of 300MHz clocks for SCK high time.	0xXX: Number of clocks for SCK high time

AUDIO_CFG_0 (0x143)

BIT	7	6	5	4	3	2	1	0
Field	AUDIO_LINK_SEL	AUDIO_PRIORITY[1:0]		I2S_MODE	AUDIO_RX_ID[1:0]		AUDIO_PRBS_CHECKER_EN	AUDIO_RX_EN
Reset	0x0	0b01		0b0	0b00		0b0	0b0
Access Type	Write, Read	Write, Read		Write, Read	Write, Read		Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
AUDIO_LINK_SEL	7	Selects the link that the adapter is connected to.	0x0: Connect to SIOA 0x1: Connect to SIOB
AUDIO_PRIORITY	6:5	Sets the priority for this channel's packet requests.	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
I2S_MODE	4	Invert WS at I²S output.	0b0: Do not invert WS (I²S mode) 0b1: Invert WS (TDM mode)
AUDIO_RX_ID	3:2	Selects audio stream for reception.	0b00: Receive stream ID 0 0b01: Receive stream ID 1 0b10: Receive stream ID 2 0b11: Receive stream ID 3
AUDIO_PRBS_CHECKER_EN	1	Enables Audio PRBS Checker	0b0: Audio PRBS checker disabled 0b1: Audio PRBS checker enabled
AUDIO_RX_EN	0	Audio Receiver Adapter Enable	0b0: Audio receiver disabled 0b1: Audio receiver enabled

GPIO_CFG_0 (0x145)

BIT	7	6	5	4	3	2	1	0
Field	GPIO_LINK_SEL	GPIO_PRIORITY[1:0]		–	–	–	–	–
Reset	0x0	0b00		–	–	–	–	–
Access Type	Write, Read	Write, Read		–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_LINK_SEL	7	Selects the link that the adapter is connected to	0x0: Connect to SIOA 0x1: Connect to SIOB
GPIO_PRIORITY	6:5	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority

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GPIO_CFG_1 (0x146)

GPIO0

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b00000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link and requires GPIO_INPUT_PIN_EN = 1. This takes the driven state of the GPIO pin and transmits it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx Compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_2 (0x147)

GPIO0

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	0b10		0x0
Access Type	Write, Read	Write, Read		–	–	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when	0b0: 40kΩ 0b1: 1MΩ

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BITFIELD	BITS	DESCRIPTION	DECODE
		GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_EDGE_RATE	2:1	Adjusts the edge rate of IO buffers when in GPIO mode.	0b00: 1ns 0b01: 2ns 0b10: 4ns 0b11: 8ns
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_3 (0x148)

GPIO0

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b00000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_4 (0x149)

GPIO1

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BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b00001				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link and requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0x0: GPIO state not transmitted over the GMSL link 0x1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx Compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_5 (0x14A)

GPIO1

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	0b10		0x0
Access Type	Write, Read	Write, Read		–	–	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when	0b0: 40kΩ 0b1: 1MΩ

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BITFIELD	BITS	DESCRIPTION	DECODE
		GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_EDGE_RATE	2:1	Adjusts the edge rate of IO buffers when in GPIO mode.	0b00: 1ns 0b01: 2ns 0b10: 4ns 0b11: 8ns
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_6 (0x14B)

GPIO1

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b00001				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_7 (0x14C)

GPIO2

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BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b00010				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0x0: GPIO state not transmitted over the GMSL link 0x1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_8 (0x14D)

GPIO2

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	0b10		0x0
Access Type	Write, Read	Write, Read		–	–	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_EDGE_RATE	2:1	Adjusts the edge rate of IO buffers when in GPIO mode.	0b00: 1ns 0b01: 2ns 0b10: 4ns 0b11: 8ns
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_9 (0x14E)

GPIO2

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b00010				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_10 (0x155)

GPIO5

BIT	7	6	5	4	3	2	1	0

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Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]
Reset	0b1	0x0	0b0	0b00101
Access Type	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_11 (0x156)

GPIO5

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_12 (0x157)

GPIO5

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b00101				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_13 (0x158)

GPIO6

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b00110				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0x0: GPIO state not transmitted over the GMSL link 0x1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_14 (0x159)

GPIO6

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_15 (0x15A)

GPIO6

BIT	7	6	5	4	3	2	1	0

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Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]
Reset	0b0	0x0	0x1	0b00110
Access Type	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_16 (0x161)

GPIO9

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b0	0x0	0b0	0b01001				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link

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BITFIELD	BITS	DESCRIPTION	DECODE
		transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_17 (0x162)

GPIO9

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b00		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_18 (0x163)

GPIO9

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b01001				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_19 (0x164)

GPIO10

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b01010				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

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GPIO_CFG_20 (0x165)

GPIO10

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_21 (0x166)

GPIO10

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b01010				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link

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BITFIELD	BITS	DESCRIPTION	DECODE
		received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_22 (0x167)

GPIO11

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b01011				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_23 (0x168)

GPIO11

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0

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Access Type	Write, Read	Write, Read	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_24 (0x169)

GPIO11

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b01011				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

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GPIO_CFG_25 (0x16A)

GPIO12

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b01100				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_26 (0x16B)

GPIO12

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_27 (0x16C)

GPIO12

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b01100				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_28 (0x16D)

GPIO13

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b01101				

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Access Type	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_29 (0x16E)

GPIO13

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	0b10		0x0
Access Type	Write, Read	Write, Read		–	–	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_EDGE_RATE	2:1	Adjusts the edge rate of IO buffers when in GPIO mode.	0b00: 1ns 0b01: 2ns 0b10: 4ns 0b11: 8ns

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_30 (0x16F)

GPIO13

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b01101				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_31 (0x170)

GPIO14

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b01110				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_32 (0x171)

GPIO14

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	0b10		0x0
Access Type	Write, Read	Write, Read		–	–	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_EDGE_RATE	2:1	Adjusts the edge rate of IO buffers when in GPIO mode.	0b00: 1ns 0b01: 2ns 0b10: 4ns 0b11: 8ns
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

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GPIO_CFG_33 (0x172)

GPIO14

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b01110				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_34 (0x173)

GPIO15

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b01111				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_35 (0x174)

GPIO15

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	00: No input resistance enabled 01: Pull-up resistor enabled 10: Pull-down resistor enabled 11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_36 (0x175)

GPIO15

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b01111				

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Access Type	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_37 (0x176)

GPIO16

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b10000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_38 (0x177)

GPIO16

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	GPIO_EDGE_RATE[1:0]		GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	0b11		0x0
Access Type	Write, Read	Write, Read		–	–	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_EDGE_RATE	2:1	Adjusts the edge rate of IO buffers when in GPIO mode.	0b00: 1ns 0b01: 2ns 0b10: 4ns 0b11: 8ns
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_39 (0x178)

GPIO16

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving,	0bXXXXX: GPIO receive ID

GPIO_CFG_40 (0x179)

GPIO17

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b0	0x0	0b0	0b10001				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

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GPIO_CFG_41 (0x17A)

GPIO17

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b00		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_42 (0x17B)

GPIO17

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10001				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link

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BITFIELD	BITS	DESCRIPTION	DECODE
		received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_43 (0x17C)

GPIO18

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b10010				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_44 (0x17D)

GPIO18

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0

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Access Type	Write, Read	Write, Read	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_45 (0x17E)

GPIO18

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10010				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

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GPIO_CFG_46 (0x17F)

GPIO19

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b10011				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO transmit ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_47 (0x180)

GPIO19

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_48 (0x181)

GPIO19

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10011				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_49 (0x182)

GPIO20

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b10100				

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Access Type	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO transmit ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_50 (0x183)

GPIO20

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

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GPIO_CFG_51 (0x184)

GPIO20

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10100				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_52 (0x185)

GPIO21

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b10101				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO Tx ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_53 (0x186)

GPIO21

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set Input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_54 (0x187)

GPIO21

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10101				

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Access Type	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_55 (0x188)

GPIO22

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b10110				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_TX_ID	4:0	Set GPIO transmit ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_56 (0x189)

GPIO22

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_57 (0x18A)

GPIO22

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10110				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output

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BITFIELD	BITS	DESCRIPTION	DECODE
		GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_58 (0x18B)

GPIO23

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b10111				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO transmit ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_59 (0x18C)

GPIO23

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BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_60 (0x18D)

GPIO23

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b10111				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

GPIO_CFG_61 (0x18E)

GPIO24

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_PIN_EN	GPIO_INPUT_TUNNEL_EN	GPIO_TX_COMPENSATION_EN	GPIO_TX_ID[4:0]				
Reset	0b1	0x0	0b0	0b11000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_PIN_EN	7	Enable GPIO pin as an input (GPI). This enables the pin's input buffer, and the driven state of the pin can be read from GPIO_VALUE_0. Also enables the pull-up or pull-down resistor as configured in GPIO_INPUT_RESISTANCE_VALUE and GPIO_INPUT_RESISTANCE_CONFIG[1:0].	0b0: This GPIO source disabled for GMSL transmission 0b1: This GPIO source enabled for GMSL transmission
GPIO_INPUT_TUNNEL_EN	6	Enables GPIO state tunneling over the GMSL link, requires GPIO_INPUT_PIN_EN = 1. This will take the driven state of the GPIO pin and transmit it over the GMSL link using the GPIO Tx ID set in GPIO_TX_ID[4:0] bitfield.	0b0: GPIO state not transmitted over the GMSL link 0b1: GPIO state transmitted over the GMSL link
GPIO_TX_COMPENSATION_EN	5	Enable Tx compensation mode to minimize GPIO jitter. This mode adds additional latency to GPIO transmission across the GMSL link.	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_TX_ID	4:0	Set GPIO transmit ID for the value of this GPIO pin. If GPIO_INPUT_PIN_EN = 1 and GPIO_INPUT_TUNNEL_EN = 1.	0bXXXXX: GPIO transmit ID

GPIO_CFG_62 (0x18F)

GPIO24

BIT	7	6	5	4	3	2	1	0
Field	GPIO_INPUT_RESISTANCE_VALUE	GPIO_INPUT_RESISTANCE_CONFIG[1:0]		–	–	–	–	GPIO_MANUAL_OUTPUT_VALUE
Reset	0b1	0b10		–	–	–	–	0x0
Access Type	Write, Read	Write, Read		–	–	–	–	Write, Read

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BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_INPUT_RESISTANCE_VALUE	7	Select the pull-up or pull-down resistance value when GPIO_INPUT_RESISTANCE_CONFIG[1:0] = 0b01 or 0b10.	0b0: 40kΩ 0b1: 1MΩ
GPIO_INPUT_RESISTANCE_CONFIG	6:5	Set input resistance for the GPIO. Only active if GPIO_INPUT_EN = 1.	0b00: No input resistance enabled—Hi-Z 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
GPIO_MANUAL_OUTPUT_VALUE	0	Controls state of local GPIO output if GPIO_OUTPUT_PIN_EN_0 = 1 and GPIO_OUTPUT_TUNNEL_EN_0 = 0.	0x0: Local GPIO driven low 0x1: Local GPIO driven high

GPIO_CFG_63 (0x190)

GPIO24

BIT	7	6	5	4	3	2	1	0
Field	GPIO_OUTPUT_PIN_EN	GPIO_OUTPUT_TUNNEL_EN	GPIO_OUTPUT_TYPE	GPIO_RX_ID[4:0]				
Reset	0b0	0x0	0x1	0b11000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

BITFIELD	BITS	DESCRIPTION	DECODE
GPIO_OUTPUT_PIN_EN	7	Enable GPIO pin as an output (GPO). Activates GPIO output driver as configured by GPIO_OUTPUT_TYPE and outputs either the value received over the link (if GPIO_OUTPUT_TUNNEL_EN = 1) or the value programmed in GPIO_MANUAL_OUTPUT_VALUE (if GPIO_OUTPUT_TUNNEL_EN = 0).	0b0: GPIO pin not enabled as an output 0b1: GPIO pin enabled as an output
GPIO_OUTPUT_TUNNEL_EN	6	Enables receiving tunneled value of the GPIO from across the GMSL link from the GPIO Rx ID set in GPIO_RX_ID[4:0] bitfield, which can be read from GPIO_VALUE. Outputs this received value on the local GPIO if GPIO_OUTPUT_PIN_EN = 1.	0b0: GPIO tunnel value is not received across GMSL link 0b1: GPIO tunnel value is received across GMSL link
GPIO_OUTPUT_TYPE	5	Configures GPIO output to push-pull or open drain.	0b0: Open drain 0b1: Push-pull
GPIO_RX_ID	4:0	GPIO ID for pin while receiving.	0bXXXXX: GPIO receive ID

OUTPUT_CFG_0 (0x192)

BIT	7	6	5	4	3	2	1	0

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Field	WMD_INT_EN	–	–	VIDEO_LOCK_OUTPUT_EN	LOCK_PIN_CONTROL[1:0]	–	RCLKOUT_EN
Reset	0b0	–	–	0b0	0b0	–	0b0
Access Type	Write, Read	–	–	Write, Read	Write, Read	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
WMD_INT_EN	7	Enables watermark detected status to be driven to GPIO pin.	0b0: Disabled 0b1: Enabled
VIDEO_LOCK_OUTPUT_EN	4	Enables VIDEO_LOCK output to the LOCK pin.	0b0: Disable VIDEO_LOCK output to LOCK pin 0b1: Enable VIDEO_LOCK output to LOCK pin
LOCK_PIN_CONTROL	3:2	Controls which GMSL link lock status type is output to LOCK pin.	0b00: LOCK_SIOA and (LOCK_SIOB | ~LINK_EN_B) 0b01: LOCK_SIOA and (LOCK_SIOB | ~LINK_EN_B) and LOCK_DCIO 0b10: LOCK_SIOA 0b11: LOCK_SIOA and LOCK_DCIO
RCLKOUT_EN	0	Enables 25MHz reference clock output to RCLKOUT pin.	0b0: RCLK output is disabled 0b1: RCLK output is enabled

CRC_CFG_0 (0x195)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	VIDEO_LINE_CRC_EN	VIDEO_PACKET_CRC_EN	FRAME_CRC_EN
Reset	–	–	–	–	–	0b1	0b0	0b1
Access Type	–	–	–	–	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
VIDEO_LINE_CRC_EN	2	Enables Video Line CRC Checking	0b0: Video line CRC checking disabled 0b1: Video line CRC checking enabled
VIDEO_PACKET_CRC_EN	1	When set, indicates that packets received at this port have appended CRC, and CRC checking should be performed at each packet.	0b0: No CRC on received packets 0b1: Received packets have CRC, and checking is enabled
FRAME_CRC_EN	0	When set, indicates that video frames received are accompanied by frame CRC in Infoframe, and frame CRC needs to be checked.	0b0: No frame CRC from Infoframe expected 0b1: Received video frame has accompanying frame CRC Infoframe

CRC_CFG_1 (0x196)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	FCRC_INFOFRAME_CC_EN	–	–	DP_SINK_RGB_FRAME_CRC_EN

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Reset	–	–	–	–	0x0	–	–	0x0
Access Type	–	–	–	–	Write, Read	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
FCRC_INFOFRAME_CC_EN	3	Enables using DP FCRC values received by infoframe CC.	0b0: Disable 0b1: Enable
DP_SINK_RGB_FRAME_CRC_EN	0	Enables frame CRC check of video displayed by DP sink.	0x0: Disable 0x1: Enable

LINEFAULT_CFG_0 (0x199)

BIT	7	6	5	4	3	2	1	0
Field	LINEFAULT_0_EN	LINEFAULT_1_EN	–	–	–	–	–	–
Reset	0b0	0b0	–	–	–	–	–	–
Access Type	Write, Read	Write, Read	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
LINEFAULT_0_EN	7	Powers Up Line-Fault Monitor 0 (LF0 pin)	0x0: Line-fault monitor 0 disabled 0x1: Line-fault monitor 0 enabled
LINEFAULT_1_EN	6	Powers Up Line-Fault Monitor 1 (LF1 pin)	0x0: Line-fault monitor 1 disabled 0x1: Line-fault monitor 1 enabled

ARQ_CFG_0 (0x19B)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	MAX_RT_ERROR_RESET
Reset	–	–	–	–	–	–	–	0x0
Access Type	–	–	–	–	–	–	–	Write Only

BITFIELD	BITS	DESCRIPTION	DECODE
MAX_RT_ERROR_RESET	0	Write 1 to clear maximum retry error. Self-clearing.	0b0: No action 0b1: Reset maximum retry error

VIDEO_PIPE_CFG_0 (0x1A0)

BIT	7	6	5	4	3	2	1	0

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Field	VID_EN_X	STREAM_SEL_X[1:0]	_	SUPPORT_BLACK_VIDEO_ENABLE	BLANK_VIDEO	COLOR_LUT_EN	VPRBS_CHECKER_EN_X
Reset	0b1	0b00	–	0x0	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
VID_EN_X	7	Enables Video on Pipe X	0b0: Video receive is disabled 0b1: Video receive is enabled
STREAM_SEL_X	6:5	Select the stream ID that Pipe X is receiving video from. Default depends on the selected I²C address through CFG pins.	0b00: Stream 0 0b01: Stream 1 0b10: Stream 2 0b11: Stream 3
SUPPORT_BLACK_VIDEO_ENABLE	3	This bit indicates whether the eDP BLACK_VIDEO_ENABLE feature is supported or not. If supported, it allows the DPTX in this device to set the DPCD BLACK_VIDEO_ENABLE register in the DP sink when there is no video present in the DPTX.	0b0: Not supported 0b1: Supported
BLANK_VIDEO	2	Blank video output from device by setting all active video pixels to black. Affects Pipe X and Pipe Y.	0b0: Do not blank video output 0b1: Blank video output
COLOR_LUT_EN	1	Enables the color LUT on Pipe X. See internal registers to configure the LUT.	0b0: Color LUT disabled 0b1: Color LUT enabled
VPRBS_CHECKER_EN_X	0	Enables video PRBS checker on Pipe X. Results reported to registers VPRBS_FAIL at 0x8C and VPRBS_ERROR_COUNT at 0x8D	0b0: Video PRBS checker disabled 0b1: Video PRBS checker enabled

VIDEO_PIPE_CFG_1 (0x1A1)

BIT	7	6	5	4	3	2	1	0
Field	VID_EN_Y	STREAM_SEL_Y[1:0]		LINK_SEL_Y	–	–	–	VPRBS_CHECKER_EN_Y
Reset	0b0	0b01		0x1	–	–	–	0b0
Access Type	Write, Read	Write, Read		Write, Read	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
VID_EN_Y	7	Enable video on Pipe Y. Set to 1 when using Even/Odd Pixel Splitter mode.	0b0: Video receive on Pipe Y disabled 0b1: Video receive on Pipe Y enabled
STREAM_SEL_Y	6:5	Select the stream ID that Pipe Y is receiving video from.	0b00: Stream 0 0b01: Stream 1

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BITFIELD	BITS	DESCRIPTION	DECODE
		Default depends on the selected I²C address through CFG pins.	0b10: Stream 2 0b11: Stream 3
LINK_SEL_Y	4	Selects which link Pipe Y receives video from.	0b0: Pipe Y receiving video from SIOA 0b1: Pipe Y receiving video from SIOB
VPRBS_CHECKER_EN_Y	0	Enable video PRBS checker on Pipe Y. Results reported to registers VPRBS_ERROR_COUNT at 0x55B, VPRBS_FAIL at 0x55c	0b0: Video PRBS checker disabled 0b1: Video PRBS checker enabled

VIDEO_PIPE_CFG_2 (0x1A2)

BIT	7	6	5	4	3	2	1	0
Field	VS_HS_PIN_EN_X	VS_HS_INPUT_MODE_X	DE_OUTPUT_EN_X	–	VS_HS_OUTPUT_EN_Y	DE_OUTPUT_EN_Y	SYNC_LOCATION_SEL[1:0]	
Reset	0x0	0x0	0x0	–	0x0	0x0	0x00	
Access Type	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
VS_HS_PIN_EN_X	7	Enable VS and HS signals from Pipe X to be output on respective GPIO pins. Overrides other GPIO configurations.	0b0: VS and HS are not output on GPIO pins 0b1: VS and HS outputs enabled to respective GPIO pins
VS_HS_INPUT_MODE_X	6	Configures VS and HS GPIO pins as input. When set, it also selects the input HS and VS signals from HS_IN_X and VS_IN_X pins at MFP13, 14 as the source of HS and VS signals used in the ALiGN_FIFO block to align the video stream from an external DES.	0b0: VS and HS pins are output 0b1: VS and HS pins are input
DE_OUTPUT_EN_X	5	Enable DE signal from Pipe X to be output on GPIO x for debug purposes. Overrides other GPIO configurations.	0x0: DE is not output to GPIO pin 0x1: DE output enabled to GPIO pin
VS_HS_OUTPUT_EN_Y	3	Enable VS and HS signals from Pipe Y to be output on their respective GPIO pins. Overrides other GPIO configurations.	0x0: VS, HS are not output to GPIO pins 0x1: VS, HS outputs enabled to GPIO pins
DE_OUTPUT_EN_Y	2	Enable DE signal from Pipe Y to be output on its GPIO pin, when RCLKOUT is not being output on the shared GPIO pin. Overrides other GPIO configurations.	0x0: DE is not output to GPIO pin 0x1: DE output enabled to GPIO pin
SYNC_LOCATION_SEL	1:0	Selects the source for video sync signals (HS, VS, DE) to be output to GPIO pin(s).	0b00: DPTX input 0b01: VTRG input 0b10: VTRG output 0b11: Align blocks input

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VIDEO_PIPE_CFG_3 (0x1A3)

BIT	7	6	5	4	3	2	1	0
Field	VS_HS_ALT_PIN_EN_X	VS_HS_ALT_INPUT_MODE_X	_	ALIGN_LOCK_PIN_LOCATION	ALIGN_LOCK_INPUT_MODE	ALIGN_LOCK_PIN_EN	SYNC_LOCATION_ALT_SEL[1:0]	
Reset	0x0	0x0	–	0b0	0b0	0b0	0b00	
Access Type	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
VS_HS_ALT_PIN_EN_X	7	Enable VS and HS signals to/from Pipe X on respective alternate GPIO pins. Direction is specified by VS_HS_ALT_INPUT_MODE_X register. Overrides other GPIO configurations.	0b0: Alternate GPIO pins not allocated for VS, HS 0b1: Alternate GPIO pins are allocated for VS, HS
VS_HS_ALT_INPUT_MODE_X	6	Configures alternate VS and HS GPIO pins as input.	0b0: Alternate VS, HS pins are output 0b1: Alternate VS, HS pins are input
ALIGN_LOCK_PIN_LOCATION	4	Selects whether MFP12 or MFP22 pin is used as align-lock when ALIGN_LOCK_PIN_EN = 1.	0x0: Use MFP12 for align-lock 0x1: Use MFP22 for align-lock
ALIGN_LOCK_INPUT_MODE	3	Used together with ALIGN_LOCK_PIN_EN to configure the MFP selected by ALIGN_LOCK_PIN_LOCATION to receive align-lock input for use in ALIGN_FIFO block as "external alignment completed in subordinate DES"	0x0: Selected MFP outputs external alignment completed signal 0x1: Selected MFP receives external alignment completed in subordinate DES
ALIGN_LOCK_PIN_EN	2	When set, configures either MFP12 or MFP22 for align-lock use.	0x0: Align-lock not configured on MFP12 or MFP22 0x1: Align-lock configured on MFP12 or MFP22
SYNC_LOCATION_ALT_SEL	1:0	Selects the source for video sync signals (HS, VS, DE) to be output to GPIO pin(s).	0b00: DPTX input 0b01: VTRG input 0b10: VTRG output 0b11: Align blocks input

DC_VIDEO_CFG_0 (0x1A5)

BIT	7	6	5	4	3	2	1	0
Field	DROP_VIDEO[3:0]				AUTO_DROP_VIDEO_EN	–	–	BLOCK_INFO_CC_FWD
Reset	0x0				0x1	–	–	0b1
Access Type	Write, Read				Write, Read	–	–	Write, Read

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BITFIELD	BITS	DESCRIPTION	DECODE
DROP_VIDEO	7:4	Selects which video stream is dropped while forwarding in the daisy-chain output. Default is selected by I²C address. The video stream shown by default on the eDP output is dropped by default. Default eDP output video stream is selected by the CFG0 pin.	0bxxx1: Drop stream ID = 0 0bxx1x: Drop stream ID = 1 0bx1xx: Drop stream ID = 2 0b1xxx: Drop stream ID = 3
AUTO_DROP_VIDEO_EN	3	Enable automatic dropping of video stream.	0b0: Disable 0b1: Enable
BLOCK_INFO_CC_FWD	0	Blocks forwarding of Infoframe CC messages from upstream device.	

DP_CFG_0 (0x1A8)

BIT	7	6	5	4	3	2	1	0
Field	AUTO_RETRAIN_ON_LOSS_OF_LINK_EN	–	DP_SINK_EDID_TIMING_EN	DP_SINK_READ_EDID_EN	DP_SPREADSPECTRUM_EN	AUTO_TIMING_EN	AUTO_DP_LINK_CONFIG	–
Reset	0x1	–	0x0	0x0	0x0	0x1	0x0	–
Access Type	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–

BITFIELD	BITS	DESCRIPTION	DECODE
AUTO_RETRAIN_ON_LOSS_OF_LINK_EN	7	Enables automatic DP link retraining upon loss of link integrity.	0b0: Not enabled 0b1: Enabled
DP_SINK_EDID_TIMING_EN	5	Reads DP sink EDID and populates video timing information.	0b0: Disable reading DP sink EDID to derive video timing information 0b1: Enable reading DP sink EDID to derive video timing information
DP_SINK_READ_EDID_EN	4	Reads DP sink EDID and sends to GMSL serializer.	0b0: Not enabled 0b1: Enable read of DP sink EDID and send to SER
DP_SPREADSPECTRUM_EN	3	Enables DP spread-spectrum clocking.	0b0: DP spread-spectrum clocking disabled 0b1: DP spread-spectrum clocking enabled
AUTO_TIMING_EN	2	Enables auto timing. Receives detected video timing from a DP serializer and automatically configures the output video timing VIDEO_TIMING_CFG_x registers.	0b0: Auto timing disabled 0b1: Auto timing enabled
AUTO_DP_LINK_CONFIG	1	Auto-sets the DP rate and lane count based on video timing and sink capability.	0b0: No action 0b1: Auto-set DP rate and lane count

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DP_CFG_1 (0x1A9)

BIT	7	6	5	4	3	2	1	0
Field	AUTO_RETRAIN_ON_DISCONNECT_EN	–	–	–	–	–	–	–
Reset	0x0	–	–	–	–	–	–	–
Access Type	Write, Read	–	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
AUTO_RETRAIN_ON_DISCONNECT_EN	7	Enables automatic retraining of DP link after HPD disconnect-reconnect event. DP link retraining is executed only if DP link was already successfully trained prior to HPD disconnect.	0b0: Disable 0b1: Enable

DP_CFG_2 (0x1AA)

BIT	7	6	5	4	3	2	1	0
Field	–	DP_LANE_COUNT[2:0]			–	DP_RATE[2:0]		
Reset	–	0x4			–	0b010		
Access Type	–	Write, Read			–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
DP_LANE_COUNT	6:4	Configures the number of DP lanes.	0b0001: One lane (lane 0 only) 0b0010: Two lanes (lanes 0 and 1 only) 0b0100: Four lanes (lanes 0, 1, 2, and 3) Others: Reserved
DP_RATE	2:0	DP Link Rate	0b0000: 1.62Gbps 0b0001: 2.7Gbps 0b0010: 5.4Gbps 0b0011: 8.1Gbps 0b0100: Reserved 0b0101: Reserved 0b0110: Reserved 0b0111: Reserved

WATERMARK_LOCK_CFG_0 (0x1B0)

BIT	7	6	5	4	3	2	1	0
Field	WATERMARK_KEY_BYTE_0[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION	DECODE
WATERMARK_KEY_BYTE_0	7:0	Write 0xBA to WATERMARK_KEY_BYTE_0 and 0xDC to WATERMARK_KEY_BYTE_1 registers to enable writing to watermark registers. Otherwise watermark registers are read-only.	0xXX: Watermark Key Byte 0 value

WATERMARK_LOCK_CFG_1 (0x1B1)

BIT	7	6	5	4	3	2	1	0
Field	WATERMARK_KEY_BYTE_1[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
WATERMARK_KEY_BYTE_1	7:0	Write 0xBA to WATERMARK_KEY_BYTE_0 and 0xDC to WATERMARK_KEY_BYTE_1 registers to enable writing to watermark registers. Otherwise watermark registers are read-only.	0xXX: Watermark Key Byte 1 value

WATERMARK_CFG_0 (0x1B2)

BIT	7	6	5	4	3	2	1	0
Field	WATERMARK_DET_EN	WATERMARK_REMOVE	WATERMARK_REGIONAL_MODE	WATERMARK_BLANK	WATERMARK_REGIONAL_BLANK_MODE	WATERMARK_REGION_BLANK_EN	–	–
Reset	0b0	0x1	0b0	0b0	0x1	0x0	–	–
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
WATERMARK_DET_EN	7	Watermark detection is enabled.	0b0: Watermark detection is disabled 0b1: Watermark detection is enabled

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BITFIELD	BITS	DESCRIPTION	DECODE
WATERMARK_REMOVE	6	Indicates if the detected watermark should be removed or not from the video.	0b0: Do not remove detected watermark from video 0b1: Remove detected watermark from video
WATERMARK_REGIONAL_MODE	5	Watermark Regional mode is enabled. Disable this bit if watermark also needs to be disabled on a downstream device.	0b0: Watermark Regional mode is disabled 0b1: Watermark Regional mode is enabled
WATERMARK_BLANK	4	Enable blanking of video when watermark error detected (frozen frame condition occurs).	0b0: Video stream not blanked on watermark error 0b1: Video stream blanked on watermark error
WATERMARK_REGIONAL_BLANK_MODE	3	Selects Blanking mode for watermark operation.	0b0: Blank full frame 0b1: Blank watermarked region
WATERMARK_REGION_BLANK_EN	2	Enable watermark manual blanking	0b0: Manual watermark blanking disabled 0b1: Manual watermark blanking enabled

WATERMARK_CFG_1 (0x1B3)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_XMIN_BYTE_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
WM_REGION_XMIN_BYTE_0	7:0	First pixel of defined regional watermark removal, measured in pixels from the beginning of the video line. (Byte 0)

WATERMARK_CFG_2 (0x1B4)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_XMIN_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
WM_REGION_XMIN_BYTE_1	7:0	First pixel of defined regional watermark removal, measured in pixels from the beginning of the video line. (Byte 1)

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WATERMARK_CFG_3 (0x1B5)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_XMAX_BYTE_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
WM_REGION_XMAX_BYTE_0	7:0	Last pixel of defined region for regional watermark removal, measured in pixels from the beginning of the video line. (Byte 0)

WATERMARK_CFG_4 (0x1B6)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_XMAX_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
WM_REGION_XMAX_BYTE_1	7:0	Last pixel of defined region for regional watermark removal, measured in pixels from the beginning of the video line. (Byte 1)

WATERMARK_CFG_5 (0x1B7)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_YMIN_BYTE_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
WM_REGION_YMIN_BYTE_0	7:0	First line of defined region for regional watermark removal, measured in lines from the beginning of the frame. (Byte 0)

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WATERMARK_CFG_6 (0x1B8)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_YMIN_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
WM_REGION_YMIN_BYTE_1	7:0	First line of defined region for regional watermark removal, measured in lines from the beginning of the frame. (Byte 1)

WATERMARK_CFG_7 (0x1B9)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_YMAX_BYTE_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
WM_REGION_YMAX_BYTE_0	7:0	Last line of defined region for regional watermark removal, measured in lines from the beginning of the frame. (Byte 0)

WATERMARK_CFG_8 (0x1BA)

BIT	7	6	5	4	3	2	1	0
Field	WM_REGION_YMAX_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
WM_REGION_YMAX_BYTE_1	7:0	Last line of defined region for regional watermark removal, measured in lines from the beginning of the frame. (Byte 1)	0xXX: Byte 1 value

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PATTERN_GENERATOR_CFG_0 (0x1BC)

BIT	7	6	5	4	3	2	1	0
Field	CUSTOM_COLOR_RED[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
CUSTOM_COLOR_RED	7:0	Video pattern generator (VPG) custom-color pixel red component value.	0xXX: LSB of custom RED color

PATTERN_GENERATOR_CFG_1 (0x1BD)

BIT	7	6	5	4	3	2	1	0
Field	CUSTOM_COLOR_GREEN[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
CUSTOM_COLOR_GREEN	7:0	Video pattern generator custom-color pixel green component value.

PATTERN_GENERATOR_CFG_2 (0x1BE)

BIT	7	6	5	4	3	2	1	0
Field	CUSTOM_COLOR_BLUE[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
CUSTOM_COLOR_BLUE	7:0	Video pattern generator custom-color pixel blue component value.

PATTERN_GENERATOR_CFG_3 (0x1BF)

BIT	7	6	5	4	3	2	1	0

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Field	CUSTOM_COLOR_RED_30BIT_LSBS[1:0]	CUSTOM_COLOR_GREEN_30BIT_LSBS[1:0]	CUSTOM_COLOR_BLUE_30BIT_LSBS[1:0]	_	MODE_30BIT_EN
Reset	0x00	0x00	0x00	–	0x0
Access Type	Write, Read	Write, Read	Write, Read	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
CUSTOM_COLOR_RED_30BIT_LSBS	7:6	Specifies the 2 LSBs for the R value for the custom RGB color if 30-bit mode is used (MODE_30BIT_EN = 1).	0bXX: Most significant 2 bits of 10-bit custom RED color
CUSTOM_COLOR_GREEN_30BIT_LSBS	5:4	Specifies the 2 LSBs for the G value for the custom RGB color if 30-bit mode is used (MODE_30BIT_EN = 1).	0bXX: Most significant 2 bits of 10-bit custom GREEN color
CUSTOM_COLOR_BLUE_30BIT_LSBS	3:2	Specifies the 2 LSBs for the B value for the custom RGB color if 30-bit mode is used (MODE_30BIT_EN = 1).	0bXX: Most significant 2 bits of 10-bit custom BLUE color
MODE_30BIT_EN	0	Enables the pattern generator in 30-bit color mode.	0x0: 24-bit pixel mode 0x1: 30-bit pixel mode

PATTERN_GENERATOR_CFG_4 (0x1C0)

BIT	7	6	5	4	3	2	1	0
Field	PATTERN_SIZE_X[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PATTERN_SIZE_X	7:0	X-dimension size of checkerboard pattern.	.

PATTERN_GENERATOR_CFG_5 (0x1C1)

BIT	7	6	5	4	3	2	1	0
Field	PATTERN_SIZE_Y[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION	DECODE
PATTERN_SIZE_Y	7:0	Y-dimension size of checkerboard pattern.	.

PATTERN_GENERATOR_CFG_6 (0x1C2)

BIT	7	6	5	4	3	2	1	0
Field	GRADIENT_COLOR_STEP_SIZE[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
GRADIENT_COLOR_STEP_SIZE	7:0	Defines the increment value to the pixel value for color gradient patterns.	0xXX: Gradient color step size value

PATTERN_GENERATOR_CFG_7 (0x1C3)

BIT	7	6	5	4	3	2	1	0
Field	PATTERN_SELECT[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PATTERN_SELECT	7:0	Select the pattern for the video pattern generator.	0x0: Color Bars (width = 1920/8) 0x1: Gray Bars (made from horizontal gray gradient with 8 steps) 0x2: Legacy Gradient Color Bars (already existing) with 4 colors (Blue, Red, Green, White) 0x3: Black and White checkerboard (RGB0=0/0/0, RGB1=255/255/255. 120 by 120) 0x4: Red and Blue checkerboard (RGB0=255/0/0, RGB1=0/0/255. 120 by 120) 0x5: Solid White 0x6: Solid Black 0x7: Solid Red (RGB=255/0/0) 0x8: Solid Green (RGB=0/255/0) 0x9: Solid Blue (RGB=0/0/255) 0xA: Solid Gray_127 (RGB=127/127/127) 0xB: Solid Gray_63 (RGB=63/63/63) 0xC: Solid_Gray_31 (RGB=31/31/31) 0xD: Solid_Gray_15 (RGB=15/15/15) 0xE: Horizontal Gradient Red (color step=1, step size=8) 0xF: Horizontal Gradient Green (color step=1, step size=8) 0x10: Horizontal Gradient Blue (color step=1, step size=8) 0x11: Horizontal Gradient White (color step=1, step size=8)

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BITFIELD	BITS	DESCRIPTION	DECODE
			0x12: Vertical Gradient Red (color step=1, step size=5) 0x13: Vertical Gradient Green (color step=1, step size=5) 0x14: Vertical Gradient Blue (color step=1, step size=5) 0x15: Vertical Gradient Whie (color step=1, step size=5) 0x16: Solid color inside white frame (frame border h=192, v=108, RGB_frame=255/255/255, RGB_box=0/255/0)

DSC_CFG_0 (0x1C5)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	DSC_DECODE_EN	MANUAL_DSC_EN	–	–	–	–	–	–
Reset	0b0	0x0	–	–	–	–	–	–
Access Type	Write, Read	Write, Read	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
DSC_DECODE_EN	7	Configures DSC decoder to be enabled.	0: Decoder is disabled 1: Decoder is enabled
MANUAL_DSC_EN	6	Indicates operation of DSC decoder with GM03 serializer, which is not capable of passing DSC PPS parameters and video timing in Infoframe.	0b0: Operation with GM13 SER with advanced Infoframe CC support 0b1: Operation with GM03 SER without advanced Infoframe CC support

HDCP_CFG_0 (0x1C7)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	HDCP_EN_X	HDCP_EN_Y	–	–	–	HDCP_GPIO_ID_Y[1:0]		HDCP_VER
Reset	0b1	0b1	–	–	–	0x0		0b1
Access Type	Write, Read	Write, Read	–	–	–	Write, Read		Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_EN_X	7	Enables HDCP decryption on SIOA.	0b0: HDCP encryption disabled 0b1: HDCP encryption enabled
HDCP_EN_Y	6	Enables HDCP decryption on SIOB.	0b0: HDCP encryption disabled 0b1: HDCP encryption enabled
HDCP_GPIO_ID_Y	2:1	Selector for hdcp2_tun_y_gpio_id.	0x0: HDCP2 Pipe Y GPIO ID = 27 0x1: HDCP2 Pipe Y GPIO ID = 26 0x2: HDCP2 Pipe Y GPIO ID = 25 0x3: HDCP2 Pipe Y GPIO ID = 28
HDCP_VER	0	Selects the HDCP version.	0b0: HDCP 1.4 0b1: HDCP 2.3

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HDCP_CFG_1 (0x1C8)

See Note §.

BIT	7	6	5	4	3	2	1	0
Field	HDCP_ADDRESS_Y[6:0]							–
Reset	0b1001010							–
Access Type	Write, Read							–

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_ADDRESS_Y	7:1	Device Address for I²C or UART addressing for Pipe Y HDCP. NOTE: Other DES in the system must NOT use the same value selected here for either DEV_ADDR or DEV_ADDR_Y. Default value is based on CFG0 pin voltage level as follows: Voltage[2:0] Device Address 000 0x94 001 0x98 010 0xD0 011 0x90 100 0xD8 101 0x50 110 0x54 111 0xD4	0b0000000: I²C write/read address is 0x00/0x01 0b0000001: I²C write/read address is 0x02/0x03 ... ... 0b1001000: I²C write/read address is 0x90/0x91 0b1001010: I²C write/read address is 0x94/0x95 0b1001100: I²C write/read address is 0x98/0x99 0b1101000: I²C write/read address is 0xD0/0xD1 0b1101010: I²C write/read address is 0xD4/0xD5 0b1101100: I²C write/read address is 0xD8/0xD9 0b0101000: I²C write/read address is 0x50/0x51 0b0101010: I²C write/read address is 0x54/0x55 ... ... 0b1111111: I²C write/read address is 0xFE/0xFF

HDCP_CFG_2 (0x1C9)

BIT	7	6	5	4	3	2	1	0
Field	HDCP_CONFIG_0[7]	HDCP_CONFIG_0[6]	HDCP_CONFIG_0[5]	HDCP_CONFIG_0[4]	HDCP_CONFIG_0[3]	HDCP_CONFIG_0[2]	HDCP_CONFIG_0[1]	HDCP_CONFIG_0[0]
Reset	0b1	0b0	0b0	0b0	0b1	0b0	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_CONFIG_0[7]	7	Writes stream ID 00/11 for AHDCP Y data to reverse SIO link	0b0: Write stream ID 00 0b1: Write stream ID 11
HDCP_CONFIG_0[6]	6	Writes stream ID 00/11 for AHDCP X data to reverse SIO link	0b0: Write stream ID 00 0b1: Write stream ID 11

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BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_CONFIG_0[5]	5	Stream overwrite select for incoming DC AHDCP Data	0b0: Write stream ID 00 0b1: Write stream ID 11
HDCP_CONFIG_0[4]	4	Stream override enable for incoming Daisy-Chain AHDCP data	0b0: Not enabled 0b1: Enabled
HDCP_CONFIG_0[3]	3	Stream identifier for incoming Daisy-Chain AHDCP Y data	0b0: Stream ID 00 0b1: Stream ID 11
HDCP_CONFIG_0[2]	2	Stream identifier for incoming Daisy-Chain AHDCP X data	0b0: Stream ID 00 0b1: Stream ID 11
HDCP_CONFIG_0[1]	1	Blocks transmission of AHDCP Y Data to Daisy-Chain link	0b0: Do not block 0b1: Block
HDCP_CONFIG_0[0]	0	Blocks transmission of AHDCP X Data to Daisy-Chain link	0b0: Do not block 0b1: Block

HDCP_CFG_3 (0x1CA)

BIT	7	6	5	4	3	2	1	0
Field	HDCP_CONFIG_1[5]	HDCP_CONFIG_1[4]	HDCP_CONFIG_1[3]	HDCP_CONFIG_1[2]	HDCP_CONFIG_1[1]	HDCP_CONFIG_1[0]	ARQ_AUTO_CFG_AHDCP_Y	ARQ_AUTO_CFG_AHDCP_X
Reset	0b1	0x0	0x1	0x0	0x0	0x0	0b1	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_CONFIG_1[5]	7	Writes stream ID 00/11 for AHDCP Y data to Daisy-Chain output	0x0: Write stream ID 00 0x1: Write stream ID 11
HDCP_CONFIG_1[4]	6	Writes stream ID 00/11 for AHDCP X data to Daisy-Chain output	0x0: Write stream ID 00 0x1: Write stream ID 11
HDCP_CONFIG_1[3]	5	Selects AHDCP Y for local data coming from link SIOA/SIOB. Register AHDCP_LINK_TO_LCL_CHK_EN should be 1 for this register to be effective.	0x0: Use AHDCP data coming from link SIOA 0x1: Use AHDCP data coming from link SIOB
HDCP_CONFIG_1[2]	4	Selects AHDCP X for local data coming from link SIOA/SIOB. Register AHDCP_LINK_TO_LCL_CHK_EN should be 1 for this register to be effective	0x0: Use AHDCP data coming from link SIOA 0x1: Reserved
HDCP_CONFIG_1[1]	3	Enables AHDCP stream check to local data in GM02 receive baseband	0x0: Disable 0x1: Enable

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BITFIELD	BITS	DESCRIPTION	DECODE
HDCP_CONFIG_1[0]	2	Disables AHDCP stamp check before stamping	0x1: Disable 0x0: Enable
ARQ_AUTO_CFG_AHDCP_Y	1	When this bit = 1, ARQ settings are automatically selected based on Splitter mode.	0b0: ARQ settings are selected based on selections in ACK_SRC_ID and MATCH_SRC_ID bitfields 0b1: ARQ settings are based on internal values
ARQ_AUTO_CFG_AHDCP_X	0	When this bit = 1, ARQ settings are automatically selected based on Splitter mode.	0b0: ARQ settings are selected based on selections in ACK_SRC_ID and MATCH_SRC_ID bitfields 0b1: ARQ settings are based on internal values

FRC_CFG_0 (0x1D2)

BIT	7	6	5	4	3	2	1	0
Field	FRC_EN	–	–	–	–	–	–	–
Reset	0b0	–	–	–	–	–	–	–
Access Type	Write, Read	–	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
FRC_EN	7	Frame-rate control enable.	0x0: Disabled 0x1: Enabled

VIDEO_TIMING_CFG_0 (0x1D6)

BIT	7	6	5	4	3	2	1	0
Field	PCLK_FREQUENCY_BYTE_0[7:0]							
Reset	0x14							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PCLK_FREQUENCY_BYTE_0	7:0	Configures pixel clock (PCLK) frequency to the nearest kHz. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Byte 0 value

VIDEO_TIMING_CFG_1 (0x1D7)

BIT	7	6	5	4	3	2	1	0
Field	PCLK_FREQUENCY_BYTE_1[7:0]							
Reset	0x44							

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Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
PCLK_FREQUENCY_BYTE_1	7:0	Configures pixel clock (PCLK) frequency to the nearest kHz. If AUTO_TIMING_EN = 1, this field will be zeroed out. Default value is for 1080p.	0xXX: Byte 1 value

VIDEO_TIMING_CFG_2 (0x1D8)

BIT	7	6	5	4	3	2	1	0
Field	PCLK_FREQUENCY_BYTE_2[7:0]							
Reset	0x02							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PCLK_FREQUENCY_BYTE_2	7:0	Configures pixel clock (PCLK) frequency to the nearest kHz. If AUTO_TIMING_EN = 1, this field will be zeroed out. Default value is for 1080p.	0xXX: Byte 2 value

VIDEO_TIMING_CFG_3 (0x1D9)

BIT	7	6	5	4	3	2	1	0
Field	H_ACTIVE_BYTE_0[7:0]							
Reset	0x80							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
H_ACTIVE_BYTE_0	7:0	Configures horizontal active (HACTIVE) resolution. Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the horizontal resolution of the mainstream video source

VIDEO_TIMING_CFG_4 (0x1DA)

BIT	7	6	5	4	3	2	1	0

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Field	H_ACTIVE_BYTE_1[7:0]
Reset	0x07
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
H_ACTIVE_BYTE_1	7:0	Configures horizontal active (HACTIVE) resolution. Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Upper 8 bits of the horizontal resolution of the mainstream video source

VIDEO_TIMING_CFG_5 (0x1DB)

BIT	7	6	5	4	3	2	1	0
Field	H_FP_BYTE_0[7:0]							
Reset	0x58							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
H_FP_BYTE_0	7:0	Configures horizontal front porch (HFP). Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the horizontal front porch of the mainstream video source

VIDEO_TIMING_CFG_6 (0x1DC)

BIT	7	6	5	4	3	2	1	0
Field	H_FP_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
H_FP_BYTE_1	7:0	Configures horizontal front porch (HFP). Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Upper 8 bits of the horizontal front porch of the mainstream video source

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VIDEO_TIMING_CFG_7 (0x1DD)

BIT	7	6	5	4	3	2	1	0
Field	H_SW_BYTE_0[7:0]							
Reset	0x2C							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
H_SW_BYTE_0	7:0	Configures horizontal sync width (HSW). Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the horizontal sync width of the mainstream video source

VIDEO_TIMING_CFG_8 (0x1DE)

BIT	7	6	5	4	3	2	1	0
Field	H_SW_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
H_SW_BYTE_1	7:0	Configures horizontal sync width (HSW). Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Upper 8 bits of the horizontal sync width of the mainstream video source

VIDEO_TIMING_CFG_9 (0x1DF)

BIT	7	6	5	4	3	2	1	0
Field	H_BP_BYTE_0[7:0]							
Reset	0x94							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
H_BP_BYTE_0	7:0	Configures horizontal back porch (HBP). Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the horizontal back porch of the mainstream video source

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VIDEO_TIMING_CFG_10 (0x1E0)

BIT	7	6	5	4	3	2	1	0
Field	H_BP_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
H_BP_BYTE_1	7:0	Configures horizontal back porch (HBP). Measured in pixels. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Upper 8 bits of the horizontal back porch of the mainstream video source

VIDEO_TIMING_CFG_11 (0x1E1)

BIT	7	6	5	4	3	2	1	0
Field	V_ACTIVE_BYTE_0[7:0]							
Reset	0x38							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
V_ACTIVE_BYTE_0	7:0	Configures vertical active (VACTIVE) resolution. Measured in lines. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the vertical resolution of the mainstream video source

VIDEO_TIMING_CFG_12 (0x1E2)

BIT	7	6	5	4	3	2	1	0
Field	V_ACTIVE_BYTE_1[7:0]							
Reset	0x04							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
V_ACTIVE_BYTE_1	7:0	Configures vertical active (VACTIVE) resolution. Measured in lines. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Upper 8 bits of the vertical resolution of the main stream video source

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VIDEO_TIMING_CFG_13 (0x1E3)

BIT	7	6	5	4	3	2	1	0
Field	V_FP[7:0]							
Reset	0x04							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
V_FP	7:0	Configures vertical front porch (VFP). Measured in lines. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the vertical front porch of the mainstream video source

VIDEO_TIMING_CFG_14 (0x1E4)

BIT	7	6	5	4	3	2	1	0
Field	V_SW[7:0]							
Reset	0x05							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
V_SW	7:0	Configures vertical sync width (VSW). Measured in lines. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the vertical sync width of the mainstream video source

VIDEO_TIMING_CFG_15 (0x1E5)

BIT	7	6	5	4	3	2	1	0
Field	V_BP[7:0]							
Reset	0x24							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
V_BP	7:0	Configures vertical back porch (VBP). Measured in lines. If AUTO_TIMING_EN = 1, this field is zeroed out. Default value is for 1080p.	0xXX: Lower 8 bits of the vertical back porch of the mainstream video source

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VIDEO_TIMING_CFG_16 (0x1E6)

BIT	7	6	5	4	3	2	1	0
Field	–	FORCE_EVENODD_PIXEL_MODE_EN	COLOR_DEPTH_MANUAL_MODE	COLOR_DEPTH_30BIT_SELECT	–	HS_VS_POLARITY_MANUAL_MODE	VS_POLARITY	HS_POLARITY
Reset	–	0x0	0x0	0x0	–	0x0	0x1	0x1
Access Type	–	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
FORCE_EVENODD_PIXEL_MODE_EN	6	Force Even/Odd Split mode.	0b0: No action 0b1: Force to Even/Odd Split mode
COLOR_DEPTH_MANUAL_MODE	5	Enables color depth override to manually program the color depth to the value in register COLOR_DEPTH_30BIT_SELECT.	0b0: No action 0b1: Color depth override enabled
COLOR_DEPTH_30BIT_SELECT	4	Selects the color depth for the video stream if register COLOR_DEPTH_OVERRIDE_EN = 0b1.	0b0: Color depth override to 24bpp (8 bits per color) 0b1: Color depth override to 30bpp (10 bits per color)
HS_VS_POLARITY_MANUAL_MODE	2	Enables manual overwrite of VS/HS polarities.	0b0: Use detected VS/HS polarities 0b1: Use VS/HS polarities specified in VS_POLARITY/HS_POLARITY bitfields
VS_POLARITY	1	VS Pulse Polarity	0b0: Negative VS polarity 0b1: Positive VS polarity
HS_POLARITY	0	HS Signal Polarity	0b0: Negative HS polarity 0b1: Positive HS polarity

PSM_STATE_CFG_0 (0x1E7)

BIT	7	6	5	4	3	2	1	0
Field	PSM_STATE[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PSM_STATE	7:0	This register contains the desired state request to the PSM state machine by the user. After the PSM starts processing the user request, it changes the value in this register to either 0x00 or to 0x01. A value of 0x00 indicates that the user request was completed successfully by the PSM without error. A	0x0: Idle/previous state completed with no error 0x1: Idle/previous state completed with error 0x2: Train/retrain link, disable video output 0x3: Train/retrain link, enable video output 0x4: Train link, set up video, disable video output 0x5: Train link, set up video, enable video output 0x6: Set up/wait video, train link, disable video output 0x7: Set up/wait video, train link, enable video output 0x8: Set up new video, no retraining, disable video output 0x9: Set up new video, no retraining, enable video output 0xA: Disable video output

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BITFIELD	BITS	DESCRIPTION	DECODE
		value of 0x01 indicates that the user request was not completed successfully by the PSM. That is, an error occurred. For more detailed status, see PSM_STATUS_0 (0x1E8). A value of 0x00 or 0x01 returned by the PSM serves as a handshake to the user to indicate that a new request can now be submitted. Rows in the table that do not have a Decode entry are reserved. Do not use these reserved values. Rows with entries in the Decode column show the user requests that are currently supported by the PSM.	0xB: Enable video output 0xF: Reinitialize (like HPD disconnect) 0x10: Disable VPG (use to stop VPG after VPG was set up) 0x11: Set up VPG, using registers or EDID for video timing (override timing received from GMSL serializer), train link if not trained 0x12: Set up VPG, using current video timing received from GMSL serializer, train link if not trained 0x20: Disable link quality test pattern setup 0x21: Set up link quality test pattern 1 0x22: Set up link quality test pattern 2 0x23: Set up link quality test pattern 3 0x24: Set up link quality test pattern 4 0x25: Set up link quality test pattern 5 0x26: Set up link quality test pattern 6 0x27: Set up link quality test pattern 7 0x80: Set up HDCP (use to reinitialize HDCP after HDCP registers have been reprogrammed) 0xE0: Safety test: APB0_M1_ERR 0xE1: Safety test: APB1_M1_ERR 0xE2: Safety test: PSM_DATA_ADDR_ERROR 0xE3: Safety test: PSM_DATA_TIMER_ERROR 0xE4: Safety test: PSM_STATE_TIMER_ERROR 0xE5: Safety test: PSM_ACCESS_ERROR 0xE6: Safety test: AUX_ERR

PSM_STATUS_0 (0x1E8)

BIT	7	6	5	4	3	2	1	0
Field	PSM_STATUS[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PSM_STATUS	7:0	Contains detailed status of the processing of the user request at PSM_STATE (0x1E7) by the PSM. This register is only valid when PSM_STATE is either 0x00 or 0x01. Values not listed in the table should be ignored.	0x00: Idle 0x01: DP link not trained. For example, user requesting PSM_STATE 0x12 without previously requesting DP link to be trained. 0x02: Invalid training parameters encountered 0x03: DP link training failed (training requests only) 0xEF: Safety test failed 0xFE: Request overrun. Proper handshake not followed and user request in PSM_STATE register updated by user before the PSM returned an idle status (0x00 or 0x01). 0xFF: Request is invalid or not supported

UART_0 (0x248)

BIT	7	6	5	4	3	2	1	0
Field	RSVD[1:0]		REM_MS_EN	LOC_MS_EN	RSVD	–	–	–
Reset	0b01		0b0	0b0	0b0	–	–	–

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Access Type	Write, Read	Write, Read	Write, Read	Write, Read	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
RSVD	7:6	UART RX source arbitration timeout duration UART RX processes packets from a single UART source at any time. When UART RX does not receive any UART packets for this duration, it will select the next UART source according to the source ID of the next following received packet.	0b00: 2 ms 0b01: 8 ms 0b10: 32 ms 0b11: never
REM_MS_EN	5	Enables UART Bypass mode control by remote GPIO pin When set, remote chip's GPIO is used as MS pin (UART mode select). When MS is high, the chip is in Bypass mode; otherwise, the chip is in Base mode.	0b0: UART Bypass mode not controlled by remote MS pin 0b1: UART Bypass mode controlled by remote MS pin
LOC_MS_EN	4	Enables UART Bypass mode control by local GPIO pin Set to use GPIO18 pin as MS pin (UART mode select). When MS is high, the chip is in Bypass mode; otherwise, the chip is in Base mode.	0b0: UART Bypass mode not controlled by local MS pin 0b1: UART Bypass mode controlled by local MS pin
RSVD	3	Selects whether to receive and send parity bit in Bypass mode	0b0: Receive and send parity bit in Bypass mode 0b1: Do not receive and send parity bit in Bypass mode

INTR13 (0x2D3)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	EOM_ERR_FLAG_DCIO	EOM_ERR_FLAG_SIOB	EOM_ERR_FLAG_SIOA
Reset	–	–	–	–	–	0b0	0b0	0b0
Access Type	–	–	–	–	–	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_ERR_FLAG_DCIO	2	Eye-opening monitor is below configured threshold for Link DCIO Read this bitfield to clear.	0b0: No EOM error on Link DCIO 0b1: EOM error on Link DCIO
EOM_ERR_FLAG_SIOB	1	Eye-opening monitor is below configured threshold for Link SIOB Read this bitfield to clear.	0b0: No EOM error on Link SIO 0b1: EOM error on Link
EOM_ERR_FLAG_SIOA	0	Eye-opening monitor is below configured threshold for Link SIOA	0b0: No EOM error on Link SIO 0b1: EOM error on Link

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BITFIELD	BITS	DESCRIPTION	DECODE
		Read this bitfield to clear.	

TR3 (0x303)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	TX_SRC_ID[2:0]		
Reset	–	–	–	–	–	0b000		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
TX_SRC_ID	2:0	Source identifier used in packets transmitted from this channel. Default value of the 2 MSbs are set by ADD2 and ADD1. Default value of the LSb is 0.	0bXXX: Source ID for packets from this channel

LUT_CTRL (0x502)

BIT	7	6	5	4	3	2	1	0
Field	–	–	LUT_MSB_DATA[1:0]		LUT_10_BIT_OUT	–	–	–
Reset	–	–	0x0		0b1	–	–	–
Access Type	–	–	Write, Read		Write, Read	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
LUT_MSB_DATA	5:4	These 2 bits are used as the MSB bits of the color LUT. When filling the LUT, write to this register first for each address and then write the LUT bits [7:0]. When LSB bits are written, these 2 bits are used as the MSB 2 bits for that address.	0x0: Write 00 to 2 MSB of 10-bit LUT 0x1: Write 01 to 2 MSB of 10-bit LUT 0x2: Write 10 to 2 MSB of 10-bit LUT 0x3: Write 11 to 2 MSB of 10-bit LUT
LUT_10_BIT_OUT	3	Selects if the color LUT should have an 8-bit or 10-bit output.	0x0: 8-bit output 0x1: 10-bit output

VRX_ASIL_INJ (0x51F)

BIT	7	6	5	4	3	2	1	0

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Field	–	–	FRC_DATAPATH_FAIL_INJ	LUT_CRC_BLUE_FAIL_INJ	LUT_CRC_GREEN_FAIL_INJ	LUT_CRC_RED_FAIL_INJ	LUT_DATAPATH_FAIL_INJ	WM_DATAPATH_FAIL_INJ
Reset	–	–	0b0	0b0	0b0	0b0	0b0	0b0
Access Type	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
FRC_DATAPATH_FAIL_INJ	5	FRC data path ASIL checker error injection.	0b0: Do not inject error 0b1: Inject error
LUT_CRC_BLUE_FAIL_INJ	4	Inject error to BLUE LUT memory CRC ASIL checker.	0b0: Do not inject error 0b1: Inject error
LUT_CRC_GREEN_FAIL_INJ	3	Inject error to GREEN LUT memory CRC ASIL checker.	0b0: Do not inject error 0b1: Inject error
LUT_CRC_RED_FAIL_INJ	2	Inject error to RED LUT memory CRC ASIL checker.	0b0: Do not inject error 0b1: Inject error
LUT_DATAPATH_FAIL_INJ	1	Inject error to LUT data path ASIL checker.	0b0: Do not inject error 0b1: Inject error
WM_DATAPATH_FAIL_INJ	0	Inject error to WM data path ASIL checker.	0b0: Do not inject error 0b1: Inject error

BLANK_REGION_CONTROL (0x522)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	BLANK_REGION_EN
Reset	–	–	–	–	–	–	–	
Access Type	–	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BLANK_REGION_EN	0	Control register to enable manual regional blanking	0x0: Disable 0x1: Enable

BLANK_REGION_XY_CFG_0 (0x523)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_XMIN_BYTE_0[7:0]							
Reset								

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Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_XMIN_BYTE_0	7:0	Byte 0 of left-side x-coordinate of manual blanking region. Value is specified with respect to x-coordinate in displayed image. For DSC-reconstructed image or image combined from 2 streams.

BLANK_REGION_XY_CFG_1 (0x524)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_XMIN_BYTE_1[7:0]							
Reset								
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_XMIN_BYTE_1	7:0	Byte 1 of left-side x-coordinate of manual blanking region. Value is specified with respect to x-coordinate in displayed image. For DSC-reconstructed image or image combined from 2 streams.

BLANK_REGION_XY_CFG_2 (0x525)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_XMAX_BYTE_0[7:0]							
Reset								
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_XMAX_BYTE_0	7:0	Byte 0 of right-side x-coordinate of manual blanking region. Value is specified with respect to x-coordinate in displayed image. For DSC-reconstructed image or image combined from 2 streams.

BLANK_REGION_XY_CFG_3 (0x526)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_XMAX_BYTE_1[7:0]							

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Reset	
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_XMAX_BYTE_1	7:0	Byte 1 of right-side x-coordinate of manual blanking region. Value is specified with respect to x-coordinate in displayed image. For DSC-reconstructed image or image combined from 2 streams.

BLANK_REGION_XY_CFG_4 (0x527)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_YMIN_BYTE_0[7:0]							
Reset								
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_YMIN_BYTE_0	7:0	Byte 0 of top-side y-coordinate of manual blanking region

BLANK_REGION_XY_CFG_5 (0x528)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_YMIN_BYTE_1[7:0]							
Reset								
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_YMIN_BYTE_1	7:0	Byte 1 of top-side y-coordinate of manual blanking region

BLANK_REGION_XY_CFG_6 (0x529)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_YMAX_BYTE_0[7:0]							
Reset								

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Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_YMAX_BYTE_0	7:0	Byte 0 of bottom-side y-coordinate of manual blanking region

BLANK_REGION_XY_CFG_7 (0x52A)

BIT	7	6	5	4	3	2	1	0
Field	BLANK_REGION_YMAX_BYTE_1[7:0]							
Reset								
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BLANK_REGION_YMAX_BYTE_1	7:0	Byte 1 of bottom-side y-coordinate of manual blanking region

LUT_CTRL (0x542)

BIT	7	6	5	4	3	2	1	0
Field	–	–	LUT_MSB_DATA[1:0]		LUT_10_BIT_OUT	–	–	–
Reset	–	–	0x0		0b1	–	–	–
Access Type	–	–	Write, Read		Write, Read	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
LUT_MSB_DATA	5:4	These 2 bits are used as the MSB bits of the color LUT. When filling the LUT, write to this register first for each address and then write the LUT bits [7:0]. When LSB bits are written, these 2 bits are used as the MSB 2 bits for that address.	0x0: Write 00 to 2 MSB of 10-bit LUT 0x1: Write 01 to 2 MSB of 10-bit LUT 0x2: Write 10 to 2 MSB of 10-bit LUT 0x3: Write 11 to 2 MSB of 10-bit LUT
LUT_10_BIT_OUT	3	Selects if the color LUT should have an 8-bit or 10-bit output.	0x0: 8-bit output 0x1: 10-bit output

PRBS_ERR (0x55B)

BIT	7	6	5	4	3	2	1	0

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Field	VPRBS_ERROR_COUNT[7:0]
Reset	0x00
Access Type	Read Clears All

BITFIELD	BITS	DESCRIPTION	DECODE
VPRBS_ERROR_COUNT	7:0	Pipe Y Video PRBS Error Counter. Clears on read.	0xXX: Number of video PRBS errors since last read

VRX_ASIL_INJ (0x55F)

BIT	7	6	5	4	3	2	1	0
Field	–	–	FRC_DATAPATH_FAIL_INJ	LUT_CRC_BLUE_FAIL_INJ	LUT_CRC_GREEN_FAIL_INJ	LUT_CRC_RED_FAIL_INJ	LUT_DATAPATH_FAIL_INJ	WM_DATAPATH_FAIL_INJ
Reset	–	–	0b0	0b0	0b0	0b0	0b0	0b0
Access Type	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
FRC_DATAPATH_FAIL_INJ	5	FRC data path ASIL checker error injection.	0b0: Do not inject error 0b1: Inject error
LUT_CRC_BLUE_FAIL_INJ	4	Inject error to BLUE LUT memory CRC ASIL checker.	0b0: Do not inject error 0b1: Inject error
LUT_CRC_GREEN_FAIL_INJ	3	Inject error to GREEN LUT memory CRC ASIL checker.	0b0: Do not inject error 0b1: Inject error
LUT_CRC_RED_FAIL_INJ	2	Inject error to RED LUT memory CRC ASIL checker.	0b0: Do not inject error 0b1: Inject error
LUT_DATAPATH_FAIL_INJ	1	Inject error to LUT data path ASIL checker.	0b0: Do not inject error 0b1: Inject error
WM_DATAPATH_FAIL_INJ	0	Inject error to WM data path ASIL checker.	0b0: Do not inject error 0b1: Inject error

AUTH0 (0x600)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CYCLE_MODULO_H[1:0]		LOCAL_CHALLENGE_MAX[1:0]		–	AUTH_GLOBAL_PERIODIC	AUTH_GLOBAL_TRIG	AUTH_EN
Reset	0b00		0b00		–	0b1	0b0	0b1

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Access Type	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CYCLE_MODULO_H	7:6	Modulo value for the frame cycle counter.	0bxx: MSB bits for frame cycle modulo value
LOCAL_CHALLENGE_MAX	5:4	Selects desired number of local challenges.	0b00: One local challenge 0b01: Two local challenges 0b10: Three local challenges 0b11: Four local challenges
AUTH_GLOBAL_PERIODIC	2	Enables video authentication feature periodic global challenge mode.	0b0: Disabled 0b1: Enabled
AUTH_GLOBAL_TRIG	1	Enables video authentication feature global challenge trigger. User needs to create a rising edge.	0b0: Disabled 0b1: Enabled
AUTH_EN	0	Enables video authentication feature.	0b0: Disabled 0b1: Enabled

AUTH1 (0x601)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CYCLE_MODULO_L[7:0]							
Reset	0x04							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CYCLE_MODULO_L	7:0	Modulo value for the frame cycle counter.	0xXX: LSB bits for frame cycle modulo value

XMINL0a (0x602)

BIT	7	6	5	4	3	2	1	0
Field	XMIN0a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMIN0a_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits. Minimum allowed value is 0x02.

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XMINH0a (0x603)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMIN0a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMIN0a_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits. Minimum allowed value is 0x02.

XMAXL0a (0x604)

BIT	7	6	5	4	3	2	1	0
Field	XMAX0a_LSB[7:0]							
Reset	0x01							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMAX0a_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

XMAXH0a (0x605)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX0a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMAX0a_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

XMINL1a (0x606)

BIT	7	6	5	4	3	2	1	0

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Field	XMIN1a_LSB[7:0]
Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
XMIN1a_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits.

XMINH1a (0x607)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMIN1a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMIN1a_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits.

XMAXL1a (0x608)

BIT	7	6	5	4	3	2	1	0
Field	XMAX1a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMAX1a_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

XMAXH1a (0x609)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX1a_MSB[5:0]					
Reset	–	–	0b000000					

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Access Type	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION
XMAX1a_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

XMINL2a (0x60A)

BIT	7	6	5	4	3	2	1	0
Field	XMIN2a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMIN2a_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits.

XMINH2a (0x60B)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMIN2a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMIN2a_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits.

XMAXL2a (0x60C)

BIT	7	6	5	4	3	2	1	0
Field	XMAX2a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
XMAX2a_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

XMAXH2a (0x60D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX2a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMAX2a_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

XMINL3a (0x60E)

BIT	7	6	5	4	3	2	1	0
Field	XMIN3a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMIN3a_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits.

XMINH3a (0x60F)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMIN3a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMIN3a_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits.

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XMAXL3a (0x610)

BIT	7	6	5	4	3	2	1	0
Field	XMAX3a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMAX3a_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

XMAXH3a (0x611)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX3a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMAX3a_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

XMINL0b (0x612)

BIT	7	6	5	4	3	2	1	0
Field	XMIN0b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMIN0b_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits. Minimum allowed value is 0x02.

XMINH0b (0x613)

BIT	7	6	5	4	3	2	1	0

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Field	–	–	XMIN0b_MSB[5:0]
Reset	–	–	0b000000
Access Type	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION
XMIN0b_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits. Minimum allowed value is 0x02.

XMAXL0b (0x614)

BIT	7	6	5	4	3	2	1	0
Field	XMAX0b_LSB[7:0]							
Reset	0x01							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMAX0b_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

XMAXH0b (0x615)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX0b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMAX0b_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

XMINL1b (0x616)

BIT	7	6	5	4	3	2	1	0
Field	XMIN1b_LSB[7:0]							
Reset	0x00							

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Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
XMIN1b_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits.

XMINH1b (0x617)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMIN1b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMIN1b_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits.

XMAXL1b (0x618)

BIT	7	6	5	4	3	2	1	0
Field	XMAX1b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMAX1b_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

XMAXH1b (0x619)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX1b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
XMAX1b_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

XMINL2b (0x61A)

BIT	7	6	5	4	3	2	1	0
Field	XMIN2b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMIN2b_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits.

XMINH2b (0x61B)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMIN2b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMIN2b_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits.

XMAXL2b (0x61C)

BIT	7	6	5	4	3	2	1	0
Field	XMAX2b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMAX2b_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

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XMAXH2b (0x61D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX2b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMAX2b_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

XMINL3b (0x61E)

BIT	7	6	5	4	3	2	1	0
Field	XMIN3b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
XMIN3b_LSB	7:0	Local Challenge X coordinate minimum value least significant 8 bits.

XMINH3b (0x61F)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMIN3b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMIN3b_MSB	5:0	Local Challenge X coordinate minimum value most significant 6 bits.

XMAXL3b (0x620)

BIT	7	6	5	4	3	2	1	0
Field	XMAX3b_LSB[7:0]							

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Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
XMAX3b_LSB	7:0	Local Challenge X coordinate maximum value least significant 8 bits.

XMAXH3b (0x621)

BIT	7	6	5	4	3	2	1	0
Field	–	–	XMAX3b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
XMAX3b_MSB	5:0	Local Challenge X coordinate maximum value most significant 6 bits.

YMINL0a (0x622)

BIT	7	6	5	4	3	2	1	0
Field	YMIN0a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMIN0a_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

YMINH0a (0x623)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN0a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
YMIN0a_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL0a (0x624)

BIT	7	6	5	4	3	2	1	0
Field	YMAX0a_LSB[7:0]							
Reset	0x01							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMAX0a_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

YMAXH0a (0x625)

BIT	7	6	5	4	3	2	1	0
Field	FRAME0a_MSB[1:0]		YMAX0a_MSB[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
FRAME0a_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).
YMAX0a_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

frame0a (0x626)

BIT	7	6	5	4	3	2	1	0
Field	FRAME0a_LSB[7:0]							
Reset	0x01							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
FRAME0a_LSB	7:0	Local challenge frame number least significant 8 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

YMINL1a (0x627)

BIT	7	6	5	4	3	2	1	0
Field	YMIN1a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMIN1a_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

YMINH1a (0x628)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN1a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
YMIN1a_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL1a (0x629)

BIT	7	6	5	4	3	2	1	0
Field	YMAX1a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMAX1a_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

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YMAXH1a (0x62A)

BIT	7	6	5	4	3	2	1	0
Field	FRAME1a_MSB[1:0]		YMAX1a_MSB[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
FRAME1a_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).
YMAX1a_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

frame1a (0x62B)

BIT	7	6	5	4	3	2	1	0
Field	FRAME1a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME1a_LSB	7:0	Local challenge frame number least significant 8 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

YMINL2a (0x62C)

BIT	7	6	5	4	3	2	1	0
Field	YMIN2a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMIN2a_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

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YMINH2a (0x62D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN2a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
YMIN2a_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL2a (0x62E)

BIT	7	6	5	4	3	2	1	0
Field	YMAX2a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMAX2a_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

YMAXH2a (0x62F)

BIT	7	6	5	4	3	2	1	0
Field	FRAME2a_MSB[1:0]		YMAX2a_MSB[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
FRAME2a_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).
YMAX2a_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

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frame2a (0x630)

BIT	7	6	5	4	3	2	1	0
Field	FRAME2a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME2a_LSB	7:0	Local challenge frame number least significant 8 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

YMINL3a (0x631)

BIT	7	6	5	4	3	2	1	0
Field	YMIN3a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMIN3a_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

YMINH3a (0x632)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN3a_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
YMIN3a_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL3a (0x633)

BIT	7	6	5	4	3	2	1	0

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Field	YMAX3a_LSB[7:0]
Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
YMAX3a_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

YMAXH3a (0x634)

BIT	7	6	5	4	3	2	1	0
Field	FRAME3a_MSB[1:0]		YMAX3a_MSB[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
FRAME3a_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).
YMAX3a_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

frame3a (0x635)

BIT	7	6	5	4	3	2	1	0
Field	FRAME3a_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME3a_LSB	7:0	Local challenge frame number least significant 8 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

YMINL0b (0x636)

BIT	7	6	5	4	3	2	1	0

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Field	YMIN0b_LSB[7:0]
Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
YMIN0b_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

YMINH0b (0x637)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN0b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
YMIN0b_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL0b (0x638)

BIT	7	6	5	4	3	2	1	0
Field	YMAX0b_LSB[7:0]							
Reset	0x01							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMAX0b_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

YMAXH0b (0x639)

BIT	7	6	5	4	3	2	1	0
Field	FRAME0b_MSB[1:0]		YMAX0b_MSB[5:0]					
Reset	0b00		0b000000					

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Access Type	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION
FRAME0b_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).
YMAX0b_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

frame0b (0x63A)

BIT	7	6	5	4	3	2	1	0
Field	FRAME0b_LSB[7:0]							
Reset	0x01							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME0b_LSB	7:0	Local challenge frame number least significant 8 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

YMINL1b (0x63B)

BIT	7	6	5	4	3	2	1	0
Field	YMIN1b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMIN1b_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

YMINH1b (0x63C)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN1b_MSB[5:0]					
Reset	–	–	0b000000					

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Access Type	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION
YMIN1b_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL1b (0x63D)

BIT	7	6	5	4	3	2	1	0
Field	YMAX1b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMAX1b_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

YMAXH1b (0x63E)

BIT	7	6	5	4	3	2	1	0
Field	FRAME1b_MSB[1:0]		YMAX1b_MSB[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
FRAME1b_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).
YMAX1b_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

frame1b (0x63F)

BIT	7	6	5	4	3	2	1	0
Field	FRAME1b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
FRAME1b_LSB	7:0	Local challenge frame number least significant 8 bit. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

YMINL2b (0x640)

BIT	7	6	5	4	3	2	1	0
Field	YMIN2b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMIN2b_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

YMINH2b (0x641)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN2b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
YMIN2b_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL2b (0x642)

BIT	7	6	5	4	3	2	1	0
Field	YMAX2b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
YMAX2b_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

YMAXH2b (0x643)

BIT	7	6	5	4	3	2	1	0
Field	FRAME2b_MSB[1:0]		YMAX2b_MSB[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
FRAME2b_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).
YMAX2b_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

frame2b (0x644)

BIT	7	6	5	4	3	2	1	0
Field	FRAME2b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME2b_LSB	7:0	Local challenge frame number least significant 8 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

YMINL3b (0x645)

BIT	7	6	5	4	3	2	1	0
Field	YMIN3b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
YMIN3b_LSB	7:0	Local Challenge Y coordinate minimum value least significant 8 bits.

YMINH3b (0x646)

BIT	7	6	5	4	3	2	1	0
Field	–	–	YMIN3b_MSB[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
YMIN3b_MSB	5:0	Local Challenge Y coordinate minimum value most significant 6 bits.

YMAXL3b (0x647)

BIT	7	6	5	4	3	2	1	0
Field	YMAX3b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
YMAX3b_LSB	7:0	Local Challenge Y coordinate maximum value least significant 8 bits.

YMAXH3b (0x648)

BIT	7	6	5	4	3	2	1	0
Field	FRAME3b_MSB[1:0]		YMAX3b_MSB[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
FRAME3b_MSB	7:6	Local challenge frame number most significant 2 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

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BITFIELD	BITS	DESCRIPTION
YMAX3b_MSB	5:0	Local Challenge Y coordinate maximum value most significant 6 bits.

frame3b (0x649)

BIT	7	6	5	4	3	2	1	0
Field	FRAME3b_LSB[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME3b_LSB	7:0	Local challenge frame number least significant 8 bits. Decimal 511 value is reserved (combined 10-bit value of the frame _MSB and _LSB).

out1 (0x64A)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CNT_MSB	–	VA_DONE_FLAG	VA_DONE_INT_EN	VA_DONE_PIN	A_OR_B	AUTH_DONE_B	AUTH_DONE_A
Reset	0b0	–	0b0	0b0	0b0	0b0	0b0	0b0
Access Type	Read Only	–	Read Only	Write, Read	Write, Read	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CNT_MSB	7	Frame counter value MSB bit.	0bx: MSB bit for frame cycle counter value
VA_DONE_FLAG	5	Video authentication done status flag. This flag is automatically cleared when the next set of authentication challenge starts.	0x0: Running 0x1: Challenge complete
VA_DONE_INT_EN	4	Enables video authentication done status to be reflected to the ERRB pin.	0b0: Disabled 0b1: Enabled
VA_DONE_PIN	3	Enables video authentication done status to be shown on GPIO pins.	0x0: Disabled 0x1: Enabled
A_OR_B	2	Shows which register bank is active.	0x0: Set A 0x1: Set B
AUTH_DONE_B	1	Challenge B is done.	0x0: Not finished 0x1: Finished
AUTH_DONE_A	0	Challenge A is done.	0x0: Not finished 0x1: Finished

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out1a (0x64B)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CNT_LSB[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CNT_LSB	7:0	Frame counter value LSB bits.	0dx: LSB bits for frame cycle counter value

out2 (0x64C)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_A_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
CRC_OUT_A_0	7:0	CRC result for Challenge A. Valid only at the rising edge of done signal.

out3 (0x64D)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_A_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
CRC_OUT_A_1	7:0	CRC result for Challenge A. Valid only at the rising edge of done signal.

out4 (0x64E)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_A_2[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
CRC_OUT_A_2	7:0	CRC result for Challenge A. Valid only at the rising edge of done signal.

out5 (0x64F)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_A_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
CRC_OUT_A_3	7:0	CRC result for Challenge A. Valid only at the rising edge of done signal.

out6 (0x650)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_B_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
CRC_OUT_B_0	7:0	CRC result for Challenge B. Valid only at the rising edge of done signal.

out7 (0x651)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_B_1[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
CRC_OUT_B_1	7:0	CRC result for Challenge B. Valid only at the rising edge of done signal.

out8 (0x652)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_B_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
CRC_OUT_B_2	7:0	CRC result for Challenge B. Valid only at the rising edge of done signal.

out9 (0x653)

BIT	7	6	5	4	3	2	1	0
Field	CRC_OUT_B_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
CRC_OUT_B_3	7:0	CRC result for Challenge B. Valid only at the rising edge of done signal.

outa (0x654)

BIT	7	6	5	4	3	2	1	0
Field	FULL_FRAME_CRC_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FULL_FRAME_CRC_0	7:0	Full-Frame CRC Result. Valid only when VIDEO_LOCK_X = 0b1.

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outb (0x655)

BIT	7	6	5	4	3	2	1	0
Field	FULL_FRAME_CRC_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FULL_FRAME_CRC_1	7:0	Full-Frame CRC Result. Valid only when VIDEO_LOCK_X = 0b1.

outc (0x656)

BIT	7	6	5	4	3	2	1	0
Field	FULL_FRAME_CRC_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FULL_FRAME_CRC_2	7:0	Full-Frame CRC Result. Valid only when VIDEO_LOCK_X = 0b1.

outd (0x657)

BIT	7	6	5	4	3	2	1	0
Field	FULL_FRAME_CRC_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FULL_FRAME_CRC_3	7:0	Full-Frame CRC Result. Valid only when VIDEO_LOCK_X = 0b1.

RLMS4 (0x804)

BIT	7	6	5	4	3	2	1	0
Field	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN

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Reset	0x4	0b10	0b1	0b1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_CHK_AMOUNT	7:4	A factor (N) used to select the order of number of observations in each eye-monitor window. N is used in the equation: Observations = 6.29 x $10^{(N+2)}$	0xX: N factor
EOM_CHK_THR	3:2	Eye-opening monitor number of error bits to allow in a measurement window.	0b00: Allow no errors 0b01: Allow one error 0b10: Allow two errors 0b11: Allow three errors
EOM_PER_MODE	1	Eye-opening monitor periodic mode enable.	0b0: Eye-opening monitor periodic mode disabled 0b1: Eye-opening monitor periodic mode enabled
EOM_EN	0	Eye-opening monitor enable. Recommended for GMSL2 use only.	0b0: Eye-opening monitor disabled 0b1: Eye-opening monitor enabled

RLMS5 (0x805)

BIT	7	6	5	4	3	2	1	0
Field	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
Reset	0b0	0b0010000						
Access Type	Write Only	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_MAN_TRG_REQ	7	Eye-opening monitor manual trigger. For use when periodic mode is disabled.	0b0: No action 0b1: EOM manual trigger request
EOM_MIN_THR	6:0	The EOM minimum threshold as defined by the equation: % eye-opening = EOM_MIN_THR/64 If the value is zero, the EOM is disabled.	0xXX: EOM minimum threshold factor

RLMS6 (0x806)

BIT	7	6	5	4	3	2	1	0
Field	EOM_PV_MODE	EOM_RST_THR[6:0]						
Reset	0b1	0b0000000						

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Access Type	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_PV_MODE	7	Eye-opening is measured vertically or horizontally.	0b0: Vertical opening mode 0b1: Horizontal opening mode
EOM_RST_THR	6:0	The EOM refresh threshold as defined by the equation: % eye-opening = EOM_MIN_THR/64. Used only with periodic EOM mode. If the value is zero, the EOM is disabled.	0xXX: EOM refresh threshold factor

RLMS7 (0x807)

BIT	7	6	5	4	3	2	1	0
Field	EOM_DONE	EOM[6:0]						
Reset	0b0	0b0000000						
Access Type	Read Only	Read Only						

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_DONE	7	Eye-opening monitor measurement done	0b0: EOM not complete 0b1: EOM complete
EOM	6:0	Last completed EOM observation	0xXX: EOM measurement result

RLMSCE (0x8CE)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	enffe
Reset	–	–	–	–	–	–	–	0b1
Access Type	–	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION
enffe	0	Feedforward Enable

CTRL0 (0xA00)

BIT	7	6	5	4	3	2	1	0

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Field	–	–	–	–	–	–	–	ENABLE_INFOFR_CC
Reset	–	–	–	–	–	–	–	0b0
Access Type	–	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
ENABLE_INFOFR_CC	0	Enables the infofr_cc (infoframe control channel) block.	0x0: Infofr_cc is not enabled 0x1: Infofr_cc is enabled

FFWD_CRC_CHK_REG0 (0xC80, 0xCA0)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_CHECK_EN	FRAME_COUNT_CHKECK_EN	EN_VOTING_FOR_FCRC	EN_VOTING_FOR_FCNT	USE_FRAME_CRC_VALID_BIT	_	FRAME_CRC_CALC_EN	SUPPRESS_LAST_LINE
Reset	0b0	0b0	0b1	0b1	0b0	–	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CRC_CHECK_EN	7	Feedforward Video Frame CRC check enable; this can also be enabled by customer-facing register FEEDFORWARD_FRAME_CRC_EN. NOTE: Do not set when DSC bytes are used.	0x0: Feedforward video frame CRC check is disabled 0x1: Feedforward video frame CRC check is enabled
FRAME_COUNT_CHKECK_EN	6	Feedforward Video Frame Count check enable	0x0: Feedforward video frame count check is disabled 0x1: Feedforward video frame count check is enabled
EN_VOTING_FOR_FCRC	5	Enable voting for Feedforward Video Frame CRC	0x0: Voting for Frame CRC is disabled. 0x1: Voting for Frame CRC is enabled.
EN_VOTING_FOR_FCNT	4	Enable voting for Feedforward Video Frame count	0x0: Voting for Frame Count is disabled. 0x1: Voting for Frame Count is enabled.
USE_FRAME_CRC_VALID_BIT	3	Use frame CRC valid bit in addional video line to enable/disable FCRC check for corresponding frame	0x0: Do not use frame CRC valid bit 0x1: Use frame CRC valid bit
FRAME_CRC_CALC_EN	1	Enables/disables calculation of Feedforward Video FCRC. This can also be enabled by customer-facing register FEEDFORWARD_FRAME_CRC_EN.	0x0: Disable calculation of FCRC. 0x1: Enable calculation of FCRC.

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BITFIELD	BITS	DESCRIPTION	DECODE
SUPPRESS_LAST_LINE	0	Suppress last video line to strip the frame CRC from the video frame. This can also be enabled via the customer-facing register FEEDFORWARD_FRAME_CRC_REMOVE.	0x0: Do not suppress last line 0x1: Suppress last line

FFWD_CRC_CHK_REG1 (0xC81, 0xCA1)

BIT	7	6	5	4	3	2	1	0
Field	–	TRIGGER_ONLY_ON_FCRC_RCVD	USE_DP_FCRC_INFO	USE_FCRC_FCNT_EXP_REG	USE_PREV_FRAME_TO_CHECK	FRAME_CRC_CHECK_SEL[2:0]		
Reset	–	0b0	0b0	0b0	0b0	0b00		
Access Type	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
TRIGGER_ONLY_ON_FCRC_RCVD	6	This register enables FCRC/FCNT check only when registers are updated or Infoframes are received, when FCRC_EXP regs or DP_FCRC Infos are used. If this register is 0, it checks FCRC/FCNT regardless of whether it is received or not.	0x0: Check FCRC/FCNT regardless of what corresponding registers are updated or infoframes are received. 0x1: Check FCRC/FCNT only when corresponding registers are updated or infoframes are received.
USE_DP_FCRC_INFO	5	Enables/disables using DP Frame CRC and Frame Count values, which are received by Infoframes (registers in DP_FCRC_INFO block). If this is set, extra line is not used to receive expected Frame CRC and Frame Count values. Note: Do not use with USE_FCRC_FCNT_EXP_REG.	0x0: Disable using DP FCRC and FCNT values, received by DP infoframes. 0x1: Enable using DP FCRC and FCNT values, received by DP infoframes.
USE_FCRC_FCNT_EXP_REG	4	Enables/disables using FRAME_CRC_EXP_* and FRAME_COUNT_EXP regs, which are manually programmed. If this is set, extra line is not used to receive expected Frame CRC and Frame Count values. Note: Do not use with USE_DP_FCRC_INFO.	0x0: Disable using FCRC_EXP_* and FCNT_EXP. 0x1: Enable using FCRC_EXP_* and FCNT_EXP.
USE_PREV_FRAME_TO_CHECK	3	Enables comparing FCRC_EXP_* and FCNT_EXP regs with CRC and FCNT values of previous frame.	0x0: Compare FCRC_EXP_* and FCNT_EXP regs with CRC/FCNT values of current frame 0x1: Compare FCRC_EXP_* and FCNT_EXP regs with CRC/FCNT values of previous frame
FRAME_CRC_CHECK_SEL	2:0	Feedforward Video Frame CRC check polynomial selector.	0x0: Use programmable CRC (24 bit data with default CRC polynomial 0x4C11DB7) 0x1: Use constant CRC polynomial (0x4C11DB7) with 24 bit data 0x2: Use constant CRC polynomial (0x4C11DB7) with 30 bit data

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BITFIELD	BITS	DESCRIPTION	DECODE
			0x3: Use default CRC (which is also used in GMSL frame CRC check) 0x4: Use DP_USER_CRC 0x5: RSVD 0x6: RSVD 0x7: RSVD

FFWD_CRC_CHK_REG2 (0xC82, 0xCA2)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_EXP_BYTE_5[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_EXP_BYTE_5	7:0	Expected FCRC value bits[47:0].

FFWD_CRC_CHK_REG3 (0xC83, 0xCA3)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_EXP_BYTE_4[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_EXP_BYTE_4	7:0	Expected FCRC value bits[39:32].

FFWD_CRC_CHK_REG4 (0xC84, 0xCA4)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_EXP_BYTE_3[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
FRAME_CRC_EXP_BYTE_3	7:0	Expected FCRC value bits[31:24].

FFWD_CRC_CHK_REG5 (0xC85, 0xCA5)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_EXP_BYTE_2[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_EXP_BYTE_2	7:0	Expected FCRC value bits[23:16].

FFWD_CRC_CHK_REG6 (0xC86, 0xCA6)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_EXP_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_EXP_BYTE_1	7:0	Expected FCRC value bits[15:8].

FFWD_CRC_CHK_REG7 (0xC87, 0xCA7)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_EXP_BYTE_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_EXP_BYTE_0	7:0	Expected FCRC value bits[7:0].

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FFWD_CRC_CHK_REG8 (0xC88, 0xCA8)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT_EXP[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
FRAME_COUNT_EXP	7:0	Expected FCNT value bits[7:0].

FFWD_CRC_CHK_REG9 (0xC89, 0xCA9)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	RESET_FRAME_CNT	UPDATE_FCNT_EXP	UPDATE_FCRC_EXP
Reset	–	–	–	–	–	0b0	0b0	0b0
Access Type	–	–	–	–	–	Write Clears All, Read	Write Clears All, Read	Write Clears All, Read

BITFIELD	BITS	DESCRIPTION
RESET_FRAME_CNT	2	Resets FRAME_CNT register. It also clears fcnt_synced register; therefore, it syncs to incoming frame_cnt value (either from fcnt_exp register or from extra line, depending on USE_FCRC_EXP_REG register).
UPDATE_FCNT_EXP	1	It is set to 1 when FCNT_EXP is written. It triggers checking FCNT at incoming VS
UPDATE_FCRC_EXP	0	It is set to 1 when FCRC_EXP_3/2/1/0 are written. It triggers checking FCRC at incoming VS

FFWD_CRC_CHK_REG10 (0xC8A, 0xCAA)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_CHECK_ERR	FRAME_COUNT_CHECK_ERR	VOTE_ERR_FCRC	VOTE_ERR_FCNT	–	–	–	–
Reset	0b0	0b0	0b0	0b0	–	–	–	–
Access Type	Write 1 to Clear, Read	Write 1 to Clear, Read	Write 1 to Clear, Read	Write 1 to Clear, Read	–	–	–	–

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BITFIELD	BITS	DESCRIPTION	DECODE
FRAME_CRC_CHECK_ERR	7	Feedforward Video FCRC check error status. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.	0x0: Check error not detected 0x1: Check error detected
FRAME_COUNT_CHECK_ERR	6	Feedforward Video Frame count check error status. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.	0x0: Check error not detected 0x1: Check error detected
VOTE_ERR_FCRC	5	Voting error occurred in received FCRC, in which all three values are different. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.	
VOTE_ERR_FCNT	4	Voting error occurred in received FCNT, in which all three values are different. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.	

FFWD_CRC_CHK_REG11 (0xC8B, 0xCAB)

BIT	7	6	5	4	3	2	1	0
Field	FCRC_CHECK_ERR_LAST	FCNT_CHECK_ERR_LAST	FCNT_SYNCED	–	–	–	–	–
Reset	0b0	0b0	0b0	–	–	–	–	–
Access Type	Read Only	Read Only	Read Only	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
FCRC_CHECK_ERR_LAST	7	Error status of last Feedforward Video frame CRC check	0x0: No error occurred in last frame CRC check 0x1: Error occurred in last frame CRC check
FCNT_CHECK_ERR_LAST	6	Error status of last Feedforward Video frame Count check	0x0: No error occurred in last frame count check 0x1: Error occurred in last frame count check
FCNT_SYNCED	5	Shows that frame count value is synced to incoming frame_cnt value (either from fcnt_exp register or from extra line, depending on USE_FCRC_EXP_REG register).	0x0: Frame_cnt is not synced 0x1: Frame_cnt is synced

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FFWD_CRC_CHK_REG12 (0xC8C, 0xCAC)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_RECEIVED_BYTE_5[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_RECEIVED_BYTE_5	7:0	Received frame CRC [47:40] in last frame.

FFWD_CRC_CHK_REG13 (0xC8D, 0xCAD)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_RECEIVED_BYTE_4[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_RECEIVED_BYTE_4	7:0	Received frame CRC [39:32] in last frame.

FFWD_CRC_CHK_REG14 (0xC8E, 0xCAE)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_RECEIVED_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_RECEIVED_BYTE_3	7:0	Received frame CRC [31:24] in last frame.

FFWD_CRC_CHK_REG15 (0xC8F, 0xCAF)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_RECEIVED_BYTE_2[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_RECEIVED_BYTE_2	7:0	Received frame CRC [23:16] in last frame.

FFWD_CRC_CHK_REG16 (0xC90, 0xCB0)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_RECEIVED_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_RECEIVED_BYTE_1	7:0	Received frame CRC [15:8] in last frame.

FFWD_CRC_CHK_REG17 (0xC91, 0xCB1)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_CRC_RECEIVED_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_CRC_RECEIVED_BYTE_0	7:0	Received frame CRC [7:0] in last frame.

FFWD_CRC_CHK_REG18 (0xC92, 0xCB2)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT_RECEIVED_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
FRAME_COUNT_RECEIVED_BYTE_3	7:0	Received frame count [31:24] in last frame.

FFWD_CRC_CHK_REG19 (0xC93, 0xCB3)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT_RECEIVED_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_COUNT_RECEIVED_BYTE_2	7:0	Received frame count [23:16] in last frame.

FFWD_CRC_CHK_REG20 (0xC94, 0xCB4)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT_RECEIVED_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_COUNT_RECEIVED_BYTE_1	7:0	Received frame count [15:8] in last frame.

FFWD_CRC_CHK_REG21 (0xC95, 0xCB5)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT_RECEIVED_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_COUNT_RECEIVED_BYTE_0	7:0	Received frame count [7:0] in last frame.

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FFWD_CRC_CHK_REG22 (0xC96, 0xCB6)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VALUE_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VALUE_BYTE_3	7:0	Feedforward Video Frame CRC value [31:24] of selected CRC. Note: DP_USER_CRC values are read from registers at DP_USER_CRC register block.

FFWD_CRC_CHK_REG23 (0xC97, 0xCB7)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VALUE_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VALUE_BYTE_2	7:0	Feedforward Video Frame CRC value [23:16] of selected CRC. Note: DP_USER_CRC values are read from registers at DP_USER_CRC register block.

FFWD_CRC_CHK_REG24 (0xC98, 0xCB8)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VALUE_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VALUE_BYTE_1	7:0	Feedforward Video Frame CRC value [15:8] of selected CRC. Note: DP_USER_CRC values are read from registers at DP_USER_CRC register block.

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FFWD_CRC_CHK_REG25 (0xC99, 0xCB9)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VALUE_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VALUE_BYTE_0	7:0	Feedforward Video Frame CRC value [7:0] of selected CRC. Note: DP_USER_CRC values are read from registers at DP_USER_CRC register block.

FFWD_CRC_CHK_REG26 (0xC9A, 0xCBA)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
FRAME_COUNT	7:0	Feedforward Video Frame count value in this block (Counts when corresponding FCRC_CALC_EN register is 1).

FFWD_CRC_CHK_REG0 (0xCC0)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_CHECK_EN	COMBINED_FRAME_COUNT_CHKECK_EN	EN_VOTING_FOR_FCRC	EN_VOTING_FOR_FCNT	USE_FRAME_CRC_VALID_BIT	–	COMBINED_FRAME_CRC_CALC_EN	SUPPRESS_LAST_LINE
Reset	0b0	0b0	0b1	0b1	0b0	–	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
COMBINED_FRAME_CRC_CHECK_EN	7	Feedforward Video Frame CRC check enable. This can also be enabled via the customer-facing	0x0: Feedforward video frame CRC check is disabled

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BITFIELD	BITS	DESCRIPTION	DECODE
		register FEEDFORWARD_COMBINED_FRAME_CRC_EN. NOTE: Do not use when DSC is enabled.	0x1: Feedforward video frame CRC check is enabled
COMBINED_FRAME_COUNT_CHKECK_EN	6	Feedforward Video Frame Count check enable	0x0: Feedforward video frame count check is disabled 0x1: Feedforward video frame count check is enabled
EN_VOTING_FOR_FCRC	5	Enable voting for Feedforward Video Frame CRC	0x0: Voting for FCRC is disabled. 0x1: Voting for FCRC is enabled.
EN_VOTING_FOR_FCNT	4	Enable voting for Feedforward video frame count	0x0: Voting for FCNT is disabled. 0x1: Voting for FCNT is enabled.
USE_FRAME_CRC_VALID_BIT	3	Use frame CRC valid bit in additional video line to enable/disable FCRC check for corresponding frame	0x0: Do not use Frame CRC valid bit 0x1: Use Frame CRC valid bit
COMBINED_FRAME_CRC_CALC_EN	1	Enables-disables calculation of Feedforward Video FCRC. This can also be enabled by customer-facing register FEEDFORWARD_FRAME_CRC_EN.	0x0: Disable calculation of FCRC 0x1: Enable calculation of FCRC
SUPPRESS_LAST_LINE	0	Suppress last video line to strip the frame CRC from the video frame. This can also be enabled via the customer-facing register FRAME_COMBINED_FRAME_CRC_REMOVE.	0x0: Do not suppress last line 0x1: Suppress last line

FFWD_CRC_CHK_REG1 (0xCC1)

BIT	7	6	5	4	3	2	1	0
Field	–	TRIGGER_ONLY_ON_FCRC_RCVD	USE_DP_FCRC_INFO	USE_FCRC_FCNT_EXP_REG	USE_PREV_FRAME_TO_CHECK	FRAME_CRC_CHECK_SEL[2:0]		
Reset	–	0b0	0b0	0b0	0b0	0x0		
Access Type	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
TRIGGER_ONLY_ON_FCRC_RCVD	6	This register enables FCRC/FCNT check only when registers are updated or infoframes are received, when FCRC_EXP regs or DP_FCRC Infos are used. If this register is 0, it checks FCRC/FCNT regardless of it is received or not.	0x0: Check FCRC/FCNT regardless of corresponding registers are updated or infoframes are received. 0x1: Check FCRC/FCNT only when corresponding registers are updated or infoframes are received.
USE_DP_FCRC_INFO	5	Enables-disables using DP FCRC and FCNT values, which are received by infoframes (Registers in DP_FCRC_INFO block). If this is set, extra line is not used to receive expected FCRC and FCNT values. Note: Do not use with USE_FCRC_FCNT_EXP_REG.	0x0: Disable using DP FCRC and FCNT values, received by DP infoframes. 0x1: Enable using DP FCRC and FCNT values, received by DP infoframes.

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BITFIELD	BITS	DESCRIPTION	DECODE
USE_FCRC_FCNT_EXP_REG	4	Enables-disables using FCRC_EXP_* and FCNT_EXP regs, which are manually programmed. If this is set, extra line is not used to receive expected FCRC regs. Note: Do not use with USE_DP_FCRC_INFO.	0x0: Disable using FCRC_EXP_* and FCNT_EXP. 0x1: Enable using FCRC_EXP_* and FCNT_EXP.
USE_PREV_FRAME_TO_CHECK	3	Enables comparing FCRC_EXP_* and FCNT_EXP regs with CRC and FCNT values of previous frame.	0x0: Compare FCRC_EXP_* and FCNT_EXP regs with CRC/FCNT values of current frame 0x1: Compare FCRC_EXP_* and FCNT_EXP regs with CRC/FCNT values of previous frame
FRAME_CRC_CHECK_SEL	2:0	Feedforward Video Frame CRC check polynomial select. This can also be written via the customer-facing register FEEDFORWARD_COMBINED_FRAME_CRC_SELECT[1:0].	0x0: Use constant CRC polynomial (0x4C11DB7) with 48-bit data 0x1: Use constant CRC polynomial (0x4C11DB7) with 60-bit data 0x2: Use constant CRC polynomial (0x4C11DB7) with 96-bit data 0x3: Use DP_USER_CRC 0x4: RSVD 0x5: RSVD 0x6: RSVD 0x7: RSVD

FFWD_CRC_CHK_REG2 (0xCC2)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_EXP_BYTE_5[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_EXP_BYTE_5	7:0	Expected FCRC value bits[47:0].

FFWD_CRC_CHK_REG3 (0xCC3)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_EXP_BYTE_4[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_EXP_BYTE_4	7:0	Expected FCRC value bits[39:32].

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FFWD_CRC_CHK_REG4 (0xCC4)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_EXP_BYTE_3[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_EXP_BYTE_3	7:0	Expected FCRC value bits[31:24].

FFWD_CRC_CHK_REG5 (0xCC5)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_EXP_BYTE_2[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_EXP_BYTE_2	7:0	Expected FCRC value bits[23:16].

FFWD_CRC_CHK_REG6 (0xCC6)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_EXP_BYTE_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_EXP_BYTE_1	7:0	Expected FCRC value bits[15:8].

FFWD_CRC_CHK_REG7 (0xCC7)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_EXP_BYTE_0[7:0]							

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Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_EXP_BYTE_0	7:0	Expected FCRC value bits[7:0].

FFWD_CRC_CHK_REG8 (0xCC8)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_COUNT_EXP[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_COUNT_EXP	7:0	Expected FCNT value bits[7:0].

FFWD_CRC_CHK_REG9 (0xCC9)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	RESET_FRAME_CNT	UPDATE_FCNT_EXP	UPDATE_FCRC_EXP
Reset	–	–	–	–	–	0b0	0b0	0b0
Access Type	–	–	–	–	–	Write Clears All, Read	Write Clears All, Read	Write Clears All, Read

BITFIELD	BITS	DESCRIPTION
RESET_FRAME_CNT	2	Resets FRAME_CNT register. It also clears fcnt_synced register, therefore it syncs to incoming frame_cnt value (either from fcnt_exp register or from extra line, depending on USE_FCRC_EXP_REG register).
UPDATE_FCNT_EXP	1	It is set to 1 when FCNT_EXP is written. It triggers checking FCNT at incoming VS
UPDATE_FCRC_EXP	0	It is set to 1 when FCRC_EXP_3/2/1/0 are written. It triggers checking FCRC at incoming VS

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FFWD_CRC_CHK_REG10 (0xCCA)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_CHECK_ERR	COMBINED_FRAME_COUNT_CHECK_ERR	VOTE_ERR_FCRC	VOTE_ERR_FCNT	–	–	–	–
Reset	0b0	0b0	0b0	0b0	–	–	–	–
Access Type	Write 1 to Clear, Read	Write 1 to Clear, Read	Write 1 to Clear, Read	Write 1 to Clear, Read	–	–	–	–

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_CHECK_ERR	7	Feedforward Video Frame CRC check error status. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.
COMBINED_FRAME_COUNT_CHECK_ERR	6	Feedforward Video Frame count check error status. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.
VOTE_ERR_FCRC	5	It shows that voting error occurred in received FCRC, in which all three values are different. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.
VOTE_ERR_FCNT	4	It shows that voting error occurred in received FCNT, in which all three values are different. Write 1 to clear. In GM02, reading customer-facing status register CRC_MISC_STATUS_0 at 0x6E also clears this status register.

FFWD_CRC_CHK_REG11 (0xCCB)

BIT	7	6	5	4	3	2	1	0
Field	FCRC_CHECK_ERR_LAST	FCNT_CHECK_ERR_LAST	FCNT_SYNCED	–	–	–	–	–
Reset	0b0	0b0	0b0	–	–	–	–	–
Access Type	Read Only	Read Only	Read Only	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
FCRC_CHECK_ERR_LAST	7	Error status of last Feedforward Video frame CRC check.	0x0: No error occurred in last frame CRC check 0x1: Error occurred in last frame CRC check
FCNT_CHECK_ERR_LAST	6	Error status of last Feedforward Video frame Count check.	0x0: No error occurred in last frame count check 0x1: Error occurred in last frame count check
FCNT_SYNCED	5	Shows that frame count value is synced to incoming frame_cnt value (either from fcnt_exp	0x0: Frame_cnt is not synced 0x1: Frame_cnt is synced

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BITFIELD	BITS	DESCRIPTION	DECODE
		register or from extra line, depending on USE_FCRC_EXP_REG register).	

FFWD_CRC_CHK_REG12 (0xCCC)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_RECEIVED_BYTE_5[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_RECEIVED_BYTE_5	7:0	Received frame CRC [47:40] in last frame.

FFWD_CRC_CHK_REG13 (0xCCD)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_RECEIVED_BYTE_4[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_RECEIVED_BYTE_4	7:0	Received frame CRC [39:32] in last frame.

FFWD_CRC_CHK_REG14 (0xCCE)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_RECEIVED_BYTE_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_RECEIVED_BYTE_3	7:0	Received frame CRC [31:24] in last frame.

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FFWD_CRC_CHK_REG15 (0xCCF)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_RECEIVED_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_RECEIVED_BYTE_2	7:0	Received frame CRC [23:16] in last frame.

FFWD_CRC_CHK_REG16 (0xCD0)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_RECEIVED_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_RECEIVED_BYTE_1	7:0	Received frame CRC [15:8] in last frame.

FFWD_CRC_CHK_REG17 (0xCD1)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_CRC_RECEIVED_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_CRC_RECEIVED_BYTE_0	7:0	Received frame CRC [7:0] in last frame.

FFWD_CRC_CHK_REG18 (0xCD2)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_COUNT_RECEIVED_BYTE_3[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_COUNT_RECEIVED_BYTE_3	7:0	Received frame count [31:24] in last frame.

FFWD_CRC_CHK_REG19 (0xCD3)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_COUNT_RECEIVED_BYTE_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_COUNT_RECEIVED_BYTE_2	7:0	Received frame count [23:16] in last frame.

FFWD_CRC_CHK_REG20 (0xCD4)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_COUNT_RECEIVED_BYTE_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_COUNT_RECEIVED_BYTE_1	7:0	Received frame count [15:8] in last frame.

FFWD_CRC_CHK_REG21 (0xCD5)

BIT	7	6	5	4	3	2	1	0
Field	COMBINED_FRAME_COUNT_RECEIVED_BYTE_0[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
COMBINED_FRAME_COUNT_RECEIVED_BYTE_0	7:0	Received frame count [7:0] in last frame.

FFWD_CRC_CHK_REG22 (0xCD6)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VAL_5[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VAL_5	7:0	Feedforward Video Frame CRC value [47:40] of selected CRC

FFWD_CRC_CHK_REG23 (0xCD7)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VAL_4[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VAL_4	7:0	Feedforward Video Frame CRC value [39:32] of selected CRC

FFWD_CRC_CHK_REG24 (0xCD8)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VAL_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VAL_3	7:0	Feedforward Video Frame CRC value [31:24] of selected CRC

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FFWD_CRC_CHK_REG25 (0xCD9)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VAL_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VAL_2	7:0	Feedforward Video Frame CRC value [23:16] of selected CRC

FFWD_CRC_CHK_REG26 (0xCDA)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VAL_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VAL_1	7:0	Feedforward Video Frame CRC value [15:8] of selected CRC

FFWD_CRC_CHK_REG27 (0xCDB)

BIT	7	6	5	4	3	2	1	0
Field	SELECTED_CRC_VAL_0[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SELECTED_CRC_VAL_0	7:0	Feedforward Video Frame CRC value [7:0] of selected CRC

FFWD_CRC_CHK_REG28 (0xCDC)

BIT	7	6	5	4	3	2	1	0
Field	FRAME_COUNT[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
FRAME_COUNT	7:0	Feedforward Video Frame count value in this block (Counts when corresponding FCRC_CALC_EN register is 1).

ALIGN_FIFO_0 (0xD80)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	RST_ALIGN_FIFO	DELAY_PIPE_X_EN	ALIGN_FIFO_EN
Reset	–	–	–	–	–	0b0	0b0	0b0
Access Type	–	–	–	–	–	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
RST_ALIGN_FIFO	2	Video combiner reset. Note: Used to reset video combiner.	0x0: Video combiner is not reset 0x1: Video combiner is reset
DELAY_PIPE_X_EN	1	Used to give a delay to Pipe X video. INITIAL_DELAY_VAL register is used to set delay.	0x0: Pipe X delay is disabled 0x1: Pipe X delay is enabled
ALIGN_FIFO_EN	0	Enables combining videos coming from Pipe X and Pipe Y.	0x0: Video combining is disabled 0x1: Video combining is enabled

ALIGN_FIFO_2 (0xD82)

BIT	7	6	5	4	3	2	1	0
Field	INITIAL_DELAY_VAL_L[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
INITIAL_DELAY_VAL_L	7:0	Lower bits (7:0) of initial delay given to video X, in terms of number of pixel clock. Write 1 to UPDATE_INITIAL_DELAY_VAL register after setting INITIAL_DELAY_VAL_L/H registers.

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BITFIELD	BITS	DESCRIPTION
		Note 1: Delay must be smaller than (2 x Horizontal Total width). Note 2: Do not set combined INITIAL_DELAY_VAL_L/H value below 5.

ALIGN_FIFO_3 (0xD83)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	INITIAL_DELAY_VAL_H[3:0]			
Reset	–	–	–	–	0x7			
Access Type	–	–	–	–	Write, Read			

BITFIELD	BITS	DESCRIPTION
INITIAL_DELAY_VAL_H	3:0	Higher bits (11:8) of initial delay given to video X, in terms of number of pixel clock. Write 1 to VIDEOCOMB_UPDATE_INITIAL_DELAY_VAL register after setting INITIAL_DELAY_VAL_L/H registers. Note 1: Delay must be smaller than (2 x Horizontal Total width). Note 2: Do not set combined INITIAL_DELAY_VAL_L/H value below 5.

ALIGN_FIFO_6 (0xD86)

BIT	7	6	5	4	3	2	1	0
Field	–	–	USE_EXT_HS_FOR_ALIGN	ENABLE_ALIGN_WITH_EXT_SIGNAL	–	–	–	–
Reset	–	–	0b1	0b0	–	–	–	–
Access Type	–	–	Write, Read	Write, Read	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
USE_EXT_HS_FOR_ALIGN	5	Enables using both Hsync and Vsync for alignment.	0x0: Does not use external HS for alignment (Only VS is used if ENABLE_ALIGN_WITH_EXT_SIGNAL=1). 0x1: Uses both external HS and VS for alignment (if ENABLE_ALIGN_WITH_EXT_SIGNAL=1).
ENABLE_ALIGN_WITH_EXT_SIGNAL	4	Enables aligning with external signals (Vsync/Hsync) coming from MFPs.	0x0: Aligning with external signals is disabled 0x1: Aligning with external signals is enabled

ALIGN_FIFO_19 (0xD93)

BIT	7	6	5	4	3	2	1	0
Field	OFFSET_DETECTED_LIVE_SIGN	OFFSET_DETECTED_LIVE_MAG_H[6:0]						

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Reset	0x0	0x00
Access Type	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION
OFFSET_DETECTED_LIVE_SIGN	7	Sign of detected offset wrt external sync signal (live value).
OFFSET_DETECTED_LIVE_MAG_H	6:0	Upper bits of magnitude of detected offset wrt external sync signal (live value).

ALIGN_FIFO_20 (0xD94)

BIT	7	6	5	4	3	2	1	0
Field	EXT_ALIGN_COMPLETED_IN_SUBORDINATE_DES	EXT_ALIGN_COMPLETED_OUT	OFFSET_DETECTED_TOGGLE	OFFSET_DETECTED_LIVE_MAG_L[4:0]				
Reset	0x0	0x0	0x0	0x00				
Access Type	Read Only	Read Only	Read Only	Read Only				

BITFIELD	BITS	DESCRIPTION
EXT_ALIGN_COMPLETED_IN_SUBORDINATE_DES	7	Shows the status of input pin named ALIGN_LOCK_PIN, which shows that alignment is completed in subordinate DES.
EXT_ALIGN_COMPLETED_OUT	6	Shows that external align is completed, which is also sent to output MFP pin to be used by main DES. Behavior is determined by register EXT_ALIGN_BLK_VID_MODE.
OFFSET_DETECTED_TOGGLE	5	Toggles when new offset detection is done.
OFFSET_DETECTED_LIVE_MAG_L	4:0	Lower bits of magnitude of detected offset wrt external sync signal (live value).

ALIGN_FIFO_21 (0xD95)

BIT	7	6	5	4	3	2	1	0
Field	OFFSET_DETECTED_MAX_SIGN	OFFSET_DETECTED_MAX_MAG_H[6:0]						
Reset	0b0	0x00						
Access Type	Read Only	Read Only						

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BITFIELD	BITS	DESCRIPTION
OFFSET_DETECTED_MAX_SIGN	7	Sign of max detected offset wrt external sync signal.
OFFSET_DETECTED_MAX_MAG_H	6:0	Upper bits of magnitude of max detected offset wrt external sync signal.

ALIGN_FIFO_22 (0xD96)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	OFFSET_DETECTED_MAX_MAG_L[4:0]				
Reset	–	–	–	0x00				
Access Type	–	–	–	Read Only				

BITFIELD	BITS	DESCRIPTION
OFFSET_DETECTED_MAX_MAG_L	4:0	Lower bits of magnitude of max detected offset wrt external sync signal.

ALIGN_FIFO_24 (0xD98)

BIT	7	6	5	4	3	2	1	0
Field	–	–	EXT_ALIGNMENT_LOCKED_LIVE	–	–	–	–	–
Reset	–	–	0x0	–	–	–	–	–
Access Type	–	–	Read Only	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION
EXT_ALIGNMENT_LOCKED_LIVE	5	This register is asserted when target offset difference is below a limit, which is set using ALIGNMENT_LOCK_SEL register. Shows live value.

ALIGN_FIFO_26 (0xD9A)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	INJECT_ERR_Y	INJECT_ERR_X	–	–
Reset	–	–	–	–	0b0	0b0	–	–
Access Type	–	–	–	–	Write, Read	Write, Read	–	–

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BITFIELD	BITS	DESCRIPTION	DECODE
INJECT_ERR_Y	3	Inject error to FIFO Y output.	0x0: Error injection is disabled 0x1: Error injection is enabled
INJECT_ERR_X	2	Inject error to FIFO X output.	0x0: Error injection is disabled 0x1: Error injection is enabled

PSM_REG_7 (0xE07)

BIT	7	6	5	4	3	2	1	0
Field	PSM_TASK_CTRL_7[7:0]							
Reset	0x08							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
PSM_TASK_CTRL_7	7:0	VC_ALIGN_VIDEO_CTRL: Programmable State Machine (PSM) task control register used for Margin Value in calculation of Initial Delay Value of Video Align block. This is used for automatic aligment of two deserializers.	0x0: b_ALIGN_VIDEO_EXT_DES1 0x1: b_ALIGN_VIDEO_EXT_DES2_REF

IFCC_CTRL_0 (0xE30)

BIT	7	6	5	4	3	2	1	0
Field	DEST_ADDRESS[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
DEST_ADDRESS	7:0	Device of address of destination device (7-bit I^2C Address)

IFCC_CTRL_1 (0xE31)

BIT	7	6	5	4	3	2	1	0
Field	REG_ADDR_L[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
REG_ADDR_L	7:0	Destination register address bit 7-0

IFCC_CTRL_2 (0xE32)

BIT	7	6	5	4	3	2	1	0
Field	REG_ADDR_H[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REG_ADDR_H	7:0	Destination register address bit 15-8

IFCC_CTRL_3 (0xE33)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_0	7:0	Read or Write Buffer 0 for IFCC transaction

IFCC_CTRL_4 (0xE34)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_1	7:0	Read or Write Buffer 1 for IFCC transaction

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IFCC_CTRL_5 (0xE35)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_2[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_2	7:0	Read or Write Buffer 2 for IFCC transaction

IFCC_CTRL_6 (0xE36)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_3[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_3	7:0	Read or Write Buffer 3 for IFCC transaction

IFCC_CTRL_7 (0xE37)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_4[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_4	7:0	Read or Write Buffer 4 for IFCC transaction

IFCC_CTRL_8 (0xE38)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_5[7:0]							

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Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_5	7:0	Read or Write Buffer 5 for IFCC transaction

IFCC_CTRL_9 (0xE39)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_6[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_6	7:0	Read or Write Buffer 6 for IFCC transaction

IFCC_CTRL_10 (0xE3A)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_7[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_7	7:0	Read or Write Buffer 7 for IFCC transaction

IFCC_CTRL_11 (0xE3B)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_8[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_8	7:0	Read or Write Buffer 8 for IFCC transaction

IFCC_CTRL_12 (0xE3C)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_9[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_9	7:0	Read or Write Buffer 9 for IFCC transaction

IFCC_CTRL_13 (0xE3D)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_10[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_10	7:0	Read or Write Buffer 10 for IFCC transaction

IFCC_CTRL_14 (0xE3E)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_11[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_11	7:0	Read or Write Buffer 11 for IFCC transaction

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IFCC_CTRL_15 (0xE3F)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_12[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_12	7:0	Read or Write Buffer 12 for IFCC transaction

IFCC_CTRL_16 (0xE40)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_13[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_13	7:0	Read or Write Buffer 13 for IFCC transaction

IFCC_CTRL_17 (0xE41)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_14[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_14	7:0	Read or Write Buffer 14 for IFCC transaction

IFCC_CTRL_18 (0xE42)

BIT	7	6	5	4	3	2	1	0
Field	IFCC_RW_BUFFER_15[7:0]							

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Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
IFCC_RW_BUFFER_15	7:0	Read or Write Buffer 15 for IFCC transaction

IFCC_CTRL_19 (0xE43)

BIT	7	6	5	4	3	2	1	0
Field	NUM_OF_BYTES[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
NUM_OF_BYTES	7:0	Number of bytes that will be written to or read from the destination device registers.

IFCC_CTRL_20 (0xE44)

BIT	7	6	5	4	3	2	1	0
Field	EXECUTE	IFCC_COMMAND[6:0]						
Reset	0x0	0x0						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION
EXECUTE	7	Execute the command specified in IFCC_COMMAND (bits 0-6), This bit auto clears when execution of command is complete. Writing Execute = 1 will auto-clear RD_DONE, RD_STATUS, WR_DONE, WR_STATUS.
IFCC_COMMAND	6:0	IFCC command to execute

IFCC_USER_CMD_STATUS_0 (0xE45)

BIT	7	6	5	4	3	2	1	0

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Field	RD_DONE	WR_DONE	–	–	RD_DIS_ERR	WR_DIS_ERR	NUM_BYTE_ERR	COMMAND_ERR
Reset	0x0	0x0	–	–	0x0	0x0	0x0	0x0
Access Type	Write, Read	Write, Read	–	–	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION
RD_DONE	7	READ COMMAND IS DONE. Auto-cleared when EXECUTE written to 1.
WR_DONE	6	WRITE COMMAND IS DONE. Auto-cleared when EXECUTE written to 1.
RD_DIS_ERR	3	An RD_DIS_ERR occurs when register reading with IFCC is disabled on remote side, and the user attempts to read a register on remote side. Auto-cleared when EXECUTE written to 1.
WR_DIS_ERR	2	An WR_DIS_ERR occurs when register reading with IFCC is disabled on remote side, and the user attempts to write a register on remote side. Auto-cleared when EXECUTE written to 1.
NUM_BYTE_ERR	1	INVALID NUMBER OF BYTES REQUESTED. Auto-cleared when EXECUTE written to 1.
COMMAND_ERR	0	Invalid Command Detected. Auto-cleared when EXECUTE written to 1.

IFCC_USER_CMD_STATUS_1 (0xE46)

BIT	7	6	5	4	3	2	1	0
Field	USER_READ_COUNT[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
USER_READ_COUNT	7:0	Number of read commands executed.

IFCC_USER_CMD_STATUS_2 (0xE47)

BIT	7	6	5	4	3	2	1	0
Field	USER_WRITE_COUNT[7:0]							
Reset	0x0							

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Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
USER_WRITE_COUNT	7:0	Number of write commands executed.

IFCC_RMT_CMD_STATUS_0 (0xE48)

BIT	7	6	5	4	3	2	1	0
Field	RMT_RD_DONE	RMT_WR_DONE	–	–	RMT_RD_DIS_ERR	RMT_WR_DIS_ERR	RMT_NUM_BYTE_ERR	RMT_COMMAND_ERR
Reset	0x0	0x0	–	–	0x0	0x0	0x0	0x0
Access Type	Write, Read	Write, Read	–	–	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION
RMT_RD_DONE	7	Remote read transaction is done.
RMT_WR_DONE	6	Remote write transaction is done.
RMT_RD_DIS_ERR	3	An RMT_RD_DIS_ERR occurs when register reading with IFCC is disabled on local side, and the remote user attempts to read a register on local side. Auto-cleared when EXECUTE written to 1
RMT_WR_DIS_ERR	2	An RMT_WR_DIS_ERR occurs when register writing with IFCC is disabled on local side, and the remote user attempts to write a register on local side. Auto-cleared when EXECUTE written to 1.
RMT_NUM_BYTE_ERR	1	INVALID NUMBER OF BYTES REQUESTED on remote transaction. Auto-cleared when EXECUTE written to 1.
RMT_COMMAND_ERR	0	INVALID REMOTE COMMAND DETECTED. Auto-cleared when EXECUTE written to 1.

IFCC_RMT_CMD_STATUS_1 (0xE49)

BIT	7	6	5	4	3	2	1	0
Field	RMT_RD_COUNT[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
RMT_RD_COUNT	7:0	Number of remote read commands executed.

IFCC_RMT_CMD_STATUS_2 (0xE4A)

BIT	7	6	5	4	3	2	1	0
Field	RMT_WR_COUNT[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RMT_WR_COUNT	7:0	Number of remote write commands executed.

OTP_ERR_INJ_CTRL_2 (0xE86)

BIT	7	6	5	4	3	2	1	0
Field	OTP_INJECT_ERROR_WORD_INDEX_MSB	–	OTP_ERROR_INJECT_EN	–	–	–	–	–
Reset	0x0	–	0x0	–	–	–	–	–
Access Type	Write, Read	–	Write, Read	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
OTP_INJECT_ERROR_WORD_INDEX_MSB	7	MSb of word index of selected OTP to inject error during CRC calculation when OTP_ERROR_INJECT_EN = 1.	
OTP_ERROR_INJECT_EN	5	Enables error injection on selected OTP	0x0: Do not enable 0x1: Enable

FRC_0 (0x10C0)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	FRCGAIN[1:0]		–
Reset	–	–	–	–	–	0x1		–
Access Type	–	–	–	–	–	Write, Read		–

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BITFIELD	BITS	DESCRIPTION	DECODE
FRCGAIN	2:1	Set frame-rate control gain	0x0: 1 bit 0x1: 2 bit 0x2: 3 bit 0x3: 4 bit

FRC_1 (0x10C1)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	FRCDATAWIDTH	–	–	–
Reset	–	–	–	–	0b1	–	–	–
Access Type	–	–	–	–	Write, Read	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
FRCDATAWIDTH	3	Set the FRC bus width	0x0: 24 bits per pixel 0x1: 30 bits per pixel

RLMS4 (0x1404, 0x1504)

BIT	7	6	5	4	3	2	1	0
Field	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
Reset	0x4				0b10		0b1	0b1
Access Type	Write, Read				Write, Read		Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_CHK_AMOUNT	7:4	A factor (N) used to select the order of number of observations in each eye-monitor window. N is used in the equation: Observations = 6.29 x $10^{(N+2)}$	0xX: N factor
EOM_CHK_THR	3:2	Eye-opening monitor number of error bits to allow in a measurement window.	0b00: Allow no errors 0b01: Allow one error 0b10: Allow two errors 0b11: Allow three errors
EOM_PER_MODE	1	Eye-opening monitor periodic mode enable.	0b0: Eye-opening monitor periodic mode disabled 0b1: Eye-opening monitor periodic mode enabled
EOM_EN	0	Eye-opening monitor enable. Recommended for GMSL2 use only.	0b0: Eye-opening monitor disabled 0b1: Eye-opening monitor enabled

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RLMS5 (0x1405, 0x1505)

BIT	7	6	5	4	3	2	1	0
Field	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
Reset	0b0	0b0010000						
Access Type	Write Only	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_MAN_TRG_REQ	7	Eye-opening monitor manual trigger. For use when periodic mode is disabled.	0b0: No action 0b1: EOM manual trigger request
EOM_MIN_THR	6:0	The EOM minimum threshold as defined by the equation: % eye-opening = EOM_MIN_THR/64 If the value is zero, the EOM is disabled.	0xXX: EOM minimum threshold factor

RLMS6 (0x1406, 0x1506)

BIT	7	6	5	4	3	2	1	0
Field	EOM_PV_MODE	EOM_RST_THR[6:0]						
Reset	0b1	0b0000000						
Access Type	Write, Read	Write, Read						

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_PV_MODE	7	Eye-opening is measured vertically or horizontally.	0b0: Vertical opening mode 0b1: Horizontal opening mode
EOM_RST_THR	6:0	The EOM refresh threshold as defined by the equation: % eye-opening = EOM_MIN_THR/64. Used only with periodic EOM mode. If the value is zero, the EOM is disabled.	0xXX: EOM refresh threshold factor

RLMS7 (0x1407, 0x1507)

BIT	7	6	5	4	3	2	1	0
Field	EOM_DONE	EOM[6:0]						
Reset	0b0	0b0000000						

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Access Type	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
EOM_DONE	7	Eye-opening monitor measurement done	0b0: EOM not complete 0b1: EOM complete
EOM	6:0	Last completed EOM observation	0xXX: EOM measurement result

SHA_1 (0x1E00)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_1[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_1	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_2 (0x1E01)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_2	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_3 (0x1E02)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_3[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
SHA_A_3	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_4 (0x1E03)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_4[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_4	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_5 (0x1E04)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_5[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_5	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_6 (0x1E05)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_6[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_6	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

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SHA_7 (0x1E06)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_7[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_7	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_8 (0x1E07)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_8[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_8	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_9 (0x1E08)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_9[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_9	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_10 (0x1E09)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_10[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
SHA_A_10	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_11 (0x1E0A)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_11[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_11	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_12 (0x1E0B)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_12[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_12	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_13 (0x1E0C)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_13[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
SHA_A_13	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_14 (0x1E0D)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_14[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_14	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_15 (0x1E0E)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_15[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_15	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_16 (0x1E0F)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_16[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_16	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

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SHA_17 (0x1E10)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_17[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_17	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_18 (0x1E11)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_18[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_18	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_19 (0x1E12)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_19[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_19	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_20 (0x1E13)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_20[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
SHA_A_20	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_21 (0x1E14)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_21[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_21	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_22 (0x1E15)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_22[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_22	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_23 (0x1E16)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_23[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
SHA_A_23	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_24 (0x1E17)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_24[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_24	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_25 (0x1E18)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_25[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_25	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_26 (0x1E19)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_26[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_26	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

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SHA_27 (0x1E1A)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_27[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_27	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_28 (0x1E1B)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_28[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_28	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_29 (0x1E1C)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_29[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_29	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_30 (0x1E1D)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_30[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
SHA_A_30	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_31 (0x1E1E)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_31[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_31	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_32 (0x1E1F)

BIT	7	6	5	4	3	2	1	0
Field	SHA_A_32[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_A_32	7:0	SHA256 signature for video authentication byte segment X. Bank A result.

SHA_1 (0x1E20)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_1[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
SHA_B_1	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_2 (0x1E21)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_2[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_2	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_3 (0x1E22)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_3[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_3	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_4 (0x1E23)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_4[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_4	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

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SHA_5 (0x1E24)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_5[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_5	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_6 (0x1E25)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_6[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_6	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_7 (0x1E26)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_7[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_7	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_8 (0x1E27)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_8[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
SHA_B_8	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_9 (0x1E28)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_9[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_9	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_10 (0x1E29)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_10[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_10	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_11 (0x1E2A)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_11[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
SHA_B_11	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_12 (0x1E2B)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_12[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_12	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_13 (0x1E2C)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_13[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_13	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_14 (0x1E2D)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_14[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_14	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

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SHA_15 (0x1E2E)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_15[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_15	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_16 (0x1E2F)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_16[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_16	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_17 (0x1E30)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_17[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_17	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_18 (0x1E31)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_18[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
SHA_B_18	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_19 (0x1E32)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_19[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_19	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_20 (0x1E33)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_20[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_20	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_21 (0x1E34)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_21[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
SHA_B_21	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_22 (0x1E35)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_22[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_22	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_23 (0x1E36)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_23[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_23	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_24 (0x1E37)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_24[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_24	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

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SHA_25 (0x1E38)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_25[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_25	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_26 (0x1E39)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_26[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_26	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_27 (0x1E3A)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_27[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_27	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_28 (0x1E3B)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_28[7:0]							

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Reset	0x00
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
SHA_B_28	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_29 (0x1E3C)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_29[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_29	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_30 (0x1E3D)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_30[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_30	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_31 (0x1E3E)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_31[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
SHA_B_31	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

SHA_32 (0x1E3F)

BIT	7	6	5	4	3	2	1	0
Field	SHA_B_32[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
SHA_B_32	7:0	SHA256 signature for video authentication byte segment X. Bank B result.

REGCRC0 (0x1F00)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	GEN_ROLLING_CRC	I2C_WR_COMPUTE	PERIODIC_COMPUTE	CHECK_CRC	RESET_CRC
Reset	–	–	–	0b0	0b0	0b0	0b0	0b0
Access Type	–	–	–	Write, Read	Write Only Clears All	Write, Read	Write, Read	Write Clears All, Read

BITFIELD	BITS	DESCRIPTION	DECODE
GEN_ROLLING_CRC	4	Calculates CRC using additional 4-bit counter, so CRC value cycles every 16 invocations.	0b0: Disabled 0b1: Enabled
I2C_WR_COMPUTE	3	CRC computing trigger. Self-clearing.	0b0: Disabled 0b1: Enabled
PERIODIC_COMPUTE	2	Enable CRC checking on periodic basis, based on CRC_PERIOD value.	0b0: Disabled 0b1: Enabled
CHECK_CRC	1	Enables compare with previous calculation upon calculation of CRC, except on first time through. On mismatch, assert ERRB.	0b0: Disabled 0b1: Enabled
RESET_CRC	0	Reset CRC value to 0xFFFF.	0b0: Reserved 0b1: One shot trigger for Reset

REGCRC1 (0x1F01)

BIT	7	6	5	4	3	2	1	0

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Field	CRC_PERIOD[7:0]
Reset	0x00
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
CRC_PERIOD	7:0	Period for CRC recomputation. Period = (value + 1) * 2ms

REGCRC2 (0x1F02)

BIT	7	6	5	4	3	2	1	0
Field	REGCRC_LSB[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGCRC_LSB	7:0	CRC Result LSB

REGCRC3 (0x1F03)

BIT	7	6	5	4	3	2	1	0
Field	REGCRC_MSB[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
REGCRC_MSB	7:0	CRC Result MSB

CC_CRC0 (0x1F20)

BIT	7	6	5	4	3	2	1	0
Field	CC_CRC_VAL[7:0]							
Reset	0x00							

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Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
CC_CRC_VAL	7:0	Calculated CRC value for the last write transaction.

CC_CRC1 (0x1F21)

BIT	7	6	5	4	3	2	1	0
Field	MSGCNTR_LSB[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
MSGCNTR_LSB	7:0	Bits [7:0] of current Message Counter value.

CC_CRC2 (0x1F22)

BIT	7	6	5	4	3	2	1	0
Field	MSGCNTR_MSB[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
MSGCNTR_MSB	7:0	Bits [15:8] of current Message Counter value.

CC_CRC3 (0x1F23)

BIT	7	6	5	4	3	2	1	0
Field	CC_CRC_ERR_CNT[7:0]							
Reset	0x00							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
CC_CRC_ERR_CNT	7:0	Number of CC CRC errors detected.

CC_CRC4 (0x1F24)

BIT	7	6	5	4	3	2	1	0
Field	MSGCNTR_ERR_CNT[7:0]							
Reset	0x00							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
MSGCNTR_ERR_CNT	7:0	Number of Message Counter errors detected.

CC_RTTN_CRC_ERR (0x1F26)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	INJECT_RTTN_CRC_ERR
Reset	–	–	–	–	–	–	–	0b0
Access Type	–	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
INJECT_RTTN_CRC_ERR	0	Set this bit before going into Sleep mode to inject error to RTTN CRC value. Use for ASIL evaluation purposes.	0b0: No action 0b1: Inject error during restore from retention memory

LUT_A (0x2000)

Start address for color lookup table COLOR_A LUT_X for Video Pipeline X.

LUT_B (0x2100)

Start address for color lookup table COLOR_B LUT_X for Video Pipeline X.

LUT_C (0x2200)

Start address for color lookup table COLOR_C LUT_X for Video Pipeline X.

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TELLTALE_CTRL_REG_0 (0x3000)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	TELLTALE_FORCE_BANK_B	TELLTALE_FORCE_BANK_A	TELLTALE_PERIODIC_MODE	TELLTALE_TRIGGER_MODE	TELLTALE_DET_EN
Reset	–	–	–	0b0	0b0	0b1	0b0	0b0
Access Type	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
TELLTALE_FORCE_BANK_B	4	Force state machine to work on Bank B only. When set, must be mutually exclusive with TELLTALE_FORCE_BANK_A.	0x0: No action 0x1: Process Bank B only
TELLTALE_FORCE_BANK_A	3	Force state machine to work on Bank A only. When set, must be mutually exclusive with TELLTALE_FORCE_BANK_B.	0x0: No action 0x1: Process Bank A only
TELLTALE_PERIODIC_MODE	2	Periodic Mode selection	0b0: Not Periodic mode 0b1: Periodic mode
TELLTALE_TRIGGER_MODE	1	Telltale Trigger Mode Control Signal, should be created as a pulse. The generated pulse must last for at least one vertical sync (VS) period but not exceed four VS periods to ensure proper timing and synchronization.	0x0: No action 0x1: Trigger telltale detection
TELLTALE_DET_EN	0	Enable telltale detection block.	0x0: Disabled 0x1: Enabled

TELLTALE_CTRL_REG_1 (0x3001)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI_INTERRUPT_EN[3:0]				BANK_A_ROI_INTERRUPT_EN[3:0]			
Reset	0xF				0xF			
Access Type	Write, Read				Write, Read			

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI_INTERRUPT_EN	7:4	Mask (Enable) error interrupt for Bank B-specific ROI operations. Each bit enables the error interrupt for the corresponding region.	0bXXX0: Region 0 error interrupt is disabled 0bXXX1: Region 0 error interrupt is enabled 0bXX0X: Region 1 error interrupt is disabled 0bXX1X: Region 1 error interrupt is enabled 0bX0XX: Region 2 error interrupt is disabled 0bX1XX: Region 2 error interrupt is enabled 0b0XXX: Region 3 error interrupt is disabled 0b1XXX: Region 3 error interrupt is enabled

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BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI_INTERRUPT_EN	3:0	Mask (Enable) error interrupt for Bank A-specific ROI operations. Each bit enables the error interrupt for the corresponding region.	0bXXX0: Region 0 error interrupt is disabled 0bXXX1: Region 0 error interrupt is enabled 0bXX0X: Region 1 error interrupt is disabled 0bXX1X: Region 1 error interrupt is enabled 0bX0XX: Region 2 error interrupt is disabled 0bX1XX: Region 2 error interrupt is enabled 0b0XXX: Region 3 error interrupt is disabled 0b1XXX: Region 3 error interrupt is enabled

TELLTALE_STATUS_REG_0 (0x3002)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI_ERROR[3:0]				BANK_A_ROI_ERROR[3:0]			
Reset	0x0				0x0			
Access Type	Read Only				Read Only			

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI_ERROR	7:4	Error indicator bits per region for Bank B.	0bXXX0: No region 0 error interrupt 0bXXX1: Region 0 error interrupt 0bXX0X: No region 1 error interrupt 0bXX1X: Region 1 error interrupt 0bX0XX: No region 2 error interrupt 0bX1XX: Region 2 error interrupt 0b0XXX: No region 3 error interrupt 0b1XXX: Region 3 error interrupt
BANK_A_ROI_ERROR	3:0	Error indicator bits per region for Bank A.	0bXXX0: No region 0 error interrupt 0bXXX1: Region 0 error interrupt 0bXX0X: No region 1 error interrupt 0bXX1X: Region 1 error interrupt 0bX0XX: No region 2 error interrupt 0bX1XX: Region 2 error interrupt 0b0XXX: No region 3 error interrupt 0b1XXX: Region 3 error interrupt

TELLTALE_STATUS_REG_1 (0x3003)

BIT	7	6	5	4	3	2	1	0
Field	TELLTALE_ACTIVE	TELLTALE_ACTIVE_BANK_A_OR_B	–	–	–	–	–	–
Reset	0b0	0b0	–	–	–	–	–	–
Access Type	Read Only	Read Only	–	–	–	–	–	–

BITFIELD	BITS	DESCRIPTION	DECODE
TELLTALE_ACTIVE	7	Telltale Active Indicator	0b0: Not active 0b1: Active
TELLTALE_ACTIVE_BANK_A_OR_B	6	Indicates Bank A or Bank B active signal.	0b0: Bank A is active 0b1: Bank B is active

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REF_COLOR_GENERAL (0x3008)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
Reset	0x1	0x0	0x0	–	–	–	0b1	0b0
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_INV_HUE	7	Reference color hue values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_SAT	6	Reference color saturation values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_LUM	5	Reference color luminance values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_HUE_MAX_H	1	Reference color hue maximum value most significant bit (bit 8).	0bX: MSb value
REF_COLOR_HUE_MIN_H	0	Reference color hue minimum value most significant bit (bit 8).	0bX: MSb value

REF_COLOR_HUE_MAX_L (0x3009)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MAX_L[7:0]							
Reset	0x4A							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_HUE_MAX_L	7:0	Reference color hue maximum value.	0xXX: Lower 8-bit value

REF_COLOR_HUE_MIN (0x300A)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MIN_L[7:0]							
Reset	0x0A							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MIN_L	7:0	Reference color hue minimum value.

REF_COLOR_SAT_MAX (0x300B)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MAX[6:0]						
Reset	–	0x64						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MAX	6:0	Reference color saturation maximum value.

REF_COLOR_SAT_MIN (0x300C)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MIN[6:0]						
Reset	–	0x28						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MIN	6:0	Reference color saturation minimum value.

REF_COLOR_LUM_MAX (0x300D)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MAX[6:0]						
Reset	–	0x4B						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MAX	6:0	Reference color luminance maximum value.

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REF_COLOR_LUM_MIN (0x300E)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MIN[6:0]						
Reset	–	0x19						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MIN	6:0	Reference color luminance minimum value.

REF_COLOR_GENERAL (0x3010)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
Reset	0x0	0x0	0x0	–	–	–	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_INV_HUE	7	Reference color hue values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_SAT	6	Reference color saturation values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_LUM	5	Reference color luminance values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_HUE_MAX_H	1	Reference color hue maximum value.	0bX: MSb value
REF_COLOR_HUE_MIN_H	0	Reference color hue minimum value.	0bX: MSb value

REF_COLOR_HUE_MAX_L (0x3011)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MAX_L[7:0]							
Reset	0x33							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MAX_L	7:0	Reference color hue maximum value.

REF_COLOR_HUE_MIN (0x3012)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MIN_L[7:0]							
Reset	0x0E							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MIN_L	7:0	Reference color hue minimum value.

REF_COLOR_SAT_MAX (0x3013)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MAX[6:0]						
Reset	–	0x64						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MAX	6:0	Reference color saturation maximum value.

REF_COLOR_SAT_MIN (0x3014)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MIN[6:0]						
Reset	–	0x28						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MIN	6:0	Reference color saturation minimum value.

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REF_COLOR_LUM_MAX (0x3015)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MAX[6:0]						
Reset	–	0x4B						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MAX	6:0	Reference color luminance maximum value.

REF_COLOR_LUM_MIN (0x3016)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MIN[6:0]						
Reset	–	0x19						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MIN	6:0	Reference color luminance minimum value.

REF_COLOR_GENERAL (0x3018)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
Reset	0x0	0x0	0x0	–	–	–	0b0	0b0
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_INV_HUE	7	Reference color hue values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_SAT	6	Reference color saturation values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_LUM	5	Reference color luminance values inversion enable.	0b0: Disable inversion 0b1: Enable inversion

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BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_HUE_MAX_H	1	Reference color hue maximum value.	0bX: MSb value
REF_COLOR_HUE_MIN_H	0	Reference color hue minimum value.	0bX: MSb value

REF_COLOR_HUE_MAX_L (0x3019)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MAX_L[7:0]							
Reset	0xA7							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MAX_L	7:0	Reference color hue maximum value.

REF_COLOR_HUE_MIN (0x301A)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MIN_L[7:0]							
Reset	0x57							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MIN_L	7:0	Reference color hue minimum value.

REF_COLOR_SAT_MAX (0x301B)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MAX[6:0]						
Reset	–	0x64						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MAX	6:0	Reference color saturation maximum value.

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REF_COLOR_SAT_MIN (0x301C)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MIN[6:0]						
Reset	–	0x28						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MIN	6:0	Reference color saturation minimum value.

REF_COLOR_LUM_MAX (0x301D)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MAX[6:0]						
Reset	–	0x4B						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MAX	6:0	Reference color luminance maximum value.

REF_COLOR_LUM_MIN (0x301E)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MIN[6:0]						
Reset	–	0x19						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MIN	6:0	Reference color luminance minimum value.

REF_COLOR_GENERAL (0x3020)

BIT	7	6	5	4	3	2	1	0

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Field	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
Reset	0x0	0x0	0x0	–	–	–	0b1	0b0
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_INV_HUE	7	Reference color hue values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_SAT	6	Reference color saturation values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_LUM	5	Reference color luminance values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_HUE_MAX_H	1	Reference color hue maximum value.	0bX: MSb value
REF_COLOR_HUE_MIN_H	0	Reference color hue minimum value.	0bX: MSb value

REF_COLOR_HUE_MAX_L (0x3021)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MAX_L[7:0]							
Reset	0x0B							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MAX_L	7:0	Reference color hue maximum value.

REF_COLOR_HUE_MIN (0x3022)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MIN_L[7:0]							
Reset	0xBC							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MIN_L	7:0	Reference color hue minimum value.

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REF_COLOR_SAT_MAX (0x3023)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MAX[6:0]						
Reset	–	0x64						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MAX	6:0	Reference color saturation maximum value.

REF_COLOR_SAT_MIN (0x3024)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MIN[6:0]						
Reset	–	0x28						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MIN	6:0	Reference color saturation minimum value.

REF_COLOR_LUM_MAX (0x3025)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MAX[6:0]						
Reset	–	0x4B						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MAX	6:0	Reference color luminance maximum value.

REF_COLOR_LUM_MIN (0x3026)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MIN[6:0]						

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Reset	–	0x19
Access Type	–	Write, Read

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MIN	6:0	Reference color luminance minimum value.

REF_COLOR_GENERAL (0x3028)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
Reset	0x0	0x0	0x0	–	–	–	0b1	0b0
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_INV_HUE	7	Reference color hue values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_SAT	6	Reference color saturation values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_LUM	5	Reference color luminance values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_HUE_MAX_H	1	Reference color hue maximum value.	0bX: MSb value
REF_COLOR_HUE_MIN_H	0	Reference color hue minimum value.	0bX: MSb value

REF_COLOR_HUE_MAX_L (0x3029)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MAX_L[7:0]							
Reset	0x68							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MAX_L	7:0	Reference color hue maximum value.

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REF_COLOR_HUE_MIN (0x302A)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MIN_L[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MIN_L	7:0	Reference color hue minimum value.

REF_COLOR_SAT_MAX (0x302B)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MAX[6:0]						
Reset	–	0x64						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MAX	6:0	Reference color saturation maximum value.

REF_COLOR_SAT_MIN (0x302C)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MIN[6:0]						
Reset	–	0x00						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MIN	6:0	Reference color saturation minimum value.

REF_COLOR_LUM_MAX (0x302D)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MAX[6:0]						

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Reset	–	0x64
Access Type	–	Write, Read

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MAX	6:0	Reference color luminance maximum value.

REF_COLOR_LUM_MIN (0x302E)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MIN[6:0]						
Reset	–	0x5F						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MIN	6:0	Reference color luminance minimum value.

REF_COLOR_GENERAL (0x3030)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_INV_HUE	REF_COLOR_INV_SAT	REF_COLOR_INV_LUM	–	–	–	REF_COLOR_HUE_MAX_H	REF_COLOR_HUE_MIN_H
Reset	0x0	0x0	0x0	–	–	–	0b1	0b0
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
REF_COLOR_INV_HUE	7	Reference color hue values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_SAT	6	Reference color saturation values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_INV_LUM	5	Reference color luminance values inversion enable.	0b0: Disable inversion 0b1: Enable inversion
REF_COLOR_HUE_MAX_H	1	Reference color hue maximum value.	0bX: MSb value
REF_COLOR_HUE_MIN_H	0	Reference color hue minimum value.	0bX: MSb value

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REF_COLOR_HUE_MAX_L (0x3031)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MAX_L[7:0]							
Reset	0x68							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MAX_L	7:0	Reference color hue maximum value.

REF_COLOR_HUE_MIN (0x3032)

BIT	7	6	5	4	3	2	1	0
Field	REF_COLOR_HUE_MIN_L[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
REF_COLOR_HUE_MIN_L	7:0	Reference color hue minimum value.

REF_COLOR_SAT_MAX (0x3033)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MAX[6:0]						
Reset	–	0x19						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MAX	6:0	Reference color saturation maximum value.

REF_COLOR_SAT_MIN (0x3034)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_SAT_MIN[6:0]						

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Reset	–	0x00
Access Type	–	Write, Read

BITFIELD	BITS	DESCRIPTION
REF_COLOR_SAT_MIN	6:0	Reference color saturation minimum value.

REF_COLOR_LUM_MAX (0x3035)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MAX[6:0]						
Reset	–	0x4B						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MAX	6:0	Reference color luminance maximum value.

REF_COLOR_LUM_MIN (0x3036)

BIT	7	6	5	4	3	2	1	0
Field	–	REF_COLOR_LUM_MIN[6:0]						
Reset	–	0x19						
Access Type	–	Write, Read						

BITFIELD	BITS	DESCRIPTION
REF_COLOR_LUM_MIN	6:0	Reference color luminance minimum value.

BANK_A_STATUS (0x3038)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_CALC_DONE	BANK_A_FRAME_COUNT[1:0]	
Reset	–	–	–	–	–	0b0	0x0	
Access Type	–	–	–	–	–	Read Only	Read Only	

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BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_CALC_DONE	2	Telltale detection completion status indicator for Bank A.	0x0: Not complete 0x1: Complete
BANK_A_FRAME_COUNT	1:0	Frame Counter value indicating active frame number for Bank A.	0bXX: 2-bit frame index

BANK_A_ROI_1_0_ERROR (0x3039)

BIT	7	6	5	4	3	2	1	0
Field	–	BANK_A_ROI_1_COLOR_2_ERROR	BANK_A_ROI_1_COLOR_1_ERROR	BANK_A_ROI_1_COLOR_0_ERROR	–	BANK_A_ROI_0_COLOR_2_ERROR	BANK_A_ROI_0_COLOR_1_ERROR	BANK_A_ROI_0_COLOR_0_ERROR
Reset	–	0x0	0x0	0x0	–	0x0	0x0	0x0
Access Type	–	Read Only	Read Only	Read Only	–	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI_1_COLOR_2_ERROR	6	Bank A color 0 ROI 1 color 2 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_1_COLOR_1_ERROR	5	Bank A color 0 ROI 1 color 1 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_1_COLOR_0_ERROR	4	Bank A color 0 ROI 1 color 0 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_0_COLOR_2_ERROR	2	Bank A color 0 ROI 0 color 2 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_0_COLOR_1_ERROR	1	Bank A color 0 ROI 0 color 1 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_0_COLOR_0_ERROR	0	Bank A color 0 ROI 0 color 0 error detected.	0b0: No error detected 0b1: Error detected

BANK_A_ROI_3_2_ERROR (0x303A)

BIT	7	6	5	4	3	2	1	0
Field	–	BANK_A_ROI_3_COLOR_2_ERROR	BANK_A_ROI_3_COLOR_1_ERROR	BANK_A_ROI_3_COLOR_0_ERROR	–	BANK_A_ROI_2_COLOR_2_ERROR	BANK_A_ROI_2_COLOR_1_ERROR	BANK_A_ROI_2_COLOR_0_ERROR
Reset	–	0x0	0x0	0x0	–	0x0	0x0	0x0
Access Type	–	Read Only	Read Only	Read Only	–	Read Only	Read Only	Read Only

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BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI_3_COLOR_2_ERROR	6	Bank A color 0 ROI 3 color 2 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_3_COLOR_1_ERROR	5	Bank A color 0 ROI 3 color 1 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_3_COLOR_0_ERROR	4	Bank A color 0 ROI 3 color 0 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_2_COLOR_2_ERROR	2	Bank A color 0 ROI 2 color 2 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_2_COLOR_1_ERROR	1	Bank A color 0 ROI 2 color 1 error detected.	0b0: No error detected 0b1: Error detected
BANK_A_ROI_2_COLOR_0_ERROR	0	Bank A color 0 ROI 2 color 0 error detected.	0b0: No error detected 0b1: Error detected

BANK_A_ROI0_XMIN_LSB (0x3040)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_XMIN_LSB	7:0	Bank A color 0 ROI 0 X Coordinates minimum value LSB.

BANK_A_ROI0_XMIN_MSB (0x3041)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI0_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_XMIN_MSB	5:0	Bank A color 0 ROI 0 X coordinates minimum value MSB.

BANK_A_ROI0_XMAX_LSB (0x3042)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_XMAX_LSB[7:0]							

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Reset	0x0
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_XMAX_LSB	7:0	Bank A color 0 ROI 0 X Coordinates maximum value LSB.

BANK_A_ROI0_XMAX_MSB (0x3043)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI0_XMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_XMAX_MSB	5:0	Bank A color 0 ROI 0 X coordinates maximum value MSB.

BANK_A_ROI0_YMIN_LSB (0x3044)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_YMIN_LSB	7:0	Bank A color 0 ROI 0 Y Coordinates minimum value LSB.

BANK_A_ROI0_YMIN_MSB (0x3045)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI0_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_YMIN_MSB	5:0	Bank A color 0 ROI 0 Y coordinates minimum value MSB.

BANK_A_ROI0_YMAX_LSB (0x3046)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_YMAX_LSB	7:0	Bank A color 0 ROI 0 Y Coordinates maximum value LSB.

BANK_A_ROI0_YMAX_MSB (0x3047)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI0_YMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_YMAX_MSB	5:0	Bank A color 0 ROI 0 Y coordinates maximum value MSB.

BANK_A_ROI0_COLOR_SELECT (0x3048)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI0_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI0_COLOR_SELECT	2:0	Bank A Color 0 ROI 0 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue

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BITFIELD	BITS	DESCRIPTION	DECODE
			0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI0_COLOR_MIN_THR_LSB (0x3049)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MIN_THR_LSB	7:0	Bank A color 0 ROI 0 minimum threshold value LSB.

BANK_A_ROI0_COLOR_MIN_THR_MSB (0x304A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MIN_THR_MSB	7:0	Bank A color 0 ROI 0 minimum threshold value MSB.

BANK_A_ROI0_COLOR_MAX_THR_LSB (0x304B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MAX_THR_LSB	7:0	Bank A color 0 ROI 0 maximum threshold value LSB.

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BANK_A_ROI0_COLOR_MAX_THR_MSB (0x304C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MAX_THR_MSB	7:0	Bank A color 0 ROI 0 maximum threshold value MSB.

BANK_A_ROI0_COLOR_COUNT_LSB (0x304D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_COUNT_LSB	7:0	Bank A Color 0 ROI 0 detected number of color (LSB).

BANK_A_ROI0_COLOR_COUNT_MSB (0x304E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_COUNT_MSB	7:0	Bank A Color 0 ROI 0 detected number of color (MSB).

BANK_A_ROI0_COLOR_SELECT (0x3050)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI0_COLOR_SELECT[2:0]		

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Reset	–	–	–	–	–	0x0
Access Type	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI0_COLOR_SELECT	2:0	Bank A Color 1 ROI 0 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI0_COLOR_MIN_THR_LSB (0x3051)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MIN_THR_LSB	7:0	Bank A color 1 ROI 0 minimum threshold value LSB.

BANK_A_ROI0_COLOR_MIN_THR_MSB (0x3052)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MIN_THR_MSB	7:0	Bank A color 1 ROI 0 minimum threshold value MSB.

BANK_A_ROI0_COLOR_MAX_THR_LSB (0x3053)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MAX_THR_LSB	7:0	Bank A color 1 ROI 0 maximum threshold value LSB.

BANK_A_ROI0_COLOR_MAX_THR_MSB (0x3054)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MAX_THR_MSB	7:0	Bank A color 1 ROI 0 maximum threshold value MSB.

BANK_A_ROI0_COLOR_COUNT_LSB (0x3055)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_COUNT_LSB	7:0	Bank A Color 1 ROI 0 detected number of color (LSB).

BANK_A_ROI0_COLOR_COUNT_MSB (0x3056)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_COUNT_MSB	7:0	Bank A Color 1 ROI 0 detected number of color (MSB).

BANK_A_ROI0_COLOR_SELECT (0x3058)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI0_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI0_COLOR_SELECT	2:0	Bank A Color 2 ROI 0 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI0_COLOR_MIN_THR_LSB (0x3059)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MIN_THR_LSB	7:0	Bank A color 2 ROI 0 minimum threshold value LSB.

BANK_A_ROI0_COLOR_MIN_THR_MSB (0x305A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MIN_THR_MSB	7:0	Bank A color 2 ROI 0 minimum threshold value MSB.

BANK_A_ROI0_COLOR_MAX_THR_LSB (0x305B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MAX_THR_LSB	7:0	Bank A color 2 ROI 0 maximum threshold value LSB.

BANK_A_ROI0_COLOR_MAX_THR_MSB (0x305C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_MAX_THR_MSB	7:0	Bank A color 2 ROI 0 maximum threshold value MSB.

BANK_A_ROI0_COLOR_COUNT_LSB (0x305D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_COUNT_LSB	7:0	Bank A Color 2 ROI 0 detected number of color (LSB).

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BANK_A_ROI0_COLOR_COUNT_MSB (0x305E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI0_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI0_COLOR_COUNT_MSB	7:0	Bank A Color 2 ROI 0 detected number of color (MSB).

BANK_A_ROI1_XMIN_LSB (0x3060)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_XMIN_LSB	7:0	Bank A color 0 ROI 1 X Coordinates minimum value LSB.

BANK_A_ROI1_XMIN_MSB (0x3061)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI1_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_XMIN_MSB	5:0	Bank A color 0 ROI 1 X Coordinates minimum value MSB.

BANK_A_ROI1_XMAX_LSB (0x3062)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_XMAX_LSB[7:0]							

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Reset	0x0
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_XMAX_LSB	7:0	Bank A color 0 ROI 1 X Coordinates maximum value LSB.

BANK_A_ROI1_XMAX_MSB (0x3063)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI1_XMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_XMAX_MSB	5:0	Bank A color 0 ROI 1 X Coordinates maximum value MSB.

BANK_A_ROI1_YMIN_LSB (0x3064)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_YMIN_LSB	7:0	Bank A color 1 ROI 0 Y Coordinates minimum value LSB.

BANK_A_ROI1_YMIN_MSB (0x3065)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI1_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_YMIN_MSB	5:0	Bank A color 0 ROI 1 Y Coordinates minimum value MSB.

BANK_A_ROI1_YMAX_LSB (0x3066)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_YMAX_LSB	7:0	Bank A color 0 ROI 1 Y Coordinates maximum value LSB.

BANK_A_ROI1_YMAX_MSB (0x3067)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI1_YMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_YMAX_MSB	5:0	Bank A color 0 ROI 1 Y Coordinates maximum value MSB.

BANK_A_ROI1_COLOR_SELECT (0x3068)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI1_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI1_COLOR_SELECT	2:0	Bank A Color 0 ROI 1 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue

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BITFIELD	BITS	DESCRIPTION	DECODE
			0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI1_COLOR_MIN_THR_LSB (0x3069)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MIN_THR_LSB	7:0	Bank A color 0 ROI 1 minimum threshold value LSB.

BANK_A_ROI1_COLOR_MIN_THR_MSB (0x306A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MIN_THR_MSB	7:0	Bank A color 0 ROI 1 minimum threshold value LSB.

BANK_A_ROI1_COLOR_MAX_THR_LSB (0x306B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MAX_THR_LSB	7:0	Bank A color 0 ROI 1 maximum threshold value LSB.

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BANK_A_ROI1_COLOR_MAX_THR_MSB (0x306C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MAX_THR_MSB	7:0	Bank A color 0 ROI 1 maximum threshold value MSB.

BANK_A_ROI1_COLOR_COUNT_LSB (0x306D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_COUNT_LSB	7:0	Bank A Color 0 ROI 1 detected number of color (LSB).

BANK_A_ROI1_COLOR_COUNT_MSB (0x306E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_COUNT_MSB	7:0	Bank A Color 0 ROI 1 detected number of color (MSB).

BANK_A_ROI1_COLOR_SELECT (0x3070)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI1_COLOR_SELECT[2:0]		

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Reset	–	–	–	–	–	0x0
Access Type	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI1_COLOR_SELECT	2:0	Bank A Color 0 ROI 1 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI1_COLOR_MIN_THR_LSB (0x3071)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MIN_THR_LSB	7:0	Bank A color 0 ROI 1 minimum threshold value LSB.

BANK_A_ROI1_COLOR_MIN_THR_MSB (0x3072)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MIN_THR_MSB	7:0	Bank A color 0 ROI 1 minimum threshold value MSB.

BANK_A_ROI1_COLOR_MAX_THR_LSB (0x3073)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MAX_THR_LSB	7:0	Bank A color 0 ROI 1 maximum threshold value LSB.

BANK_A_ROI1_COLOR_MAX_THR_MSB (0x3074)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MAX_THR_MSB	7:0	Bank A color 0 ROI 1 maximum threshold value MSB.

BANK_A_ROI1_COLOR_COUNT_LSB (0x3075)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_COUNT_LSB	7:0	Bank A Color 0 ROI 1detected number of color (LSB).

BANK_A_ROI1_COLOR_COUNT_MSB (0x3076)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_COUNT_MSB	7:0	Bank A Color 0 ROI 1 detected number of color (MSB).

BANK_A_ROI1_COLOR_SELECT (0x3078)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI1_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI1_COLOR_SELECT	2:0	Bank A Color 2 ROI 1 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI1_COLOR_MIN_THR_LSB (0x3079)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MIN_THR_LSB	7:0	Bank A color 2 ROI 1 minimum threshold value LSB.

BANK_A_ROI1_COLOR_MIN_THR_MSB (0x307A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MIN_THR_MSB	7:0	Bank A color 2 ROI 1 minimum threshold value MSB.

BANK_A_ROI1_COLOR_MAX_THR_LSB (0x307B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MAX_THR_LSB	7:0	Bank A color 2 ROI 1 maximum threshold value LSB.

BANK_A_ROI1_COLOR_MAX_THR_MSB (0x307C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_MAX_THR_MSB	7:0	Bank A color 2 ROI 1 maximum threshold value MSB.

BANK_A_ROI1_COLOR_COUNT_LSB (0x307D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_COUNT_LSB	7:0	Bank A Color 2 ROI 1 detected number of color (LSB).

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BANK_A_ROI1_COLOR_COUNT_MSB (0x307E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI1_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI1_COLOR_COUNT_MSB	7:0	Bank A Color 2 ROI 1 detected number of color (MSB).

BANK_A_ROI2_XMIN_LSB (0x3080)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_XMIN_LSB	7:0	Bank A color 0 ROI 2 X coordinates minimum value LSB.

BANK_A_ROI2_XMIN_MSB (0x3081)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI2_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_XMIN_MSB	5:0	Bank A color 0 ROI 2 X coordinates minimum value MSB.

BANK_A_ROI2_XMAX_LSB (0x3082)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_XMAX_LSB[7:0]							

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Reset	0x0
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_XMAX_LSB	7:0	Bank A color 0 ROI 2 X coordinates maximum value LSB.

BANK_A_ROI2_XMAX_MSB (0x3083)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI2_XMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_XMAX_MSB	5:0	Bank A color 0 ROI 2 X coordinates maximum value MSB.

BANK_A_ROI2_YMIN_LSB (0x3084)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_YMIN_LSB	7:0	Bank A color 1 ROI 2 Y coordinates minimum value LSB.

BANK_A_ROI2_YMIN_MSB (0x3085)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI2_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_YMIN_MSB	5:0	Bank A color 0 ROI 2 Y coordinates minimum value MSB.

BANK_A_ROI2_YMAX_LSB (0x3086)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_YMAX_LSB	7:0	Bank A color 0 ROI 2 Y coordinates maximum value LSB.

BANK_A_ROI2_COLOR_SELECT (0x3088)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI2_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI2_COLOR_SELECT	2:0	Bank A Color 0 ROI 2 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI2_COLOR_MIN_THR_LSB (0x3089)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MIN_THR_LSB	7:0	Bank A color 0 ROI 2 minimum threshold value LSB.

BANK_A_ROI2_COLOR_MIN_THR_MSB (0x308A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MIN_THR_MSB	7:0	Bank A color 0 ROI 2 minimum threshold value MSB.

BANK_A_ROI2_COLOR_MAX_THR_LSB (0x308B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MAX_THR_LSB	7:0	Bank A color 0 ROI 2 maximum threshold value LSB.

BANK_A_ROI2_COLOR_MAX_THR_MSB (0x308C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MAX_THR_MSB	7:0	Bank A color 0 ROI 2 maximum threshold value MSB.

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BANK_A_ROI2_COLOR_COUNT_LSB (0x308D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_COUNT_LSB	7:0	Bank A Color 0 ROI 2 detected number of color (LSB).

BANK_A_ROI2_COLOR_COUNT_MSB (0x308E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_COUNT_MSB	7:0	Bank A Color 0 ROI 2 detected number of color (MSB).

BANK_A_ROI2_COLOR_SELECT (0x3090)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI2_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI2_COLOR_SELECT	2:0	Bank A Color 0 ROI 2 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

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BANK_A_ROI2_COLOR_MIN_THR_LSB (0x3091)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MIN_THR_LSB	7:0	Bank A color 0 ROI 2 minimum threshold value LSB.

BANK_A_ROI2_COLOR_MIN_THR_MSB (0x3092)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MIN_THR_MSB	7:0	Bank A color 0 ROI 2 minimum threshold value MSB.

BANK_A_ROI2_COLOR_MAX_THR_LSB (0x3093)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MAX_THR_LSB	7:0	Bank A color 0 ROI 2 maximum threshold value LSB.

BANK_A_ROI2_COLOR_MAX_THR_MSB (0x3094)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MAX_THR_MSB	7:0	Bank A color 0 ROI 2 maximum threshold value MSB.

BANK_A_ROI2_COLOR_COUNT_LSB (0x3095)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_COUNT_LSB	7:0	Bank A Color 0 ROI 2 detected number of color (LSB).

BANK_A_ROI2_COLOR_COUNT_MSB (0x3096)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_COUNT_MSB	7:0	Bank A Color 0 ROI 2 detected number of color (MSB).

BANK_A_ROI2_COLOR_SELECT (0x3098)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI2_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

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BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI2_COLOR_SELECT	2:0	Bank A Color 2 ROI 2 Color Set selection bits.	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI2_COLOR_MIN_THR_LSB (0x3099)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MIN_THR_LSB	7:0	Bank A color 2 ROI 2 minimum threshold value LSB.

BANK_A_ROI2_COLOR_MIN_THR_MSB (0x309A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MIN_THR_MSB	7:0	Bank A color 2 ROI 2 minimum threshold value MSB.

BANK_A_ROI2_COLOR_MAX_THR_LSB (0x309B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MAX_THR_LSB	7:0	Bank A color 2 ROI 2 maximum threshold value LSB.

BANK_A_ROI2_COLOR_MAX_THR_MSB (0x309C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_MAX_THR_MSB	7:0	Bank A color 2 ROI 2 maximum threshold value MSB.

BANK_A_ROI2_COLOR_COUNT_LSB (0x309D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_COUNT_LSB	7:0	Bank A Color 2 ROI 2 detected number of color (LSB).

BANK_A_ROI2_COLOR_COUNT_MSB (0x309E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI2_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI2_COLOR_COUNT_MSB	7:0	Bank A Color 2 ROI 2 detected number of color (MSB).

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BANK_A_ROI3_XMIN_LSB (0x30A0)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_XMIN_LSB	7:0	Bank A color 0 ROI 3 X coordinates minimum value LSB

BANK_A_ROI3_XMIN_MSB (0x30A1)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI3_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_XMIN_MSB	5:0	Bank A color 0 ROI 3 X coordinates minimum value MSB

BANK_A_ROI3_XMAX_LSB (0x30A2)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_XMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_XMAX_LSB	7:0	Bank A color 0 ROI 3 X coordinates maximum value LSB

BANK_A_ROI3_XMAX_MSB (0x30A3)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI3_XMAX_MSB[5:0]					

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Reset	–	–	0x0
Access Type	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_XMAX_MSB	5:0	Bank A color 0 ROI 3 X coordinates maximum value MSB

BANK_A_ROI3_YMIN_LSB (0x30A4)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_YMIN_LSB	7:0	Bank A color 1 ROI 3 Y coordinates minimum value LSB

BANK_A_ROI3_YMIN_MSB (0x30A5)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI3_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_YMIN_MSB	5:0	Bank A color 0 ROI 3 Y coordinates minimum value MSB.

BANK_A_ROI3_YMAX_LSB (0x30A6)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_YMAX_LSB	7:0	Bank A color 0 ROI 3 Y coordinates maximum value LSB

BANK_A_ROI3_YMAX_MSB (0x30A7)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_A_ROI3_YMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_YMAX_MSB	5:0	Bank A color 0 ROI 1 Y coordinates maximum value MSB

BANK_A_ROI3_COLOR_SELECT (0x30A8)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI3_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI3_COLOR_SELECT	2:0	Bank A Color 0 ROI 3 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI3_COLOR_MIN_THR_LSB (0x30A9)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MIN_THR_LSB	7:0	Bank A color 0 ROI 3 minimum threshold value LSB

BANK_A_ROI3_COLOR_MIN_THR_MSB (0x30AA)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MIN_THR_MSB	7:0	Bank A color 0 ROI 3 minimum threshold value MSB

BANK_A_ROI3_COLOR_MAX_THR_LSB (0x30AB)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MAX_THR_LSB	7:0	Bank A color 0 ROI 3 maximum threshold value LSB

BANK_A_ROI3_COLOR_MAX_THR_MSB (0x30AC)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MAX_THR_MSB	7:0	Bank A color 0 ROI 3 maximum threshold value MSB

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BANK_A_ROI3_COLOR_COUNT_LSB (0x30AD)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_COUNT_LSB	7:0	Bank A Color 0 ROI 3 detected number of color (LSB)

BANK_A_ROI3_COLOR_COUNT_MSB (0x30AE)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_COUNT_MSB	7:0	Bank A Color 0 ROI 3 detected number of color (MSB)

BANK_A_ROI3_COLOR_SELECT (0x30B0)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI3_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI3_COLOR_SELECT	2:0	Bank A Color 0 ROI 3 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

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BANK_A_ROI3_COLOR_MIN_THR_LSB (0x30B1)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MIN_THR_LSB	7:0	Bank A color 0 ROI 2 minimum threshold value LSB

BANK_A_ROI3_COLOR_MIN_THR_MSB (0x30B2)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MIN_THR_MSB	7:0	Bank A color 0 ROI 3 minimum threshold value MSB

BANK_A_ROI3_COLOR_MAX_THR_LSB (0x30B3)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MAX_THR_LSB	7:0	Bank A color 0 ROI 3 maximum threshold value LSB

BANK_A_ROI3_COLOR_MAX_THR_MSB (0x30B4)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MAX_THR_MSB	7:0	Bank A color 0 ROI 3 maximum threshold value MSB

BANK_A_ROI3_COLOR_COUNT_LSB (0x30B5)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_COUNT_LSB	7:0	Bank A Color 0 ROI 3 detected number of color (LSB)

BANK_A_ROI3_COLOR_COUNT_MSB (0x30B6)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_COUNT_MSB	7:0	Bank A Color 0 ROI 3 detected number of color (MSB)

BANK_A_ROI3_COLOR_SELECT (0x30B8)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_A_ROI3_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

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BITFIELD	BITS	DESCRIPTION	DECODE
BANK_A_ROI3_COLOR_SELECT	2:0	Bank A Color 2 ROI 3 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_A_ROI3_COLOR_MIN_THR_LSB (0x30B9)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MIN_THR_LSB	7:0	Bank A color 2 ROI 2 minimum threshold value LSB

BANK_A_ROI3_COLOR_MIN_THR_MSB (0x30BA)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MIN_THR_MSB	7:0	Bank A color 2 ROI 3 minimum threshold value MSB

BANK_A_ROI3_COLOR_MAX_THR_LSB (0x30BB)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MAX_THR_LSB	7:0	Bank A color 2 ROI 3 maximum threshold value LSB

BANK_A_ROI3_COLOR_MAX_THR_MSB (0x30BC)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_MAX_THR_MSB	7:0	Bank A color 2 ROI 3 maximum threshold value MSB

BANK_A_ROI3_COLOR_COUNT_LSB (0x30BD)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_COUNT_LSB	7:0	Bank A Color 2 ROI 3 detected number of color (LSB)

BANK_A_ROI3_COLOR_COUNT_MSB (0x30BE)

BIT	7	6	5	4	3	2	1	0
Field	BANK_A_ROI3_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_A_ROI3_COLOR_COUNT_MSB	7:0	Bank A Color 2 ROI 3 detected number of color (MSB)

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BANK_B_STATUS (0x30C0)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_CALC_DONE	BANK_B_FRAME_COUNT[1:0]	
Reset	–	–	–	–	–	0b0	0x0	
Access Type	–	–	–	–	–	Read Only	Read Only	

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_CALC_DONE	2	Telltale detection completion indicator for Bank B	0b0: Not complete 0b1: Complete
BANK_B_FRAME_COUNT	1:0	Frame Counter value indicating active frame number for Bank B	

BANK_B_ROI_1_0_ERROR (0x30C1)

BIT	7	6	5	4	3	2	1	0
Field	–	BANK_B_ROI_1_COLOR_2_ERROR	BANK_B_ROI_1_COLOR_1_ERROR	BANK_B_ROI_1_COLOR_0_ERROR	–	BANK_B_ROI_0_COLOR_2_ERROR	BANK_B_ROI_0_COLOR_1_ERROR	BANK_B_ROI_0_COLOR_0_ERROR
Reset	–	0x0	0x0	0x0	–	0x0	0x0	0x0
Access Type	–	Read Only	Read Only	Read Only	–	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI_1_COLOR_2_ERROR	6	Bank B color 0 ROI 1 color 2 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_1_COLOR_1_ERROR	5	Bank B color 0 ROI 1 color 1 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_1_COLOR_0_ERROR	4	Bank B color 0 ROI 1 color 0 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_0_COLOR_2_ERROR	2	Bank B color 0 ROI 0 color 2 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_0_COLOR_1_ERROR	1	Bank B color 0 ROI 0 color 1 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_0_COLOR_0_ERROR	0	Bank B color 0 ROI 0 color 0 error detected	0x0: No error detected 0x1: Error detected

BANK_B_ROI_3_2_ERROR (0x30C2)

BIT	7	6	5	4	3	2	1	0

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Field	–	BANK_B_ROI_3_COLOR_2_ERROR	BANK_B_ROI_3_COLOR_1_ERROR	BANK_B_ROI_3_COLOR_0_ERROR	–	BANK_B_ROI_2_COLOR_2_ERROR	BANK_B_ROI_2_COLOR_1_ERROR	BANK_B_ROI_2_COLOR_0_ERROR
Reset	–	0x0	0x0	0x0	–	0x0	0x0	0x0
Access Type	–	Read Only	Read Only	Read Only	–	Read Only	Read Only	Read Only

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI_3_COLOR_2_ERROR	6	Bank B color 0 ROI 3 color 2 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_3_COLOR_1_ERROR	5	Bank B color 0 ROI 3 color 1 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_3_COLOR_0_ERROR	4	Bank B color 0 ROI 3 color 0 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_2_COLOR_2_ERROR	2	Bank B color 0 ROI 2 color 2 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_2_COLOR_1_ERROR	1	Bank B color 0 ROI 2 color 1 error detected	0x0: No error detected 0x1: Error detected
BANK_B_ROI_2_COLOR_0_ERROR	0	Bank B color 0 ROI 2 color 0 error detected	0x0: No error detected 0x1: Error detected

BANK_B_ROI0_XMIN_LSB (0x30C8)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_XMIN_LSB	7:0	Bank B color 0 ROI 0 X coordinates minimum value LSB

BANK_B_ROI0_XMIN_MSB (0x30C9)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI0_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_XMIN_MSB	5:0	Bank B color 0 ROI 0 X coordinates minimum value MSB

BANK_B_ROI0_XMAX_LSB (0x30CA)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_XMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_XMAX_LSB	7:0	Bank B color 0 ROI 0 X coordinates maximum value LSB

BANK_B_ROI0_XMAX_MSB (0x30CB)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI0_XMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_XMAX_MSB	5:0	Bank B color 0 ROI 0 X coordinates maximum value MSB

BANK_B_ROI0_YMIN_LSB (0x30CC)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_YMIN_LSB	7:0	Bank B color 0 ROI 0 Y coordinates minimum value LSB

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BANK_B_ROI0_YMIN_MSB (0x30CD)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI0_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_YMIN_MSB	5:0	Bank B color 0 ROI 0 Y coordinates minimum value MSB

BANK_B_ROI0_YMAX_LSB (0x30CE)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_YMAX_LSB	7:0	Bank B color 0 ROI 0 Y coordinates maximum value LSB

BANK_B_ROI0_YMAX_MSB (0x30CF)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI0_YMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_YMAX_MSB	5:0	Bank B color 0 ROI 0 Y coordinates maximum value MSB

BANK_B_ROI0_COLOR_SELECT (0x30D0)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI0_COLOR_SELECT[2:0]		

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Reset	–	–	–	–	–	0x0
Access Type	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI0_COLOR_SELECT	2:0	Bank B Color 0 ROI 0 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI0_COLOR_MIN_THR_LSB (0x30D1)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MIN_THR_LSB	7:0	Bank B color 0 ROI 0 minimum threshold value LSB

BANK_B_ROI0_COLOR_MIN_THR_MSB (0x30D2)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MIN_THR_MSB	7:0	Bank B color 0 ROI 0 minimum threshold value MSB

BANK_B_ROI0_COLOR_MAX_THR_LSB (0x30D3)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MAX_THR_LSB	7:0	Bank B color 0 ROI 0 maximum threshold value LSB

BANK_B_ROI0_COLOR_MAX_THR_MSB (0x30D4)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MAX_THR_MSB	7:0	Bank B color 0 ROI 0 maximum threshold value MSB

BANK_B_ROI0_COLOR_COUNT_LSB (0x30D5)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_COUNT_LSB	7:0	Bank B Color 0 ROI 0 detected number of color (LSB)

BANK_B_ROI0_COLOR_COUNT_MSB (0x30D6)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_COUNT_MSB	7:0	Bank B Color 0 ROI 0 detected number of color (MSB)

BANK_B_ROI0_COLOR_SELECT (0x30D8)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI0_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI0_COLOR_SELECT	2:0	Bank B Color 1 ROI 0 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI0_COLOR_MIN_THR_LSB (0x30D9)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MIN_THR_LSB	7:0	Bank B color 1 ROI 0 minimum threshold value LSB

BANK_B_ROI0_COLOR_MIN_THR_MSB (0x30DA)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MIN_THR_MSB	7:0	Bank B color 1 ROI 0 minimum threshold value MSB

BANK_B_ROI0_COLOR_MAX_THR_LSB (0x30DB)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MAX_THR_LSB	7:0	Bank B color 1 ROI 0 maximum threshold value LSB

BANK_B_ROI0_COLOR_MAX_THR_MSB (0x30DC)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MAX_THR_MSB	7:0	Bank B color 1 ROI 0 maximum threshold value MSB

BANK_B_ROI0_COLOR_COUNT_LSB (0x30DD)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_COUNT_LSB	7:0	Bank B Color 1 ROI 0 detected number of color (LSB)

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BANK_B_ROI0_COLOR_COUNT_MSB (0x30DE)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_COUNT_MSB	7:0	Bank B Color 1 ROI 0 detected number of color (MSB)

BANK_B_ROI0_COLOR_SELECT (0x30E0)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI0_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI0_COLOR_SELECT	2:0	Bank B Color 2 ROI 0 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI0_COLOR_MIN_THR_LSB (0x30E1)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MIN_THR_LSB	7:0	Bank B color 2 ROI 0 minimum threshold value LSB

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BANK_B_ROI0_COLOR_MIN_THR_MSB (0x30E2)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MIN_THR_MSB	7:0	Bank B color 2 ROI 0 minimum threshold value MSB

BANK_B_ROI0_COLOR_MAX_THR_LSB (0x30E3)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MAX_THR_LSB	7:0	Bank B color 2 ROI 0 maximum threshold value LSB

BANK_B_ROI0_COLOR_MAX_THR_MSB (0x30E4)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_MAX_THR_MSB	7:0	Bank B color 2 ROI 0 maximum threshold value MSB

BANK_B_ROI0_COLOR_COUNT_LSB (0x30E5)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_COUNT_LSB[7:0]							

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Reset	0x0
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_COUNT_LSB	7:0	Bank B Color 2 ROI 0 detected number of color (LSB)

BANK_B_ROI0_COLOR_COUNT_MSB (0x30E6)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI0_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI0_COLOR_COUNT_MSB	7:0	Bank B Color 2 ROI 0 detected number of color (MSB)

BANK_B_ROI1_XMIN_LSB (0x30E8)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_XMIN_LSB	7:0	Bank B color 0 ROI 1 X coordinates minimum value LSB

BANK_B_ROI1_XMIN_MSB (0x30E9)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI1_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_XMIN_MSB	5:0	Bank B color 0 ROI 1 X coordinates minimum value MSB

BANK_B_ROI1_XMAX_LSB (0x30EA)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_XMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_XMAX_LSB	7:0	Bank B color 0 ROI 1 X coordinates maximum value LSB

BANK_B_ROI1_XMAX_MSB (0x30EB)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI1_XMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_XMAX_MSB	5:0	Bank B color 0 ROI 1 X coordinates maximum value MSB

BANK_B_ROI1_YMIN_LSB (0x30EC)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_YMIN_LSB	7:0	Bank B color 1 ROI 1 Y coordinates minimum value LSB

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BANK_B_ROI1_YMIN_MSB (0x30ED)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI1_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_YMIN_MSB	5:0	Bank B color 0 ROI 1 Y coordinates minimum value MSB

BANK_B_ROI1_YMAX_LSB (0x30EE)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_YMAX_LSB	7:0	Bank B color 0 ROI 1 Y coordinates maximum value LSB

BANK_B_ROI1_YMAX_MSB (0x30EF)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI1_YMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_YMAX_MSB	5:0	Bank B color 0 ROI 1 Y coordinates maximum value MSB

BANK_B_ROI1_COLOR_SELECT (0x30F0)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI1_COLOR_SELECT[2:0]		

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Reset	–	–	–	–	–	0x0
Access Type	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI1_COLOR_SELECT	2:0	Bank B Color 0 ROI 1 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI1_COLOR_MIN_THR_LSB (0x30F1)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MIN_THR_LSB	7:0	Bank B color 0 ROI 1 minimum threshold value LSB

BANK_B_ROI1_COLOR_MIN_THR_MSB (0x30F2)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MIN_THR_MSB	7:0	Bank B color 0 ROI 1 minimum threshold value MSB

BANK_B_ROI1_COLOR_MAX_THR_LSB (0x30F3)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MAX_THR_LSB	7:0	Bank B color 0 ROI 1 maximum threshold value LSB

BANK_B_ROI1_COLOR_MAX_THR_MSB (0x30F4)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MAX_THR_MSB	7:0	Bank B color 0 ROI 1 maximum threshold value MSB

BANK_B_ROI1_COLOR_COUNT_LSB (0x30F5)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_COUNT_LSB	7:0	Bank B Color 0 ROI 1 detected number of color (LSB)

BANK_B_ROI1_COLOR_COUNT_MSB (0x30F6)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_COUNT_MSB	7:0	Bank B Color 0 ROI 1 detected number of color (MSB)

BANK_B_ROI1_COLOR_SELECT (0x30F8)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI1_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI1_COLOR_SELECT	2:0	Bank B Color 0 ROI 1 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI1_COLOR_MIN_THR_LSB (0x30F9)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MIN_THR_LSB	7:0	Bank B color 0 ROI 1 minimum threshold value LSB

BANK_B_ROI1_COLOR_MIN_THR_MSB (0x30FA)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MIN_THR_MSB	7:0	Bank B color 0 ROI 1 minimum threshold value MSB

BANK_B_ROI1_COLOR_MAX_THR_LSB (0x30FB)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MAX_THR_LSB	7:0	Bank B color 0 ROI 1 maximum threshold value LSB

BANK_B_ROI1_COLOR_MAX_THR_MSB (0x30FC)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MAX_THR_MSB	7:0	Bank B color 0 ROI 1 maximum threshold value MSB

BANK_B_ROI1_COLOR_COUNT_LSB (0x30FD)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_COUNT_LSB	7:0	Bank B Color 0 ROI 1detected number of color (LSB)

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BANK_B_ROI1_COLOR_COUNT_MSB (0x30FE)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_COUNT_MSB	7:0	Bank B Color 0 ROI 1 detected number of color (MSB)

BANK_B_ROI1_COLOR_SELECT (0x3100)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI1_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI1_COLOR_SELECT	2:0	Bank B Color 2 ROI 1 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI1_COLOR_MIN_THR_LSB (0x3101)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MIN_THR_LSB	7:0	Bank B color 2 ROI 1 minimum threshold value LSB

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BANK_B_ROI1_COLOR_MIN_THR_MSB (0x3102)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MIN_THR_MSB	7:0	Bank B color 2 ROI 1 minimum threshold value MSB

BANK_B_ROI1_COLOR_MAX_THR_LSB (0x3103)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MAX_THR_LSB	7:0	Bank B color 2 ROI 1 maximum threshold value LSB

BANK_B_ROI1_COLOR_MAX_THR_MSB (0x3104)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_MAX_THR_MSB	7:0	Bank B color 2 ROI 1 maximum threshold value MSB

BANK_B_ROI1_COLOR_COUNT_LSB (0x3105)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_COUNT_LSB[7:0]							

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Reset	0x0
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_COUNT_LSB	7:0	Bank B Color 2 ROI 1 detected number of color (LSB)

BANK_B_ROI1_COLOR_COUNT_MSB (0x3106)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI1_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI1_COLOR_COUNT_MSB	7:0	Bank B Color 2 ROI 1 detected number of color (MSB)

BANK_B_ROI2_XMIN_LSB (0x3108)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_XMIN_LSB	7:0	Bank B color 0 ROI 2 X coordinates minimum value LSB

BANK_B_ROI2_XMIN_MSB (0x3109)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI2_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_XMIN_MSB	5:0	Bank B color 0 ROI 2 X coordinates minimum value MSB

BANK_B_ROI2_XMAX_LSB (0x310A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_XMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_XMAX_LSB	7:0	Bank B color 0 ROI 2 X coordinates maximum value LSB

BANK_B_ROI2_XMAX_MSB (0x310B)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI2_XMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_XMAX_MSB	5:0	Bank B color 0 ROI 2 X coordinates maximum value MSB

BANK_B_ROI2_YMIN_LSB (0x310C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_YMIN_LSB	7:0	Bank B color 1 ROI 2 Y coordinates minimum value LSB

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BANK_B_ROI2_YMIN_MSB (0x310D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI2_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_YMIN_MSB	5:0	Bank B color 0 ROI 2 Y coordinates minimum value MSB

BANK_B_ROI2_YMAX_LSB (0x310E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_YMAX_LSB	7:0	Bank B color 0 ROI 2 Y coordinates maximum value LSB

BANK_B_ROI2_YMAX_MSB (0x310F)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI2_YMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_YMAX_MSB	5:0	Bank B color 0 ROI 2 Y coordinates maximum value MSB

BANK_B_ROI2_COLOR_SELECT (0x3110)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI2_COLOR_SELECT[2:0]		

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Reset	–	–	–	–	–	0x0
Access Type	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI2_COLOR_SELECT	2:0	Bank B Color 0 ROI 2 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI2_COLOR_MIN_THR_LSB (0x3111)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MIN_THR_LSB	7:0	Bank B color 0 ROI 2 minimum threshold value LSB

BANK_B_ROI2_COLOR_MIN_THR_MSB (0x3112)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MIN_THR_MSB	7:0	Bank B color 0 ROI 2 minimum threshold value MSB

BANK_B_ROI2_COLOR_MAX_THR_LSB (0x3113)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MAX_THR_LSB	7:0	Bank B color 0 ROI 2 maximum threshold value LSB

BANK_B_ROI2_COLOR_MAX_THR_MSB (0x3114)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MAX_THR_MSB	7:0	Bank B color 0 ROI 2 maximum threshold value MSB

BANK_B_ROI2_COLOR_COUNT_LSB (0x3115)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_COUNT_LSB	7:0	Bank B Color 0 ROI 2 detected number of color (LSB)

BANK_B_ROI2_COLOR_COUNT_MSB (0x3116)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_COUNT_MSB	7:0	Bank B Color 0 ROI 2 detected number of color (MSB)

BANK_B_ROI2_COLOR_SELECT (0x3118)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI2_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI2_COLOR_SELECT	2:0	Bank B Color 0 ROI 2 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI2_COLOR_MIN_THR_LSB (0x3119)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MIN_THR_LSB	7:0	Bank B color 0 ROI 2 minimum threshold value LSB

BANK_B_ROI2_COLOR_MIN_THR_MSB (0x311A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MIN_THR_MSB	7:0	Bank B color 0 ROI 2 minimum threshold value MSB

BANK_B_ROI2_COLOR_MAX_THR_LSB (0x311B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MAX_THR_LSB	7:0	Bank B color 0 ROI 2 maximum threshold value LSB

BANK_B_ROI2_COLOR_MAX_THR_MSB (0x311C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MAX_THR_MSB	7:0	Bank B color 0 ROI 2 maximum threshold value MSB

BANK_B_ROI2_COLOR_COUNT_LSB (0x311D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_COUNT_LSB	7:0	Bank B Color 0 ROI 2 detected number of color (LSB)

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BANK_B_ROI2_COLOR_COUNT_MSB (0x311E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_COUNT_MSB	7:0	Bank B Color 0 ROI 2 detected number of color (MSB)

BANK_B_ROI2_COLOR_SELECT (0x3120)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI2_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI2_COLOR_SELECT	2:0	Bank B Color 2 ROI 2 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI2_COLOR_MIN_THR_LSB (0x3121)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MIN_THR_LSB	7:0	Bank B color 2 ROI 2 minimum threshold value LSB

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BANK_B_ROI2_COLOR_MIN_THR_MSB (0x3122)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MIN_THR_MSB	7:0	Bank B color 2 ROI 2 minimum threshold value MSB

BANK_B_ROI2_COLOR_MAX_THR_LSB (0x3123)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MAX_THR_LSB	7:0	Bank B color 2 ROI 2 maximum threshold value LSB

BANK_B_ROI2_COLOR_MAX_THR_MSB (0x3124)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_MAX_THR_MSB	7:0	Bank B color 2 ROI 2 maximum threshold value MSB

BANK_B_ROI2_COLOR_COUNT_LSB (0x3125)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_COUNT_LSB[7:0]							

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Reset	0x0
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_COUNT_LSB	7:0	Bank B Color 2 ROI 2 detected number of color (LSB)

BANK_B_ROI2_COLOR_COUNT_MSB (0x3126)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI2_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI2_COLOR_COUNT_MSB	7:0	Bank B Color 2 ROI 2 detected number of color (MSB)

BANK_B_ROI3_XMIN_LSB (0x3128)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_XMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_XMIN_LSB	7:0	Bank B color 0 ROI 3 X coordinates minimum value LSB

BANK_B_ROI3_XMIN_MSB (0x3129)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI3_XMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_XMIN_MSB	5:0	Bank B color 0 ROI 3 X coordinates minimum value MSB

BANK_B_ROI3_XMAX_LSB (0x312A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_XMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_XMAX_LSB	7:0	Bank B color 0 ROI 3 X coordinates maximum value LSB

BANK_B_ROI3_XMAX_MSB (0x312B)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI3_XMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_XMAX_MSB	5:0	Bank B color 0 ROI 3 X coordinates maximum value MSB

BANK_B_ROI3_YMIN_LSB (0x312C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_YMIN_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_YMIN_LSB	7:0	Bank B color 1 ROI 3 Y coordinates minimum value LSB

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BANK_B_ROI3_YMIN_MSB (0x312D)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI3_YMIN_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_YMIN_MSB	5:0	Bank B color 0 ROI 3 Y coordinates minimum value MSB

BANK_B_ROI3_YMAX_LSB (0x312E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_YMAX_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_YMAX_LSB	7:0	Bank B color 0 ROI 3 Y coordinates maximum value LSB

BANK_B_ROI3_YMAX_MSB (0x312F)

BIT	7	6	5	4	3	2	1	0
Field	–	–	BANK_B_ROI3_YMAX_MSB[5:0]					
Reset	–	–	0x0					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_YMAX_MSB	5:0	Bank B color 0 ROI 1 Y coordinates maximum value MSB

BANK_B_ROI3_COLOR_SELECT (0x3130)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI3_COLOR_SELECT[2:0]		

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Reset	–	–	–	–	–	0x0
Access Type	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI3_COLOR_SELECT	2:0	Bank B Color 0 ROI 3 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI3_COLOR_MIN_THR_LSB (0x3131)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MIN_THR_LSB	7:0	Bank B color 0 ROI 3 minimum threshold value LSB

BANK_B_ROI3_COLOR_MIN_THR_MSB (0x3132)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MIN_THR_MSB	7:0	Bank B color 0 ROI 3 minimum threshold value MSB

BANK_B_ROI3_COLOR_MAX_THR_LSB (0x3133)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MAX_THR_LSB[7:0]							

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Reset	0xFF
Access Type	Write, Read

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MAX_THR_LSB	7:0	Bank B color 0 ROI 3 maximum threshold value LSB

BANK_B_ROI3_COLOR_MAX_THR_MSB (0x3134)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MAX_THR_MSB	7:0	Bank B color 0 ROI 3 maximum threshold value MSB

BANK_B_ROI3_COLOR_COUNT_LSB (0x3135)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_COUNT_LSB	7:0	Bank B Color 0 ROI 3 detected number of color (LSB)

BANK_B_ROI3_COLOR_COUNT_MSB (0x3136)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_COUNT_MSB	7:0	Bank B Color 0 ROI 3 detected number of color (MSB)

BANK_B_ROI3_COLOR_SELECT (0x3138)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI3_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI3_COLOR_SELECT	2:0	Bank B Color 0 ROI 3 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI3_COLOR_MIN_THR_LSB (0x3139)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MIN_THR_LSB	7:0	Bank B color 0 ROI 2 minimum threshold value LSB

BANK_B_ROI3_COLOR_MIN_THR_MSB (0x313A)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MIN_THR_MSB	7:0	Bank B color 0 ROI 3 minimum threshold value MSB

BANK_B_ROI3_COLOR_MAX_THR_LSB (0x313B)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MAX_THR_LSB	7:0	Bank B color 0 ROI 3 maximum threshold value LSB

BANK_B_ROI3_COLOR_MAX_THR_MSB (0x313C)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MAX_THR_MSB	7:0	Bank B color 0 ROI 3 maximum threshold value MSB

BANK_B_ROI3_COLOR_COUNT_LSB (0x313D)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_COUNT_LSB	7:0	Bank B Color 0 ROI 3 detected number of color (LSB)

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BANK_B_ROI3_COLOR_COUNT_MSB (0x313E)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_COUNT_MSB	7:0	Bank B Color 0 ROI 3 detected number of color (MSB)

BANK_B_ROI3_COLOR_SELECT (0x3140)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	BANK_B_ROI3_COLOR_SELECT[2:0]		
Reset	–	–	–	–	–	0x0		
Access Type	–	–	–	–	–	Write, Read		

BITFIELD	BITS	DESCRIPTION	DECODE
BANK_B_ROI3_COLOR_SELECT	2:0	Bank B Color 2 ROI 3 Color Set selection bits	0x0: Color 0 - Red 0x1: Color 1 - Orange 0x2: Color 2 - Green 0x3: Color 3 - Blue 0x4: Color 4 - White 0x5: Color 5 - Gray 0x6: Illegal 0x7: Illegal

BANK_B_ROI3_COLOR_MIN_THR_LSB (0x3141)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MIN_THR_LSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MIN_THR_LSB	7:0	Bank B color 2 ROI 2 minimum threshold value LSB

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BANK_B_ROI3_COLOR_MIN_THR_MSB (0x3142)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MIN_THR_MSB[7:0]							
Reset	0x0							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MIN_THR_MSB	7:0	Bank B color 2 ROI 3 minimum threshold value MSB

BANK_B_ROI3_COLOR_MAX_THR_LSB (0x3143)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MAX_THR_LSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MAX_THR_LSB	7:0	Bank B color 2 ROI 3 maximum threshold value LSB

BANK_B_ROI3_COLOR_MAX_THR_MSB (0x3144)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_MAX_THR_MSB[7:0]							
Reset	0xFF							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_MAX_THR_MSB	7:0	Bank B color 2 ROI 3 maximum threshold value MSB

BANK_B_ROI3_COLOR_COUNT_LSB (0x3145)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_COUNT_LSB[7:0]							

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Reset	0x0
Access Type	Read Only

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_COUNT_LSB	7:0	Bank B Color 2 ROI 3 detected number of color (LSB)

BANK_B_ROI3_COLOR_COUNT_MSB (0x3146)

BIT	7	6	5	4	3	2	1	0
Field	BANK_B_ROI3_COLOR_COUNT_MSB[7:0]							
Reset	0x0							
Access Type	Read Only							

BITFIELD	BITS	DESCRIPTION
BANK_B_ROI3_COLOR_COUNT_MSB	7:0	Bank B Color 2 ROI 3 detected number of color (MSB)

MAIN_STREAM_MISC0_S0 (0x6944)

This 8-bit value contains information about the main link video stream clocking and color representation. These bits are mapped from the DisplayPort specification MISC0 register definitions.

BIT	7	6	5	4	3	2	1	0
Field	MS_BIT_DEPTH_S0[2:0]			–	–	–	–	–
Reset	0b000			–	–	–	–	–
Access Type	Write, Read			–	–	–	–	–

BITFIELD	BITS	DESCRIPTION
MS_BIT_DEPTH_S0	7:5	Number of bits per color 000 = 6 bits 001 = 8 bits 010 = 10 bits 011 = 12 bits 100 = 16 bits 101, 110, 111 = Reserved

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USER_DATA_COUNT_PER_LANE_S0_B0 (0x695C)

Lower 8 bits to determine the total data count for the transmitter framing logic to read from the user FIFO before sending a blanking start symbol. In other words, this value is the total number of 16-bit words in a line of active data. The calculated value should be rounded up.

BIT	7	6	5	4	3	2	1	0
Field	USER_DATA_COUNT_PER_LANE_S0_L[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
USER_DATA_COUNT_PER_LANE_S0_L	7:0	The equation for setting the value of this register is: SYMBOL_COUNT = ((HRES * bits per pixel) + 7) / 8 UDC = (SYMBOL_COUNT + lane_count – 1) / lane_count.

USER_DATA_COUNT_PER_LANE_S0_B1 (0x695D)

Middle 8 bits to determine the total data count for the transmitter framing logic to read from the user FIFO before sending a blanking start symbol. In other words, this value is the total number of 16-bit words in a line of active data. The calculated value should be rounded up.

BIT	7	6	5	4	3	2	1	0
Field	USER_DATA_COUNT_PER_LANE_S0_M[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
USER_DATA_COUNT_PER_LANE_S0_M	7:0	The equation for setting the value of this register is: SYMBOL_COUNT = ((HRES * bits per pixel) + 7) / 8 UDC = (SYMBOL_COUNT + lane_count – 1) / lane_count.

VTRG_CTRL_B0 (0x7000)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	vtrg_en
Reset	–	–	–	–	–	–	–	0b1
Access Type	–	–	–	–	–	–	–	Write, Read

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BITFIELD	BITS	DESCRIPTION	DECODE
vtrg_en	0	Video Timing Regenerator (VTRG) enable	0: VTRG disabled 1: VTRG enabled (recommended)

VTRG_CTRL_B0 (0x7080)

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	vtrg_en
Reset	–	–	–	–	–	–	–	0b1
Access Type	–	–	–	–	–	–	–	Write, Read

BITFIELD	BITS	DESCRIPTION
vtrg_en	0	VTRG Enable

DEC_PPS0_3_B0 (0xA260)

DSC Decoder PPS 0-3 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	dsc_version_major[3:0]				dsc_version_minor[3:0]			
Reset	0x0				0x0			
Access Type	Write, Read				Write, Read			

BITFIELD	BITS	DESCRIPTION
dsc_version_major	7:4	Contains the major version of DSC
dsc_version_minor	3:0	Contains the minor version of DSC

DEC_PPS0_3_B1 (0xA261)

DSC Decoder PPS 0-3 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	pps_identifier[7:0]							
Reset	0x00							
Access Type	Write, Read							

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BITFIELD	BITS	DESCRIPTION
pps_identifier	7:0	Application-specific identifier that may be used to differentiate between different PPS tables. If PPS transmission is not defined by an application specification, the value should be 0x00.

DEC_PPS0_3_B2 (0xA262)

DSC Decoder PPS 0-3 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	RSVD[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RSVD	7:0	RESERVED: DSC Decoder PPS 0-3 Register Byte 2

DEC_PPS0_3_B3 (0xA263)

DSC Decoder PPS0-3 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	bits_per_component[3:0]				linebuf_depth[3:0]			
Reset	0x0				0x0			
Access Type	Write, Read				Write, Read			

BITFIELD	BITS	DESCRIPTION	DECODE
bits_per_component	7:4	Bits per component 0x0 = 16bpc (allowed only when dsc_version_minor = 0x2). 0x8 = 8bpc 0xA = 10bpc 0xC = 12bpc 0xE = 14bpc (allowed only when dsc_version_minor = 0x2)	0x8: 8bpc 0xA: 10bpc 0xC: 12bpc 0xE: 14bpc
linebuf_depth	3:0	Contains the line buffer bit depth used to generate the bitstream. 0x0 = NOT SUPPORTED	0x0: 16 bits (NOT supported) 0x8: 8 bits 0x9: 9 bits 0xA: 10 bits 0xB: 11 bits 0xC: 12 bits

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BITFIELD	BITS	DESCRIPTION	DECODE
			0xD: 13 bits 0xE: 14 bits (Not supported) 0xF: 15 bits (Not supported)

DEC_PPS4_7_B0 (0xA264)

DSC Decoder Core PPS4-7 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	–	–	block_pred_enable	convert_rgb	simple_422	vbr_enable	bits_per_pixel_h[1:0]	
Reset	–	–	0b0	0b0	0b0	0b0	0b00	
Access Type	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	

BITFIELD	BITS	DESCRIPTION	DECODE
block_pred_enable	5	Active-High Input Block Prediction Enable	0b0: Block Prediction is not used to code any group within the picture 0b1: Block Prediction is used
convert_rgb	4	Indicates whether DSC color space conversion is active.	0b0: Color space is YCbCv. 0b1: Encoder converts RGB to YCoCg-R, and Decoder converts YCoCg-R to RGB
simple_422	3	Indicates that the decoder will recreate 4:2:2 sub-sampled stream. NOT supported. Must be set to 0.	
vbr_enable	2	VBR Enable mode	0x0: VBR mode is disabled 0x1: VBR mode is enabled
bits_per_pixel_h	1:0	Higher bits of target bits per pixel [9:8]. Value is in 6.4 format (4 LSBs for fractional part).	

DEC_PPS4_7_B1 (0xA265)

DSC Decoder Core PPS4-7 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	bits_per_pixel_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
bits_per_pixel_l	7:0	Lower bits of target bits per pixel [7:0]. Value is in 6.4 format.

DEC_PPS4_7_B2 (0xA266)

DSC Decoder PPS4-7 Register Byte 2

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BIT	7	6	5	4	3	2	1	0
Field	pic_height_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
pic_height_h	7:0	Picture Height (vertical active) High Byte [15:8]

DEC_PPS4_7_B3 (0xA267)

DSC Decoder PPS4-7 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	pic_height_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
pic_height_l	7:0	Picture Height (vertical active) Low Byte [7:0]

DEC_PPS8_11_B0 (0xA268)

DSC Decoder PPS8-11 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	pic_width_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
pic_width_h	7:0	Picture Width (horizontal active) High Byte [15:8]

DEC_PPS8_11_B1 (0xA269)

DSC Decoder PPS8-11 Register Byte 1

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BIT	7	6	5	4	3	2	1	0
Field	pic_width_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
pic_width_l	7:0	Picture Width (horizontal active) Low Byte [7:0]

DEC_PPS8_11_B2 (0xA26A)

DSC Decoder PPS8-11 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	slice_height_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
slice_height_h	7:0	Slice Height High Byte [15:8]

DEC_PPS8_11_B3 (0xA26B)

DSC Decoder PPS8-11 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	slice_height_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
slice_height_l	7:0	Slice Height Low Byte [7:0]

DEC_PPS12_15_B0 (0xA26C)

DSC Decoder PPS12-15 Register Byte 0

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BIT	7	6	5	4	3	2	1	0
Field	slice_width_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
slice_width_h	7:0	Slice Width High Byte [15:8]

DEC_PPS12_15_B1 (0xA26D)

DSC Decoder PPS12-15 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	slice_width_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
slice_width_l	7:0	Slice Width Low Byte [7:0]

DEC_PPS12_15_B2 (0xA26E)

DSC Decoder PPS12-15 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	chunk_size_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
chunk_size_h	7:0	Chunk Size High Byte [15:8]

DEC_PPS12_15_B3 (0xA26F)

DSC Decoder PPS12-15 Register Byte 3

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BIT	7	6	5	4	3	2	1	0
Field	chunk_size_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
chunk_size_l	7:0	Chunk Size Low Byte [7:0]

DEC_PPS16_19_B0 (0xA270)

DSC Decoder PPS16_19 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	initial_xmit_delay_h[1:0]	
Reset	–	–	–	–	–	–	0b00	
Access Type	–	–	–	–	–	–	Write, Read	

BITFIELD	BITS	DESCRIPTION
initial_xmit_delay_h	1:0	Initial Encoder Transmit Delay High Bits [9:8].

DEC_PPS16_19_B1 (0xA271)

DSC Decoder PPS16_19 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	initial_xmit_delay_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
initial_xmit_delay_l	7:0	Initial Encoder Transmit Delay High Byte [9:8]

DEC_PPS16_19_B2 (0xA272)

DSC Decoder PPS16_19 Register Byte 2

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BIT	7	6	5	4	3	2	1	0
Field	initial_dec_delay_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
initial_dec_delay_h	7:0	Initial Decoding Delay High Byte [15:8]

DEC_PPS16_19_B3 (0xA273)

DSC Decoder PPS16_19 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	initial_dec_delay_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
initial_dec_delay_l	7:0	Initial Decoding Delay High Byte [15:8]

DEC_PPS20_23_B0 (0xA274)

DSC Decoder PPS20-23 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	RSVD[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RSVD	7:0	RESERVED: DSC Decoder PPS 20-23 Register Byte 0

DEC_PPS20_23_B1 (0xA275)

DSC Decoder PPS20-23 Register Byte 1

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BIT	7	6	5	4	3	2	1	0
Field	–	–	initial_scale_value[5:0]					
Reset	–	–	0b000000					
Access Type	–	–	Write, Read					

BITFIELD	BITS	DESCRIPTION
initial_scale_value	5:0	Initial Scale Value

DEC_PPS20_23_B2 (0xA276)

DSC Decoder PPS20-23 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	scale_increment_interval_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
scale_increment_interval_h	7:0	Scale Increment Interval High Byte [15:8]

DEC_PPS20_23_B3 (0xA277)

DSC Decoder PPS20-23 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	scale_increment_interval_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
scale_increment_interval_l	7:0	Scale Increment Interval Low Byte [7:0]

DEC_PPS24_27_B0 (0xA278)

DSC Decoder PPS24-27 Register Byte 0

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BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	scale_decrement_interval_h[3:0]			
Reset	–	–	–	–	0x0			
Access Type	–	–	–	–	Write, Read			

BITFIELD	BITS	DESCRIPTION
scale_decrement_interval_h	3:0	Scale Decrement Interval High Bits [11:8]

DEC_PPS24_27_B1 (0xA279)

DSC Decoder PPS24-27 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	scale_decrement_interval_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
scale_decrement_interval_l	7:0	Scale Decrement Interval Low Byte [7:0]

DEC_PPS24_27_B2 (0xA27A)

Scale Decrement Interval Byte 2

BIT	7	6	5	4	3	2	1	0
Field	RSVD[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
RSVD	7:0	RESERVED: DSC Decoder PPS 24_27 Register Byte 2

DEC_PPS24_27_B3 (0xA27B)

DSC Decoder PPS24-27 Register Byte 3

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BIT	7	6	5	4	3	2	1	0
Field	–	–	–	first_line_bpg_offset[4:0]				
Reset	–	–	–	0b00000				
Access Type	–	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION
first_line_bpg_offset	4:0	Number of additional bits that are allocated for each group on the first line of a slice.

DEC_PPS28_31_B0 (0xA27C)

DSC Decoder PPS28-31 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	nfl_bpg_offset_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
nfl_bpg_offset_h	7:0	Non-First Line (NFL) BPG Offset High Byte. Value includes 11 fractional bits.

DEC_PPS28_31_B1 (0xA27D)

DSC Decoder PPS28-31 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	nfl_bpg_offset_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
nfl_bpg_offset_l	7:0	NFL BPG Offset Low Byte

DEC_PPS28_31_B2 (0xA27E)

DSC Decoder PPS28-31 Register Byte 2

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BIT	7	6	5	4	3	2	1	0
Field	slice_bpg_offset_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
slice_bpg_offset_h	7:0	Slice BPG Offset High Byte [15:8]

DEC_PPS28_31_B3 (0xA27F)

DSC Decoder PPS28-31 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	slice_bpg_offset_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
slice_bpg_offset_l	7:0	Slice BPG Offset Low Byte [7:0]

DEC_PPS32_35_B0 (0xA280)

DSC Decoder PPS32-35 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	initial_offset_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
initial_offset_h	7:0	Initial Offset High Byte [15:8]

DEC_PPS32_35_B1 (0xA281)

DSC Decoder PPS 32-35 Register Byte 1

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BIT	7	6	5	4	3	2	1	0
Field	initial_offset_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
initial_offset_l	7:0	Initial Offset Low Byte [7:0]

DEC_PPS32_35_B2 (0xA282)

DSC Decoder PPS 32-35 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	final_offset_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
final_offset_h	7:0	Final Offset High Byte [15:0]

DEC_PPS32_35_B3 (0xA283)

DSC Decoder PPS 32-35 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	final_offset_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
final_offset_l	7:0	Final Offset Low Byte [7:0]

DEC_PPS36_39_B0 (0xA284)

DSC Decoder PPS 36-39 Register Byte 0

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BIT	7	6	5	4	3	2	1	0
Field	–	–	–	flatness_min_qp[4:0]				
Reset	–	–	–	0b00000				
Access Type	–	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION
flatness_min_qp	4:0	Flatness Minimum QP

DEC_PPS36_39_B1 (0xA285)

DSC Decoder PPS 36-39 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	flatness_max_qp[4:0]				
Reset	–	–	–	0b00000				
Access Type	–	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION
flatness_max_qp	4:0	Flatness Maximum QP

DEC_PPS36_39_B2 (0xA286)

DSC Decoder PPS 36-39 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	rc_model_size_h[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_model_size_h	7:0	RC Model Size High Byte [15:8]

DEC_PPS36_39_B3 (0xA287)

DSC Decoder PPS 36-39 Register Byte 3

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BIT	7	6	5	4	3	2	1	0
Field	rc_model_size_l[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_model_size_l	7:0	RC Model Size Low Byte [7:0]

DEC_PPS40_43_B0 (0xA288)

DSC Decoder PPS40-43 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	–	rc_edge_factor[3:0]			
Reset	–	–	–	–	0x0			
Access Type	–	–	–	–	Write, Read			

BITFIELD	BITS	DESCRIPTION
rc_edge_factor	3:0	RC Edge Factor

DEC_PPS40_43_B1 (0xA289)

DSC Decoder PPS40-43 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	–	–	–	rc_quant_incr_limit0[4:0]				
Reset	–	–	–	0b00000				
Access Type	–	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION
rc_quant_incr_limit0	4:0	RC Quant Increment Limit 0

DEC_PPS40_43_B2 (0xA28A)

DSC Decoder PPS40-43 Register Byte 2

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BIT	7	6	5	4	3	2	1	0
Field	–	–	–	rc_quant_incr_limit1[4:0]				
Reset	–	–	–	0b00000				
Access Type	–	–	–	Write, Read				

BITFIELD	BITS	DESCRIPTION
rc_quant_incr_limit1	4:0	RC Quant Increment Limit 1

DEC_PPS40_43_B3 (0xA28B)

DSC Decoder PPS40-43 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	rc_tgt_offset_hi[3:0]				rc_tgt_offset_lo[3:0]			
Reset	0x0				0x0			
Access Type	Write, Read				Write, Read			

BITFIELD	BITS	DESCRIPTION
rc_tgt_offset_hi	7:4	RC Target Offset High
rc_tgt_offset_lo	3:0	RC Target Offset Low

DEC_PPS44_47_B0 (0xA28C)

DSC Decoder PPS44-47 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	rc_buff_thresh0[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buff_thresh0	7:0	RC Buffer Threshold 0

DEC_PPS44_47_B1 (0xA28D)

DSC Decoder PPS44-47 Register Byte 1

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BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh1[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh1	7:0	RC Buffer Threshold 1

DEC_PPS44_47_B2 (0xA28E)

DSC Decoder PPS44-47 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh2[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh2	7:0	RC Buffer Threshold 2

DEC_PPS44_47_B3 (0xA28F)

DSC Decoder PPS44-47 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh3[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh3	7:0	RC Buffer Threshold 3

DEC_PPS48_51_B0 (0xA290)

DSC Decoder PPS48-51 Register Byte 0

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BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh4[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh4	7:0	RC Buffer Threshold 4

DEC_PPS48_51_B1 (0xA291)

DSC Decoder PPS48-51 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh5[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh5	7:0	RC Buffer Threshold 5

DEC_PPS48_51_B2 (0xA292)

DSC Decoder PPS48-51 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh6[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh6	7:0	RC Buffer Threshold 6

DEC_PPS48_51_B3 (0xA293)

DSC Decoder PPS48-51 Register Byte 3

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BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh7[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh7	7:0	RC Buffer Threshold 7

DEC_PPS52_55_B0 (0xA294)

DSC Decoder PPS52-55 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh8[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh8	7:0	RC Buffer Threshold 8

DEC_PPS52_55_B1 (0xA295)

DSC Decoder PPS52-55 Register byte 1

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh9[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh9	7:0	RC Buffer Threshold 9

DEC_PPS52_55_B2 (0xA296)

DSC Decoder PPS52-55 Register Byte 2

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BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh10[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh10	7:0	RC Buffer Threshold 10

DEC_PPS52_55_B3 (0xA297)

DSC Decoder PPS52-55 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh11[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh11	7:0	RC Buffer Threshold 11

DEC_PPS56_59_B0 (0xA298)

DSC Decoder PPS56-59 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh12[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh12	7:0	RC Buffer Threshold 12

DEC_PPS56_59_B1 (0xA299)

DSC Decoder PPS56-59 Register Byte 1

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BIT	7	6	5	4	3	2	1	0
Field	rc_buf_thresh13[7:0]							
Reset	0x00							
Access Type	Write, Read							

BITFIELD	BITS	DESCRIPTION
rc_buf_thresh13	7:0	RC Buffer Threshold 13

DEC_PPS56_59_B2 (0xA29A)

DSC Decoder PPS56-59 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp0[4:0]					range_max_qp0_h[2:0]		
Reset	0b000					0b00000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp0	7:3	Range Minimum QP0
range_max_qp0_h	2:0	Range Maximum QP0 High Bits [4:2]

DEC_PPS56_59_B3 (0xA29B)

DSC Decoder PPS56-59 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp0_l[1:0]		range_bpg_offset0[5:0]					
Reset	0b000000		0b00					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp0_l	7:6	Range Maximum QP0 Low Bits [1:0]
range_bpg_offset0	5:0	Range BPG Offset 0

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DEC_PPS60_63_B0 (0xA29C)

DSC Decoder PPS60-63 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp1[4:0]					range_max_qp1_h[2:0]		
Reset	0b000					0b00000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp1	7:3	Range Minimum QP1
range_max_qp1_h	2:0	Range Maximum QP1 High Bits [4:2]

DEC_PPS60_63_B1 (0xA29D)

DSC Decoder PPS60-63 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp1_l[1:0]		range_bpg_offset1[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp1_l	7:6	Range Maximum Low Bits [1:0]
range_bpg_offset1	5:0	Range BPG Offset 1

DEC_PPS60_63_B2 (0xA29E)

DSC Decoder PPS60-63 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp2[4:0]					range_max_qp2_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

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BITFIELD	BITS	DESCRIPTION
range_min_qp2	7:3	Range Minimum QP2
range_max_qp2_h	2:0	Range Maximum QP2 High Bits [4:2]

DEC_PPS60_63_B3 (0xA29F)

DSC Decoder PPS60-63 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp2_l[1:0]		range_bpg_offset2[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp2_l	7:6	Range Maximum QP2 Low Bits [1:0]
range_bpg_offset2	5:0	Range BPG Offset 2

DEC_PPS64_67_B0 (0xA2A0)

DSC Decoder PPS64-67 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp3[4:0]					range_max_qp3_h[2:0]		
Reset	0b00000					0b00		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp3	7:3	Range Minimum QP3
range_max_qp3_h	2:0	Range Maximum QP3 Low Bits [1:0]

DEC_PPS64_67_B1 (0xA2A1)

DSC Decoder PPS64-67 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp3_l[1:0]		range_bpg_offset3[5:0]					

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Reset	0b00	0b000000
Access Type	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION
range_max_qp3_l	7:6	Range Maximum QP3 Low Bits [1:0]
range_bpg_offset3	5:0	Range BPG Offset 3

DEC_PPS64_67_B2 (0xA2A2)

DSC Decoder PPS64-67 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp4[4:0]					range_max_qp4_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp4	7:3	Range Minimum QP4
range_max_qp4_h	2:0	Range Maximum QP4 High Bits [4:2]

DEC_PPS64_67_B3 (0xA2A3)

DSC Decoder PPS64-67 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp4_l[1:0]		range_bpg_offset4[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp4_l	7:6	Range Maximum QP4 Low Bits [1:0]
range_bpg_offset4	5:0	Range BPG Offset 4

DEC_PPS68_71_B0 (0xA2A4)

DSC Decoder PPS68-71 Register Byte 0

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BIT	7	6	5	4	3	2	1	0
Field	range_min_qp5[4:0]					range_max_qp5_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp5	7:3	Range Minimum QP5
range_max_qp5_h	2:0	Range Maximum QP5 High Bits [4:2]

DEC_PPS68_71_B1 (0xA2A5)

DSC Decoder PPS68-71 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp5_l[1:0]		range_bpg_offset5[5:0]					
Reset	0b000		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp5_l	7:6	Range Maximum QP5 Low Bits [1:0]
range_bpg_offset5	5:0	Range BPG Offset 5

DEC_PPS68_71_B2 (0xA2A6)

DSC Decoder PPS68-71 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp6[4:0]					range_max_qp6_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp6	7:3	Range Minimum QP6
range_max_qp6_h	2:0	Range Maximum QP6 High Bits [4:2]

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DEC_PPS68_71_B3 (0xA2A7)

DSC Decoder PPS68-71 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp6_l[1:0]		range_bpg_offset6[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp6_l	7:6	Range Maximum QP6 Low Bits [1:0]
range_bpg_offset6	5:0	Range BPG Offset 6

DEC_PPS72_75_B0 (0xA2A8)

DSC Decoder PPS72-75 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp7[4:0]					range_max_qp7_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp7	7:3	Range Minimum QP7
range_max_qp7_h	2:0	Range Maximum QP7 High Bits [4:2]

DEC_PPS72_75_B1 (0xA2A9)

DSC Decoder PPS72-75 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp7_l[1:0]		range_bpg_offset7[5:0]					
Reset	0b000		0b000000					
Access Type	Write, Read		Write, Read					

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BITFIELD	BITS	DESCRIPTION
range_max_qp7_l	7:6	Range Maximum QP7 Low Bits [1:0]
range_bpg_offset7	5:0	Range BPG Offset 7

DEC_PPS72_75_B2 (0xA2AA)

DSC Decoder PPS72-75 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp8[4:0]					range_max_qp8_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp8	7:3	Range Minimum QP8
range_max_qp8_h	2:0	Range Maximum QP8 High Bits [4:2]

DEC_PPS72_75_B3 (0xA2AB)

DSC Decoder PPS72-75 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp8_l[1:0]		range_bpg_offset8[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp8_l	7:6	Range Maximum QP8 Low Bits [1:0]
range_bpg_offset8	5:0	Rang BPG Offset 8

DEC_PPS76_79_B0 (0xA2AC)

DSC Decoder PPS76-79 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp9[4:0]					range_max_qp9_h[2:0]		

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Reset	0b00000	0b000
Access Type	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION
range_min_qp9	7:3	Range Minimum QP9
range_max_qp9_h	2:0	Range Maximum QP9 High Bits [4:2]

DEC_PPS76_79_B1 (0xA2AD)

DSC Decoder PPS76-79 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp9_l[1:0]		range_bpg_offset9[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp9_l	7:6	Range Maximum QP9 Low Bits [1:0]
range_bpg_offset9	5:0	Range BPG Offset 9

DEC_PPS76_79_B2 (0xA2AE)

DSC Decoder PPS76-79 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp10[4:0]					range_max_qp10_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp10	7:3	Range Minimum QP10
range_max_qp10_h	2:0	Range Maximum QP10 High Bits [4:2]

DEC_PPS76_79_B3 (0xA2AF)

DSC Decoder PPS76-79 Register Byte 3

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BIT	7	6	5	4	3	2	1	0
Field	range_max_qp10_l[1:0]		range_bpg_offset10[5:0]					
Reset	0b000		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp10_l	7:6	Range Maximum QP10 Low Bits [1:0]
range_bpg_offset10	5:0	Range BPG Offset 10

DEC_PPS80_83_B0 (0xA2B0)

DSC Decoder PPS80-83 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp11[4:0]					range_max_qp11_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp11	7:3	Range Minimum QP11
range_max_qp11_h	2:0	Range Maximum QP11 High Bit [4:2]

DEC_PPS80_83_B1 (0xA2B1)

DSC Decoder PPS80-83 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp11_l[1:0]		range_bpg_offset11[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp11_l	7:6	Range Maximum QP11 Low Bits [1:0]
range_bpg_offset11	5:0	Range BPG Offset 11

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DEC_PPS80_83_B2 (0xA2B2)

DSC Decoder PPS80-83 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp12[4:0]					range_max_qp12_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp12	7:3	Range Minimum QP12
range_max_qp12_h	2:0	Range Maximum QP12 High Bits [4:2]

DEC_PPS80_83_B3 (0xA2B3)

DSC Decoder PPS80-83 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp12_l[1:0]		range_bpg_offset12[5:0]					
Reset	0b000		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp12_l	7:6	Range Maximum Qp12 Low Bits[1:0]
range_bpg_offset12	5:0	Range BPG Offset 12

DEC_PPS84_87_B0 (0xA2B4)

DSC Decoder PPS84_87 Register Byte 0

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp13[4:0]					range_max_qp13_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

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BITFIELD	BITS	DESCRIPTION
range_min_qp13	7:3	Range Minimum QP13
range_max_qp13_h	2:0	Range Maximum QP13 High Bits [4:2]

DEC_PPS84_87_B1 (0xA2B5)

DSC Decoder PPS84_87 Register Byte 1

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp13_l[1:0]		range_bpg_offset13[5:0]					
Reset	0b00		0b000000					
Access Type	Write, Read		Write, Read					

BITFIELD	BITS	DESCRIPTION
range_max_qp13_l	7:6	Range Maximum QP13 Low Bits [1:0]
range_bpg_offset13	5:0	Range BPG Offset 13

DEC_PPS84_87_B2 (0xA2B6)

DSC Decoder PPS84_87 Register Byte 2

BIT	7	6	5	4	3	2	1	0
Field	range_min_qp14[4:0]					range_max_qp14_h[2:0]		
Reset	0b00000					0b000		
Access Type	Write, Read					Write, Read		

BITFIELD	BITS	DESCRIPTION
range_min_qp14	7:3	Range Minimum QP14
range_max_qp14_h	2:0	Range Maximum QP14 High Bits [4:2]

DEC_PPS84_87_B3 (0xA2B7)

DSC Decoder PPS84_87 Register Byte 3

BIT	7	6	5	4	3	2	1	0
Field	range_max_qp14_l[1:0]		range_bpg_offset14[5:0]					

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Reset	0b00	0b000000
Access Type	Write, Read	Write, Read

BITFIELD	BITS	DESCRIPTION
range_max_qp14_l	7:6	Range Maximum QP14 Low Bits [1:0]
range_bpg_offset14	5:0	Range BPG Offset 14

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Revision History

REVISION NUMBER	REVISION DATE	DESCRIPTION	PAGES CHANGED
0	8/24	Initial release	—
1	11/24	**Global:** Moved sections for better understanding. Corrected grammatical, formats/styles, and spelling. **Data Sheet** Added GMSL Protocol Layer, Link Integrity Features, Even/Odd Pixel Splitter Mode, GMSL Reference Clock, Video Pipeline, High-Bandwidth Digital Content Protection, Video Stream Selection, Stream ID to I2C/UART Address Association, Video Timing Status, GMSL Pin Function Map, MFP Pin Equivalent Circuits, Functional Safety, Video PRBS, and Audio PRBS, Updated value of PCLK from 1230MHz to 1200MHz on bullet on first page. Added DP Tx Features Supported table to eDp Transmitter section. EC Table: DC ELECTRICAL CHARACTERISTICS / POWER SUPPLY CURRENTS (MAX96856RC); Updated “Dual GMSL3 inputs …, even/odd pixel mode… 4xHBR3 lanes” condition and in VDD row, updated Typ column from 495mA to 580mA and Max column from 1210mA to 1270mA. Added AC ELECTRICAL CHARACTERISTICS / REFERENCE CLOCK OUTPUT (RCLKOUT), AC ELECTRICAL CHARACTERISTICS / REFERENCE CLOCK REQUIREMENTS (EXTERNAL CLOCK INPUT ON X1/OSC, X2 UNCONNECTED) rows. Pin Descriptions: Updated pin 12 from MFP8/ERRB to ERRB and GPIO8 to ERRB in Figure and table, deleted description for GPIO8. Updated description for MFP12 and MFP22 function. Updated pin 11 from MFP7/LOCK to LOCK and GPIO7 to LOCK in Figure and table, deleted description for GPIO7. Updated bullets in Link Start-Up section. ERRB Pin Operation: Added additional information and figure. Added Pull-up resistors in STP and Coax Typical Applications and changed MFP pin 12 from MFP8/ERRB to ERRB and deleted GPIO8 from GPIO column.. **Register Map** Global: Added missing decode information. Opened bitfield WM_DETECT (0x88). Opened register blocks DSC_PPS PMX_RAM (0xA260-0xA2B7), RLMS_SIO DC (0x8CE). Opened registers in PSM_TASK_REGS block 0xE30 – 0xE4A. Opened register DP_CFG_1 (0x1A9). Added missing decode information.	—

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2	2/25	Added Minimum Blanking and Other Video Limits section Updating Ordering Information denotation of Future Product Electrical Characteristics: Added Power-Down Current and Sleep Current specifications Register: Updated Register Decode for DEV_ID16_BYTE_1 Exposed Registers: 0x2D3, 0xA00, 0x804, 0x805, 0x806, 0x807, 0x1404, 0x1405, 0x1406, 0x1407, 0x1504, 0x1505, 0x1506, 0x1507	

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